# UM-2: a 4+4 model of the Universe

Einstein-Lovelock gravity, the dirac16complex field, three generations of quarks and leptons with three neutrino flavours, their mixing and their flavour conversion, photons, gluons, and the $W^+$, $W^-$ and $Z$ bosons without a Higgs particle; every mass from the $x_0$ dependence of wave functions

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# 1 How to read this document

This document DEFINES the model universe UM-2: its space and coordinates, its metric (a power series of terms of decreasing magnitude), its fields exactly as the author lists them, every interaction among them, the mechanism that gives the particles their masses through the hidden coordinate $x_0$, and what follows for gravity, for the dirac16complex field, for the three neutrino flavours, for the dark sector and for the cosmology (CMB, CNB, BAO). It is self-contained: every equation it uses is written here, and every symbol of every display equation is defined right below that equation. The computations that produced its numbers are the scripts in `universe_model_2/code/` and their outputs in `universe_model_2/outputs/` (287 named checks, all passing); they are named only so that the results can be reproduced.

**Status labels.** Every statement carries one of these labels:

- **AUTHOR**: taken from the author's notebooks (the coordinates, the metric, the Lovelock weights, the gamma matrices, the mass parameter of dirac16complex);
- **RECORD**: taken from the Revision record of this repository (a choice made there and kept, such as the regular tip and the self-interaction of dirac16complex);
- **STANDARD**: textbook physics or a measured value used as an input;
- **ASSUMED**: a choice made for this model;
- **INVENTED**: a new phenomenological ingredient of this model (an interaction that the Standard Model does not contain);
- **DERIVED**: computed and verified in this work (the verification is stated with the result);
- **ESTIMATE**: an order-of-magnitude evaluation, not a precision computation;
- **OPEN**: not resolved in this version.

**Every equation is followed by its symbols.** Every display equation is followed by the list of all of its symbols ("Symbols in (FE-n)." or "Symbols in this equation."), each with its meaning, its unit, its value and its status where these apply. A FIELD EQUATION carries the tag (FE-n) and a term of the ACTION (an interaction term) the tag (IN-n); the numbers are those of the one inventory of UM-2 (FE-1 to FE-136, IN-1 to IN-18), shared by all documents of UM-2. Whenever a field equation or an interaction term is used, it is written out again, with its tag and its complete symbol list, so that a reader never has to search for a definition. One symbol has one meaning throughout (version 3 used several letters for several things; version 4 gives every meaning its own symbol).

**Units and conventions.** Natural units $\hbar = c = 1$ (the uncertainty argument keeps $\hbar$ visible). The coordinates are the author's $X = \{x_0, x_1, \ldots, x_7\}$, always with the hidden direction first: $x_0$ the hidden space-like direction, $x_1$, $x_2$, $x_3$ ordinary space, $x_4$ the time, $x_5$, $x_6$, $x_7$ three extra time-like directions. Capital indices $M, N, P, Q$ run over $x_0, \ldots, x_7$, frame indices $a, b, c, d$ over $0, \ldots, 7$, Greek indices $\mu, \nu$ over the observable coordinates $x_1, \ldots, x_4$. A dot is $\mathrm{d}/\mathrm{d}x_4$. $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$. There is no Higgs particle.

**Versions.** Version 1 (2026-10-17) contained one Standard-Model generation. Version 2 (2026-10-18) contains three full generations of quarks and leptons, anomaly free generation by generation, with three neutrino flavours, their masses, their mixing and their flavour conversion; the chiral tip $Z$ portal of version 1 was removed. Version 3 (2026-10-18) added the portal I13 (which gives the dark matter its relic abundance), definite benchmark strengths of the portals I11 and I12, the coupled Kohn-Sham solution in ten environments, the thermal history, the magnitudes of the terms of the metric series, $\sigma_8$, the Majorana alternative and the corrections of an independent review. Version 4 (2026-10-18), by the author's order of 2026-10-18 (15:42) to organise and present the solution better, contains the physics of version 3, re-presented: every field equation and every interaction term is written out with its tag and its complete symbol list wherever it is used; every interaction is explained (what it couples and why it is consistent with QCD and the electroweak model) and analysed (its strength and its bounds); the $x_0$ mass mechanism is presented first as the author's uncertainty argument and then through the exact $x_0$ eigenvalue equations; and

every symbol has one meaning. Version 4 quotes the refined outputs of the precision fixes made with the residual analysis of 2026-10-18 (52 further checks): the response of the $W$ eigenvalue to the I12 mean field of the dark matter is $-0.11125$ per unit amplitude (version 3: $-0.1126$, limited by the rounding of finite elements) and that of the photon exactly zero. The residual analysis also quantifies the places where two equations use the same quantity with different approximations and the neglected terms that are not small (stated where they occur and collected in the last section).

# 2 Summary

1. **Spacetime.** Eight dimensions with the author's coordinates $X = \{x_0, x_1, \dots, x_7\}$ and the frame signature $(+,+,+,+,-,-,-,-)$ in this order. The metric depends on $x_0$ only through $z = 6Hx_0$, periodically with the period $\pi$ in $z$, i.e. with the period $\pi/(6H)$ in $x_0$ ($H$ the author's constant, $H = 14.296$ TeV in UM-2).
2. **Gravity.** Einstein-Lovelock gravity in eight dimensions with the author's weights $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$ and the cosmological constant $\Lambda = -10H^2$, kept unchanged; it couples to every field. A block stabiliser (a vacuum energy of the hidden block, ASSUMED) gives a stable static block at $K_* = 12 + \frac{4}{3}\sqrt{51}$ on which four-dimensional gravity is attractive: its Einstein-Hilbert coefficient is $\mathcal{G}(K_*) = \frac{20}{3} + \sqrt{51} = 13.81 > 0$ (DERIVED).
3. **The deflating extra times.** In the primordial era the extra times $x_5$, $x_6$, $x_7$ (and $x_0$ with them) deflate at rates up to $|\mathcal{H}_f| = 2.63H$ into the stabilised block; then the block oscillates with the radion frequency $3.387H$ while 3-space keeps expanding (DERIVED, full nonlinear equations).
4. **dirac16complex** $\Psi$: a 16-component complex anticommuting spinor of the Clifford algebra of signature $(4,4)$ (Pin(4,4) irreducible; two inequivalent 8-dimensional irreducible representations of Spin(4,4)), coupled to gravity through the canonical spin connection, which is non-trivial, $\gamma^M \Omega_M = \frac{3}{2}(H/f)\gamma^{x_0} + (\dot{b}_4/b_4 - \frac{1}{2}\dot{a}_4)\gamma^{x_4}$, and to every other field through its self-interaction and the three portals. It splits into two Lorentz-invariant halves, $\Psi_A$ ($\mathcal{B} = -1$) and $\Psi_B$ ($\mathcal{B} = +1$), with $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$ the author's (Krein) operator.
5. **Standard-Model content (three generations, no Higgs particle).** Photon, gluons, $W^+$ (charge $+e$), $W^-$ (charge $-e$), $Z$ (neutral); three generations of quarks (up and down, charm and strange, top and bottom); three generations of leptons, the left-handed doublets $\ell_1$, $\ell_2$, $\ell_3$ (the three-flavour left-handed neutrinos $\nu_e$, $\nu_\mu$, $\nu_\tau$ with the left-handed electron, muon and tau), the right-handed singlets of the electron, muon and tau, and a right-handed neutrino in each generation. The gauge anomalies cancel generation by generation. Electroweak symmetry is broken at the regular tip of the hidden direction by a non-linear sigma field $\Sigma \in SU(2)$ that has no radial mode, hence no Higgs particle.
6. **Masses and mixing.** Every four-dimensional mass is the eigenvalue of an $x_0$ operator acting on a wave function periodic in $x_0$ with the period $\pi/(6H)$. Massless: the graviton, the photon, the gluons (flat $x_0$ profiles) and the fold bands of $\Psi$ (chiral zero modes). Massive: $W^\pm$ and $Z$ (profiles bent at the tip), the twelve fermions (a doublet zero mode and a singlet zero mode of each fermion mixed at the tip by a $3 \times 3$ tip matrix in each sector), the bulk band of $\Psi$ and every Kaluza-Klein mode (DERIVED spectra, fitted to the measured masses with order-one parameters). The doublet bulk mass is common to the three generations of a sector; then the mixing matrices are $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$ and $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$ (DERIVED, exact); with $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$ (ASSUMED) they are the tip matrices $\mathcal{K}_\nu$ and $\mathcal{K}_d$, whose values are the measured PMNS and CKM matrices (STANDARD inputs). The author's uncertainty argument explains the heavy masses; the light masses come from the tip of the hidden cell.
7. **Three neutrino flavours.** Dirac neutrinos with the masses 1.000, 8.712 and 50.140 meV (normal ordering, the NuFIT 6.0 splittings, the lightest mass ASSUMED) and the PMNS mixing of NuFIT 6.0 (Dirac phase 212°). Their flavour conversion is computed in vacuum, in matter (the MSW effect), in the Sun, at accelerators, in the early Universe (where it accompanies neutrino decoupling, $N_{\mathrm{eff}} = 3.0432$), in the cosmic neutrino background (three mass eigenstates) and in supernovae (the MSW resonances of the envelope, the fast flavour instability of the dense neutrino gas and its effect on the explosions).
8. **New interactions (INVENTED).** Besides the tip sigma field, three portals couple $\Psi$ to the Standard Model: the gauge-kinetic portal I12 couples $\bar{\Psi}\Psi$ to pairs of gauge bosons ($W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$, gluon pairs; benchmark strength $(38.2\ \mathrm{TeV})^{-3}$, unitarity safe); the weak vector contact portal I11 couples the whole field to the $B - L$ fermion currents (benchmark $G_{\mathrm{V}} = 4.38 \times 10^{-17}\ \mathrm{GeV}^{-2}$: it produces the massless fold bands by freeze-in, $\Delta N_{\mathrm{eff}} = 0.001$); and the $x_0$-chiral vector portal I13 couples the current $\mathrm{i}\bar{\Psi}\gamma^M P_b \Psi$ ($M \neq x_0$), which vanishes on the fold bands, to the baryon-number current ($G_{\mathrm{A}} = 1.278 \times 10^{-9}\ \mathrm{GeV}^{-2}$). Every new interaction is invariant under $SU(3)_c \times SU(2)_L \times U(1)_Y$.

9. **The dark sector.** The lightest odd states of the two halves $\Psi_A$ and $\Psi_B$, two degenerate Dirac fermions of mass $3.1022\,m_\Psi = 44.35$ TeV ($m_\Psi = H$), are electrically and weakly neutral, stable by the fold parity, cold, and together the dark matter. I12 alone cannot produce their relic abundance within perturbative unitarity; with I13 they annihilate in the s-wave into quark pairs and freeze out with $\Omega_{\rm DM}h^2_{100} = 0.060$ each, 0.120 together (Boltzmann equation with the thermal history of the model; DERIVED); their scattering on xenon nuclei, $4.0 \times 10^{-46}$ cm$^2$ per nucleon, is 0.16 of the present LZ limit. The self-interaction of $\Psi$ must be weak, $G_\lambda \leq 2.3 \times 10^{-15}$ GeV$^{-2}$ (ASSUMED), or the dark matter would thermalise the fold bands. No state of $\Psi$ is dark energy; the dark energy of UM-2 is a tuned four-dimensional cosmological term.
10. **Cosmology.** The four-dimensional effective theory of the stabilised block is exact general relativity, so the CMB, the CNB and the BAO follow from the standard Einstein-Boltzmann equations with the UM-2 particle content; $\sigma_8 = 0.812$. The metric is a power series of terms of decreasing magnitude: the order-one cosmological potentials are $2.4 \times 10^{-5}$ and smaller, the order-two terms $6 \times 10^{-10}$, the radion $10^{-34}$, and the Kaluza-Klein gravitons act only below $10^{-21}$ m.
11. **The coupled solution.** The Kohn-Sham equations of dirac16complex, coupled to gravity (the block response) and to the mean fields of every Standard-Model field (I11, I12, I13), are solved self-consistently from the cosmic mean density to neutron-star cores and through the early Universe: the back-reactions are small (the dark-matter level moves by at most $1.5 \times 10^{-6}$ of its mass, at the dark-matter freeze-out; the block by $\Delta \ln f \leq 1.1 \times 10^{-34}$, pushed by the hot plasma at the freeze-out, and by at most $10^{-48}$ from nucleosynthesis on), so the solution is the stabilised background with the $x_0$ spectra, the Kohn-Sham states and the linear cosmological perturbations, and the coupled corrections are computed.

# 3 Space, coordinates and the periodic hidden direction

## 3.1 The coordinates (AUTHOR)

The author's coordinates are $X = \{x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7\}$ (in the author's notebook `X = {x0[], x1[], ..., x7[]}`):

- $x_0$: the hidden space-like direction; the angle variable $z = 6Hx_0$ takes every real value;
- $x_1$, $x_2$, $x_3$: ordinary 3-space;
- $x_4$: the time;
- $x_5$, $x_6$, $x_7$: three extra time-like directions.

$H > 0$ is the author's constant (an inverse length; in UM-2 $H = 14.296$ TeV, fixed by the measured masses). Frame indices $a = 0, \ldots, 7$ are aligned with $x_0, \ldots, x_7$, and the frame metric is

$$\eta_{ab} = \mathrm{diag}(+1,\ +1,\ +1,\ +1,\ -1,\ -1,\ -1,\ -1)$$

**Symbols in this equation.**

- $\eta_{ab}$: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ ($\eta^{ab}$ the same matrix). Unit: dimensionless. Status: AUTHOR.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.

in the order $x_0, x_1, \ldots, x_7$: four space-like directions ($x_0$, $x_1$, $x_2$, $x_3$) and four time-like ones ($x_4$, $x_5$, $x_6$, $x_7$).

## 3.2 The hidden coordinate $y$, the fold and the tips (DERIVED)

The metric depends on $x_0$ only through $|\sin z|$ and $\cot^2 z$. On every interval of $z$ on which $\sin z \neq 0$ the hidden coordinate $y$ of the $y$ chart, the scaled coordinate $\xi$, the block scale $f$, the block variable $K$ and the two rates of the background are defined by

$$
\begin{aligned}
&y=\frac{\ln|\sin z|}{6H}, \qquad \mathrm{d}y=\cot z\,\mathrm{d}x_0, \qquad \xi=\frac{y}{L}\in[-1,0],\\
&\bar{g}_{MN}\,\mathrm{d}X^M\mathrm{d}X^N=\mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big)-\mathrm{d}x_4^2+f^2\Big[\mathrm{d}y^2-\mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big)\Big],\\
&f=\mathrm{e}^{-a_4}b_4^{1/2}, \qquad K=\frac{1}{f^2}, \qquad \mathcal{H}=\dot{a}_4, \qquad \mathcal{H}_f=\frac{\dot{f}}{f}=\frac{\dot{b}_4}{2b_4}-\dot{a}_4, \qquad \dot{K}=-2K\mathcal{H}_f,\\
&\sqrt{|\bar{g}|}=\mathrm{e}^{3a_4}f^4|\cot z|(\sin^2 z)^{1/4}=\mathrm{e}^{3a_4}f^4\,\mathrm{e}^{3Hy}\Big|\frac{\mathrm{d}y}{\mathrm{d}x_0}\Big|
\end{aligned}
\tag{FE-32}
$$

**Symbols in (FE-32).**

- $y$: The hidden coordinate of the $y$ chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- $z$: The angle variable of the hidden direction, $z=6Hx_0$; one period of $x_0$ is $0<z<\pi$. Unit: dimensionless. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4>0$. Unit: dimensionless. Status: AUTHOR.
- $K$: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot{a}_4$ (the Hubble rate of the observer; in the hot Universe

$\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $\dot{f}$: The time derivative $\mathrm{d}f/\mathrm{d}x_4$ of the block scale. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.

One period $0 < z < \pi$ of $x_0$ (length $\pi/(6H)$) consists of two halves, $0 < z < \pi/2$ and $\pi/2 < z < \pi$, joined at the **fold** $z = \pi/2$, where $\cos z = 0$ and $y = 0$. Each half ends at a **tip** ($z \to 0^+$ or $z \to \pi^-$, where $y \to -\infty$), at infinite proper distance from the fold. As in the Kohn-Sham record of this repository the tip is replaced by a regular cutoff at $y = -L$ (RECORD), with

$$|\sin z| = \mathrm{e}^{6Hy}, \qquad -L \le y \le 0, \qquad HL = 3$$

**Symbols in this equation.**

- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

so that $\xi = y/L$ runs from $-1$ (the tip) to 0 (the fold) on each half. The **fold reflection** $z \mapsto \pi - z$ (that is, $x_0 \mapsto \pi/(6H) - x_0$) maps the two halves onto each other; the modes of a field on one period are even or odd under it. Because $y$ is a function of $|\sin z|$,

$$y(z + \pi) = y(z), \qquad y(\pi - z) = y(z), \qquad z = 6Hx_0 :$$

**Symbols in this equation.**

- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

every function of $y(z)$ is periodic in $x_0$ with the period $\pi/(6H)$, which is the author's requirement on every wave function of the model. The proper length of one half, $fL$, is $8.926 \times 10^{-21}$ m on the stabilised block, and one coordinate period of $x_0$ is $\pi/(6H) = 7.227 \times 10^{-21}$ m (DERIVED).

# 4 The metric as a power series of terms of decreasing magnitude

The author asks for a metric general enough to model the CMB, the CNB and the BAO, perhaps expressible as a power series of terms of decreasing magnitude. UM-2 writes the metric as such a series (DERIVED for the sizes of the terms):

$$g_{MN} = \bar{g}_{MN} + \epsilon\, h^{(1)}_{MN} + \epsilon^2\, h^{(2)}_{MN} + \ldots, \qquad h^{(1)}_{MN}(X^\mu, x_0) = \sum_{n\geq 0} h^{(1,n)}_{MN}(X^\mu)\, F_n(x_0), \qquad \epsilon \sim 10^{-5} \qquad \text{(FE-87)}$$

**Symbols in (FE-87).**

- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- $\epsilon$: The bookkeeping parameter of the power series of the metric: the term of order $q$ is multiplied by $\epsilon^q$ (the size of the first-order potentials). Unit: dimensionless. Value: $\epsilon \sim 10^{-5}$. Status: DERIVED.
- $h^{(1)}_{MN}$, $h^{(2)}_{MN}$: The first- and second-order terms of the power series of the metric. Unit: dimensionless. Status: DERIVED.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $h^{(1,n)}_{MN}$: The four-dimensional coefficient of the $n$-th $x_0$ profile $F_n$ in the first-order term ($n = 0$: the flat zero mode, the perturbations of four-dimensional gravity; $n \geq 1$: the Kaluza-Klein gravitons). Unit: dimensionless. Status: DERIVED.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.

The bookkeeping parameter $\epsilon \sim 10^{-5}$ is the size of the CMB temperature anisotropy (STANDARD); the first-order term is expanded in the $x_0$ profiles $F_n$ of the hidden direction (the flat profile $F_0 = 1$ and the Kaluza-Klein profiles $F_n$, $n \geq 1$, below).

## 4.1 Order 0: the corrected primordial metric (AUTHOR coordinates, DERIVED properties)

In the author's chart, with two general functions $a_4(x_4)$ and $b_4(x_4) > 0$ of the time, the order-0 metric is

$$\begin{aligned}\bar{g} = \mathrm{diag}\Big(&\mathrm{e}^{-2a_4} b_4 \cot^2 z,\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ \mathrm{e}^{2a_4},\ -1,\\ &-\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4} b_4 (\sin^2 z)^{1/6}\Big), \qquad z = 6Hx_0\end{aligned} \qquad \text{(FE-31)}$$

**Symbols in (FE-31).**

- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

where $(\sin^2 z)^{1/6}$ is the positive branch. This is the metric of the Revision record of 2026-10-16 (version 2), whose sources are all diagonal. The notebook metric

$$g_\mathrm{nb} = \mathrm{diag}\big(\cot^2 z,\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\ \mathrm{e}^{2a_4}(\sin^2 z)^{1/6},\\ -1,\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6},\ -\mathrm{e}^{-2a_4}(\sin^2 z)^{1/6}\big),$$

**Symbols in this equation.**

- $g_\mathrm{nb}$: The metric of the author's notebook (before the author's order of 2026-10-14 added the factor $b_4$ to the extra-time components), written as the diagonal matrix of its components in the order $x_0, x_1, \ldots, x_7$; it is quoted for comparison only, the model uses the corrected metric $\bar{g}_{MN}$. Unit: dimensionless. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.

to whose extra-time components the author's order of 2026-10-14 added the factor $b_4$, differs from $\bar{g}_{MN}$ in the hidden component and in the warp of 3-space. In the $y$ chart (written out again with its definitions):

$$\begin{aligned}
&y = \frac{\ln|\sin z|}{6H}, \qquad \mathrm{d}y = \cot z\,\mathrm{d}x_0, \qquad \xi = \frac{y}{L} \in [-1, 0],\\
&\bar{g}_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{e}^{2a_4}\big(\mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2\big) - \mathrm{d}x_4^2 + f^2\Big[\mathrm{d}y^2 - \mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2 + \mathrm{d}x_6^2 + \mathrm{d}x_7^2\big)\Big],\\
&f = \mathrm{e}^{-a_4}b_4^{1/2}, \qquad K = \frac{1}{f^2}, \qquad \mathcal{H} = \dot{a}_4, \qquad \mathcal{H}_f = \frac{\dot{f}}{f} = \frac{\dot{b}_4}{2b_4} - \dot{a}_4, \qquad \dot{K} = -2K\mathcal{H}_f,\\
&\sqrt{|\bar{g}|} = \mathrm{e}^{3a_4}f^4|\cot z|(\sin^2 z)^{1/4} = \mathrm{e}^{3a_4}f^4\,\mathrm{e}^{3Hy}\Big|\frac{\mathrm{d}y}{\mathrm{d}x_0}\Big|
\end{aligned} \tag{FE-32}$$

**Symbols in (FE-32).**

- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\bar{g}_{MN}$: The order-0 (background) metric of the power series: the corrected primordial metric in the author's coordinates, determined by the two functions $a_4(x_4)$ and $b_4(x_4)$; $|\bar{g}|$ is the absolute value of its determinant. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2} T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $\dot{f}$: The time derivative $\mathrm{d}f/\mathrm{d}x_4$ of the block scale. Unit: GeV. Status: DERIVED.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $\dot{K}$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.

The block $\{x_0, x_5, x_6, x_7\}$ is $f^2$ times a four-dimensional space of constant curvature $-H^2$ and signature $(1, 3)$; $f$ is its common scale factor ("the block"), $\mathcal{H}$ is the observer's Hubble rate, and $\mathcal{H}_f < 0$ means that the extra times and $x_0$ deflate. The vielbein of the order-0 metric is

$$\mathbf{e}^0{}_{x_0} = f \cot z, \qquad \mathbf{e}^1{}_{x_1} = \mathbf{e}^2{}_{x_2} = \mathbf{e}^3{}_{x_3} = \mathrm{e}^{a_4}, \qquad \mathbf{e}^4{}_{x_4} = 1, \qquad \mathbf{e}^5{}_{x_5} = \mathbf{e}^6{}_{x_6} = \mathbf{e}^7{}_{x_7} = f\,(\sin^2 z)^{1/12}$$

**Symbols in this equation.**

- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M \mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

(all other components zero). **Curvature (DERIVED; sympy, identical in both charts, 25 checks).** The Riemann tensor is pair-diagonal, $R^{ab}{}_{cd} = 0$ unless $\{a,b\} = \{c,d\}$, and every sectional curvature is one of

$$R^{x_1x_2}{}_{x_1x_2} = \mathcal{H}^2, \qquad R^{x_1x_4}{}_{x_1x_4} = \mathcal{H}^2 + \dot{\mathcal{H}}, \qquad R^{x_1x_0}{}_{x_1x_0} = \mathcal{H}\mathcal{H}_f, \qquad R^{x_4x_0}{}_{x_4x_0} = \mathcal{H}_f^2 + \dot{\mathcal{H}}_f,$$
$$R^{x_0x_5}{}_{x_0x_5} = \mathcal{H}_f^2 - H^2K; \tag{FE-33}$$

the same for every pair of the same kind; all other $R^{ab}{}_{cd}$ with $\{a,b\} \neq \{c,d\}$ vanish

**Symbols in (FE-33).**

- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a,b)$. Unit: $1/\text{length}^2$. Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: $\text{GeV}^2$. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: $\text{GeV}^2$. Status: DERIVED.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.

None depends on $x_0$; the curvature is the same on both halves of every period.

## 4.2 Order 1: the observable perturbations and the hidden tower

The $n = 0$ term of the first-order metric has the flat profile $F_0 = 1$; in the Newtonian gauge its line element is

$$g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N = \mathrm{e}^{2a_4}\Big[\big(1-2\Psi_{\mathrm{N}}\big)\big(\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2\big) + \mathrm{d}\vec{x}\cdot h^{\mathrm{TT}}\cdot\mathrm{d}\vec{x}\Big] - \big(1+2\Phi\big)\,\mathrm{d}x_4^2 + f^2\big(1+2\sigma_{\mathrm{r}}\big)\Big[\mathrm{d}y^2 - \mathrm{e}^{2Hy}\big(\mathrm{d}x_5^2+\mathrm{d}x_6^2+\mathrm{d}x_7^2\big)\Big] \tag{FE-88}$$

**Symbols in (FE-88).**

- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1-2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4\times10^{-5}$. Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $\vec{x}$: The position in ordinary 3-space, $\vec{x}=(x_1,x_2,x_3)$; $\mathrm{d}\vec{x}^{\,2}=\mathrm{d}x_1^2+\mathrm{d}x_2^2+\mathrm{d}x_3^2$. Unit: length. Status: STANDARD.
- $h^\mathrm{TT}$: The transverse-traceless (tensor) part of the first-order metric of 3-space (gravitational waves). Unit: dimensionless. Value: an input (tensor-to-scalar ratio below 0.036). Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1+2\Phi$). Unit: dimensionless. Value: at most $2.4\times10^{-5}$. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_\mathrm{r}$: The radion perturbation: the first-order change of the block scale, $f^2\to f^2(1+2\sigma_\mathrm{r})$. Unit: dimensionless. Value: at most $1.1\times10^{-34}$ (static response). Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le0$. Unit: length. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.

with the lapse potential $\Phi$, the curvature potential $\Psi_\mathrm{N}$ (the factor $\epsilon$ is absorbed into them), the transverse-traceless tensor $h^\mathrm{TT}$ and the radion $\sigma_\mathrm{r}$ (the relative fluctuation of the block size). The scalar and tensor sectors carry the CMB, the CNB and the BAO; the radion is massive (48.42 TeV, the section on gravity) and does not propagate at cosmological distances. The terms $n\ge1$ are Kaluza-Klein modes with masses above the gap $3H/(2f_*)=99.5$ TeV. Order 2 contains the products of first-order terms and the back-reaction of $x_0$-localised sources; its magnitude is estimated below, its detailed form is not computed (OPEN).

## 4.3 The magnitudes of the terms (DERIVED; the order-2 sizes are ESTIMATES and the tensor bound is STANDARD)

The solution of the coupled equations fixes the size of every term (4 checks):

**Order 0**, $\bar{g}_{MN}$: magnitude 1.

**Order 1,** $n = 0$ (the Newtonian potentials). On cosmological scales the size is the root-mean-square Weyl potential per logarithmic interval of the comoving wavenumber $k$ today, from the linear solution (the section on cosmology):

$$\Phi_{\rm rms}(k) = \Big[\frac{k^3 P_{\Phi_{\rm W}}(k)}{2\pi^2}\Big]^{1/2}, \qquad \Phi_{\rm W} = \frac{\Phi + \Psi_{\rm N}}{2}, \qquad \Phi_{\rm W} = -\frac{3}{2}\,\Omega_{\rm m}\mathcal{H}_0^2\,\frac{\delta_{\rm m}}{{\rm e}^{a_4}k^2} \quad \text{(inside the horizon)},$$
$$\Phi_{\rm W} = -\frac{3}{5}\,\mathcal{R}\,{\rm e}^{-a_4}D_+ \quad \text{(largest scales)}, \qquad D_+ = \frac{5}{2}\,\Omega_{\rm m}\,\frac{\mathcal{H}}{\mathcal{H}_0}\int_{-\infty}^{a_4}\Big(\frac{\mathcal{H}_0}{\mathcal{H}}\Big)^3 {\rm e}^{-2\tilde{a}_4}\,{\rm d}\tilde{a}_4, \qquad \text{(FE-98)}$$
$$\frac{\mathcal{H}}{\mathcal{H}_0} = \big(\Omega_{\rm m}{\rm e}^{-3a_4} + 1 - \Omega_{\rm m}\big)^{1/2}$$

**Symbols in (FE-98).**

- $\Phi_{\rm rms}(k)$: The root-mean-square amplitude of the Weyl potential per logarithmic interval of $k$. Unit: dimensionless. Value: $2.42 \times 10^{-5}$ at $k = 10^{-4}$ Mpc$^{-1}$. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum ${\rm e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $P_{\Phi_{\rm W}}(k)$: The power spectrum of the Weyl potential. Unit: length$^3$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Phi_{\rm W}$: The Weyl potential, $\Phi_{\rm W} = (\Phi + \Psi_{\rm N})/2$ (it deflects light). Unit: dimensionless. Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\Psi_{\rm N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_{\rm N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\Omega_{\rm m}$: The matter density today in units of the critical density. Unit: dimensionless. Value: 0.3150. Status: STANDARD.
- $\mathcal{H}_0$: The Hubble rate today (the Hubble constant). Unit: GeV (km/s/Mpc). Value: 67.38 km/s/Mpc. Status: DERIVED.
- $\delta_{\rm m}$: The density contrast of the total matter (dark matter and baryons). Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; ${\rm e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is ${\rm e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{R}$: The primordial comoving curvature perturbation (conserved outside the horizon), counted positive where the density is above its mean, so that the Newtonian potentials $\Phi$, $\Psi_{\rm N}$ and $\Phi_{\rm W}$ are negative there (on the largest scales in the matter era $\Phi_{\rm W} = -\frac{3}{5}\mathcal{R}\,{\rm e}^{-a_4}D_+$). Unit: dimensionless. Status: STANDARD.
- $D_+$: The linear growth factor of the matter perturbations, normalised as $D_+ = {\rm e}^{a_4}$ in the matter era. Unit: dimensionless. Value: $D_+{\rm e}^{-a_4} = 0.7878$ today. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- $\tilde{a}_4$: An earlier value of $a_4$ (the integration variable of the growth integral). Unit: dimensionless. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of $X$, ${\rm d}X/{\rm d}Y$ the derivative of $X$ with respect to $Y$, ${\rm d}^nX$ the volume element of $n$ variables. Status: STANDARD.

$\Phi_{\rm rms}$ is $2.42 \times 10^{-5}$, $2.28 \times 10^{-5}$, $1.51 \times 10^{-5}$, $4.12 \times 10^{-6}$, $1.74 \times 10^{-6}$, $3.55 \times 10^{-7}$ and $5.03 \times 10^{-8}$ at $k = 10^{-4}$, $10^{-3}$, $10^{-2}$, 0.05, 0.1, 0.3 and 1 Mpc$^{-1}$. It agrees with the Poisson form ($\Omega_{\rm m} = 0.3150$, $\mathcal{H}_0 = 67.38$ km/s/Mpc) applied to the linear matter density to $1.5 \times 10^{-6}$ for $k \geq 10^{-2}$ Mpc$^{-1}$, and on the largest scales with the matter-era form ($\mathcal{R}$ from $A_{\rm s}$ and $n_{\rm s}$, $k_0 = 0.05$ Mpc$^{-1}$) to $4.4 \times 10^{-5}$, where $D_+{\rm e}^{-a_4} = 0.7878$ today is the growth suppression by $\Lambda_4$. Bound objects of mass $M_1$ and radius $r$ have the surface potential

$$\Phi = \frac{G_{\mathrm{N}} M_1}{r} :$$

**Symbols in this equation.**

- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $G_{\mathrm{N}}$: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $M_1,\ M_2$: The masses of the two static sources in the potential $V_{\mathrm{N}}$. Unit: GeV. Status: STANDARD.
- $r$: A distance: between the two masses in $V_{\mathrm{N}}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.

the Earth $7.0 \times 10^{-10}$, the Sun $2.1 \times 10^{-6}$, a white dwarf (0.6 solar masses, 8700 km) $1.0 \times 10^{-4}$, a neutron star (1.4 solar masses, 12 km) 0.17; in galaxies and clusters $\Phi$ is of the order of the squared velocity dispersion (in units of the speed of light): $5.4 \times 10^{-7}$ (220 km/s) and $1.1 \times 10^{-5}$ (1000 km/s). The series converges quickly everywhere except near neutron stars and black holes, where the full nonlinear equations apply.

**Order 1,** $n \geq 1$ (the Kaluza-Klein gravitons). The graviton has no tip term ($\beta = 0$), so its even levels are

$$F_n(\xi) = \mathrm{e}^{-3HL\xi/2}\Big(\cos\varpi_n\xi + \frac{3HL}{2\varpi_n}\sin\varpi_n\xi\Big), \qquad \varpi_n = \pi,\ 2\pi,\ 3\pi,\ \ldots,$$
$$\tilde{m}_n^2 = \frac{9H^2L^2}{4} + \varpi_n^2, \qquad m_n^{\mathrm{KK}} = \frac{\tilde{m}_n}{fL}, \qquad N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi = \frac{1}{2}\Big(1 + \frac{9H^2L^2}{4\varpi_n^2}\Big), \qquad N_0 = \frac{1 - \mathrm{e}^{-3HL}}{3HL} \tag{FE-53}$$

**Symbols in (FE-53).**

- $F(\xi),\ F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\varpi,\ \varpi_n$: The wave number of an oscillating $x_0$ profile on $-1 \leq \xi \leq 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); $\varpi_n$ that of the $n$-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde{m},\ \tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- $m_n^{\mathrm{KK}},\ m_{\mathrm{KK}}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $N_n,\ N_0$: The normalisation integral of the profile $F_n$ of the $n$-th Kaluza-Klein level, $N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi$; $N_0$ that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1 - \mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.

with $\tilde{m}_n^2 = 30.1196,\ 59.7284,\ 109.0764$ for $n = 1, 2, 3$ (finite elements agree to $3.9 \times 10^{-9}$). Two static masses $M_1$ and $M_2$ with the normalised $x_0$ profiles $\varrho_1$ and $\varrho_2$ attract at the distance $r$ with

$$V_{\mathrm{N}}(r) = -\frac{G_{\mathrm{N}} M_1 M_2}{r}\Big[1 + \sum_{n\geq 1} \mathcal{A}_n\, \mathrm{e}^{-m_n^{\mathrm{KK}} r}\Big], \qquad \mathcal{A}_n = \frac{N_0}{N_n}\Big(\int_{-1}^{0} \varrho_1 F_n \,\mathrm{d}\xi\Big)\Big(\int_{-1}^{0} \varrho_2 F_n \,\mathrm{d}\xi\Big) \tag{FE-54}$$

**Symbols in (FE-54).**

- $V_{\mathrm{N}}(r)$: The potential energy of two static masses at the distance $r$ (Newton potential with the Kaluza-Klein corrections). Unit: GeV. Status: DERIVED.
- $r$: A distance: between the two masses in $V_{\mathrm{N}}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.
- $G_{\mathrm{N}}$: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $M_1,\ M_2$: The masses of the two static sources in the potential $V_{\mathrm{N}}$. Unit: GeV. Status: STANDARD.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $\mathcal{A}_n$: The relative strength of the Yukawa force of the $n$-th Kaluza-Klein graviton between two sources with the profiles $\varrho_1$, $\varrho_2$. Unit: dimensionless. Value: 0.0904, 0.0136, 0.0057 ($n = 1, 2, 3$, two quark-doublet profiles). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $m_n^{\mathrm{KK}},\ m_{\mathrm{KK}}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $N_n,\ N_0$: The normalisation integral of the profile $F_n$ of the $n$-th Kaluza-Klein level, $N_n = \int_{-1}^{0} \mathrm{e}^{3HL\xi} F_n^2 \,\mathrm{d}\xi$; $N_0$ that of the flat zero mode. Unit: dimensionless. Value: $N_0 = (1 - \mathrm{e}^{-3HL})/(3HL) = 0.1111$. Status: DERIVED.
- $\varrho_1,\ \varrho_2$: The $x_0$ profiles of the two sources of $V_{\mathrm{N}}$ as probability densities in $\xi$ ($\int_{-1}^{0} \varrho_1 \,\mathrm{d}\xi = 1$): $L$ times the profile $\varrho_f$ of the zero mode of each source. Unit: dimensionless. Status: DERIVED.
- $F(\xi),\ F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

For two masses with the same profile the strength is $(N_0/N_n)(\int \varrho_1 F_n \,\mathrm{d}\xi)^2$: $\mathcal{A}_n = 0.0728,\ 0.1469,\ 0.1809$ for two sources at the fold, 0.0904, 0.0136, 0.0057 for the quark-doublet zero-mode profile and 0.0208, 0.0197, 0.0105 for the lepton doublet; a source with the flat profile of the gauge zero modes (the gluon field energy, which makes up most of the nucleon mass) excites none ($\int \mathrm{e}^{3HL\xi} F_n \,\mathrm{d}\xi = 0$, the orthogonality of $F_n$ to the flat profile, verified to $2 \times 10^{-10}$). The first level is the Kaluza-Klein scale, $m_1^{\mathrm{KK}} = m_{\mathrm{KK}} = 121.32$ TeV; its range $1/m_{\mathrm{KK}} = 1.63 \times 10^{-21}$ m makes the Yukawa factor $\mathrm{e}^{-3.1\times 10^{16}}$ at 50 micrometres: Newton's law is exact at every measurable distance.

**The radion** $\sigma_{\mathrm{r}}$: its static response to matter is at most $\Delta \ln f = 1.1 \times 10^{-34}$ (at the dark-matter freeze-out, from the hidden pressure of the hot plasma; $10^{-62}$ for the Earth; the section on the coupled solution); its range is $4.08 \times 10^{-21}$ m; the primordial oscillation decays (lifetime $0.43\,\mathrm{s}/C_{\mathrm{r}}$, the last section).

**Order 2** (ESTIMATE: the squares of the first-order terms):

$$\epsilon^2 h^{(2)} \sim \Phi_{\mathrm{rms}}^2 = 5.8 \times 10^{-10} \quad \big(k = 10^{-4}\ \mathrm{Mpc}^{-1}\big)$$

**Symbols in this equation.**

- $\epsilon$: The bookkeeping parameter of the power series of the metric: the term of order $q$ is multiplied by $\epsilon^q$ (the size of the first-order potentials). Unit: dimensionless. Value: $\epsilon \sim 10^{-5}$. Status: DERIVED.

- $h^{(1)}_{MN}$, $h^{(2)}_{MN}$: The first- and second-order terms of the power series of the metric. Unit: dimensionless. Status: DERIVED.
- $\Phi_{\mathrm{rms}}(k)$: The root-mean-square amplitude of the Weyl potential per logarithmic interval of $k$. Unit: dimensionless. Value: $2.42 \times 10^{-5}$ at $k = 10^{-4}$ Mpc$^{-1}$. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

on the largest cosmological scales (the ratio of order 2 to order 1 is $2.4 \times 10^{-5}$, far below the $10^{-3}$ accuracy at which the first-order cosmology is used; DERIVED in the residual analysis), $4.5 \times 10^{-12}$ at the Sun, 0.03 at a neutron star.

**Tensor modes** $h^{\mathrm{TT}}$: their primordial amplitude is not predicted (the primordial spectrum is an input of UM-2); the bound on the tensor-to-scalar ratio (STANDARD, from memory) gives

$$h^{\mathrm{TT}}_{\mathrm{rms}} < \left(r_{\mathrm{ts}} A_{\mathrm{s}}\right)^{1/2} = 8.7 \times 10^{-6}, \qquad r_{\mathrm{ts}} < 0.036.$$

**Symbols in this equation.**

- $h^{\mathrm{TT}}_{\mathrm{rms}}$: The root-mean-square amplitude of the primordial transverse-traceless (tensor) metric perturbation per logarithmic interval of the wavenumber, the size of the gravitational-wave term of the first-order metric. Unit: dimensionless. Value: below $8.7 \times 10^{-6}$. Status: STANDARD.
- $r_{\mathrm{ts}}$: The tensor-to-scalar ratio: the power of the primordial tensor perturbations divided by the power $A_{\mathrm{s}}$ of the primordial curvature perturbations at the pivot scale. Unit: dimensionless. Value: $r_{\mathrm{ts}} < 0.036$ (95 %). Status: STANDARD.
- $A_{\mathrm{s}}$: The amplitude of the primordial spectrum at $k_0$. Unit: dimensionless. Value: $2.099 \times 10^{-9}$ ($\ln(10^{10} A_{\mathrm{s}}) = 3.044$, Planck 2018). Status: STANDARD.

On cosmological scales the terms therefore decrease as 1, $2.4 \times 10^{-5}$ (order 1), $5.8 \times 10^{-10}$ (order 2), $10^{-34}$ (radion) and 0 (Kaluza-Klein gravitons at any measurable distance): the $n = 0$ term of order 1 carries the CMB, the CNB and the BAO.

# 5 The fields of UM-2 and the action

The author's order of 2026-10-18 lists the coupled fields: gravity (Einstein-Lovelock gravity, which couples to all fields), the dirac16complex quantized fermion spinor field (which couples to all other fields), the three-flavour left-handed neutrinos, the left-handed electrons, the right-handed electron singlet, three quark generations, gluons, photons and the three weak gauge bosons $W^+$, $W^-$ and $Z$, and no Higgs particle. UM-2 realises this list as follows (the other two lepton generations and their right-handed partners, the right-handed neutrinos, the tip sigma field and the block stabiliser are those of version 3):

| the author's field | the field of UM-2 | content | couples to | four-dimensional masses |
|---|---|---|---|---|
| Einstein-Lovelock gravity | the metric $g_{MN}$ (vielbein $\mathbf{e}^a{}_M$) | the graviton, the Kaluza-Klein gravitons, the radion | every field, through $\sqrt{\lvert g\rvert}$, $g^{MN}$ and the spin connection in every action; the stabiliser | graviton 0; Kaluza-Klein gravitons from 121.32 TeV; radion 48.42 TeV |
| dirac16complex | $\Psi$, 16 complex Grassmann components, halves $\Psi_A$ and $\Psi_B$ | fold bands (massless), bulk band (dark matter) | gravity (spin connection), itself ($\lambda$), every fermion (I11), the quarks (I13), gluons, $W$ and $B$ (I12) | fold bands 0; dark matter 44.35 TeV |

| the author's field | the field of UM-2 | content | couples to | four-dimensional masses |
|---|---|---|---|---|
| three-flavour left-handed neutrinos | the upper components of the lepton doublets $\ell_i$ | $\nu_e$, $\nu_\mu$, $\nu_\tau$ (flavour states) $= U(\nu_1, \nu_2, \nu_3)$ (mass states) | $W^\pm$, $Z$, gravity, $\Psi$ (I11), $\Sigma$ at the tip | 1.000, 8.712, 50.140 meV |
| left-handed electrons | the lower component of $\ell_1$ | the left-handed electron, charge $-1$ | photon, $W^\pm$, $Z$, gravity, $\Psi$ (I11), $\Sigma$ | 0.51100 MeV (with $e_R$) |
| right-handed electron singlet | $e_{R1}$ | the right-handed electron, charge $-1$ | photon, $Z$, gravity, $\Psi$ (I11), $\Sigma$ | (the electron) |
| the other two lepton generations and their right-handed partners | $\ell_2$, $\ell_3$, $e_{R2}$, $e_{R3}$ | muon and tau, left- and right-handed | as the electron | 105.658 MeV, 1776.93 MeV |
| right-handed neutrinos | $\nu_{R1}$, $\nu_{R2}$, $\nu_{R3}$ | gauge singlets | gravity, $\Psi$ (I11), $\Sigma$ at the tip | (Dirac partners of the neutrinos) |
| three quark generations | $Q_i$, $u_{Ri}$, $d_{Ri}$ | up, down, charm, strange, top, bottom, three colours each | gluons, photon, $W^\pm$, $Z$, gravity, $\Psi$ (I11, I13), $\Sigma$ | from 2.16 MeV (up) to 172.56 GeV (top) |
| gluons | $G^A_M$, $A = 1, \ldots, 8$ | eight massless gluons | quarks, themselves, $\Psi$ (I12), gravity | 0 |
| photons | $A_M$ (a combination of $W^3_M$ and $B_M$) | the photon | charged fermions, $W^\pm$, $\Psi$ (I12), gravity | 0 |
| $W^+$, $W^-$, $Z$ | $W^\pm_M$ (from $W^1_M$, $W^2_M$), $Z_M$ (from $W^3_M$, $B_M$) | $W^+$ charge $+e$, $W^-$ charge $-e$, $Z$ neutral | fermions, photon, each other, $\Sigma$ at the tip, $\Psi$ (I12), gravity | 80.3625, 80.3625, 91.1879 GeV |
| no Higgs particle | the tip sigma field $\Sigma \in SU(2)$, without a radial mode | three would-be Goldstone modes, the longitudinal $W^+$, $W^-$, $Z$ | $W$, $B$, the fermions (tip Yukawa terms), gravity | no particle |
| (version 3) | the block stabiliser $\rho_{\mathrm{stab}}(f)$ | a vacuum energy of the hidden block | gravity | none |

The action of UM-2 is the sum of all terms (IN-1); varying it gives every field equation of the model:

$$S = S_{\mathrm{EL}} + S_{\mathrm{stab}} + S_\Psi + S_{\mathrm{gauge}} + S_{\mathrm{f}} + S_{\mathrm{tip}} + S_{\mathrm{I11}} + S_{\mathrm{I12}} + S_{\mathrm{I13}} \tag{IN-1}$$

**Symbols in (IN-1).**

- $S$: The action of UM-2 (the integral of the Lagrangian density over spacetime); its variation with respect to every field gives the field equations. Unit: dimensionless ($\hbar = 1$). Status: ASSUMED.
- $S_{\mathrm{EL}}$: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- $S_{\mathrm{stab}}$: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- $S_\Psi$: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- $S_{\mathrm{gauge}}$: The action of the gauge fields of SU(3), SU(2) and $\mathrm{U}(1)_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- $S_{\mathrm{f}}$: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- $S_{\mathrm{tip}}$: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.

- $S_{\mathrm{I11}}$: The action of the portal I11 (dirac16complex vector current times the $B-L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- $S_{\mathrm{I12}}$: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- $S_{\mathrm{I13}}$: The action of the portal I13 (the $x_0$-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.

The following sections write out every term and the field equation of every field; the section on the interactions then writes out, explains and analyses every interaction.

# 6 Gravity and the block stabiliser

## 6.1 Einstein-Lovelock gravity, which couples to all fields (AUTHOR weights)

The gravitational action is the Einstein-Lovelock action with the author's weights:

$$\begin{aligned} S_{\mathrm{EL}} &= \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)}-2\Lambda\Big), \qquad \kappa = 8\pi G_8, \\ \alpha_{(p)} &= w_{(p)}\,H^{2-2p}, \qquad w_{(1)}=1, \quad w_{(2)}=\frac{1}{32}, \quad w_{(3)}=\frac{1}{1152}, \qquad \Lambda=-10H^2 \end{aligned} \tag{IN-2}$$

**Symbols in (IN-2).**

- $S_{\mathrm{EL}}$: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $p$: The Lovelock order $p=1,2,3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order $p$, $\alpha_{(p)}=w_{(p)}H^{2-2p}$ ($\alpha_{(1)}=1$). Unit: $\mathrm{GeV}^{2-2p}$. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)}=R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda=-10H^2$. Unit: $\mathrm{GeV}^2$. Value: $-10H^2$. Status: AUTHOR.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_8$: The eight-dimensional Newton constant ($\kappa=8\pi G_8$). Unit: $\mathrm{GeV}^{-6}$. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)}=1$, $w_{(2)}=1/32$, $w_{(3)}=1/1152$. Unit: dimensionless. Value: $(1,1/32,1/1152)$. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

with the Lovelock densities (generalised Kronecker delta: the determinant of the matrix of ordinary Kronecker deltas)

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2-4R_{MN}R^{MN}+R_{MNPQ}R^{MNPQ} \end{aligned} \tag{IN-3}$$

**Symbols in (IN-3).**

- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: 1/length$^{2p}$. Status: STANDARD.
- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: 1/length$^2$. Status: STANDARD.

$\mathcal{L}_{(3)}$ is the cubic Lovelock density. (The author's notebook writes $\Lambda = 10$ with the opposite sign convention for $\Lambda$.) UM-2 keeps these values. Gravity couples to every field through the volume element $\sqrt{|g|}$, the inverse metric in every kinetic term and the canonical spin connection of every spinor (IN-7 in the section on dirac16complex). Varying the action with respect to the metric gives the Einstein-Lovelock equations,

$$\begin{aligned} &\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{M}{}_{N} + \Lambda\, \delta^M_N = \kappa\, \mathcal{T}^M{}_N, \\ &\mathcal{E}_{(p)}{}^{M}{}_{N} = -\frac{1}{2^{p+1}}\, \delta^{M M_1 N_1 \cdots M_p N_p}_{N P_1 Q_1 \cdots P_p Q_p}\, R^{P_1 Q_1}{}_{M_1 N_1} \cdots R^{P_p Q_p}{}_{M_p N_p} \end{aligned} \tag{FE-1}$$

**Symbols in (FE-1).**

- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order $p$, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: GeV$^{2-2p}$. Status: AUTHOR.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^M{}_N - \frac{1}{2}\delta^M_N R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV$^2$. Value: $-10H^2$. Status: AUTHOR.
- $\delta$: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV$^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV$^{-2}$. Status: AUTHOR.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV$^8$. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- ${R^{MN}}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = {R^P}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; ${R^{ab}}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: $1/\text{length}^2$. Status: STANDARD.

$\mathcal{E}_{(1)}$ being the Einstein tensor, with the energy-momentum tensor of all matter defined and decomposed as

$$\begin{aligned} &\Delta S_\mathrm{m} = -\frac{1}{2}\int \mathrm{d}^8X\,\sqrt{|g|}\,\mathcal{T}_{MN}\,\Delta g^{MN}, \qquad \rho = -{\mathcal{T}^{x_4}}_{x_4}, \qquad p_M = {\mathcal{T}^M}_M \quad (M \neq x_4, \text{ no sum}), \\ &{\mathcal{T}^M}_N = \mathrm{diag}\big(p_0,\ p_3,\ p_3,\ p_3,\ -\rho,\ p_5,\ p_5,\ p_5\big) \quad \text{(a diagonal source)}, \qquad \bar{p} = \frac{p_0 + 3p_5}{4} \end{aligned} \tag{FE-2}$$

**Symbols in (FE-2).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_\mathrm{eff}$ an addition to $N_\mathrm{eff}$, $\Delta S_\mathrm{m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $S_\mathrm{m}$: The matter action: all terms of the action except $S_\mathrm{EL}$ (the stabiliser included). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- ${\mathcal{T}^M}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -{\mathcal{T}^{x_4}}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $p$: A pressure: $p_M = {\mathcal{T}^M}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = {\mathcal{T}^{x_0}}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = {\mathcal{T}^{x_1}}_{x_1} = {\mathcal{T}^{x_2}}_{x_2} = {\mathcal{T}^{x_3}}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = {\mathcal{T}^{x_5}}_{x_5} = {\mathcal{T}^{x_6}}_{x_6} = {\mathcal{T}^{x_7}}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

The total source is the sum of the energy-momentum tensors of every term of the action (each written out with its field below):

$$\mathcal{T}^{M}{}_{N} = \mathcal{T}_{\mathrm{stab}}{}^{M}{}_{N} + \mathcal{T}_{\Psi}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{gauge}}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{f}}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{tip}}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{I11}}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{I12}}{}^{M}{}_{N} + \mathcal{T}_{\mathrm{I13}}{}^{M}{}_{N} \tag{FE-3}$$

**Symbols in (FE-3).**

- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{T}_{\mathrm{stab}}$: The energy-momentum tensor of the stabiliser. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\Psi}$: The energy-momentum tensor of the dirac16complex field. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\mathrm{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the Standard-Model fermions. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\mathrm{tip}}$: The energy-momentum tensor of the tip terms (non-zero only for the tip directions). Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\mathrm{I11}}$: The energy-momentum tensor of the portal I11. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\mathrm{I12}}$: The energy-momentum tensor of the portal I12. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mathcal{T}_{\mathrm{I13}}$: The energy-momentum tensor of the portal I13. Unit: $\mathrm{GeV}^8$. Status: DERIVED.

Every Lovelock tensor is divergence-free identically, so the field equations imply the conservation of the source, which holds on the matter field equations (DERIVED for the order-0 metric for $p = 1, 2, 3$):

$$\nabla_M \mathcal{T}^{M}{}_{N} = 0, \qquad \text{because} \qquad \nabla_M \mathcal{E}_{(p)}{}^{M}{}_{N} = 0 \quad (p = 1, 2, 3) \tag{FE-11}$$

**Symbols in (FE-11).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^{N}{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.

**For a pair-diagonal curvature (DERIVED).** When the Riemann tensor is pair-diagonal (the order-0 metric), the Lovelock densities and the diagonal Lovelock tensors are sums of products of sectional curvatures,

$$\begin{aligned}
&\mathcal{L}_{(1)} = 2\sum_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(2)} = 8\sum_{\text{2 disjoint pairs}}\ \prod_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(3)} = 48\sum_{\text{3 disjoint pairs}}\ \prod_{\text{pairs}} R^{ab}{}_{ab},\\
&\mathcal{E}_{(1)}{}^{c}{}_{c} = -\sum_{\text{pairs avoiding } c} R^{ab}{}_{ab}, \qquad \mathcal{E}_{(2)}{}^{c}{}_{c} = -4\sum_{\text{2 disjoint pairs avoiding } c}\ \prod_{\text{pairs}} R^{ab}{}_{ab},\\
&\mathcal{E}_{(3)}{}^{c}{}_{c} = -24\sum_{\text{3 disjoint pairs avoiding } c}\ \prod_{\text{pairs}} R^{ab}{}_{ab}
\end{aligned} \tag{FE-34}$$

**Symbols in (FE-34).**

- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: $1/\text{length}^2$. Status: STANDARD.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- $c$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.

which was checked against the Einstein tensor from the Ricci tensor ($p = 1$), against the explicit Gauss-Bonnet tensor ($p = 2$), against the Bianchi identities ($p = 1, 2, 3$), against the Euler-Lagrange equations of the action with a lapse ($p = 1, 2, 3$) and against the record's equations (all exact).

## 6.2 The block stabiliser (ASSUMED)

A source term, a vacuum energy of the hidden block, stabilises the block:

$$S_{\text{stab}} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\rho_{\text{stab}}(f), \qquad \kappa\rho_{\text{stab}}(f) = \kappa\rho_{\text{stab}}(f_*)\Big(\frac{f_*}{f}\Big)^8, \qquad f = \mathrm{e}^{-a_4}\,b_4^{1/2} \tag{IN-4}$$

**Symbols in (IN-4).**

- $S_{\text{stab}}$: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^{M}{}_{N} = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\rho_{\text{stab}}$: The energy density of the block stabiliser, a function of the block scale, $\rho_{\text{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\text{stab}}(f_*)$. Unit: $\text{GeV}^8$. Value: $\kappa\rho_{\text{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\text{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\text{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \text{GeV}^{-2}$. Status: AUTHOR.
- $f_*$: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.

Its energy-momentum tensor is

$$\mathcal{T}_{\text{stab}}{}^{M}{}_{N} = \rho_{\text{stab}}\,\text{diag}(1,\ -1,\ -1,\ -1,\ -1,\ 1,\ 1,\ 1) : \qquad \rho = \rho_{\text{stab}}, \qquad p_3 = -\rho_{\text{stab}}, \qquad p_0 = p_5 = \rho_{\text{stab}} \tag{FE-4}$$

**Symbols in (FE-4).**

- $\mathcal{T}_{\text{stab}}$: The energy-momentum tensor of the stabiliser. Unit: $\text{GeV}^8$. Status: DERIVED.

- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\rho_{\mathrm{stab}}$: The energy density of the block stabiliser, a function of the block scale, $\rho_{\mathrm{stab}} \propto f^{-8}$; its value at the static block is $\rho_{\mathrm{stab}}(f_*)$. Unit: $\mathrm{GeV}^8$. Value: $\kappa\rho_{\mathrm{stab}}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.

It is conserved: the conservation law of the order-0 source,

$$\dot\rho = -3\mathcal{H}(\rho + p_3) - \mathcal{H}_f(\rho + p_0) - 3\mathcal{H}_f(\rho + p_5) = -3\mathcal{H}(\rho + p_3) - 4\mathcal{H}_f(\rho + \bar p) \tag{FE-41}$$

**Symbols in (FE-41).**

- $\dot\rho$: The time derivative $\mathrm{d}\rho/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^9$. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot f/f = \dot b_4/(2b_4) - \dot a_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\bar p$: The hidden pressure, the pressure averaged over the block, $\bar p = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar p = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

gives, with $p_3 = -\rho_{\mathrm{stab}}$ and $\bar p = \rho_{\mathrm{stab}}$, $\dot\rho = -8\mathcal{H}_f\rho$, i.e. $\rho_{\mathrm{stab}} \propto f^{-8}$. It is four-dimensionally Lorentz invariant ($p_3 = -\rho$), isotropic in the block ($p_0 = p_5$), as the order-0 field equations require (the section on gravity), and independent of $x_0$. The $f^{-8}$ law is that of an eight-dimensional energy density whose only scale is the size of the block (a Casimir-type or flux energy). Its normalisation is fixed by the static block, $\kappa\rho_{\mathrm{stab}}(f_*) = (26 + 4\sqrt{51})H^2$ at $f_* = K_*^{-1/2}$. Its microscopic origin is OPEN (a Casimir energy of the fields of the cell or a flux are natural candidates; neither was computed in UM-2).

# 7 The dirac16complex field, which couples to all other fields

## 7.1 The Clifford algebra of the author (AUTHOR, DERIVED verification)

The author's real $16 \times 16$ matrices are built from real $4 \times 4$ matrices $\mathsf{s}_h$ and $\mathsf{t}_h$ ($h = 1, 2, 3$; matrix indices $k, q = 1, \ldots, 4$; $\epsilon_{\mathrm{LC}}$ the Levi-Civita symbol):

$$\big(\mathsf{s}_h\big)_{kq} = \epsilon_{\mathrm{LC}}^{hkq4} - \delta_{k4}\delta_{qh} + \delta_{kh}\delta_{q4}, \qquad \big(\mathsf{t}_h\big)_{kq} = \epsilon_{\mathrm{LC}}^{hkq4} + \delta_{k4}\delta_{qh} - \delta_{kh}\delta_{q4},$$

**Symbols in this equation.**

- $\mathsf{s}_h$: The author's real antisymmetric $4\times4$ matrices $\mathsf{s}_h$ ($h=1,2,3$; in the notebook $s_h = Q_a - Q_b$), the building blocks of the matrices $\mathsf{Z}_1$, $\mathsf{Z}_2$, $\mathsf{Z}_3$. Unit: dimensionless. Status: AUTHOR.
- $h$: The label $h=1,2,3$ of the author's real $4\times4$ matrices $\mathsf{s}_h$, $\mathsf{t}_h$ and of the $8\times8$ matrices $\mathsf{Z}_h$ and $\mathsf{Z}_{7-h}$ built from them. Status: AUTHOR.
- $k$, $q$: The row and column indices $k,q=1,2,3,4$ of the author's $4\times4$ matrices $\mathsf{s}_h$ and $\mathsf{t}_h$. Status: AUTHOR.
- $\epsilon^{hkq4}_{\mathrm{LC}}$: The Levi-Civita symbol of four indices: $+1$ if $(h,k,q,4)$ is an even permutation of $(1,2,3,4)$, $-1$ if it is an odd permutation, and 0 if two indices are equal. Unit: dimensionless. Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M=N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n\times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $\mathsf{t}_h$: The author's real antisymmetric $4\times4$ matrices $\mathsf{t}_h$ ($h=1,2,3$; in the notebook $t_h = Q_a + Q_b$), the building blocks of the matrices $\mathsf{Z}_6$, $\mathsf{Z}_5$, $\mathsf{Z}_4$. Unit: dimensionless. Status: AUTHOR.

then the real $8\times8$ matrices $\mathsf{Z}_a$ ($a=0,\ldots,7$; $\mathbf{1}$ the identity of the size of the block) and the $16\times16$ gamma matrices:

$$\mathsf{Z}_0 = \mathbf{1}, \qquad \mathsf{Z}_h = \begin{pmatrix} 0 & \mathsf{s}_h \\ \mathsf{s}_h & 0 \end{pmatrix}, \qquad \mathsf{Z}_{7-h} = \begin{pmatrix} 0 & \mathsf{t}_h \\ -\mathsf{t}_h & 0 \end{pmatrix} \quad (h=1,2,3), \qquad \mathsf{Z}_7 = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3\mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6,$$
$$\mathsf{S} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}, \qquad \gamma^a = \begin{pmatrix} 0 & \bar{\mathsf{Z}}_a \\ \mathsf{Z}_a & 0 \end{pmatrix} \quad (a=0,\ldots,7).$$

**Symbols in this equation.**

- $\mathsf{Z}_a$: The author's real $8\times8$ matrices $\mathsf{Z}_a$ ($a=0,\ldots,7$; the notebook's $\zeta_A = \tau_A$): $\mathsf{Z}_0$ is the identity, $\mathsf{Z}_1,\mathsf{Z}_2,\mathsf{Z}_3$ and $\mathsf{Z}_6,\mathsf{Z}_5,\mathsf{Z}_4$ are built from $\mathsf{s}_h$ and $\mathsf{t}_h$, and $\mathsf{Z}_7 = \mathsf{Z}_1\cdots\mathsf{Z}_6$; they are the lower-left $8\times8$ blocks of the gamma matrices $\gamma^a$. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $h$: The label $h=1,2,3$ of the author's real $4\times4$ matrices $\mathsf{s}_h$, $\mathsf{t}_h$ and of the $8\times8$ matrices $\mathsf{Z}_h$ and $\mathsf{Z}_{7-h}$ built from them. Status: AUTHOR.
- $\mathsf{s}_h$: The author's real antisymmetric $4\times4$ matrices $\mathsf{s}_h$ ($h=1,2,3$; in the notebook $s_h = Q_a - Q_b$), the building blocks of the matrices $\mathsf{Z}_1$, $\mathsf{Z}_2$, $\mathsf{Z}_3$. Unit: dimensionless. Status: AUTHOR.
- $\mathsf{t}_h$: The author's real antisymmetric $4\times4$ matrices $\mathsf{t}_h$ ($h=1,2,3$; in the notebook $t_h = Q_a + Q_b$), the building blocks of the matrices $\mathsf{Z}_6$, $\mathsf{Z}_5$, $\mathsf{Z}_4$. Unit: dimensionless. Status: AUTHOR.
- $h$: The block Hamiltonian: the $2\times2$ differential operator of one block that generates the time evolution of $\chi$, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies $\varepsilon$. Unit: GeV. Status: DERIVED.
- $\mathsf{S}$: The author's real $8\times8$ matrix with the $4\times4$ identity in its two off-diagonal blocks and zeros elsewhere (the notebook's $\sigma$); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.
- $\bar{\mathsf{Z}}_a$: The partner matrices $\bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}$ of the author ($a=0,\ldots,7$; the notebook's $\bar\tau_A$), the upper-right $8\times8$ blocks of the gamma matrices $\gamma^a$. Unit: dimensionless. Status: AUTHOR.
- $a$: A frame (orthonormal) index running over $0,1,\ldots,7$, aligned with $x_0,\ldots,x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $a$, $b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k=0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

The frame matrix along $x_0$ is written $\gamma^{x_0} = \gamma^0$, and so on up to $\gamma^{x_7} = \gamma^7$ (the notebook's `T16[A]`). They satisfy (DERIVED, exact; 31 checks)

$$\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \tfrac{1}{2}\big(\mathsf{Z}_a\bar{\mathsf{Z}}_b + \mathsf{Z}_b\bar{\mathsf{Z}}_a\big) = \eta_{ab}\,\mathbf{1}, \qquad \mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7.$$

**Symbols in this equation.**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $\eta_{ab}$: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ ($\eta^{ab}$ the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\mathsf{Z}_a$: The author's real $8 \times 8$ matrices $\mathsf{Z}_a$ ($a = 0, \ldots, 7$; the notebook's $\zeta_A = \tau_A$): $\mathsf{Z}_0$ is the identity, $\mathsf{Z}_1, \mathsf{Z}_2, \mathsf{Z}_3$ and $\mathsf{Z}_6, \mathsf{Z}_5, \mathsf{Z}_4$ are built from $\mathsf{s}_h$ and $\mathsf{t}_h$, and $\mathsf{Z}_7 = \mathsf{Z}_1 \cdots \mathsf{Z}_6$; they are the lower-left $8 \times 8$ blocks of the gamma matrices $\gamma^a$. Unit: dimensionless. Status: AUTHOR.
- $\bar{\mathsf{Z}}_a$: The partner matrices $\bar{\mathsf{Z}}_a = \mathsf{S}\,\mathsf{Z}_a^T\,\mathsf{S}$ of the author ($a = 0, \ldots, 7$; the notebook's $\bar{\tau}_A$), the upper-right $8 \times 8$ blocks of the gamma matrices $\gamma^a$. Unit: dimensionless. Status: AUTHOR.
- $\mathsf{S}$: The author's real $8 \times 8$ matrix with the $4 \times 4$ identity in its two off-diagonal blocks and zeros elsewhere (the notebook's $\sigma$); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.

The Clifford algebra, the Dirac adjoint, the four-dimensional chirality and the author's operator $\mathcal{B}$, whose eigenspaces are the two halves of dirac16complex, are

$$\begin{aligned}
&\gamma^a\gamma^b + \gamma^b\gamma^a = 2\eta^{ab}\,\mathbf{1}, \qquad \eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1), \qquad \gamma^M = \mathbf{e}_a{}^M\gamma^a,\\
&\bar\Psi = \Psi^\dagger C, \qquad C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}, \qquad \Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}, \qquad \mathcal{B} = -\mathrm{i}C\gamma^{x_4},\\
&\Psi = \Psi_A + \Psi_B, \qquad \mathcal{B}\Psi_A = -\Psi_A, \qquad \mathcal{B}\Psi_B = +\Psi_B
\end{aligned} \tag{IN-6}$$

**Symbols in (IN-6).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $\eta_{ab}$: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1,+1,+1,+1,-1,-1,-1,-1)$ ($\eta^{ab}$ the same matrix). Unit: dimensionless. Status: AUTHOR.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $C$: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.

- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues $\mp 1$ define the halves $\Psi_A$ and $\Psi_B$, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- $\Psi_A$, $\Psi_B$: The two halves of dirac16complex, $\Psi = \Psi_A + \Psi_B$, the eigenspaces of $\mathcal{B}$ with the eigenvalues $-1$ and $+1$; each half carries one dark-matter species. Unit: $\mathrm{GeV}^{7/2}$. Status: AUTHOR.

With the author's matrices

$$C = \begin{pmatrix} -\mathsf{S} & 0 \\ 0 & \mathsf{S} \end{pmatrix}, \qquad \Gamma_8 = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7} = \mathrm{diag}\big(-\mathbf{1},\ \mathbf{1}\big),$$

$$C^T = C, \quad C^2 = \mathbf{1}, \quad \big(C\gamma^a\big)^T = -C\gamma^a, \quad \mathcal{B}^\dagger = \mathcal{B}, \quad \mathcal{B}^2 = \mathbf{1},$$

**Symbols in this equation.**

- $C$: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- $\mathsf{S}$: The author's real $8\times 8$ matrix with the $4\times 4$ identity in its two off-diagonal blocks and zeros elsewhere (the notebook's $\sigma$); $\mathsf{S} = \mathsf{Z}_1\mathsf{Z}_2\mathsf{Z}_3 = \mathsf{Z}_4\mathsf{Z}_5\mathsf{Z}_6\mathsf{Z}_7$. Unit: dimensionless. Status: AUTHOR.
- $\Gamma_8$: The chirality of eight dimensions, $\Gamma_8 = \gamma^{x_0}\gamma^{x_1}\cdots\gamma^{x_7}$, equal to $\mathrm{diag}(-\mathbf{1},\mathbf{1})$ for the author's matrices; its eigenspaces (eigenvalues $-1$ and $+1$) are the two inequivalent 8-dimensional irreducible representations of Spin(4,4). Unit: dimensionless. Status: DERIVED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues $\mp 1$ define the halves $\Psi_A$ and $\Psi_B$, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

and $\mathcal{B}$ has the signature $(8,8)$. Since every $C\gamma^a$ is real and antisymmetric, $\bar\Psi\gamma^a\Psi$ is anti-Hermitian and the Hermitian vector currents are $\mathrm{i}\bar\Psi\gamma^a\Psi$; $\bar\Psi\Psi$ is Hermitian. The 16-dimensional representation is irreducible under Pin(4,4) (commutant of dimension 1), the double cover of $O(4,4)$, and it splits under Spin(4,4), the double cover of $SO(4,4)$, into two inequivalent irreducible 8-dimensional representations, the eigenspaces $\Gamma_8 = \mp 1$ (commutant of dimension 2, intertwiner of dimension 0).

## 7.2 The Lagrangian and the canonical spin connection

$\Psi$ has 16 complex anticommuting (Grassmann) components. It couples to gravity through the canonical (metric-compatible, torsion-free) spin connection, which couples every spinor of UM-2 to gravity:

$$
\begin{aligned}
&D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \qquad \Omega_M = \tfrac{1}{2}\,\omega_{Mab}\,\mathcal{S}^{ab}, \qquad \mathcal{S}^{ab} = \tfrac{1}{4}\big[\gamma^a,\gamma^b\big],\\
&\omega_M{}^a{}_b = \mathbf{e}^a{}_N\big(\partial_M \mathbf{e}_b{}^N + \Gamma^N{}_{MP}\,\mathbf{e}_b{}^P\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b, \qquad g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\,\mathbf{e}^b{}_N,\\
&\Gamma^N{}_{MP} = \tfrac{1}{2}\,g^{NQ}\big(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP}\big)
\end{aligned}
\tag{IN-7}
$$

**Symbols in (IN-7).**

- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- $\omega_M{}^a{}_b$: The coefficients of the spin connection of the vielbein (the Levi-Civita connection in the frame basis); $\omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b$. Unit: 1/length. Status: STANDARD.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $\mathcal{S}^{ab}$: The generators of frame rotations on spinors, $\mathcal{S}^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ (the commutator). Unit: dimensionless. Status: STANDARD.
- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}^a{}_M,\ \mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^N{}_{MP}$: The Christoffel symbols (the Levi-Civita connection) of the metric, $\Gamma^N{}_{MP} = \frac{1}{2}g^{NQ}(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP})$. Unit: 1/length. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\eta_{ab}$: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ ($\eta^{ab}$ the same matrix). Unit: dimensionless. Status: AUTHOR.
- $c$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

it makes $\gamma^N$ covariantly constant, $\partial_M\gamma^N + \Gamma^N{}_{MP}\gamma^P + \Omega_M\gamma^N - \gamma^N\Omega_M = 0$ (DERIVED). The action of dirac16complex is the record's:

$$
\begin{aligned}
S_\Psi &= \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2\Big],\\
m_\Psi &= 3H\mathcal{M} = H \qquad (\mathcal{M} = 1/3)
\end{aligned}
\tag{IN-5}
$$

**Symbols in (IN-5).**

- $S_\Psi$: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\mathcal{M}$: The record's dimensionless mass parameter of dirac16complex, $\mathcal{M} = 1/3$ ($m_\Psi = 3H\mathcal{M}$). Unit: dimensionless. Value: 1/3. Status: RECORD.

The mass parameter is the author's $m_\Psi = 3H\mathcal{M}$; UM-2 uses $\mathcal{M} = 1/3$, i.e. $m_\Psi = H$ (RECORD, the value of the Kohn-Sham record). $\lambda$ is the contact self-coupling. It lets the dark matter annihilate into the fold bands, so the cosmology bounds it: $G_\lambda = \lambda H\tilde I_\lambda/(2f^4v_{\mathrm{t}}) \le 2.3 \times 10^{-15}\ \mathrm{GeV}^{-2}$ (ASSUMED; the dark sector), i.e. $\lambda \le 1.4 \times 10^{-10}H^{-6}$ for $v_{\mathrm{t}} = H^{-3}$; the value $\lambda_{\mathrm{test}} = 7 \times 10^{-5}H^{-6}$ of the Kohn-Sham tests is a numerical test value. Varying the action (with the portals of the section on the interactions) gives the field equation of dirac16complex,

$$\gamma^M D_M\Psi = \Big[m_\Psi + \lambda\,\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - \mathrm{i}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\,J^{B-L}_M\gamma^M\Psi - \mathrm{i}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0} J^B_M\gamma^M P_b\Psi \qquad \text{(FE-12)}$$

**Symbols in (FE-12).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.

- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7\times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times 10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}}G^A_{MN}G^{AMN} + c_{\mathrm{W}}W^I_{MN}W^{IMN} + c_{\mathrm{B}}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: $\mathrm{GeV}^4$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_{\mathrm{V}}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times 10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $J^{B-L}_M$: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $J^B_M$: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $P_a,\ P_b$: The projectors on the two $x_0$-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.

and its conjugate. Its gravitational term is $\gamma^M\Omega_M$; for the order-0 metric (DERIVED, exact for every $a_4$, $b_4$ and on both halves of a period)

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4} \tag{FE-13}$$

**Symbols in (FE-13).**

- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\dot b_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\dot a_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

which vanishes only in flat 4+4 space ($H = 0$ with constant $a_4$, $b_4$): the coupling of $\Psi$ to gravity is non-trivial. $\Psi$ is a singlet of the Standard-Model group, so it couples to the Standard Model only through the portals I11, I12 and I13. Its energy-momentum tensor (the source of $\Psi$ in the gravitational equations) is

$$(\mathcal{T}_\Psi)_{MN} = -\frac{1}{4}\Big[\bar\Psi\gamma_M D_N\Psi + \bar\Psi\gamma_N D_M\Psi - (D_M\bar\Psi)\gamma_N\Psi - (D_N\bar\Psi)\gamma_M\Psi\Big] + g_{MN}\,\mathcal{L}_\Psi,$$
$$\mathcal{L}_\Psi = \tfrac{1}{2}\big(\bar\Psi\gamma^P D_P\Psi - (D_P\bar\Psi)\gamma^P\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2 \qquad \text{(FE-5)}$$

**Symbols in (FE-5).**

- $\mathcal{T}_\Psi$: The energy-momentum tensor of the dirac16complex field. Unit: GeV$^8$. Status: DERIVED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_\Psi$: The Lagrangian density of dirac16complex (the bracket of $S_\Psi$). Unit: GeV$^8$. Status: RECORD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: GeV$^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.

and its boundary conditions at the fold ($y = 0$) and at the regular tip ($y = -L$) are

$$\big(1-\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \;\;\text{(even)} \quad\text{or}\quad \big(1+\gamma^{x_0}\big)\Psi\big|_{y=0} = 0 \;\;\text{(odd)}, \qquad \big(1-\gamma^{x_0}\big)\Psi\big|_{y=-L} = 0 \qquad \text{(FE-28)}$$

**Symbols in (FE-28).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.

(both parities are modes of the full period; the conditions are self-adjoint, DERIVED).

### 7.3 The two halves $\Psi_A$ and $\Psi_B$ and the fold parity (DERIVED)

$\mathcal{B}$ commutes with every four-dimensional Lorentz generator $\gamma^a\gamma^b$ ($a,b \in \{1,2,3,4\}$) and with $C$, so the split $\Psi = \Psi_A + \Psi_B$ of IN-6 ($\mathcal{B}\Psi_A = -\Psi_A$, $\mathcal{B}\Psi_B = +\Psi_B$) gives two four-dimensionally Lorentz-invariant halves of 8 components, and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ does not mix them. In the good sector (no extra-time momenta) every interaction of UM-2 commutes with $\mathcal{B}$ (DERIVED with the author's matrices: $\mathcal{B}$ commutes with $C$, with $\gamma^a$ and with $C\gamma^a$ for $a = 0,\dots,4$ and anticommutes with them for $a = 5,6,7$, and $\gamma^M\Omega_M$ contains $\gamma^{x_0}$ and $\gamma^{x_4}$ only), so the numbers of quanta of the two halves are separately conserved; extra-time momenta would mix the halves. Under the fold reflection $z \mapsto \pi - z$ the modes are even or odd; the bulk Lagrangian is symmetric under it.

## 8 The gauge fields: gluons, photons, $W^+$, $W^-$ and $Z$ (ASSUMED structure, STANDARD content)

The gauge group is $SU(3)_c \times SU(2)_L \times U(1)_Y$, with eight-dimensional gauge fields: the gluons $G^A_M$ ($A = 1,\dots,8$), the $SU(2)$ fields $W^I_M$ ($I = 1,2,3$) and the hypercharge field $B_M$:

$$S_{\text{gauge}} = -\frac{1}{4}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\frac{1}{\hat g_3^2}\,G^A_{MN}G^{AMN} + \frac{1}{\hat g_2^2}\,W^I_{MN}W^{IMN} + \frac{1}{\hat g_1^2}\,B_{MN}B^{MN}\Big) \tag{IN-8}$$

**Symbols in (IN-8).**

- $S_{\text{gauge}}$: The action of the gauge fields of SU(3), SU(2) and U(1)$_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0,\dots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0,x_1,\dots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\dots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\dots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\hat g_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat g_3^2$); $g_3^2 = \hat g_3^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1,\dots,8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1,\dots,8$; a repeated index is summed. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \dots$): it runs over $x_0,x_1,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \dots$), running over $x_0,\dots,x_7$; a repeated index is summed. Status: STANDARD.
- $\hat g_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat g_2^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1,2,3$; $W^3$ the third component), the matrix $W_M = W^I_M\tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- $\hat g_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat g_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.

with the field strengths

$$
\begin{aligned}
&G_M = G_M^A\, t^A, \qquad G_{MN} = \partial_M G_N - \partial_N G_M - \mathrm{i}\big[G_M, G_N\big], \qquad G_{MN}^A = 2\,\mathrm{tr}\big(t^A G_{MN}\big),\\
&W_M = W_M^I\,\frac{\tau^I}{2}, \qquad W_{MN} = \partial_M W_N - \partial_N W_M - \mathrm{i}\big[W_M, W_N\big], \qquad W_{MN}^I = \mathrm{tr}\big(\tau^I W_{MN}\big),\\
&B_{MN} = \partial_M B_N - \partial_N B_M
\end{aligned}
\tag{IN-9}
$$

**Symbols in (IN-9).**

- $G_M^A$, $G_M$, $G_{MN}$, $G_{MN}^A$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G_M^A$ ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength $G_{MN}$ (matrix) and its components $G_{MN}^A$; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

In this geometric normalisation the couplings stand in front of the kinetic terms and not in the covariant derivatives; the fields have the dimension of a mass, the eight-dimensional couplings $\hat{g}_3$, $\hat{g}_2$, $\hat{g}_1$ the dimension $\mathrm{GeV}^{-2}$, and the four-dimensional couplings are $g_3^2 = \hat{g}_3^2/V_\perp$, $g_2^2 = \hat{g}_2^2/V_\perp$, $g_1^2 = \hat{g}_1^2/V_\perp$, where $V_\perp$ is the proper volume of the block per unit 4-volume (the section on dimensional reduction). The physical electroweak fields are canonically normalised (divided by the coupling):

$$
\begin{aligned}
&W_M^\pm = \frac{W_M^1 \mp \mathrm{i} W_M^2}{\sqrt{2}\,\hat{g}_2}, \qquad Z_M = \frac{W_M^3 - B_M}{\hat{g}_Z}, \qquad A_M = \frac{\hat{g}_1^2\, W_M^3 + \hat{g}_2^2\, B_M}{\hat{g}_1 \hat{g}_2 \hat{g}_Z},\\
&\hat{g}_Z = \sqrt{\hat{g}_2^2 + \hat{g}_1^2}, \qquad \cos\theta_\mathrm{W} = \frac{\hat{g}_2}{\hat{g}_Z}, \qquad \sin\theta_\mathrm{W} = \frac{\hat{g}_1}{\hat{g}_Z}
\end{aligned}
\tag{IN-10}
$$

**Symbols in (IN-10).**

- $W_M^\pm$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i} W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.

- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV$^{-2}$. Status: STANDARD.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV$^3$. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $\theta_\mathrm{W}$: The weak mixing angle of the couplings, $\cos\theta_\mathrm{W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_\mathrm{W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_\mathrm{W} = 0.776368$. Status: DERIVED.

**$W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$** ($W^-$ is the antiparticle of $W^+$; the charge follows from the isospin raising and lowering components of $W^1 \mp \mathrm{i}W^2$ and $Y = 0$ of the $W$ fields); $Z$ and the photon $A$ are neutral (DERIVED). The quadratic part of the gauge action in terms of the canonical fields is (DERIVED; ${W^\pm}_{MN}$, $Z_{MN}$, $A_{MN}$ the abelian field strengths of FE-25 below)

$$\frac{1}{\hat{g}_2^2}\, W^I_{MN} W^{IMN} + \frac{1}{\hat{g}_1^2}\, B_{MN}B^{MN} = 2\,{W^+}_{MN}W^{-MN} + Z_{MN}Z^{MN} + A_{MN}A^{MN} + \big(\text{cubic and quartic terms}\big),$$

**Symbols in this equation.**

- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $W^\pm_M$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W^\pm_M = (W^1_M \mp \mathrm{i}W^2_M)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV$^3$. Status: STANDARD.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV$^3$. Status: STANDARD.

and the non-abelian (cubic and quartic) terms of $W^I_{MN}W^{IMN}$ contain the vertices $W^+W^-\gamma$ (strength $e$), $W^+W^-Z$ (strength $g_2\cos\theta_\mathrm{W}$) and the quartic vertices $W^+W^-W^+W^-$, $W^+W^-\gamma\gamma$, $W^+W^-Z\gamma$, $W^+W^-ZZ$; every vertex conserves the electric charge, so a single $W^+$ or $W^-$ always comes with a charged partner (STANDARD). With the fermion currents of the three generations,

$$\begin{aligned} J_3^{AN} &= \sum_{i=1}^{3} \big(\mathrm{i}\bar{Q}_i\gamma^N t^A Q_i + \mathrm{i}\bar{u}_{Ri}\gamma^N t^A u_{Ri} + \mathrm{i}\bar{d}_{Ri}\gamma^N t^A d_{Ri}\big), \qquad J_2^{IN} = \sum_{i=1}^{3} \Big(\mathrm{i}\bar{Q}_i\gamma^N \frac{\tau^I}{2} Q_i + \mathrm{i}\bar{\ell}_i\gamma^N \frac{\tau^I}{2}\ell_i\Big), \\ J_1^N &= \sum_{i=1}^{3}\sum_{\psi} Y_\psi\, \mathrm{i}\bar{\psi}_i\gamma^N\psi_i, \qquad J^N_\mathrm{em} = J_2^{3N} + J_1^N, \qquad J_2^{\pm N} = \frac{J_2^{1N} \mp \mathrm{i}J_2^{2N}}{\sqrt{2}} \end{aligned} \tag{FE-20}$$

**Symbols in (FE-20).**

- $J_3^{AN}$: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: GeV$^7$. Status: STANDARD.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.

- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $J_2^{IN}$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $J_1^N$: The hypercharge current of all fermions (the source of $B$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $Y$: The hypercharge (the generator of $\mathrm{U}(1)_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $J_{\mathrm{em}}^N$: The electromagnetic current, $J_{\mathrm{em}} = J_2^3 + J_1$ (in units of the elementary charge). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.

and the effective inverse couplings produced by the portal I12 (a field-dependent coupling, the section on the interactions),

$$\frac{1}{\hat{g}_{3,\mathrm{eff}}^2} = \frac{1}{\hat{g}_3^2} + \frac{4c_{\mathrm{G}}\bar{\Psi}\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat{g}_{2,\mathrm{eff}}^2} = \frac{1}{\hat{g}_2^2} + \frac{4c_{\mathrm{W}}\bar{\Psi}\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat{g}_{1,\mathrm{eff}}^2} = \frac{1}{\hat{g}_1^2} + \frac{4c_{\mathrm{B}}\bar{\Psi}\Psi}{\Lambda_\star^3} \tag{FE-21}$$

**Symbols in (FE-21).**

- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}_{i,\mathrm{eff}}^2 = 1/\hat{g}_i^2 + 4c\,\bar{\Psi}\Psi/\Lambda_\star^3$ with $c = c_{\mathrm{G}}, c_{\mathrm{W}}, c_{\mathrm{B}}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_{\mathrm{G}}$: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.

- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_\mathrm{W}$: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{W}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_\mathrm{B}$: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_\mathrm{B}/\Lambda_\star^3 = c_{\mathrm{I}12}/\Lambda_\star^3$. Status: INVENTED.

the gauge-field equations are (in the unitary gauge $\Sigma = \mathbf{1}$ of the tip sigma field, geometric fields):

$$\nabla_M\Big(\frac{G^{MN}}{\hat{g}_{3,\mathrm{eff}}^2}\Big) - \mathrm{i}\Big[G_M,\ \frac{G^{MN}}{\hat{g}_{3,\mathrm{eff}}^2}\Big] = J_3^{AN}\, t^A \tag{FE-22}$$

**Symbols in (FE-22).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $G_M^A$, $G_M$, $G_{MN}$, $G_{MN}^A$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G_M^A$ ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength $G_{MN}$ (matrix) and its components $G_{MN}^A$; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}_{i,\mathrm{eff}}^2 = 1/\hat{g}_i^2 + 4c\,\bar{\Psi}\Psi/\Lambda_\star^3$ with $c = c_\mathrm{G}, c_\mathrm{W}, c_\mathrm{B}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $J_3^{AN}$: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.

$$\nabla_M\Big(\frac{W^{MN}}{\hat{g}_{2,\mathrm{eff}}^2}\Big) - \mathrm{i}\Big[W_M,\ \frac{W^{MN}}{\hat{g}_{2,\mathrm{eff}}^2}\Big] = J_2^{IN}\,\frac{\tau^I}{2} + \delta_\mathrm{tip}\,\frac{v_\Sigma^2}{4}\Big(W^N - B^N\,\frac{\tau^3}{2}\Big) \qquad (\Sigma = \mathbf{1}) \tag{FE-23}$$

**Symbols in (FE-23).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_{3,\mathrm{eff}}$, $\hat{g}_{2,\mathrm{eff}}$, $\hat{g}_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat{g}_{i,\mathrm{eff}}^2 = 1/\hat{g}_i^2 + 4c\,\bar{\Psi}\Psi/\Lambda_\star^3$ with $c = c_\mathrm{G}, c_\mathrm{W}, c_\mathrm{B}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $J_2^{IN}$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status:

DERIVED.

- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

$$\nabla_M\Big(\frac{B^{MN}}{\hat g_{1,\mathrm{eff}}^2}\Big) = J_1^N - \delta_{\mathrm{tip}}\,\frac{v_\Sigma^2}{4}\,\big(W^{3N} - B^N\big) \qquad (\Sigma = \mathbf{1}) \tag{FE-24}$$

**Symbols in (FE-24).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat g_{3,\mathrm{eff}}$, $\hat g_{2,\mathrm{eff}}$, $\hat g_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat g_{i,\mathrm{eff}}^2 = 1/\hat g_i^2 + 4c\,\bar\Psi\Psi/\Lambda_\star^3$ with $c = c_{\mathrm{G}}, c_{\mathrm{W}}, c_{\mathrm{B}}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $J_1^N$: The hypercharge current of all fermions (the source of $B$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\delta_{\mathrm{tip}}$: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I\tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

To linear order in the fields (I12 neglected) they become the equations of the physical fields:

$$\begin{aligned}
&\nabla_M W^{\pm MN} = \hat g_2\,J_2^{\pm N} + \delta_{\mathrm{tip}}\,\frac{v_\Sigma^2}{4}\,\hat g_2^2\,W^{\pm N}, \qquad \nabla_M A^{MN} = \hat e\,J_{\mathrm{em}}^N, \qquad \hat e = \frac{\hat g_2\hat g_1}{\hat g_Z},\\
&\nabla_M Z^{MN} = \hat g_Z\big(J_2^{3N} - \sin^2\theta_{\mathrm{W}}\,J_{\mathrm{em}}^N\big) + \delta_{\mathrm{tip}}\,\frac{v_\Sigma^2}{4}\,\hat g_Z^2\,Z^N,\\
&W^{\pm MN} = \nabla^M W^{\pm N} - \nabla^N W^{\pm M}, \qquad Z^{MN} = \nabla^M Z^N - \nabla^N Z^M, \qquad A^{MN} = \nabla^M A^N - \nabla^N A^M
\end{aligned} \tag{FE-25}$$

**Symbols in (FE-25).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

- $W_M^{\pm}$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^{\pm} = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^{\pm}$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\delta_{\mathrm{tip}}$: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W_M^3 + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\hat{e}$: The eight-dimensional electric coupling, $\hat{e} = \hat{g}_2\hat{g}_1/\hat{g}_Z$ ($e^2 = \hat{e}^2/V_\perp$). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $J_{\mathrm{em}}^N$: The electromagnetic current, $J_{\mathrm{em}} = J_2^3 + J_1$ (in units of the elementary charge). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $J_2^{IN}$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.

The tip terms are mass terms for $W^{\pm}$ and $Z$ only: the photon receives none and stays massless; for an $x_0$ profile these tip terms become the Robin conditions of the $x_0$ eigenvalue problem (the section on the $x_0$ mass mechanism). The boundary conditions of gravity and of the gauge fields at the fold and the tip are

$$\begin{aligned} &z \to \pi - z: \qquad \partial_y F\big|_{y=0} = 0 \ \ \text{(even)} \quad \text{or} \quad F\big|_{y=0} = 0 \ \ \text{(odd)};\\ &\text{at } y = -L\text{: the natural (Robin) condition of the tip term} \end{aligned} \tag{FE-27}$$

**Symbols in (FE-27).**

- $z$: The angle variable of the hidden direction, $z = 6Hx_0$; one period of $x_0$ is $0 < z < \pi$. Unit: dimensionless. Status: AUTHOR.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

($F$ the $x_0$ profile of the field). The energy-momentum tensor of the gauge fields is

$$\begin{aligned} (\mathcal{T}_{\mathrm{gauge}})_{MN} = &\frac{1}{\hat{g}_3^2}\Big(G^A_{MP}G^A{}_N{}^P - \frac{1}{4}g_{MN}G^A_{PQ}G^{APQ}\Big) + \frac{1}{\hat{g}_2^2}\Big(W^I_{MP}W^I{}_N{}^P - \frac{1}{4}g_{MN}W^I_{PQ}W^{IPQ}\Big)\\ &+ \frac{1}{\hat{g}_1^2}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big) \end{aligned} \tag{FE-7}$$

**Symbols in (FE-7).**

- $\mathcal{T}_{\text{gauge}}$: The energy-momentum tensor of the gauge fields. Unit: GeV$^8$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.

# 9 The fermions: the three-flavour left-handed neutrinos, the electrons, two more lepton generations and three quark generations (ASSUMED structure, INVENTED boundary assignment)

Each multiplet is an eight-dimensional spinor of the same Clifford algebra (16 components times its gauge indices). There are three generations, $i = 1, 2, 3$, with the same quantum numbers:

| field ($i = 1, 2, 3$) | members | $SU(3)_c$ | $SU(2)_L$ | $Y$ | electric charges $Q_{\text{em}} = T_3 + Y$ | zero mode |
|---|---|---|---|---|---|---|
| $Q_i$ | left-handed up and down, charm and strange, top and bottom quarks | 3 | 2 | 1/6 | up-type 2/3, down-type $-1/3$ | $\Gamma_4 = +1$ |
| $u_{Ri}$ | right-handed up, charm, top quarks | 3 | 1 | 2/3 | 2/3 | $\Gamma_4 = -1$ |
| $d_{Ri}$ | right-handed down, strange, bottom quarks | 3 | 1 | $-1/3$ | $-1/3$ | $\Gamma_4 = -1$ |
| $\ell_i$ | $\nu_e$ with the left-handed electron, $\nu_\mu$ with the left-handed muon, $\nu_\tau$ with the left-handed tau | 1 | 2 | $-1/2$ | neutrino 0, charged lepton $-1$ | $\Gamma_4 = +1$ |
| $e_{Ri}$ | right-handed electron, muon, tau | 1 | 1 | $-1$ | $-1$ | $\Gamma_4 = -1$ |

| field ($i = 1, 2, 3$) | members | $SU(3)_c$ | $SU(2)_L$ | $Y$ | electric charges $Q_{\mathrm{em}} = T_3 + Y$ | zero mode |
|---|---|---|---|---|---|---|
| $\nu_{Ri}$ | three right-handed neutrinos | 1 | 1 | 0 | 0 | $\Gamma_4 = -1$ |

The author's list is contained in this table: the **three-flavour left-handed neutrinos** are the upper (neutral) components of the three lepton doublets $\ell_i$; the **left-handed electrons** are the lower component of $\ell_1$ and the **right-handed electron singlet** is $e_{R1}$; the other two lepton generations and their right-handed partners ($\ell_2$, $\ell_3$, $e_{R2}$, $e_{R3}$) and the three right-handed neutrinos are those of version 3; the **three quark generations** are $Q_i$, $u_{Ri}$, $d_{Ri}$. The action is

$$S_{\mathrm{f}} = \sum_{i=1}^{3} \sum_{\psi} \int \mathrm{d}^8X \, \sqrt{|g|} \left[ \tfrac{1}{2} \big( \bar{\psi}_i \gamma^M D_M \psi_i - (D_M \bar{\psi}_i) \gamma^M \psi_i \big) - \mu_{\psi i} \, \bar{\psi}_i \psi_i \right], \qquad \text{(IN-11)}$$
$$\psi \in \{Q,\ u_R,\ d_R,\ \ell,\ e_R,\ \nu_R\}, \qquad \mu_{\psi i} = \mu_Q \ (\psi = Q), \qquad \mu_{\psi i} = \mu_\ell \ (\psi = \ell)$$

**Symbols in (IN-11).**

- $S_{\mathrm{f}}$: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0 \, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN} \, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|} \, \mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN} \gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu \Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304 \, H$, $\mu_\ell = 9.2055 \, H$). Status: ASSUMED.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: GeV$^{7/2}$. Status: ASSUMED.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $\mu_Q$: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- $\mu_\ell$: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell = 5.95288$). Status: DERIVED.

with the covariant derivative (the same canonical spin connection as for $\Psi$, and the gauge fields) and the electric charge

$$D_M = \partial_M + \Omega_M - \mathrm{i}G^A_M\, t^A - \mathrm{i}W^I_M\,\frac{\tau^I}{2} - \mathrm{i}\,Y B_M, \qquad Q_{\mathrm{em}} = T_3 + Y \tag{IN-12}$$

**Symbols in (IN-12).**

- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $Y$: The hypercharge (the generator of U(1)$_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $Q_{\mathrm{em}}$: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; $Q^f_{\mathrm{em}}$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $T_3$: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); $T^f_3$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.

The bulk masses are diagonal in the generation index, and the doublet bulk mass is the same for the three generations of a sector ($\mu_\ell$ for the leptons, $\mu_Q$ for the quarks; ASSUMED), while the singlet bulk masses of $e_{Ri}$, $\nu_{Ri}$, $u_{Ri}$, $d_{Ri}$ differ from generation to generation. Varying the action gives the Dirac equations of the six multiplets, each with its gravitational coupling (in $D_M$), its gauge couplings (in $D_M$), its portal terms (I11 for every multiplet,

I13 for the quarks) and its tip Yukawa term (with $\delta_{\mathrm{tip}}$, the density that places a term on the tip):

$$\gamma^M D_M Q_i = \mu_Q\, Q_i - \mathrm{i}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\, q_Q\, \big(\mathrm{i}\bar\Psi\gamma_M\Psi\big)\gamma^M Q_i - \frac{\mathrm{i}}{3}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0}\big(\mathrm{i}\bar\Psi\gamma_M P_b\Psi\big)\gamma^M Q_i + \delta_{\mathrm{tip}}\sum_{j=1}^{3}\Sigma\begin{pmatrix}(\mathcal{Y}_u)_{ij}\,u_{Rj}\\(\mathcal{Y}_d)_{ij}\,d_{Rj}\end{pmatrix} \tag{FE-14}$$

**Symbols in (FE-14).**

- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $Q_i,\ \bar Q_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mu_Q$: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_{\mathrm{V}}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi,\ q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi,\ \bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $P_a,\ P_b$: The projectors on the two $x_0$-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- $\delta_{\mathrm{tip}}$: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu,\ \mathcal{Y}_e,\ \mathcal{Y}_u,\ \mathcal{Y}_d$: The complex $3\times3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $u_R,\ u_{Ri},\ \bar u_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar u_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R,\ d_{Ri},\ \bar d_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar d_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

$$\gamma^M D_M u_{Ri} = \mu_{\psi i}\, u_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{u_R}\,\big(\mathrm{i}\bar\Psi\gamma_M\Psi\big)\gamma^M u_{Ri} - \frac{\mathrm{i}}{3}\,\frac{c_\mathrm{A}}{\Lambda_\star^6}\sum_{M\neq x_0}\big(\mathrm{i}\bar\Psi\gamma_M P_b\Psi\big)\gamma^M u_{Ri}$$
$$+\,\delta_\mathrm{tip}\sum_{j=1}^{3}(\mathcal{Y}_u)^*_{ji}\,\big(\Sigma^\dagger Q_j\big)_1 \qquad (\psi = u_R) \tag{FE-15}$$

**Symbols in (FE-15).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $u_R$, $u_{Ri}$, $\bar u_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar u_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $c_\mathrm{A}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $P_a$, $P_b$: The projectors on the two $x_0$-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3\times3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $^*$: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.

- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

$$\gamma^M D_M d_{Ri} = \mu_{\psi i}\, d_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{d_R} \left(\mathrm{i}\bar{\Psi}\gamma_M \Psi\right)\gamma^M d_{Ri} - \frac{\mathrm{i}}{3}\,\frac{c_\mathrm{A}}{\Lambda_\star^6} \sum_{M \neq x_0} \left(\mathrm{i}\bar{\Psi}\gamma_M P_b \Psi\right)\gamma^M d_{Ri} + \delta_\mathrm{tip} \sum_{j=1}^{3} (\mathcal{Y}_d)^*_{ji} \left(\Sigma^\dagger Q_j\right)_2 \qquad (\psi = d_R) \tag{FE-16}$$

**Symbols in (FE-16).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38 \times 10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi$, $q_f$: The $B - L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $c_\mathrm{A}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $P_a$, $P_b$: The projectors on the two $x_0$-chiralities, $P_a = (1 + \gamma^{x_0})/2$ and $P_b = (1 - \gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3 \times 3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.

- $^*$: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma=\mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma=\mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $Q_i,\ \bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i,\ \bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi\in\{Q,u_R,d_R,\ell,e_R,\nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi}=\psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

$$\gamma^M D_M \ell_i = \mu_\ell\,\ell_i - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\,q_\ell\,\big(\mathrm{i}\bar{\Psi}\gamma_M\Psi\big)\gamma^M\ell_i + \delta_\mathrm{tip}\sum_{j=1}^{3}\Sigma\begin{pmatrix}(\mathcal{Y}_\nu)_{ij}\,\nu_{Rj}\\(\mathcal{Y}_e)_{ij}\,e_{Rj}\end{pmatrix} \tag{FE-17}$$

**Symbols in (FE-17).**

- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M=\mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M=g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M=\partial_M+\Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\ell_i,\ \bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $i$: The generation index ($i=1,2,3$), which also labels the neutrino mass states $\nu_1,\nu_2,\nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mu_\ell$: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell=5.95288$). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V}=4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi,\ q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q=q_{u_R}=q_{d_R}=\frac{1}{3}$, $q_\ell=q_{e_R}=q_{\nu_R}=-1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi,\ \bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi}=\Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip}=\delta_\mathrm{D}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j=1,2,3$). Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma=\mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma=\mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu,\ \mathcal{Y}_e,\ \mathcal{Y}_u,\ \mathcal{Y}_d$: The complex $3\times 3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i,j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $\nu_R,\ \nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R,\ e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

$$\gamma^M D_M e_{Ri} = \mu_{\psi i}\, e_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\, q_{e_R}\left(\mathrm{i}\bar\Psi\gamma_M\Psi\right)\gamma^M e_{Ri} + \delta_\mathrm{tip}\sum_{j=1}^{3}(\mathcal{Y}_e)^*_{ji}\left(\Sigma^\dagger \ell_j\right)_2 \qquad (\psi = e_R) \tag{FE-18}$$

**Symbols in (FE-18).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3\times3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $^*$: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\ell_i$, $\bar\ell_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i$, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

$$\gamma^M D_M \nu_{Ri} = \mu_{\psi i}\,\nu_{Ri} - \mathrm{i}\,\frac{c_\mathrm{V}}{\Lambda_\star^6}\,q_{\nu_R}\left(\mathrm{i}\bar\Psi\gamma_M\Psi\right)\gamma^M\nu_{Ri} + \delta_\mathrm{tip}\sum_{j=1}^{3}(\mathcal{Y}_\nu)^*_{ji}\left(\Sigma^\dagger\ell_j\right)_1 \qquad (\psi=\nu_R) \tag{FE-19}$$

**Symbols in (FE-19).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\delta_\mathrm{tip}$: The density that places a tip term in the field equations of eight dimensions, $\delta_\mathrm{tip} = \delta_\mathrm{D}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3\times3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $^*$: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\ell_i$, $\bar\ell_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i$, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

Their energy-momentum tensor is

$$(\mathcal{T}_{\mathrm{f}})_{MN} = \sum_{i=1}^{3}\sum_{\psi}\Big\{ -\frac{1}{4}\Big[\bar{\psi}_i\gamma_M D_N\psi_i + \bar{\psi}_i\gamma_N D_M\psi_i - (D_M\bar{\psi}_i)\gamma_N\psi_i - (D_N\bar{\psi}_i)\gamma_M\psi_i\Big] + g_{MN}\,\mathcal{L}_{\psi i}\Big\},$$
$$\mathcal{L}_{\psi i} = \tfrac{1}{2}\big(\bar{\psi}_i\gamma^P D_P\psi_i - (D_P\bar{\psi}_i)\gamma^P\psi_i\big) - \mu_{\psi i}\,\bar{\psi}_i\psi_i \qquad \text{(FE-6)}$$

**Symbols in (FE-6).**

- $\mathcal{T}_{\mathrm{f}}$: The energy-momentum tensor of the Standard-Model fermions. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\psi i}$: The Lagrangian density of the fermion multiplet $\psi$ of the generation $i$ (the bracket of $S_{\mathrm{f}}$). Unit: $\mathrm{GeV}^8$. Status: ASSUMED.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.

**Boundary conditions.** With the selection projector $P_{\mathrm{sel}}$, which commutes with every four-dimensional Lorentz generator (DERIVED), the conditions at the fold ($y = 0$) and at the tip ($y = -L$) are

$$P_{\mathrm{sel}} = \frac{1 + \Gamma_4\gamma^{x_0}}{2}\,\frac{1 - \mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2};$$
$$\text{doublets } (Q, \ell): \quad P_{\mathrm{sel}}\big(1 - \gamma^{x_0}\big)\psi\big|_{y=0} = \big(1 - P_{\mathrm{sel}}\big)\big(1 + \gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1 - \gamma^{x_0}\big)\psi\big|_{y=-L} = 0;$$
$$\text{singlets } (u_R, d_R, e_R, \nu_R): \quad P_{\mathrm{sel}}\big(1 + \gamma^{x_0}\big)\psi\big|_{y=0} = \big(1 - P_{\mathrm{sel}}\big)\big(1 - \gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1 + \gamma^{x_0}\big)\psi\big|_{y=-L} = 0 \qquad \text{(FE-29)}$$

**Symbols in (FE-29).**

- $P_{\mathrm{sel}}$: The projector that selects the two blocks of the half $\mathcal{B} = -1$ in which the Standard-Model zero modes live, $P_{\mathrm{sel}} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-\mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2}$. Unit: dimensionless. Status: INVENTED.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

These are self-adjoint boundary conditions of the Dirac operator (DERIVED). At the tip they are modified by the tip Yukawa terms, which turn into the self-adjoint mixing conditions (DERIVED: the boundary form vanishes identically for every complex tip matrix, sympy, 42 real variables)

$$\begin{aligned}
&\text{one generation:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}\, a_L(-L), \qquad \chi = (a,\ \mathrm{i}b);\\
&\text{three generations:}\quad b_L(-L) = \mathcal{K}\, b_R(-L), \qquad a_R(-L) = -\mathcal{K}^\dagger a_L(-L), \qquad \mathcal{K} \in \{\mathcal{K}_\nu, \mathcal{K}_e, \mathcal{K}_u, \mathcal{K}_d\}; \qquad \text{(FE-30)}\\
&\text{UM-2:}\quad \mathcal{K}_e = \mathcal{K}_u = \mathbf{1}, \qquad \mathcal{K}_\nu = U, \qquad \mathcal{K}_d = V_{\mathrm{CKM}}
\end{aligned}$$

**Symbols in (FE-30).**

- $a$, $b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3 \times 3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \, \mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.

- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $V_{\mathrm{CKM}}$: The CKM matrix, the unitary $3 \times 3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

(the choice of the tip matrices is ASSUMED; it is explained in the section on the $x_0$ mass mechanism). The result (DERIVED): every doublet field has exactly one four-dimensional Weyl zero mode with $\Gamma_4 = +1$ and every singlet field exactly one with $\Gamma_4 = -1$, all localised at the fold and all in the two blocks selected by $P_{\mathrm{sel}}$, which lie in the half $\mathcal{B} = -1$. The zero-mode spectrum is three Standard-Model generations, each with its right-handed neutrino.

**Anomaly cancellation generation by generation (DERIVED, exact fractions).** Writing every zero mode as a left-handed Weyl field (a right-handed field of hypercharge $Y$ as a left-handed one of hypercharge $-Y$), the gauge anomaly coefficients of one generation are

$$\mathcal{A}_{\mathrm{SU(3)^2U(1)}} = \sum_{\text{triplets}} N_{\mathrm{dbl}}\, Y = 2\cdot\frac{1}{6} - \frac{2}{3} + \frac{1}{3} = 0, \qquad \mathcal{A}_{\mathrm{SU(2)^2U(1)}} = \sum_{\text{doublets}} N_{\mathrm{col}}\, Y = 3\cdot\frac{1}{6} - \frac{1}{2} = 0,$$

**Symbols in this equation.**

- $\mathcal{A}_{\mathrm{SU(3)^2U(1)}}$, $\mathcal{A}_{\mathrm{SU(2)^2U(1)}}$, $\mathcal{A}_{\mathrm{U(1)^3}}$: The gauge anomaly coefficients of one generation, with every zero mode written as a left-handed Weyl field (a right-handed one with the hypercharge $-Y$): the sum over the colour triplets of $N_{\mathrm{dbl}}Y$ (mixed $SU(3)^2U(1)_Y$ anomaly), the sum over the weak doublets of $N_{\mathrm{col}}Y$ (mixed $SU(2)^2U(1)_Y$ anomaly) and the sum over all components of $N_{\mathrm{col}}N_{\mathrm{dbl}}Y^3$ (the $U(1)_Y^3$ anomaly); the theory is consistent only if they vanish. Unit: dimensionless. Value: 0 for each generation of UM-2. Status: DERIVED.
- $N_{\mathrm{dbl}}$: The number of weak-isospin components of a fermion multiplet: 2 for an $SU(2)$ doublet, 1 for a singlet. Unit: dimensionless. Status: STANDARD.
- $Y$: The hypercharge (the generator of $\mathrm{U}(1)_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $N_{\mathrm{col}}$: The number of colour components of a fermion multiplet: 3 for a quark, 1 for a lepton. Unit: dimensionless. Status: STANDARD.

$$\mathcal{A}_{\mathrm{U(1)^3}} = \sum N_{\mathrm{col}}N_{\mathrm{dbl}}\, Y^3 = \frac{1}{36} - \frac{8}{9} + \frac{1}{9} - \frac{1}{4} + 1 = 0, \qquad \mathcal{A}_{\mathrm{grav^2U(1)}} = \sum N_{\mathrm{col}}N_{\mathrm{dbl}}\, Y = 1 - 2 + 1 - 1 + 1 = 0,$$

**Symbols in this equation.**

- $\mathcal{A}_{\mathrm{SU(3)^2U(1)}}$, $\mathcal{A}_{\mathrm{SU(2)^2U(1)}}$, $\mathcal{A}_{\mathrm{U(1)^3}}$: The gauge anomaly coefficients of one generation, with every zero mode written as a left-handed Weyl field (a right-handed one with the hypercharge $-Y$): the sum over the colour triplets of $N_{\mathrm{dbl}}Y$ (mixed $SU(3)^2U(1)_Y$ anomaly), the sum over the weak doublets of $N_{\mathrm{col}}Y$ (mixed $SU(2)^2U(1)_Y$ anomaly) and the sum over all components of $N_{\mathrm{col}}N_{\mathrm{dbl}}Y^3$ (the $U(1)_Y^3$ anomaly); the theory is consistent only if they vanish. Unit: dimensionless. Value: 0 for each generation of UM-2. Status: DERIVED.
- $N_{\mathrm{col}}$: The number of colour components of a fermion multiplet: 3 for a quark, 1 for a lepton. Unit: dimensionless. Status: STANDARD.
- $N_{\mathrm{dbl}}$: The number of weak-isospin components of a fermion multiplet: 2 for an $SU(2)$ doublet, 1 for a singlet. Unit: dimensionless. Status: STANDARD.
- $Y$: The hypercharge (the generator of $\mathrm{U}(1)_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $\mathcal{A}_{\mathrm{grav^2U(1)}}$, $\mathcal{A}_{\mathrm{SU(3)^3}}$: Two further anomaly coefficients of one generation (every zero mode written as a left-handed Weyl field, a right-handed one with the hypercharge $-Y$): the mixed gravitational-hypercharge anomaly $\mathcal{A}_{\mathrm{grav^2U(1)}} = \sum N_{\mathrm{col}}N_{\mathrm{dbl}}Y$ over all components, and the cubic colour anomaly $\mathcal{A}_{\mathrm{SU(3)^3}}$ (the number of left-handed minus right-handed colour triplets, weighted with their isospin multiplicity); the theory is consistent only if they vanish. Unit: dimensionless. Value: 0 (each generation). Status: DERIVED.

and $\mathcal{A}_{\mathrm{SU(3)^3}}$ (left-handed minus right-handed colour triplets) $= 2 - 1 - 1 = 0$; the twelve $SU(2)$ doublets of three generations are an even number (no Witten anomaly). The anomalies cancel generation by generation; three lepton generations with a single quark generation would leave $\mathcal{A}_{\mathrm{SU(2)^2U(1)}} = -1$ and $\mathcal{A}_{\mathrm{U(1)^3}} = 3/2$, so the second and third quark generations are required together with the second and third lepton generations (DERIVED).

**QCD (STANDARD).** The gluons are the massless $x_0$ zero modes of $G^A_M$. Below the QCD scale the quarks are confined into hadrons; the top quark decays (into $W^+$ and a bottom quark) before it can form one. The baryons of the cosmology are the protons (two up quarks and a down quark, charge +1) and the neutrons (an up quark and two down quarks, charge 0), whose masses $m_{\rm p} = 938.3$ MeV and $m_{\rm n} = 939.6$ MeV come from QCD; they are inputs of the cosmology. The QCD scale depends on the block through $g_3^2 = \hat{g}_3^2/V_\perp$ (the section on the coupled solution).

# 10 No Higgs particle: the tip sigma field (INVENTED for this geometry)

At the regular tip a non-linear sigma field $\Sigma(x) \in SU(2)$ carries the electroweak symmetry breaking, and the tip Yukawa terms couple the doublets to the singlets:

$$
\begin{aligned}
S_{\rm tip} &= \int_{y=-L} \mathrm{d}^7 X_{\rm tip} \sqrt{|g_{\rm tip}|}\, \Big[ -\frac{v_\Sigma^2}{4}\, \mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu \Sigma\big) - \mathcal{L}_{\rm Y} \Big], \\
\mathcal{L}_{\rm Y} &= \sum_{i,j=1}^{3} \Big[ \bar{\ell}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} + \bar{Q}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_u)_{ij}\, u_{Rj} \\ (\mathcal{Y}_d)_{ij}\, d_{Rj} \end{pmatrix} \Big] + \text{h.c.}
\end{aligned}
\tag{IN-13}
$$

**Symbols in (IN-13).**

- $S_{\rm tip}$: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X_{\rm tip}$: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\rm tip}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- $g_{\rm tip}$: The metric induced on the seven-dimensional tip $y = -L$ (the metric $g_{MN}$ restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\rm tip}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\rm W}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\rm W} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathcal{L}_{\rm Y}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: $\mathrm{GeV}^7$. Status: INVENTED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3 \times 3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.

- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

with the covariant derivative and the gauge transformation

$$D_\mu \Sigma = \partial_\mu \Sigma - \mathrm{i} W^I_\mu \frac{\tau^I}{2} \Sigma + \mathrm{i} B_\mu \Sigma \frac{\tau^3}{2}, \qquad \Sigma^\dagger \Sigma = \mathbf{1}, \qquad \Sigma \to U_L \Sigma U_Y^\dagger, \qquad U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2} \tag{IN-14}$$

**Symbols in (IN-14).**

- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $U_L$: An SU(2) gauge transformation (a unitary $2 \times 2$ matrix acting on doublets from the left). Unit: dimensionless. Status: STANDARD.
- $U_Y$: A $\mathrm{U}(1)_Y$ gauge transformation of the sigma field, $U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\vartheta$: The angle (gauge parameter) of a $\mathrm{U}(1)_Y$ transformation. Unit: dimensionless. Status: STANDARD.

with four complex $3 \times 3$ tip Yukawa matrices $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$ (the upper component of $\Sigma^\dagger \ell_i$ couples to $\nu_R$, the lower one to $e_R$). $\Sigma$ has no radial mode, so there is no Higgs particle; its three would-be Goldstone modes become the longitudinal $W^+$, $W^-$ and $Z$. Every term is $SU(2)_L \times U(1)_Y$ invariant, the Yukawa terms for arbitrary complex generation matrices (DERIVED with random group elements and random matrices, deviation $1.1 \times 10^{-14}$). In the unitary gauge $\Sigma = \mathbf{1}$ the first term is

$$-\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big)\Big|_{\Sigma=\mathbf{1}} = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}_2^2\,W_\mu^+ W^{-\mu} + \hat{g}_Z^2\,Z_\mu Z^\mu\Big] \tag{IN-15}$$

**Symbols in (IN-15).**

- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2\times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $W_M^\pm,\ W^+,\ W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.

which gives $W^+$ and $W^-$ one common mass (they are particle and antiparticle), gives the $Z$ a mass larger by $1/\cos\theta_{\mathrm{W}}$ at leading order (custodial symmetry: $\hat{g}_Z^2 = \hat{g}_2^2/\cos^2\theta_{\mathrm{W}}$), and leaves the photon massless. The Yukawa terms mix the doublet and singlet zero modes at the tip and give the fermions their masses and their mixing (the tip mixing conditions FE-30). The equations of motion of $\Sigma$ are the Noether identities of the gauge symmetry (STANDARD for a gauged non-linear sigma model):

$$\begin{aligned}
&\nabla^\mu j_\mu - \mathrm{i}\big[W^\mu,\ j_\mu\big] = -\mathrm{i}\,\frac{\tau^I}{2}\sum_{i,j=1}^{3}\Big[\bar{\ell}_i\,\frac{\tau^I}{2}\,\Sigma\begin{pmatrix}(\mathcal{Y}_\nu)_{ij}\,\nu_{Rj}\\(\mathcal{Y}_e)_{ij}\,e_{Rj}\end{pmatrix} + \bar{Q}_i\,\frac{\tau^I}{2}\,\Sigma\begin{pmatrix}(\mathcal{Y}_u)_{ij}\,u_{Rj}\\(\mathcal{Y}_d)_{ij}\,d_{Rj}\end{pmatrix} - \mathrm{h.c.}\Big],\\
&j_\mu = j_\mu^I\,\frac{\tau^I}{2},\qquad j_\mu^I = \frac{\mathrm{i}v_\Sigma^2}{4}\,\mathrm{tr}\big(\tau^I(D_\mu\Sigma)\Sigma^\dagger\big),\qquad j_\mu^I\big|_{\Sigma=\mathbf{1}} = \frac{v_\Sigma^2}{4}\big(W_\mu^I - \delta^{I3}B_\mu\big)
\end{aligned} \tag{FE-26}$$

**Symbols in (FE-26).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $j_\mu,\ j_\mu^I$: The current of the sigma field on the tip, $j_\mu^I = \frac{\mathrm{i}v_\Sigma^2}{4}\mathrm{tr}(\tau^I(D_\mu\Sigma)\Sigma^\dagger)$, and the matrix $j_\mu = j_\mu^I\tau^I/2$. Unit: $\mathrm{GeV}^6$. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $W_M^I,\ W_M,\ W_{MN},\ W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1,2,3$; $W^3$ the third component), the matrix $W_M = W_M^I\tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1,2,3$; a repeated index is summed. Status: STANDARD.
- $i$: The generation index ($i = 1,2,3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.

- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3 \times 3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

in the unitary gauge they are not independent of the gauge-field equations. The energy-momentum tensor of the tip terms is

$$(\mathcal{T}_{\mathrm{tip}})_{\mu\nu} = \delta_{\mathrm{tip}}\Big[\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D_\nu\Sigma + (D_\nu\Sigma)^\dagger D_\mu\Sigma\big) + g_{\mu\nu}\,\mathcal{L}_{\mathrm{tip}}\Big], \qquad \delta_{\mathrm{tip}} = \frac{\delta_{\mathrm{D}}(y+L)}{f},$$
$$\mathcal{L}_{\mathrm{tip}} = -\frac{v_\Sigma^2}{4}\,\mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu\Sigma\big) - \mathcal{L}_{\mathrm{Y}} \tag{FE-8}$$

**Symbols in (FE-8).**

- $\mathcal{T}_{\mathrm{tip}}$: The energy-momentum tensor of the tip terms (non-zero only for the tip directions). Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- $\delta_{\mathrm{tip}}$: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y+L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.

- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{L}_{\mathrm{tip}}$: The Lagrangian density of the tip action (the bracket of $S_{\mathrm{tip}}$). Unit: $\mathrm{GeV}^7$. Status: INVENTED.
- $\delta_{\mathrm{D}}$: The Dirac delta function: $\delta_{\mathrm{D}}(y + L)$ is zero except at $y = -L$ and has the unit integral over $y$. Unit: 1/length. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{L}_{\mathrm{Y}}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: $\mathrm{GeV}^7$. Status: INVENTED.

(zero for the hidden components; its gravitational effect is part of the $W^\pm$, $Z$ and fermion rest energies counted in the plasma).

# 11 The interactions: every interaction in equation form, explained and analysed

The author's order: "You must invent/create and document new phenomenological interactions for these standard model particles that should be consistent with qcd and electro-weak models", and "Put each and every interaction in equation form, and explain and analyze it". This section does so for every interaction of UM-2. Each subsection gives (1) the interaction as an equation (the interaction term of the action with its tag IN-n, or the form that follows from it), (2) what it couples and why it is consistent with QCD and the electroweak model, and (3) its strength and its bounds. The catalogue at the end lists them all.

## 11.1 Gravity with itself and with every field

**The equations.** The self-interaction of gravity is the Einstein-Lovelock action with the author's weights, written out again:

$$S_{\mathrm{EL}} = \frac{1}{2\kappa}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda\Big), \qquad \kappa = 8\pi G_8,$$
$$\alpha_{(p)} = w_{(p)}\,H^{2-2p}, \qquad w_{(1)} = 1, \quad w_{(2)} = \frac{1}{32}, \quad w_{(3)} = \frac{1}{1152}, \qquad \Lambda = -10H^2 \tag{IN-2}$$

**Symbols in (IN-2).**

- $S_{\mathrm{EL}}$: The Einstein-Lovelock action of gravity. Unit: dimensionless. Status: AUTHOR.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.

- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order $p$, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: $\mathrm{GeV}^{2-2p}$. Status: AUTHOR.
- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: $\mathrm{GeV}^2$. Value: $-10H^2$. Status: AUTHOR.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_8$: The eight-dimensional Newton constant ($\kappa = 8\pi G_8$). Unit: $\mathrm{GeV}^{-6}$. Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

with the Lovelock densities (written out again)

$$\begin{aligned} \mathcal{L}_{(p)} &= \frac{1}{2^p}\,\delta^{M_1N_1\cdots M_pN_p}_{P_1Q_1\cdots P_pQ_p}\,R^{P_1Q_1}{}_{M_1N_1}\cdots R^{P_pQ_p}{}_{M_pN_p}, \\ \mathcal{L}_{(1)} &= R, \qquad \mathcal{L}_{(2)} = R^2 - 4R_{MN}R^{MN} + R_{MNPQ}R^{MNPQ} \end{aligned} \tag{IN-3}$$

**Symbols in (IN-3).**

- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.
- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta^M_N = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: $1/\mathrm{length}^2$. Status: STANDARD.

and gravity couples to every field through $\sqrt{|g|}$, the inverse metric of every kinetic term and, for every spinor (dirac16complex and the Standard-Model fermions), through the canonical spin connection:

$$D_M\Psi = \partial_M\Psi + \Omega_M\Psi, \qquad D_M\bar\Psi = \partial_M\bar\Psi - \bar\Psi\Omega_M, \qquad \Omega_M = \tfrac{1}{2}\,\omega_{Mab}\,\mathcal{S}^{ab}, \qquad \mathcal{S}^{ab} = \tfrac{1}{4}\big[\gamma^a,\gamma^b\big],$$
$$\omega_M{}^a{}_b = \mathbf{e}^a{}_N\big(\partial_M \mathbf{e}_b{}^N + \Gamma^N{}_{MP}\,\mathbf{e}_b{}^P\big), \qquad \omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b, \qquad g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\,\mathbf{e}^b{}_N, \tag{IN-7}$$
$$\Gamma^N{}_{MP} = \tfrac{1}{2}\,g^{NQ}\big(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP}\big)$$

**Symbols in (IN-7).**

- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- $\omega_M{}^a{}_b$: The coefficients of the spin connection of the vielbein (the Levi-Civita connection in the frame basis); $\omega_{Mab} = \eta_{ac}\,\omega_M{}^c{}_b$. Unit: 1/length. Status: STANDARD.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature $R^{ab}{}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $\mathcal{S}^{ab}$: The generators of frame rotations on spinors, $\mathcal{S}^{ab} = \frac{1}{4}[\gamma^a, \gamma^b]$ (the commutator). Unit: dimensionless. Status: STANDARD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mathbf{e}^a{}_M$, $\mathbf{e}_a{}^M$: The vielbein (frame field) of the metric: $g_{MN} = \eta_{ab}\,\mathbf{e}^a{}_M\mathbf{e}^b{}_N$; $\mathbf{e}_a{}^M$ is the inverse matrix. Unit: dimensionless. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Gamma^N{}_{MP}$: The Christoffel symbols (the Levi-Civita connection) of the metric, $\Gamma^N{}_{MP} = \frac{1}{2}g^{NQ}(\partial_M g_{QP} + \partial_P g_{QM} - \partial_Q g_{MP})$. Unit: 1/length. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\eta_{ab}$: The flat metric of the frame, $\eta_{ab} = \mathrm{diag}(+1, +1, +1, +1, -1, -1, -1, -1)$ ($\eta^{ab}$ the same matrix). Unit: dimensionless. Status: AUTHOR.
- $c$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.

On the order-0 metric the gravitational coupling of every spinor is the non-zero term

$$\gamma^M\Omega_M = \frac{3}{2}\,\frac{H}{f}\,\gamma^{x_0} + \Big(\frac{\dot b_4}{b_4} - \frac{\dot a_4}{2}\Big)\gamma^{x_4} \tag{FE-13}$$

**Symbols in (FE-13).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$,

…), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\dot{b}_4$: The time derivative $\mathrm{d}b_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\dot{a}_4$: The time derivative $\mathrm{d}a_4/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

**Why it is consistent.** The Lovelock densities are the only scalar densities built from the curvature whose field equations are of second order in the derivatives of the metric in eight dimensions (no ghost from higher derivatives), and every Lovelock tensor is divergence-free, so the source is conserved; the minimal coupling through $\sqrt{|g|}$, $g^{MN}$ and $\Omega_M$ is the unique coupling that respects general covariance and local frame rotations. Gravity does not distinguish the gauge charges, so it is consistent with every gauge symmetry of the Standard Model.

**Strength and bounds.** On the stabilised block four-dimensional gravity is exactly general relativity with Newton's constant $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$ (the section on dimensional reduction); the Kaluza-Klein gravitons add Yukawa forces of relative strength $\mathcal{A}_n < 0.2$ and range below $1.63 \times 10^{-21}$ m (FE-54 in the section on the metric), invisible at every measurable distance. The sourceless block of the author's weights has repulsive four-dimensional gravity ($\mathcal{G} = -1$); the stabiliser of the next subsection is what makes it attractive. Every source pushes on the block through its hidden pressure; the largest push in the history of the model moves the block by $\Delta \ln f = 1.1 \times 10^{-34}$ (the coupled solution).

## 11.2 The block stabiliser

**The equation.** A vacuum energy of the block, written out again:

$$S_\mathrm{stab} = -\int \mathrm{d}^8X\,\sqrt{|g|}\,\rho_\mathrm{stab}(f), \qquad \kappa\rho_\mathrm{stab}(f) = \kappa\rho_\mathrm{stab}(f_*)\Big(\frac{f_*}{f}\Big)^8, \qquad f = \mathrm{e}^{-a_4}\,b_4^{1/2} \tag{IN-4}$$

**Symbols in (IN-4).**

- $S_\mathrm{stab}$: The action of the block stabiliser (a vacuum energy of the block). Unit: dimensionless. Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, ${g^M}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\rho_\mathrm{stab}$: The energy density of the block stabiliser, a function of the block scale, $\rho_\mathrm{stab} \propto f^{-8}$; its value at the static block is $\rho_\mathrm{stab}(f_*)$. Unit: GeV$^8$. Value: $\kappa\rho_\mathrm{stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status:

DERIVED.

- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ $\mathrm{GeV}^{-2}$. Status: AUTHOR.
- $f_*$: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.

**Why it is consistent.** It is a function of the metric only (of the block volume), four-dimensionally Lorentz invariant and isotropic in the block, so it respects every symmetry of the observable four dimensions and is blind to the gauge charges. **Strength.** $\kappa\rho_\mathrm{stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566H^2$, fixed by the static block; it gives the radion its mass $\omega_\mathrm{r} = 3.387H = 48.42$ TeV. Its microscopic origin is OPEN.

## 11.3 The self-interaction of dirac16complex

**The equation.** The contact term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the action of dirac16complex, written out again:

$$S_\Psi = \int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi\big) - m_\Psi\,\bar\Psi\Psi - \frac{\lambda}{2}\big(\bar\Psi\Psi\big)^2\Big],$$
$$m_\Psi = 3H\mathcal{M} = H \qquad (\mathcal{M} = 1/3) \tag{IN-5}$$

**Symbols in (IN-5).**

- $S_\Psi$: The action of the dirac16complex field (kinetic term with the canonical spin connection, mass term and contact self-interaction). Unit: dimensionless. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10}H^{-6}$ (for $\Delta N_\mathrm{eff} \le 0.001$). Status: RECORD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296

TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\mathcal{M}$: The record's dimensionless mass parameter of dirac16complex, $\mathcal{M} = 1/3$ ($m_\Psi = 3H\mathcal{M}$). Unit: dimensionless. Value: 1/3. Status: RECORD.

**What it couples, and why it is consistent.** $\Psi$ with itself; $\Psi$ is a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$, so the term is gauge invariant; $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$ commutes with $\mathcal{B}$, so it does not mix the halves. In the Kohn-Sham description it produces the mean fields $\frac{15}{16}\lambda n^{\mathrm{sc}}$ and $-\frac{1}{16}\lambda n$ (FE-64 in the section on dirac16complex in the cell).

**Strength and bounds.** Through the transition scalar density of the dark-matter orbital and a fold zero mode it lets the dark matter annihilate into the massless fold bands; keeping their radiation below $\Delta N_{\mathrm{eff}} = 0.001$ requires $G_\lambda \le 2.3 \times 10^{-15}$ GeV$^{-2}$ (ASSUMED bound, derived with FE-135 and FE-136 in the section on the dark sector), i.e. $\lambda \le 1.4 \times 10^{-10} H^{-6}$ for $v_{\mathrm{t}} = H^{-3}$. The value $\lambda_{\mathrm{test}} = 7 \times 10^{-5} H^{-6}$ with which the Kohn-Sham machinery was tested would thermalise the fold bands ($\Delta N_{\mathrm{eff}} = 0.19$) and is therefore only a test value.

## 11.4 The strong interaction (QCD)

**The equations.** The gauge fields enter through their action and their field strengths, and the fermions through their action and their covariant derivative (all written out again):

$$S_{\mathrm{gauge}} = -\frac{1}{4}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(\frac{1}{\hat g_3^2}\,G^A_{MN}G^{AMN} + \frac{1}{\hat g_2^2}\,W^I_{MN}W^{IMN} + \frac{1}{\hat g_1^2}\,B_{MN}B^{MN}\Big) \tag{IN-8}$$

**Symbols in (IN-8).**

- $S_{\mathrm{gauge}}$: The action of the gauge fields of SU(3), SU(2) and U(1)$_Y$ in eight dimensions. Unit: dimensionless. Status: ASSUMED.
- d: The differential: d$X$ is an infinitesimal change of $X$, d$X$/d$Y$ the derivative of $X$ with respect to $Y$, d$^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\hat g_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat g_3^2$); $g_3^2 = \hat g_3^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat g_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat g_2^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\hat g_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat g_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.

$$
\begin{aligned}
&G_M = G_M^A\, t^A, \qquad G_{MN} = \partial_M G_N - \partial_N G_M - \mathrm{i}\big[G_M, G_N\big], \qquad G_{MN}^A = 2\,\mathrm{tr}\big(t^A G_{MN}\big),\\
&W_M = W_M^I\,\frac{\tau^I}{2}, \qquad W_{MN} = \partial_M W_N - \partial_N W_M - \mathrm{i}\big[W_M, W_N\big], \qquad W_{MN}^I = \mathrm{tr}\big(\tau^I W_{MN}\big),\\
&B_{MN} = \partial_M B_N - \partial_N B_M
\end{aligned}
\tag{IN-9}
$$

**Symbols in (IN-9).**

- $G_M^A,\ G_M,\ G_{MN},\ G_{MN}^A$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G_M^A$ ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength $G_{MN}$ (matrix) and its components $G_{MN}^A$; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $W_M^I,\ W_M,\ W_{MN},\ W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $B_M,\ B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

$$
\begin{aligned}
&S_\mathrm{f} = \sum_{i=1}^{3}\sum_{\psi}\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\tfrac{1}{2}\big(\bar\psi_i\gamma^M D_M\psi_i - (D_M\bar\psi_i)\gamma^M\psi_i\big) - \mu_{\psi i}\,\bar\psi_i\psi_i\Big],\\
&\psi \in \{Q,\ u_R,\ d_R,\ \ell,\ e_R,\ \nu_R\}, \qquad \mu_{\psi i} = \mu_Q\ (\psi = Q), \qquad \mu_{\psi i} = \mu_\ell\ (\psi = \ell)
\end{aligned}
\tag{IN-11}
$$

**Symbols in (IN-11).**

- $S_\mathrm{f}$: The action of the Standard-Model fermions of three generations (kinetic and bulk-mass terms). Unit: dimensionless. Status: ASSUMED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+, +, +, +, -, -, -, -$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.

- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, ... the matrices along $x_0$, $x_4$, ...), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c = f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i} = c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q = 5.2304\,H$, $\mu_\ell = 9.2055\,H$). Status: ASSUMED.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mu_Q$: The common bulk mass of the three quark doublets. Unit: GeV. Value: $5.2304\,H$ ($c_Q = 3.38230$). Status: DERIVED.
- $\mu_\ell$: The common bulk mass of the three lepton doublets. Unit: GeV. Value: $9.2055\,H$ ($c_\ell = 5.95288$). Status: DERIVED.

$$D_M = \partial_M + \Omega_M - \mathrm{i}G^A_M\, t^A - \mathrm{i}W^I_M\, \frac{\tau^I}{2} - \mathrm{i}\, Y B_M, \qquad Q_{\mathrm{em}} = T_3 + Y \tag{IN-12}$$

**Symbols in (IN-12).**

- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $\Omega_M$: The spin connection acting on spinors, the matrix $\Omega_M = \frac{1}{2}\omega_{Mab}\mathcal{S}^{ab}$. Unit: 1/length. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$;

$W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $Y$: The hypercharge (the generator of $\mathrm{U}(1)_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $Q_{\mathrm{em}}$: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; $Q^f_{\mathrm{em}}$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $T_3$: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); $T^f_3$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.

Inserting the covariant derivative IN-12 into the fermion action IN-11 gives the couplings of the fermions to the gauge fields through the currents (written out again)

$$J^{AN}_3 = \sum_{i=1}^{3} \left(\mathrm{i}\bar{Q}_i \gamma^N t^A Q_i + \mathrm{i}\bar{u}_{Ri}\gamma^N t^A u_{Ri} + \mathrm{i}\bar{d}_{Ri}\gamma^N t^A d_{Ri}\right), \qquad J^{IN}_2 = \sum_{i=1}^{3} \left(\mathrm{i}\bar{Q}_i \gamma^N \frac{\tau^I}{2} Q_i + \mathrm{i}\bar{\ell}_i \gamma^N \frac{\tau^I}{2} \ell_i\right),$$
$$J^N_1 = \sum_{i=1}^{3} \sum_{\psi} Y_\psi \, \mathrm{i}\bar{\psi}_i \gamma^N \psi_i, \qquad J^N_{\mathrm{em}} = J^{3N}_2 + J^N_1, \qquad J^{\pm N}_2 = \frac{J^{1N}_2 \mp \mathrm{i} J^{2N}_2}{\sqrt{2}} \tag{FE-20}$$

**Symbols in (FE-20).**

- $J^{AN}_3$: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $t^A$: The generators of SU(3) in the fundamental representation (the Gell-Mann matrices divided by 2), $\mathrm{tr}(t^A t^B) = \frac{1}{2}\delta^{AB}$. Unit: dimensionless. Status: STANDARD.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $J^{IN}_2$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $J^N_1$: The hypercharge current of all fermions (the source of $B$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.

- $Y$: The hypercharge (the generator of $\mathrm{U}(1)_Y$); $Y_\psi$ the hypercharge of the multiplet $\psi$ ($Q$: $\frac{1}{6}$, $u_R$: $\frac{2}{3}$, $d_R$: $-\frac{1}{3}$, $\ell$: $-\frac{1}{2}$, $e_R$: $-1$, $\nu_R$: $0$). Unit: dimensionless. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $J^N_{\mathrm{em}}$: The electromagnetic current, $J_{\mathrm{em}} = J^3_2 + J_1$ (in units of the elementary charge). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $J^{\pm N}_2$: The charged combinations of the isospin current, $J^\pm_2 = (J^1_2 \mp \mathrm{i}J^2_2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.

so the quarks couple to the gluons through the term

$$-\int \mathrm{d}^8X\, \sqrt{|g|}\, G^A_M\, J_3^{AM},$$

**Symbols in this equation.**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $J_3^{AN}$: The colour current of the quarks (the source of the gluons), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.

and the commutator in the gluon field strength of IN-9, inserted into the gluon term of IN-8, gives the three- and four-gluon vertices.

**Why it is consistent with QCD.** This is QCD in eight dimensions: the gauge group $SU(3)_c$ is never broken (the tip carries no colour, so the gluon profiles have no tip term), the gluons are exactly massless $x_0$ zero modes with flat profiles, and the four-dimensional theory of the zero modes is QCD with six quark flavours, asymptotically free ($b_3 = 11 - \frac{2}{3}N_{\mathrm{f}} > 0$), confining below its scale, with protons and neutrons as the baryons of the cosmology. No interaction of UM-2 carries colour except the gauge couplings; the portals couple to colour-singlet currents (I11, I13) or to the colour-singlet invariant $G^A_{MN}G^{AMN}$ (I12), so they leave the colour structure of QCD intact.

**Strength.** $g_3^2 = \hat{g}_3^2/V_\perp$ with the measured $\alpha_3(m_{\mathrm{Z}}) = 0.1180$ (STANDARD); at the Kaluza-Klein scale $\alpha_3(m_{\mathrm{KK}}) = 0.0606$ (one loop, six flavours). Because the four-dimensional couplings scale as $f^{-4}$, the QCD scale is the most block-sensitive quantity of UM-2 ($\mathrm{d}\ln\Lambda^{(3)}_{\mathrm{QCD}}/\mathrm{d}\ln f = -49.0$, the nucleon mass $-46.1$; the coupled solution).

## 11.5 Electromagnetism and the weak interactions of the three generations

**The equations.** The physical electroweak fields are the combinations (written out again)

$$\begin{aligned} W^\pm_M &= \frac{W^1_M \mp \mathrm{i}W^2_M}{\sqrt{2}\,\hat{g}_2}, \qquad Z_M = \frac{W^3_M - B_M}{\hat{g}_Z}, \qquad A_M = \frac{\hat{g}_1^2\,W^3_M + \hat{g}_2^2\,B_M}{\hat{g}_1\hat{g}_2\hat{g}_Z}, \\ \hat{g}_Z &= \sqrt{\hat{g}_2^2 + \hat{g}_1^2}, \qquad \cos\theta_{\mathrm{W}} = \frac{\hat{g}_2}{\hat{g}_Z}, \qquad \sin\theta_{\mathrm{W}} = \frac{\hat{g}_1}{\hat{g}_Z} \end{aligned} \tag{IN-10}$$

**Symbols in (IN-10).**

- $W_M^\pm$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i} W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W_M^3 + \hat{g}_2^2 B_M)/(\hat{g}_1 \hat{g}_2 \hat{g}_Z)$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $\theta_\mathrm{W}$: The weak mixing angle of the couplings, $\cos\theta_\mathrm{W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_\mathrm{W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_\mathrm{W} = 0.776368$. Status: DERIVED.

The gauge couplings of the fermions (from IN-11 and IN-12 of the previous subsection), in terms of the currents FE-20 and of these physical fields, are (DERIVED)

$$
\begin{aligned}
&-\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big(W_M^I\,J_2^{IM} + B_M\,J_1^M\Big)\\
&= -\int \mathrm{d}^8X\,\sqrt{|g|}\,\Big[\hat{g}_2\big(W_M^+ J_2^{-M} + W_M^- J_2^{+M}\big) + \hat{g}_Z\,Z_M\big(J_2^{3M} - \sin^2\theta_\mathrm{W}\,J_\mathrm{em}^M\big) + \hat{e}\,A_M\,J_\mathrm{em}^M\Big],
\end{aligned}
$$

**Symbols in this equation.**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta_N^M$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $J_2^{IN}$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field U(1)$_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $J_1^N$: The hypercharge current of all fermions (the source of $B$). Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $W_M^\pm$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i} W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.

- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $J^N_{\mathrm{em}}$: The electromagnetic current, $J_{\mathrm{em}} = J^3_2 + J_1$ (in units of the elementary charge). Unit: GeV$^7$. Status: STANDARD.
- $\hat{e}$: The eight-dimensional electric coupling, $\hat{e} = \hat{g}_2\hat{g}_1/\hat{g}_Z$ ($e^2 = \hat{e}^2/V_\perp$). Unit: GeV$^{-2}$. Status: STANDARD.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV$^3$. Status: STANDARD.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: GeV$^7$. Status: STANDARD.

with $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ and $J_{\mathrm{em}} = J^3_2 + J_1$ (FE-20). These currents are the sources of the linearised field equations of $W^+$, $W^-$, $Z$ and the photon (written out again), which also define $\hat{e}$:

$$
\begin{aligned}
&\nabla_M W^{\pm MN} = \hat{g}_2\, J_2^{\pm N} + \delta_{\mathrm{tip}}\,\frac{v_\Sigma^2}{4}\,\hat{g}_2^2\, W^{\pm N}, \qquad \nabla_M A^{MN} = \hat{e}\, J^N_{\mathrm{em}}, \qquad \hat{e} = \frac{\hat{g}_2\hat{g}_1}{\hat{g}_Z},\\
&\nabla_M Z^{MN} = \hat{g}_Z\big(J_2^{3N} - \sin^2\theta_{\mathrm{W}}\, J^N_{\mathrm{em}}\big) + \delta_{\mathrm{tip}}\,\frac{v_\Sigma^2}{4}\,\hat{g}_Z^2\, Z^N,\\
&W^{\pm MN} = \nabla^M W^{\pm N} - \nabla^N W^{\pm M}, \qquad Z^{MN} = \nabla^M Z^N - \nabla^N Z^M, \qquad A^{MN} = \nabla^M A^N - \nabla^N A^M
\end{aligned}
\tag{FE-25}
$$

**Symbols in (FE-25).**

- $\nabla$: The covariant derivative of a tensor with the Levi-Civita connection of the metric $g_{MN}$ (Christoffel symbols $\Gamma^N{}_{MP}$); $\nabla_M$ acts on the index $M$, $\nabla^M = g^{MN}\nabla_N$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $W^\pm_M$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W^\pm_M = (W^1_M \mp \mathrm{i}W^2_M)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV$^3$. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $J_2^{\pm N}$: The charged combinations of the isospin current, $J_2^\pm = (J_2^1 \mp \mathrm{i}J_2^2)/\sqrt{2}$ (the sources of $W^\pm$). Unit: GeV$^7$. Status: STANDARD.
- $\delta_{\mathrm{tip}}$: The density that places a tip term in the field equations of eight dimensions, $\delta_{\mathrm{tip}} = \delta_{\mathrm{D}}(y + L)/f$ (the Dirac delta function at the tip with the proper length of the hidden direction). Unit: 1/length. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: GeV$^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV$^3$. Status: STANDARD.
- $\hat{e}$: The eight-dimensional electric coupling, $\hat{e} = \hat{g}_2\hat{g}_1/\hat{g}_Z$ ($e^2 = \hat{e}^2/V_\perp$). Unit: GeV$^{-2}$. Status: STANDARD.
- $J^N_{\mathrm{em}}$: The electromagnetic current, $J_{\mathrm{em}} = J^3_2 + J_1$ (in units of the elementary charge). Unit: GeV$^7$. Status: STANDARD.
- $\hat{g}_1$: The eight-dimensional U(1)$_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: GeV$^{-2}$. Status: ASSUMED.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: GeV$^{-2}$. Status: STANDARD.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W^3_M - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.
- $J_2^{IN}$: The weak-isospin current of the doublets $Q_i$, $\ell_i$ (the source of $W^I$). Unit: GeV$^7$. Status: STANDARD.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.

On the zero modes (four-dimensional fields; the couplings divided by $V_\perp^{1/2}$) these are the charged current of the three generations, with the PMNS matrix in the neutrino fields $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ and the CKM matrix between the up-type components $(Q_i)_1$ and the down-type components $(Q_j)_2$ of the quark doublets in the mass basis,

$$-\frac{g_2}{\sqrt{2}}\Big[W_\mu^+\Big(\sum_\alpha \bar\nu_\alpha\gamma^\mu P_L\,\ell_\alpha^- + \sum_{i,j=1}^{3}\big(V_{\rm CKM}\big)_{ij}\,\big(\bar Q_i\big)_1\gamma^\mu P_L\big(Q_j\big)_2\Big) + \text{h.c.}\Big],$$

**Symbols in this equation.**

- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $W_M^\pm$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat g_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV$^3$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour $\alpha$; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: GeV$^{3/2}$. Status: STANDARD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $P_L$, $P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $\ell_\alpha^-$, $\bar\ell_\alpha^-$: The charged lepton of the flavour $\alpha$ ($e^-$, $\mu^-$, $\tau^-$; the lower member of the lepton doublet) at low energy, and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $V_{\rm CKM}$: The CKM matrix, the unitary $3\times3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\rm CKM} = \mathcal{K}_u^\dagger\mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.
- $Q_i$, $\bar Q_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

and the neutral currents of the $Z$ and of the photon, summed over the fermions $f$ (STANDARD form; the couplings of FE-102),

$$-\frac{g_2}{\cos\theta_{\rm W}}\,Z_\mu\sum_f \bar\psi_f\gamma^\mu\big(\tilde g_L^f P_L + \tilde g_R^f P_R\big)\psi_f - e\,A_\mu\sum_f Q_{\rm em}^f\,\bar\psi_f\gamma^\mu\psi_f, \qquad e = \frac{g_2 g_1}{\big(g_2^2+g_1^2\big)^{1/2}}.$$

**Symbols in this equation.**

- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W} = \hat g_2/\hat g_Z$, $\sin\theta_{\rm W} = \hat g_1/\hat g_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W} = 0.776368$. Status: DERIVED.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat g_Z$. Unit: GeV$^3$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\psi_i$, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: GeV$^{7/2}$ (eight dimensions), GeV$^{3/2}$ (four dimensions). Status: ASSUMED.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$,

$\dots$), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.

- $\tilde{g}^f_L,\ \tilde{g}^f_R,\ \tilde{g}^f_V$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}$. Unit: dimensionless. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $P_L,\ P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $e$: The elementary charge (the charge of the positron), $e^2 = 4\pi\alpha_{\rm em}$; $W^+$ has the charge $+e$ and $W^-$ the charge $-e$. Unit: dimensionless. Value: $e^2 = 4\pi/137.036$ at low energy. Status: STANDARD.
- $A_M$: The photon field, canonically normalised, $A_M = (\hat{g}_1^2 W^3_M + \hat{g}_2^2 B_M)/(\hat{g}_1\hat{g}_2\hat{g}_Z)$. Unit: GeV$^3$. Status: STANDARD.
- $Q_{\rm em}$: The electric charge in units of the elementary charge, $Q_{\rm em} = T_3 + Y$; $Q^f_{\rm em}$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $g_1$: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\rm DM}$. Status: DERIVED.

A fermion of weak isospin $T_3$ and hypercharge $Y$ couples to the photon with $eQ_{\rm em} = e(T_3+Y)$ and to the $Z$ with $(g_2/\cos\theta_{\rm W})(T_3 P_L - Q_{\rm em}\sin^2\theta_{\rm W})$ (DERIVED).

**Why it is consistent with the electroweak model.** These are exactly the couplings of the electroweak model of three generations: the doublets $Q_i$, $\ell_i$ and the singlets carry the Standard-Model quantum numbers (the table of the fermion section), every term is invariant under $SU(2)_L \times U(1)_Y$, the electric charge is $Q_{\rm em} = T_3 + Y$, and the gauge anomalies cancel generation by generation (the fermion section). Because the three doublets of a sector share one bulk mass, the $W$ and $Z$ zero modes couple to every generation with the same strength (lepton universality), and the neutral currents are flavour diagonal in every basis: the first even Kaluza-Klein gauge mode couples to a zero mode of localisation $c$ through $\int|\chi_c|^2 F_1\,{\rm d}\xi$, which relative to the zero mode is 0.14434 for every left-handed lepton and $-0.30066$ for every left-handed quark, and the right-handed rotations are trivial, so in the mass basis the largest flavour-changing neutral coupling is $10^{-17}$ (round-off; DERIVED). With different doublet bulk masses the PMNS rotation would create flavour-changing couplings up to 0.033; the shared bulk mass is what makes UM-2 free of tree-level flavour-changing neutral currents, as the Standard Model is. The PMNS and CKM matrices carry the whole flavour structure of the charged currents.

**Strength and bounds.** $g_2$ and $g_1$ are fixed by the measured couplings ($\overline{\rm MS}$ value $\sin^2\theta_{\overline{\rm MS}} = 0.23122$ for the couplings to the fermions, $\alpha_{\rm em} = 1/127.95$ at $m_{\rm Z}$); at low energy the exchange of $W^\pm$ and $Z$ becomes the four-fermion interactions with the measured Fermi constant (next subsection). The invisible width of the $Z$ counts three neutrinos exactly, against the measured $2.9963 \pm 0.0074$ (0.5 standard deviations), and the tree-level total width with the QCD factor, 2485.2 MeV, lies 0.41 % below the measured 2495.5 MeV, the size of the electroweak loop corrections that UM-2 does not compute (the section on the neutrinos). Charged-lepton flavour violation through the neutrino masses (the decay of a muon into an electron and a photon, of rate proportional to $|\sum_i U^*_{\mu i}U_{ei}m_i^2/m_{\rm W}^2|^2$) has the branching ratio $2.8\times10^{-55}$: unobservable.

## 11.6 The weak interactions at low energy and the forward scattering of neutrinos

**The equations.** At momentum transfers far below $m_{\rm W}$ and $m_{\rm Z}$ the exchange of $W^\pm$ and $Z$ between the zero-mode fermions reduces to current-current interactions (STANDARD):

$$\mathcal{L}_{\rm CC} = -2\sqrt{2}\,G_{\rm F}\big(\bar{\nu}_\alpha\gamma^\mu P_L\,\ell^-_\alpha\big)\big(\bar{\ell}^-_\beta\gamma_\mu P_L\,\nu_\beta\big), \qquad \mathcal{L}_{\rm NC} = -2\sqrt{2}\,G_{\rm F}\big(\bar{\nu}_\alpha\gamma^\mu P_L\,\nu_\alpha\big)\sum_f \bar{\psi}_f\gamma_\mu\big(\tilde{g}^f_L P_L + \tilde{g}^f_R P_R\big)\psi_f,$$

$$\nu_\alpha = \sum_{i=1}^{3} U_{\alpha i}\,\nu_i, \qquad P_L = \frac{1+\Gamma_4}{2}, \qquad P_R = \frac{1-\Gamma_4}{2}$$

(FE-101)

**Symbols in (FE-101).**

- $\mathcal{L}_{\rm CC}$, $\mathcal{L}_{\rm NC}$: The charged-current and neutral-current four-fermion Lagrangians (exchange of $W^\pm$ and of $Z$ far below their masses). Unit: GeV$^4$. Status: STANDARD.
- $G_{\rm F}$: The Fermi constant. Unit: GeV$^{-2}$. Value: $1.1663788 \times 10^{-5}$ GeV$^{-2}$ (measured). Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour $\alpha$; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: GeV$^{3/2}$. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, ... the matrices along $x_0$, $x_4$, ...), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $P_L$, $P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $\ell^-_\alpha$, $\bar\ell^-_\alpha$: The charged lepton of the flavour $\alpha$ ($e^-$, $\mu^-$, $\tau^-$; the lower member of the lepton doublet) at low energy, and its Dirac adjoint. Unit: GeV$^{3/2}$. Status: STANDARD.
- $\beta$: A second flavour index running over $e, \mu, \tau$. Status: STANDARD.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: GeV$^{3/2}$. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\psi_i$, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar\psi = \psi^\dagger C$ the Dirac adjoint. Unit: GeV$^{7/2}$ (eight dimensions), GeV$^{3/2}$ (four dimensions). Status: ASSUMED.
- $\tilde g^f_L$, $\tilde g^f_R$, $\tilde g^f_V$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde g^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}$. Unit: dimensionless. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\rm CP} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

$$\frac{G_{\rm F}}{\sqrt{2}} = \frac{g_2^2}{8 m_{\rm W}^2}, \qquad \tilde g^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}, \qquad \tilde g^f_R = -Q^f_{\rm em}\sin^2\theta_{\rm W}, \qquad \tilde g^f_V = \tilde g^f_L + \tilde g^f_R = T^f_3 - 2Q^f_{\rm em}\sin^2\theta_{\rm W} \tag{FE-102}$$

**Symbols in (FE-102).**

- $G_{\rm F}$: The Fermi constant. Unit: GeV$^{-2}$. Value: $1.1663788 \times 10^{-5}$ GeV$^{-2}$ (measured). Status: STANDARD.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\rm DM}$. Status: DERIVED.
- $m_{\rm W}$: The mass of $W^+$ and $W^-$. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $\tilde g^f_L$, $\tilde g^f_R$, $\tilde g^f_V$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde g^f_L = T^f_3 - Q^f_{\rm em}\sin^2\theta_{\rm W}$. Unit: dimensionless. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.

- $T_3$: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); $T_3^f$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $Q_{\mathrm{em}}$: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; $Q_{\mathrm{em}}^f$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.
- $V$: The label V of the vector coupling of a fermion to the $Z$, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.

A neutrino crossing a medium scatters forward coherently on it; averaged over an unpolarised medium at rest the interactions add the flavour-diagonal potentials

$$
\begin{aligned}
&V_\alpha = \sqrt{2}\,G_{\mathrm{F}}\Big[\delta_{\alpha e}\big(n_{e^-} - n_{e^+}\big) + \sum_f \tilde{g}_V^f\big(n_f - n_{\bar{f}}\big)\Big],\\
&V = \mathrm{diag}\big(V_{\mathrm{CC}},\ 0,\ 0\big) + V_{\mathrm{NC}}\,\mathbf{1}, \qquad V_{\mathrm{CC}} = \sqrt{2}\,G_{\mathrm{F}}\,n_{\mathrm{e}}, \qquad V_{\mathrm{NC}} = -\frac{G_{\mathrm{F}}\,n_{\mathrm{n}}}{\sqrt{2}}
\end{aligned}
\tag{FE-103}
$$

**Symbols in (FE-103).**

- $V_\alpha$, $V$: The matter potential of forward scattering: $V_\alpha$ of the flavour $\alpha$ ($V_e$, $V_\mu$), $V$ the $3\times 3$ matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788\times 10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1\cdots N_n}^{M_1\cdots M_n}$, the determinant of the $n\times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.
- $n_{e^-}$, $n_{e^+}$: The number densities of electrons and of positrons. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_{\mathrm{em}}^f \sin^2\theta_{\mathrm{W}}$. Unit: dimensionless. Status: STANDARD.
- $V$: The label V of the vector coupling of a fermion to the $Z$, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $n_{\bar{f}}$: The number density of the antiparticles of the fermion $f$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $V_{\mathrm{CC}}$, $V_{\mathrm{NC}}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $n_{\mathrm{e}}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_{\mathrm{n}}$: The number density of neutrons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.

**Why it is consistent.** This is the Fermi theory of the electroweak model with the PMNS matrix in the flavour fields; UM-2 uses the measured $G_{\mathrm{F}} = 1.1663788\times 10^{-5}$ $\mathrm{GeV}^{-2}$. **Use.** These potentials drive the MSW conversion in the Sun, the Earth and supernovae, and the neutrino-neutrino term drives the collective flavour evolution of dense neutrino gases (the section on the neutrinos).

## 11.7 Electroweak symmetry breaking at the tip without a Higgs particle (INVENTED)

**The equations.** The tip sigma field and the tip Yukawa couplings, written out again:

$$
\begin{aligned}
S_{\mathrm{tip}} &= \int_{y=-L} \mathrm{d}^7 X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|}\, \Big[ -\frac{v_\Sigma^2}{4}\, \mathrm{tr}\big((D_\mu \Sigma)^\dagger D^\mu \Sigma\big) - \mathcal{L}_{\mathrm{Y}} \Big], \\
\mathcal{L}_{\mathrm{Y}} &= \sum_{i,j=1}^{3} \Big[ \bar{\ell}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_\nu)_{ij}\, \nu_{Rj} \\ (\mathcal{Y}_e)_{ij}\, e_{Rj} \end{pmatrix} + \bar{Q}_i\, \Sigma \begin{pmatrix} (\mathcal{Y}_u)_{ij}\, u_{Rj} \\ (\mathcal{Y}_d)_{ij}\, d_{Rj} \end{pmatrix} \Big] + \mathrm{h.c.}
\end{aligned}
\tag{IN-13}
$$

**Symbols in (IN-13).**

- $S_{\mathrm{tip}}$: The action on the tip $y = -L$: the sigma field and the tip Yukawa couplings. Unit: dimensionless. Status: INVENTED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X_{\mathrm{tip}}$: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- $g_{\mathrm{tip}}$: The metric induced on the seven-dimensional tip $y = -L$ (the metric $g_{MN}$ restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathcal{L}_{\mathrm{Y}}$: The tip Yukawa Lagrangian: the couplings of the doublets to the singlets through the sigma field and the tip matrices $\mathcal{Y}$ (plus the Hermitian conjugate, h.c.). Unit: $\mathrm{GeV}^7$. Status: INVENTED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$: The complex $3 \times 3$ tip Yukawa matrices of the neutrino, charged-lepton, up-quark and down-quark sectors (generation indices $i, j$); on the static block they become the tip mixing matrices $\mathcal{K}$ (the normalisation of that relation is not fixed: OPEN). Unit: dimensionless. Status: INVENTED.
- $\nu_R$, $\nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R$, $e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$, $\tau_R$). Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $Q_i$, $\bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

- $d_R$, $d_{Ri}$, $\bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

$$D_\mu \Sigma = \partial_\mu \Sigma - \mathrm{i} W^I_\mu \frac{\tau^I}{2} \Sigma + \mathrm{i} B_\mu \Sigma \frac{\tau^3}{2}, \qquad \Sigma^\dagger \Sigma = \mathbf{1}, \qquad \Sigma \to U_L \Sigma U_Y^\dagger, \qquad U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2} \tag{IN-14}$$

**Symbols in (IN-14).**

- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M \tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $U_L$: An SU(2) gauge transformation (a unitary $2 \times 2$ matrix acting on doublets from the left). Unit: dimensionless. Status: STANDARD.
- $U_Y$: A $\mathrm{U}(1)_Y$ gauge transformation of the sigma field, $U_Y = \mathrm{e}^{\mathrm{i}\vartheta\tau^3/2}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\vartheta$: The angle (gauge parameter) of a $\mathrm{U}(1)_Y$ transformation. Unit: dimensionless. Status: STANDARD.

and in the unitary gauge the mass term of the weak bosons:

$$-\frac{v_\Sigma^2}{4} \mathrm{tr}\big((D_\mu\Sigma)^\dagger D^\mu \Sigma\big)\Big|_{\Sigma=\mathbf{1}} = -\frac{v_\Sigma^2}{8}\Big[2\hat{g}_2^2\, W^+_\mu W^{-\mu} + \hat{g}_Z^2\, Z_\mu Z^\mu\Big] \tag{IN-15}$$

**Symbols in (IN-15).**

- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu\Sigma)$); on the static block it fixes the Robin parameter $\beta_\mathrm{W}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_\mathrm{W} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.

- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2 / V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $W_M^\pm$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^\pm = (W_M^1 \mp \mathrm{i} W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\hat{g}_Z$: The eight-dimensional coupling of the $Z$, $\hat{g}_Z = (\hat{g}_2^2 + \hat{g}_1^2)^{1/2}$. Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.

On the static block the tip term becomes the Robin condition of the $W^\pm$ and $Z$ profiles; the measured masses then fix its strength (the section on the $x_0$ mass mechanism):

$$\rho_{\mathrm{EW}} = \frac{\tilde{m}_{\mathrm{W}}^2}{\tilde{m}_{\mathrm{Z}}^2 \cos^2\theta_{\mathrm{W}}} = \rho_{\mathrm{EW},0}, \qquad \cos^2\theta_{\mathrm{W}} = \frac{m_{\mathrm{W}}^2/m_{\mathrm{Z}}^2}{\rho_{\mathrm{EW},0}}, \qquad m_{\mathrm{KK}} = m_{\mathrm{W}}\,\frac{\tilde{m}_1}{\tilde{m}_{\mathrm{W}}}, \qquad m_{\mathrm{Z}} = m_{\mathrm{W}}\,\frac{\tilde{m}_{\mathrm{Z}}}{\tilde{m}_{\mathrm{W}}} \tag{FE-55}$$

**Symbols in (FE-55).**

- $\rho_{\mathrm{EW}}$: The electroweak parameter, $\rho_{\mathrm{EW}} = \tilde{m}_{\mathrm{W}}^2/(\tilde{m}_{\mathrm{Z}}^2 \cos^2\theta_{\mathrm{W}})$. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}_{\mathrm{W}}$: The dimensionless $x_0$ eigenvalue of the lowest $W^\pm$ mode ($\beta = \beta_{\mathrm{W}}$). Unit: dimensionless. Value: $\tilde{m}_{\mathrm{W}}^2 = 1.32149 \times 10^{-5}$. Status: DERIVED.
- $\tilde{m}_{\mathrm{Z}}$: The dimensionless $x_0$ eigenvalue of the lowest $Z$ mode ($\beta = \beta_{\mathrm{Z}}$). Unit: dimensionless. Value: $\tilde{m}_{\mathrm{Z}}^2 = 1.70150 \times 10^{-5}$. Status: DERIVED.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $\rho_{\mathrm{EW},0}$: The measured value of the electroweak parameter (PDG fit). Unit: dimensionless. Value: 1.00038. Status: STANDARD.
- $m_{\mathrm{W}}$: The mass of $W^+$ and $W^-$. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $m_{\mathrm{Z}}$: The mass of the $Z$. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $m_n^{\mathrm{KK}}$, $m_{\mathrm{KK}}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

**What it couples, and why it is consistent with the electroweak model.** $\Sigma$ couples to $W^I$ and $B$ through its covariant derivative and to the fermions through the Yukawa terms. It is the gauged non-linear sigma model of $SU(2)_L \times U(1)_Y \to U(1)_{\mathrm{em}}$ (the electroweak chiral Lagrangian at lowest order), placed on the tip: every term is invariant under $\Sigma \to U_L \Sigma U_Y^\dagger$ with the matching transformations of the gauge fields and fermions (DERIVED, deviation $1.1 \times 10^{-14}$ with random group elements and random complex tip matrices); the three would-be Goldstone modes of $\Sigma$ become the longitudinal $W^+$, $W^-$ and $Z$; the unbroken combination is the photon, which receives no tip term and stays exactly massless; the custodial symmetry of the kinetic term gives $m_{\mathrm{Z}}/m_{\mathrm{W}} = 1/\cos\theta_{\mathrm{W}}$ at leading order. The difference from the electroweak model is that $\Sigma$ has no radial mode: there is no Higgs particle, as the author orders.

**Strength and bounds.** $\beta_{\mathrm{W}} = 0.011912$ reproduces both measured masses ($m_{\mathrm{W}} = 80.3625$ GeV, $m_{\mathrm{Z}} = 91.1879$ GeV) and the measured tree-level custodial violation $\rho_{\mathrm{EW},0} - 1 = 0.00038$, which comes from the bending of the profiles at the tip, and gives the Kaluza-Klein scale $m_{\mathrm{KK}} = 121.32$ TeV. A positive tip term cannot make the $W$ heavier than the Dirichlet limit, $m_{\mathrm{W}} \le 0.01824\, m_{\mathrm{KK}}$. Without a Higgs particle the scattering of longitudinal $W^+W^-$ pairs violates perturbative unitarity near $\sqrt{8\pi} \times 246$ GeV $\approx 1.2$ TeV, far below the Kaluza-Klein modes that could restore it: a CONFLICT WITH DATA inherent in the order, as is the absence of the observed Higgs boson of 125 GeV. The oblique parameter $S$ of the Higgsless tip is not computed (OPEN).

## 11.8 Fermion masses and mixing from the tip (INVENTED)

**The equations.** The Yukawa part $\mathcal{L}_{\rm Y}$ of IN-13 (written out above) places, in each fermion field equation (FE-14 to FE-19), a tip term that is equivalent to the self-adjoint tip mixing conditions, written out again:

$$\begin{aligned}
&\text{one generation:}\quad b_L(-L)=\mathcal{K}\,b_R(-L),\qquad a_R(-L)=-\mathcal{K}\,a_L(-L),\qquad \chi=(a,\ \mathrm{i}b);\\
&\text{three generations:}\quad b_L(-L)=\mathcal{K}\,b_R(-L),\qquad a_R(-L)=-\mathcal{K}^\dagger a_L(-L),\qquad \mathcal{K}\in\{\mathcal{K}_\nu,\mathcal{K}_e,\mathcal{K}_u,\mathcal{K}_d\};\\
&\text{UM-2:}\quad \mathcal{K}_e=\mathcal{K}_u=\mathbf{1},\qquad \mathcal{K}_\nu=U,\qquad \mathcal{K}_d=V_{\rm CKM}
\end{aligned}\tag{FE-30}$$

**Symbols in (FE-30).**

- $a$, $b$: The upper and lower components of a block spinor, $\chi=(a,\mathrm{i}b)$, real at $k=0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L=(1+\Gamma_4)/2$. Status: STANDARD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3\times3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e=\mathcal{K}_u=\mathbf{1}$, $\mathcal{K}_\nu=U$, $\mathcal{K}_d=V_{\rm CKM}$. Unit: dimensionless. Status: ASSUMED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R=(1-\Gamma_4)/2$. Status: STANDARD.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi=(\chi_1,\chi_2)^T=(a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y=1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha=\sum_i U_{\alpha i}\nu_i$ (rows $\alpha=e,\mu,\tau$, columns the mass states $i$); in UM-2 $U=\mathcal{K}_e^\dagger\mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12}=0.308$, $\sin^2\theta_{13}=0.02215$, $\sin^2\theta_{23}=0.470$, $\delta_{\rm CP}=212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $V_{\rm CKM}$: The CKM matrix, the unitary $3\times3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\rm CKM}=\mathcal{K}_u^\dagger\mathcal{K}_d$. Unit: dimensionless. Value: $s_{12}=0.22501$, $s_{13}=0.003732$, $s_{23}=0.04183$, $\delta=1.147$ (PDG). Status: STANDARD.

**What it couples, and why it is consistent.** Each lepton doublet with the singlets $\nu_{Rj}$ and $e_{Rj}$, each quark doublet with $u_{Rj}$ and $d_{Rj}$, through $\Sigma$, at the tip only. The terms are gauge invariant for arbitrary complex tip matrices; with one doublet bulk mass per sector and unitary tip matrices the masses are the one-generation roots of the exact $x_0$ mass condition and the mixing matrices are $U=\mathcal{K}_e^\dagger\mathcal{K}_\nu$ and $V_{\rm CKM}=\mathcal{K}_u^\dagger\mathcal{K}_d$ (DERIVED; the section on the $x_0$ mass mechanism): unitary, as in the Standard Model, with no tree-level flavour-changing neutral currents.

**Strength.** Tip matrices of order one ($\mathcal{K}_e=\mathcal{K}_u=\mathbf{1}$, $\mathcal{K}_\nu=U$, $\mathcal{K}_d=V_{\rm CKM}$) and bulk masses from $5.2H$ to $54.2H$ in magnitude give the whole fermion hierarchy, fourteen orders of magnitude from the lightest neutrino to the top quark, because the tip values of exponentially localised zero modes depend exponentially on their localisations.

## 11.9 Portal I11: the vector contact portal to the $B-L$ currents (INVENTED)

**The equation.**

$$\begin{aligned}
&S_{\rm I11}=\int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_{\rm V}}{\Lambda_\star^6}\left(\mathrm{i}\bar\Psi\gamma^M\Psi\right)J_M^{B-L},\qquad J_M^{B-L}=\sum_{i=1}^{3}\sum_{\psi}q_\psi\left(\mathrm{i}\bar\psi_i\gamma_M\psi_i\right),\\
&q_Q=q_{u_R}=q_{d_R}=\frac{1}{3},\qquad q_\ell=q_{e_R}=q_{\nu_R}=-1
\end{aligned}\tag{IN-16}$$

**Symbols in (IN-16).**

- $S_{\mathrm{I11}}$: The action of the portal I11 (dirac16complex vector current times the $B-L$ current of the fermions). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, ${g^M}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $c_{\mathrm{V}}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}}=4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi=\Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M={\mathbf{e}_a}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M=g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1,M_2,\ldots$): it runs over $x_0,x_1,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- $J^{B-L}_M$: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $i$: The generation index ($i=1,2,3$), which also labels the neutrino mass states $\nu_1,\nu_2,\nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi\in\{Q,u_R,d_R,\ell,e_R,\nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q=q_{u_R}=q_{d_R}=\frac{1}{3}$, $q_\ell=q_{e_R}=q_{\nu_R}=-1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\psi_i$, $\bar\psi_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi\in\{Q,u_R,d_R,\ell,e_R,\nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar\psi=\psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.

It enters the field equation of $\Psi$ (the term with $J^{B-L}_M$ in FE-12), the Dirac equation of every fermion (FE-14 to FE-19) and the energy-momentum tensor

$$(\mathcal{T}_{\mathrm{I11}})_{MN}=g_{MN}\,\frac{c_{\mathrm{V}}}{\Lambda_\star^6}\big(\mathrm{i}\bar\Psi\gamma^P\Psi\big)J^{B-L}_P,\qquad (\mathcal{T}_{\mathrm{I13}})_{MN}=g_{MN}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{P\neq x_0}\big(\mathrm{i}\bar\Psi\gamma^P P_b\Psi\big)J^B_P \tag{FE-9}$$

**Symbols in (FE-9).**

- $\mathcal{T}_{\mathrm{I11}}$: The energy-momentum tensor of the portal I11. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1,M_2,\ldots$): it runs over $x_0,x_1,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1,N_2,\ldots$), running over $x_0,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, ${g^M}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.

- $c_{\mathrm{V}}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $J_M^{B-L}$: The $B-L$ current of the Standard-Model fermions (baryon minus lepton number), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.
- $\mathcal{T}_{\mathrm{I13}}$: The energy-momentum tensor of the portal I13. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $P_a$, $P_b$: The projectors on the two $x_0$-chiralities, $P_a = (1+\gamma^{x_0})/2$ and $P_b = (1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- $J_M^B$: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: $\mathrm{GeV}^7$. Status: STANDARD.

On the zero modes it becomes the four-dimensional contact interaction (with I13)

$$\mathcal{L}_4 = \big(G_{\mathrm{V}}\, q_f\, \tilde I_{of} + G_{\mathrm{A}}\, q_f^B\, \tilde I_{of}^{(b)}\big)\big(\mathrm{i}\bar\Psi_o\gamma^\mu\Psi_o\big)\big(\mathrm{i}\bar\psi_f\gamma_\mu\psi_f\big),$$
$$\tilde I_{of} = \frac{1}{H}\int_{-L}^{0}\big(a_o^2+b_o^2\big)\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad \tilde I_{of}^{(b)} = \frac{1}{H}\int_{-L}^{0} b_o^2\,\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad G_{\mathrm{V}} = \frac{c_{\mathrm{V}}H}{2f^4 v_{\mathrm{t}}\Lambda_\star^6}, \qquad G_{\mathrm{A}} = \frac{c_{\mathrm{A}}H}{2f^4 v_{\mathrm{t}}\Lambda_\star^6},$$
$$n_{\psi i}(y) = \frac{n_f\,\varrho_f(y)\,\mathrm{e}^{-3Hy}}{2f^4 v_{\mathrm{t}}}, \qquad \varrho(y;c) = \frac{2c}{L}\,\frac{\mathrm{e}^{2cy/L}}{1-\mathrm{e}^{-2c}} \tag{FE-70}$$

**Symbols in (FE-70).**

- $\mathcal{L}_4$: The four-dimensional contact Lagrangian of I11 and I13 between a state $o$ of dirac16complex and a Standard-Model fermion component $f$. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\tilde I_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde I_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde I_Q$, $\tilde I_{u_R}$, $\tilde I_{d_R}$, $\tilde I_\ell$, $\tilde I_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde I_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde I_Q = 8.2722$, $\tilde I_\ell = 1.2524$, $\tilde I_{u_R} = 0.3854$, $\tilde I_{d_R} = 0.3961$, $\tilde I_{e_R} = 0.3972$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.

- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_{Q}$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_{Q} = 2.6135$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $a$, $b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $\varrho_f(y)$, $\varrho(y;c)$: The $x_0$ profile of a zero mode as a probability density on one half, $\int_{-L}^{0}\varrho\,\mathrm{d}y = 1$: $\varrho_f$ that of the fermion component $f$, $\varrho(y;c)$ its form for the localisation $c$. Unit: 1/length (GeV). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_\mathrm{t}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $c_\mathrm{A}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_\mathrm{A} = 1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value:

doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.

**What it couples, and why it is consistent.** The vector current of the whole field $\Psi$ (both halves, fold bands included) with the $B-L$ currents of the fifteen multiplets of three generations and the three right-handed neutrinos. Each current is a gauge singlet, so the term is invariant under $SU(3)_c \times SU(2)_L \times U(1)_Y$; it is generation blind (no flavour-changing coupling); and $B-L$ is free of gauge and gravitational anomalies when the right-handed neutrinos are present ($\sum(B-L)^3 = 3\cdot 2\cdot\frac{1}{27} - 3\cdot\frac{1}{27} - 3\cdot\frac{1}{27} - 2 + 1 + 1 = 0$ and $\sum(B-L) = 0$ per generation, left-handed Weyl components), so the current is conserved and could even be gauged. It is a vector coupling, so it conserves parity in four dimensions and the chirality of every fermion.

**Strength and bounds.** Its strength must be small, or it would keep the massless fold bands in equilibrium (too much dark radiation). The benchmark (ASSUMED) is fixed by their freeze-in: the plasma produces fold-band pairs at the rate (FE-129 in the section on the dark sector) that adds $\Delta N_{\text{eff}} = 5.21 \times 10^{29}\, G_{\text{V}}^2$ GeV$^4$ from $T_{\max} = 10$ TeV, so $G_{\text{V}} = 4.38 \times 10^{-17}$ GeV$^{-2}$ ($3.8 \times 10^{-12}\, G_{\text{F}}$) gives $\Delta N_{\text{eff}} = 0.001$ without ever reaching equilibrium (the production rate per fold quantum is 0.012 of the Hubble rate at $T_{\max}$). Its other effects are negligible: its mean field moves the dark-matter level by at most $7.3 \times 10^{-17}$ GeV ($1.6 \times 10^{-21}$ of its mass, at the freeze-out; $4.3 \times 10^{-19}$ GeV in a neutron-star core), contributes $3.9 \times 10^{-60}$ cm$^2$ to the direct-detection cross-section, lets the dark matter annihilate into right-handed neutrinos at only $2.0 \times 10^{-17}$ of its I13 rate into quarks, and the fold band stays massless under every I11 mean field (its first-order change of the gap is at most $3.0 \times 10^{-9}$ per unit amplitude, rounding; zero in exact arithmetic).

## 11.10 Portal I12: the gauge-kinetic scalar portal (INVENTED)

**The equation.**

$$S_{\text{I12}} = -\int \mathrm{d}^8X\, \sqrt{|g|}\, \frac{\bar\Psi\Psi}{\Lambda_\star^3}\, \mathcal{F}, \qquad \mathcal{F} = c_{\text{G}}\, G^A_{MN} G^{AMN} + c_{\text{W}}\, W^I_{MN} W^{IMN} + c_{\text{B}}\, B_{MN} B^{MN} \tag{IN-17}$$

**Symbols in (IN-17).**

- $S_{\text{I12}}$: The action of the portal I12 (scalar density of dirac16complex times the gauge kinetic invariants). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\, \mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\, \mathrm{d}X^M \mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\, \mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\text{G}} G^A_{MN} G^{AMN} + c_{\text{W}} W^I_{MN} W^{IMN} + c_{\text{B}} B_{MN} B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: GeV$^4$. Status: INVENTED.
- $c_{\text{G}}$: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\text{G}}/\Lambda_\star^3 = c_{\text{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1, \ldots, 8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), GeV$^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.

- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $c_{\mathrm{W}}$: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $c_{\mathrm{B}}$: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

It enters the field equation of $\Psi$ (the term $\mathcal{F}/\Lambda_\star^3$ in FE-12), the energy-momentum tensor

$$(\mathcal{T}_{\mathrm{I12}})_{MN} = \frac{4\bar\Psi\Psi}{\Lambda_\star^3}\Big[c_{\mathrm{G}}\Big(G^A_{MP}G^A{}_N{}^P - \frac{1}{4}g_{MN}G^A_{PQ}G^{APQ}\Big) + c_{\mathrm{W}}\Big(W^I_{MP}W^I{}_N{}^P - \frac{1}{4}g_{MN}W^I_{PQ}W^{IPQ}\Big) + c_{\mathrm{B}}\Big(B_{MP}B_N{}^P - \frac{1}{4}g_{MN}B_{PQ}B^{PQ}\Big)\Big] \tag{FE-10}$$

**Symbols in (FE-10).**

- $\mathcal{T}_{\mathrm{I12}}$: The energy-momentum tensor of the portal I12. Unit: $\mathrm{GeV}^8$. Status: DERIVED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $c_{\mathrm{G}}$: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $G_M^A$, $G_M$, $G_{MN}$, $G_{MN}^A$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G_M^A$ ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength $G_{MN}$ (matrix) and its components $G_{MN}^A$; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N = \delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $c_{\mathrm{W}}$: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.

- $c_{\mathrm{B}}$: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

and acts on the gauge fields as a field-dependent coupling (written out again):

$$\frac{1}{\hat g_{3,\mathrm{eff}}^2} = \frac{1}{\hat g_3^2} + \frac{4c_{\mathrm{G}}\bar\Psi\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat g_{2,\mathrm{eff}}^2} = \frac{1}{\hat g_2^2} + \frac{4c_{\mathrm{W}}\bar\Psi\Psi}{\Lambda_\star^3}, \qquad \frac{1}{\hat g_{1,\mathrm{eff}}^2} = \frac{1}{\hat g_1^2} + \frac{4c_{\mathrm{B}}\bar\Psi\Psi}{\Lambda_\star^3} \tag{FE-21}$$

**Symbols in (FE-21).**

- $\hat g_{3,\mathrm{eff}}$, $\hat g_{2,\mathrm{eff}}$, $\hat g_{1,\mathrm{eff}}$: The field-dependent eight-dimensional couplings produced by I12: $1/\hat g_{i,\mathrm{eff}}^2 = 1/\hat g_i^2 + 4c\,\bar\Psi\Psi/\Lambda_\star^3$ with $c = c_{\mathrm{G}}, c_{\mathrm{W}}, c_{\mathrm{B}}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\hat g_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat g_3^2$); $g_3^2 = \hat g_3^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_{\mathrm{G}}$: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $\hat g_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat g_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_{\mathrm{W}}$: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $\hat g_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat g_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $c_{\mathrm{B}}$: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.

On the dark matter it becomes the four-dimensional operator

$$\mathcal{L}_4^{\mathrm{I12}} = -\mathcal{O}_{\mathrm{DM}}\,\frac{\bar\Psi_{\mathrm{DM}}\Psi_{\mathrm{DM}}}{\Lambda_\star^3}\big(c_{\mathrm{G}}\,G^A_{\mu\nu}G^{A\mu\nu} + c_{\mathrm{W}}\,W^I_{\mu\nu}W^{I\mu\nu} + c_{\mathrm{B}}\,B_{\mu\nu}B^{\mu\nu}\big), \qquad \mathcal{O}_{\mathrm{DM}} = \frac{\partial\varepsilon_{\mathrm{DM}}}{\partial m_\Psi},$$
$$G_{\mathrm{g}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{G}}\,g_3^2}{\Lambda_\star^3}, \qquad G_{\mathrm{W}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{W}}\,g_2^2}{\Lambda_\star^3}, \qquad G_{\mathrm{B}} = \mathcal{O}_{\mathrm{DM}}\,\frac{c_{\mathrm{B}}\,g_1^2}{\Lambda_\star^3} \tag{FE-71}$$

**Symbols in (FE-71).**

- $\mathcal{L}_4^{\mathrm{I12}}$: The four-dimensional Lagrangian of I12 on the dark matter. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- $\Psi_{\mathrm{DM}}$, $\bar\Psi_{\mathrm{DM}}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $c_{\mathrm{G}}$: The coefficient of the gluon term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{G}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A = 1,\dots,8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1,\dots,8$; a repeated index is summed. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1,\dots,x_4$ (as $\mu$). Status: STANDARD.

- $c_{\mathrm{W}}$: The coefficient of the SU(2) term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{W}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $c_{\mathrm{B}}$: The coefficient of the hypercharge term of the portal I12. Unit: dimensionless. Value: $c_{\mathrm{B}}/\Lambda_\star^3 = c_{\mathrm{I12}}/\Lambda_\star^3$. Status: INVENTED.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $\varepsilon_{\mathrm{DM}}$: The energy of the dark-matter level at rest (its mass $M_{\mathrm{DM}}$ on the static block). Unit: GeV. Value: $3.102204\, m_\Psi = 44.35$ TeV. Status: DERIVED.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $G_{\mathrm{g}}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $6.09 \times 10^{-15}$ $\mathrm{GeV}^{-3}$ (benchmark). Status: DERIVED.
- $g_3$: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $G_{\mathrm{W}}$: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $2.96 \times 10^{-15}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $G_{\mathrm{B}}$: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $1.08 \times 10^{-15}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $g_1$: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.

with $\mathcal{O}_{\mathrm{DM}} = \partial\varepsilon_{\mathrm{DM}}/\partial m_\Psi = 0.7072$ (DERIVED, Hellmann-Feynmann, checked to $10^{-8}$).

**What it couples, and why it is consistent.** The scalar density $\bar\Psi\Psi$ with the three gauge invariants $G_{MN}^A G^{AMN}$, $W_{MN}^I W^{IMN}$, $B_{MN}B^{MN}$; with $W_{MN}^I W^{IMN}$ expanded in the physical fields (the gauge section) it couples $\bar\Psi\Psi$ to the pairs $W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$ and to gluon pairs. Each invariant is gauge invariant and Lorentz invariant, so I12 breaks no symmetry of QCD or of the electroweak model; it changes only the strength of the gauge couplings where $\Psi$ has a scalar density, and in the QCD vacuum the gluon condensate makes it a uniform mass shift of $\Psi$. The scalar density of every chiral zero mode vanishes identically (its lower component is zero), so **I12 does not touch the massless fold bands**; it couples the massive states, in particular the dark matter, which it lets annihilate into gauge-boson pairs in the p-wave.

**Strength and bounds.** Its benchmark strength (ASSUMED) is set by unitarity, $G_{\mathrm{I12}} = \pi/(2M_{\mathrm{DM}}^3) = (38.2\ \mathrm{TeV})^{-3}$, which keeps its annihilation cross-section below the partial-wave bound of total angular momentum 0 ($^3P_0$) up to $E_{\mathrm{cm}} = 2.40\, M_{\mathrm{DM}}$:

$$\begin{aligned}
\sigma_{\mathrm{I12}} &= \frac{G_{\mathrm{I12}}^2\, E_{\mathrm{cm}}^3 \big(E_{\mathrm{cm}}^2 - 4M_{\mathrm{DM}}^2\big)^{1/2}}{8\pi}, \qquad G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2, \\
\sigma_{\mathrm{I13}} &= \sum_f \frac{N_{\mathrm{c}}\, G_f^2\, E_{\mathrm{cm}}^2 \big(1 + 2M_{\mathrm{DM}}^2/E_{\mathrm{cm}}^2\big)}{24\pi\beta_{\mathrm{cm}}}, \qquad G_f = G_{\mathrm{A}}\, q_f^B\, \tilde I_{of}^{(b)}, \\
\beta_{\mathrm{cm}} &= \sqrt{1 - \frac{4M_{\mathrm{DM}}^2}{E_{\mathrm{cm}}^2}}, \qquad k_{\mathrm{cm}} = \sqrt{\frac{E_{\mathrm{cm}}^2}{4} - M_{\mathrm{DM}}^2}
\end{aligned} \tag{FE-127}$$

**Symbols in (FE-127).**

- $\sigma_{\mathrm{I12}}$, $\sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.

- $G_{\mathrm{I12}}$: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: $\mathrm{GeV}^{-3}$. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $E_{\mathrm{cm}}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\mathrm{g}}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $6.09 \times 10^{-15}$ $\mathrm{GeV}^{-3}$ (benchmark). Status: DERIVED.
- $G_{\mathrm{W}}$: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $2.96 \times 10^{-15}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $G_{\mathrm{B}}$: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $1.08 \times 10^{-15}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\mathrm{c}}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- $G_f$: The I13 coupling of the dark matter to the quark component $f$, $G_f = G_{\mathrm{A}} q_f^B \tilde{I}_{of}^{(b)}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\beta_{\mathrm{cm}}$: The velocity of each particle of the pair in the centre-of-mass frame. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}_{of}^{(b)} = H^{-1} \int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the overlap of I13); $\tilde{I}_{Q}^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_{Q}^{(b)} = 2.6135$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $k_{\mathrm{cm}}$: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

$$\sigma_{\mathrm{I12}} \leq \frac{\pi}{4k_{\mathrm{cm}}^2} \text{ (total angular momentum 0)}, \qquad \sigma_{\mathrm{I13}} \leq \frac{3\pi}{4k_{\mathrm{cm}}^2} \text{ (total angular momentum 1)} \tag{FE-128}$$

**Symbols in (FE-128).**

- $\sigma_{\mathrm{I12}}$, $\sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $k_{\mathrm{cm}}$: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

With universal coefficients $c_{\mathrm{G}} = c_{\mathrm{W}} = c_{\mathrm{B}} = c_{\mathrm{I12}}$ and the gauge couplings at the annihilation scale $2M_{\mathrm{DM}}$ (one loop: $g_3^2 = 0.779$, $g_2^2 = 0.378$, $g_1^2 = 0.138$) this is $c_{\mathrm{I12}}/\Lambda_\star^3 = 1.106 \times 10^{-14}$ $\mathrm{GeV}^{-3}$, i.e. $\Lambda_\star/c_{\mathrm{I12}}^{1/3} = 44.9$ TeV, and $G_{\mathrm{g}}$, $G_{\mathrm{W}}$, $G_{\mathrm{B}} = 6.09$, 2.96, $1.08 \times 10^{-15}$ $\mathrm{GeV}^{-3}$. I12 alone cannot produce the relic abundance within perturbative unitarity (it would leave $\Omega_{\mathrm{DM}} h_{100}^2 = 8.0$ per species at the benchmark; the dark sector). Its other effects: the dark-matter density shifts the gauge couplings by at most $1.7 \times 10^{-16}$ (at the freeze-out), its direct-detection cross-section through the trace anomaly is at most $6 \times 10^{-53}$ $\mathrm{cm}^2$, and the benchmark changes the strong coupling by 10 % through the thermal dark-matter density only at $T = 27$ TeV ($M_{\mathrm{DM}}/T = 1.6$): the effective description holds through the freeze-out and everything later.

## 11.11 Portal I13: the $x_0$-chiral vector portal to the baryon-number current (INVENTED)

**The equation.**

$$S_{\mathrm{I13}} = \int \mathrm{d}^8X\,\sqrt{|g|}\,\frac{c_{\mathrm{A}}}{\Lambda_\star^6}\sum_{M\neq x_0}\left(\mathrm{i}\bar\Psi\gamma^M P_b\Psi\right)J^B_M,\qquad P_b=\frac{1-\gamma^{x_0}}{2},\qquad P_a=\frac{1+\gamma^{x_0}}{2},$$
$$J^B_M=\frac{1}{3}\sum_{i=1}^{3}\left(\mathrm{i}\bar Q_i\gamma_M Q_i+\mathrm{i}\bar u_{Ri}\gamma_M u_{Ri}+\mathrm{i}\bar d_{Ri}\gamma_M d_{Ri}\right) \tag{IN-18}$$

**Symbols in (IN-18).**

- $S_{\mathrm{I13}}$: The action of the portal I13 (the $x_0$-chiral vector current of dirac16complex times the baryon-number current). Unit: dimensionless. Status: INVENTED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X=\{x_0,\ldots,x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X=\{x_0,x_1,\ldots,x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X=\mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1,\ldots,x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0,\ldots,x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^M{}_N=\delta^M_N$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}}=1.278\times10^{-9}$ GeV$^{-2}$ (dark-matter relic). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1,M_2,\ldots$): it runs over $x_0,x_1,\ldots,x_7$; a repeated index is summed. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi=\Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M=\mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M=g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $P_a$, $P_b$: The projectors on the two $x_0$-chiralities, $P_a=(1+\gamma^{x_0})/2$ and $P_b=(1-\gamma^{x_0})/2$ (in a block: the upper and the lower component). Unit: dimensionless. Status: INVENTED.
- $J^B_M$: The baryon-number current of the quarks (one third per quark), summed over the generations. Unit: GeV$^7$. Status: STANDARD.
- $i$: The generation index ($i=1,2,3$), which also labels the neutrino mass states $\nu_1,\nu_2,\nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $Q_i$, $\bar Q_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $u_R$, $u_{Ri}$, $\bar u_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar u_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.
- $d_R$, $d_{Ri}$, $\bar d_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar d_{Ri}$ its Dirac adjoint. Unit: GeV$^{7/2}$. Status: ASSUMED.

($\gamma^{x_0}$ the frame matrix; the sum runs over the seven directions $M=x_1,\ldots,x_7$). It enters the field equation of $\Psi$ (the $J^B_M$ term of FE-12), the Dirac equations of the quarks (FE-14 to FE-16) and the energy-momentum tensor FE-9 (above); on the zero modes it is the $G_{\mathrm{A}}$ term of the contact interaction FE-70 (above), and its annihilation cross-section is $\sigma_{\mathrm{I13}}$ of FE-127 (above), bounded by FE-128.

**What it couples, and why it is consistent** (DERIVED with the author's matrices, exact):

- for $M \neq x_0$ the bilinear $\mathrm{i}\bar{\Psi}\gamma^M P_b \Psi$ is Hermitian: $\gamma^{x_0}$ anticommutes with every $\gamma^M$, $M \neq x_0$, so $P_a \gamma^M = \gamma^M P_b$, and $C$ anticommutes with $\gamma^{x_0}$, so $C P_b C^{-1} = P_a$; for $M = x_0$ it is not ($\gamma^{x_0}$ commutes with $P_b$), so that component is left out (it would make the action complex; $J^B_{x_0}$ vanishes on the quark zero modes anyway);
- its components along $x_1, \ldots, x_4$ form a four-dimensional vector ($\gamma^{x_0}$ commutes with the four-dimensional Lorentz generators), and $C\gamma^M P_b$ commutes with $\mathcal{B}$ for $M = x_1, \ldots, x_4$, so I13 does not mix the halves $\Psi_A$, $\Psi_B$ in four dimensions; for the extra times it anticommutes with $\mathcal{B}$, which is harmless because $J^B_{x_5}$, $J^B_{x_6}$, $J^B_{x_7}$ vanish on the zero modes (which lie in the half $\mathcal{B} = -1$);
- in the block form of the Kohn-Sham reduction $\gamma^{x_0} = \sigma_3$, so $P_b$ projects on the lower component of an orbital; the fold bands have no lower component, so **I13 vanishes identically on the fold bands** and cannot produce dark radiation; it couples the massive states, in particular the dark matter (the lowest odd level, whose lower component is non-zero);
- $J^B$ is the baryon-number current, a singlet of $SU(3)_c \times SU(2)_L \times U(1)_Y$ term by term and conserved in perturbation theory, so I13 is gauge invariant and respects the symmetries of QCD and of the electroweak model; it is a vector coupling to quarks only.

**Why the baryon-number current** (DERIVED; the elastic rate is an ESTIMATE). With the $B - L$ current instead, the annihilation into right-handed neutrino pairs would bring the three right-handed neutrinos into equilibrium above the freeze-out (they reach 0.9997 of the equilibrium number by $M_{\mathrm{DM}}/T = 40$) and they would decouple as three Weyl species adding $\Delta N_{\mathrm{eff}} = 0.144$ ($N_{\mathrm{eff}} = 3.19$, 2.5 standard deviations above the ACT DR6 value $2.86 \pm 0.13$). With $J^B$ the right-handed neutrinos keep only their tip coupling and their I11 coupling and are never thermalised (the dark sector).

**Strength and bounds.** The strength is fixed by the dark-matter relic: each species freezes out with $\Omega_{\mathrm{DM}} h^2_{100} = 0.060$ for $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV$^{-2}$ ($1.1 \times 10^{-4}\, G_{\mathrm{F}}$), the s-wave cross-section capped at its unitarity bound of total angular momentum 1, which it reaches only above the relative velocity 0.51 (22 % of the freeze-out rate comes from collisions at the bound; without the cap $G_{\mathrm{A}} = 1.269 \times 10^{-9}$ GeV$^{-2}$). It predicts the direct-detection cross-section $4.0 \times 10^{-46}$ cm$^2$ per nucleon on xenon, 0.16 of the LZ limit extrapolated to 44.35 TeV (testable), and an s-wave annihilation into quark pairs today with $\sigma v_{\mathrm{rel}} = 9.8 \times 10^{-26}$ cm$^3$/s (the thermal value of a heavy relic). The contact form is an effective description: for the quark doublet $(G_{\mathrm{A}} q^B_Q \tilde{I}^{(b)}_Q)^{-1/2} = 30$ TeV lies below the energies $E_{\mathrm{cm}} \approx 89$ TeV of the annihilations, so a mediator at or below about 90 TeV must complete it (OPEN). The physics accuracy of $G_{\mathrm{A}}$ is about 1 %: the ideal-gas thermal history leaves out the order-$\alpha$ interaction pressure of the plasma ($-5.4$ % at the freeze-out), which would raise $\Omega_{\mathrm{DM}} h^2_{100}$ by about 2.7 % and $G_{\mathrm{A}}$ by about 1.4 % (ESTIMATE; residual analysis).

## 11.12 The radion couplings

**The equation.** A change of the block scale, $f \to f(1 + \sigma_{\mathrm{r}})$, changes every mass of a species $s$ by

$$\Delta m_s = \frac{\mathrm{d} \ln m_s}{\mathrm{d} \ln f}\, m_s\, \sigma_{\mathrm{r}},$$

**Symbols in this equation.**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $s$: The label of a particle species of the plasma or of matter (photon, gluons, $W^{\pm}$, $Z$, quarks, leptons, nucleons, dark matter); a sum over $s$ runs over the species present. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- $\sigma_{\mathrm{r}}$: The radion perturbation: the first-order change of the block scale, $f^2 \to f^2(1+2\sigma_{\mathrm{r}})$. Unit: dimensionless. Value: at most $1.1 \times 10^{-34}$ (static response). Status: DERIVED.

so the radion couples to every massive field with the strength of its block sensitivity ($-1$ for the Kaluza-Klein masses, $-0.501$ for $m_{\mathrm{W}}$, $-0.772$ for $M_{\mathrm{DM}}$, $-6.8$ to $-41$ for the fermions, $-46.1$ for the nucleon; the section on dimensional reduction). **Why it is consistent.** It is a gravitational (metric) mode: its couplings follow from general covariance and are blind to the gauge charges. **Strength and bounds.** Its mass is $\omega_{\mathrm{r}} = 48.42$ TeV (range $4.08 \times 10^{-21}$ m), its static response to matter is at most $\Delta \ln f = 1.1 \times 10^{-34}$, and the oscillation left over from the deflation decays with the Planck-suppressed rate of the last section (lifetime $0.43\,\mathrm{s}/C_{\mathrm{r}}$; the moduli problem, OPEN).

## 11.13 Removed in version 2: the chiral tip $Z$ portal

Version 1 contained a fourth portal on the tip,

$$S_{\mathrm{I9}} = \int_{y=-L} \mathrm{d}^7 X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|}\, \big(c_{\mathrm{Z}} + \tilde{c}_{\mathrm{Z}} \varsigma_{\mathrm{t}}\big) \big(\mathrm{i}\bar{\Psi}\gamma^{\mu} P_L \Psi\big)\, \frac{\mathrm{i} v_\Sigma^2}{2}\, \mathrm{tr}\big(\Sigma^{\dagger} D_{\mu} \Sigma\, \tau^3\big),$$

**Symbols in this equation.**

- $S_{\mathrm{I9}}$: The action of the chiral tip $Z$ portal I9 of version 1, which coupled the four-dimensional left-handed current of dirac16complex to the $Z$ part of the current of the tip sigma field; removed from UM-2 in version 2 (its coefficients are set to zero). Unit: dimensionless (an action, $\hbar = 1$). Status: INVENTED (removed).
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X_{\mathrm{tip}}$: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- $g_{\mathrm{tip}}$: The metric induced on the seven-dimensional tip $y = -L$ (the metric $g_{MN}$ restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $c_{\mathrm{Z}}$, $\tilde{c}_{\mathrm{Z}}$: The coefficients of the removed portal I9: $c_{\mathrm{Z}}$ of its part that is even under the fold reflection, $\tilde{c}_{\mathrm{Z}}$ of its oriented (odd) part; UM-2 sets both to zero. Unit: $\mathrm{GeV}^{-6}$. Value: 0. Status: ASSUMED.
- $\varsigma_{\mathrm{t}}$: The sign ($+1$ or $-1$) that distinguishes the two sides of a tip (the tips at $z \to 0^+$ and $z \to \pi^-$ of neighbouring periods), used only in the oriented part of the removed portal I9. Unit: dimensionless. Status: INVENTED (removed).
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^{\dagger} C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^{\mu}$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^{\mu}$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $P_L$, $P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $v_\Sigma$: The scale of the tip sigma field (its kinetic term is $-\frac{v_\Sigma^2}{4}\mathrm{tr}((D_\mu\Sigma)^\dagger D^\mu \Sigma)$); on the static block it fixes the Robin parameter $\beta_{\mathrm{W}}$. Unit: $\mathrm{GeV}^{5/2}$. Value: fixed through $\beta_{\mathrm{W}} = 0.011912$. Status: INVENTED.
- tr: The trace of a matrix: the sum of its diagonal entries. Status: STANDARD.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger\Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless.

Status: INVENTED.

- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $D_M,\ D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $\tau^I$: The Pauli matrices acting on weak-isospin doublets ($\tau^I/2$ the generators of SU(2); $\tau^3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.

whose even part $c_\mathrm{Z}$ made the $\Psi_A$ fold band (two Weyl species) couple to the $Z$ like two neutrinos and so stand in for the missing $\nu_\mu$, $\nu_\tau$ of one generation. Its algebra is DERIVED and remains true: $P_L$ is a Hermitian projector commuting with the four-dimensional Lorentz generators and with $C\gamma^\mu$; $\Gamma_4 = -\mathcal{B}\gamma^{x_0}$, so $P_L = (1-\mathcal{B}\gamma^{x_0})/2$, which is $(1+\gamma^{x_0})/2$ on $\Psi_A$ and $(1-\gamma^{x_0})/2$ on $\Psi_B$ and annihilates $\Psi_B$ at the tip (every orbital satisfies $(1-\gamma^{x_0})\Psi = 0$ there): **no $P_L$ tip current and no tip operator built on $\bar\Psi\Psi$ reaches $\Psi_B$.** With three neutrinos the portal has no role left, and both of its parts are bounded or disfavoured by the data (DERIVED): its even part gives the $Z$ an invisible width $N_\nu = 3 + 2r_\mathrm{Z}^2$, bounded by the measured $N_\nu = 2.9963 \pm 0.0074$ to $r_\mathrm{Z}^2 < 0.0056$; even at that bound the fold band would stay in equilibrium down to 14.7 MeV and add, as a species of $N_\mathrm{W}$ Weyl fermions that decouples early,

$$\Delta N_\mathrm{eff} = N_\mathrm{W}\Big(\frac{10.75}{g_s}\Big)^{4/3} F_\mathrm{h}, \qquad F_\mathrm{h} = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/\big(1+h_\nu\big)\Big]^4 = 1.0095 \tag{FE-133}$$

**Symbols in (FE-133).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_\mathrm{eff}$ an addition to $N_\mathrm{eff}$, $\Delta S_\mathrm{m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $N_\mathrm{eff}$: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); $\Delta N_\mathrm{eff}$ an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $N_\mathrm{W}$: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.
- $g_\rho,\ g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $F_\mathrm{h}$: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $h_\nu$: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

($N_\mathrm{W} = 2$, $g_s = 10.75$), i.e. $\Delta N_\mathrm{eff} = 2.02$, far above the measured $N_\mathrm{eff} = 2.99 \pm 0.17$ (Planck 2018) and $2.86 \pm 0.13$ (ACT DR6) (STANDARD); its oriented part $\tilde c_\mathrm{Z}$ (odd under the fold reflection) would let the heavy odd $\Psi_A$ states decay into the $\Psi_A$ fold band and a $Z$, and by the inverse decays and the exchange process keep the fold band in equilibrium until the odd states or the $Z$ bosons disappear (at $g_s = 105.75$ or 86.25), adding $\Delta N_\mathrm{eff} = 0.096$ to 0.126 (ESTIMATE), i.e. $N_\mathrm{eff} = 3.14$ to 3.17, 2.1 to 2.4 standard deviations above the ACT DR6 value (disfavoured). UM-2 therefore sets $c_\mathrm{Z} = \tilde c_\mathrm{Z} = 0$ (ASSUMED). Consequences: $N_\nu = 3$ exactly; neither fold band is ever thermalised (no tip operator, zero scalar density for I12, the weak benchmark of I11, the bound on the self-interaction); both halves keep an exact fold parity, so the lightest odd states of $\Psi_A$ and of $\Psi_B$ are both stable.

**Not included, and why.** A dipole portal $\bar\Psi\mathcal{S}^{MN}\Psi B_{MN}$ would couple the fold bands to the photon and thermalise them (too much dark radiation). No gauge-invariant operator couples the neutral $\Psi$ to a single $W^+$ or $W^-$; $W^\pm$ enter only in pairs (I12) or with a charged fermion pair (the covariant derivatives of the fermions in I11).

## 11.14 Consistency of the new interactions with QCD and the electroweak model: summary

- **Gauge invariance.** Every term of the action is invariant under $SU(3)_c \times SU(2)_L \times U(1)_Y$: $\Psi$ is a gauge singlet and the portals couple it to gauge-invariant composites (the $B - L$ and baryon-number currents, the field-strength invariants); the tip terms are invariant under $\Sigma \to U_L \Sigma U_Y^\dagger$ for arbitrary complex tip matrices (DERIVED, $1.1 \times 10^{-14}$).
- **Anomalies.** The gauge anomalies of the fermions cancel generation by generation (exact fractions, the fermion section); the currents of I11 ($B - L$, with the right-handed neutrinos) and I13 ($B$, perturbatively) are conserved.
- **Electroweak symmetry breaking without a Higgs particle.** The breaking is carried by the tip sigma field, a non-linear realisation without a radial mode: the $W^\pm$ and $Z$ masses and the custodial ratio are those of the electroweak model at tree level, the photon stays massless, and the price is the loss of perturbative unitarity of $W^+W^-$ scattering near 1.2 TeV and the absence of the observed Higgs boson (CONFLICTS WITH DATA, inherent in the order).
- **QCD.** Colour is never broken; the gluons are exactly massless; the portals couple only to colour singlets.
- **Flavour.** The shared doublet bulk mass makes the neutral currents flavour diagonal (no tree-level flavour-changing neutral currents); the PMNS and CKM matrices are unitary tip matrices.
- **Validity.** The portals are contact operators of an effective description valid below their scale; I13 needs a mediator at or below about 90 TeV (OPEN).

## 11.15 The catalogue of interactions

| # | interaction | fields | equation | role in UM-2 |
|---|---|---|---|---|
| 1 | Einstein-Lovelock self-interaction | $g_{MN}$ | IN-2, IN-3, FE-1 | background, block dynamics, four-dimensional Newton constant $\propto 1/\mathcal{G}(K_*)$ |
| 2 | gravity of every field | $g_{MN}$ with all | IN-7 ($\sqrt{\lvert g\rvert}$, $g^{MN}$, $\Omega_M$ in each action), FE-13 | universal coupling; $\Psi$ through the canonical spin connection |
| 3 | block stabiliser | $g_{MN}$ | IN-4 | static, stable block with attractive four-dimensional gravity |
| 4 | $\Psi$ self-interaction | $\Psi$ | IN-5 ($-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$) | Kohn-Sham mean fields; bounded by the fold-band radiation |
| 5 | strong interaction | gluons, the six quarks | IN-8, IN-9, IN-11, IN-12, FE-20; $-G^A_M J_3^{AM}$ | confinement, protons and neutrons |
| 6 | electromagnetic interaction | photon, charged fermions, $W^\pm$ | IN-10, IN-11, IN-12, FE-25; $-eQ_{\rm em}A_\mu\bar{\psi}\gamma^\mu\psi$; $W^+W^-\gamma$ | CMB photons, recombination, Thomson drag |
| 7 | weak charged current | $W^\pm$, three lepton and quark generations | $-\frac{g_2}{\sqrt{2}}W^+_\mu(\dots) + \text{h.c.}$ with $U$, $V_{\rm CKM}$ | beta decay, neutron-proton conversion, neutrinos produced and detected in flavour states |

| # | interaction | fields | equation | role in UM-2 |
|---|---|---|---|---|
| 8 | weak neutral current | $Z$, all fermions | $-\frac{g_2}{\cos\theta_{\mathrm{W}}} Z_\mu \bar{\psi}\gamma^\mu(\tilde{g}_L P_L + \tilde{g}_R P_R)\psi$ | neutrino decoupling, $Z$ widths |
| 9 | gauge self-interactions | $W^\pm$, $Z$, $\gamma$, gluons | IN-9 (commutators) | electroweak and QCD consistency |
| 10 | $W^\pm$, $Z$ masses | $\Sigma$, $W^I$, $B$ | IN-13, IN-14, IN-15, FE-55 | $x_0$ profiles bent at the tip |
| 11 | fermion masses and mixing | $\Sigma$, $\ell_i$, $\nu_{Rj}$, $e_{Rj}$, $Q_i$, $u_{Rj}$, $d_{Rj}$ | IN-13 ($\mathcal{L}_{\mathrm{Y}}$), FE-30 | tip mixing of the chiral zero modes: masses, PMNS, CKM |
| 12 | low-energy weak interactions, forward scattering | $\nu$ with electrons, nucleons, $\nu$ | FE-101, FE-102, FE-103 | MSW conversion, early-Universe potentials, fast flavour instability |
| 13 | $\Psi$ with the fermion currents | $\Psi$, all fermions | IN-16 (I11), FE-70 | freeze-in of the fold bands ($\Delta N_{\mathrm{eff}} = 0.001$); Kohn-Sham mean fields |
| 14 | $\Psi$ with gauge pairs | $\Psi$, $W^+W^-$, $ZZ$, $Z\gamma$, $\gamma\gamma$, gluon pairs | IN-17 (I12), FE-21, FE-71 | p-wave annihilation (subdominant); condensate mass shifts; field-dependent couplings |
| 15 | radion couplings | $\sigma_{\mathrm{r}}$ with all massive fields | $\Delta m_s = (\mathrm{d}\ln m_s/\mathrm{d}\ln f)\, m_s\, \sigma_{\mathrm{r}}$ | the masses track the block size |
| 16 | $\Psi$ ($x_0$-chiral) with the baryon current | massive states of $\Psi$, the six quarks | IN-18 (I13), FE-70 | s-wave annihilation: the dark-matter relic; direct detection |

# 12 The $x_0$ mass mechanism

The author's requirement (order of 2026-10-18): "Mass will be given to all non-massless particles (except the massless particles: graviton, photon and perhaps the quanta of the dirac16complex field), by appropriate wave function dependence on the space coordinate x0 (either explicit, derived, or assumed), which must be periodic in x0 with period pi/(6H)." This section presents the mechanism in three steps: the author's uncertainty argument, written as equations; the exact $x_0$ eigenvalue equations of UM-2, which give every mass, the mixing matrices and the massless zero modes; and precisely what the uncertainty relation does and does not bound.

## 12.1 The author's argument: localisation in $x_0$ and the uncertainty relation (AUTHOR)

**Step 1: every wave function is periodic in $x_0$.** Every field of UM-2 depends on $x_0$ through $z = 6Hx_0$ and $y(z)$ (the section on space and coordinates): the profile $F$ of a boson and the block spinor $\chi$ of a fermion are functions of $y$, so

$$F\big(y(6Hx_0+\pi)\big) = F\big(y(6Hx_0)\big), \qquad \chi\big(y(6Hx_0+\pi)\big) = \chi\big(y(6Hx_0)\big)\,:$$

**Symbols in this equation.**

- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.

every wave function is periodic in $x_0$ with the period $\pi/(6H)$.

**Step 2: localisation and the uncertainty relation.** A wave function localised in $x_0$ (well inside one period) with the spread $\Delta x_0$ has a spread $\Delta k_{x_0}$ of the momentum along $x_0$ (the author's $(p)_{x_0}$; in UM-2 the letter $p$ is a pressure and $k$ a momentum) that obeys the Heisenberg uncertainty relation

$$\Delta x_0 \, \Delta k_{x_0} \ge \frac{\hbar}{2}, \qquad \big(\Delta x_0\big)^2 = \langle x_0^2 \rangle - \langle x_0 \rangle^2, \qquad \big(\Delta k_{x_0}\big)^2 = \langle k_{x_0}^2 \rangle - \langle k_{x_0} \rangle^2, \qquad k_{x_0} = -\mathrm{i}\hbar \, \frac{\partial}{\partial x_0}$$

**Symbols in this equation.**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_\mathrm{eff}$ an addition to $N_\mathrm{eff}$, $\Delta S_\mathrm{m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0 \, \Delta p_{x_0} \ge \hbar/2$). Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $k_{x_0}$: The component of the momentum along the hidden coordinate $x_0$, the momentum conjugate to $x_0$ (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the $x_0$ dependence. Unit: GeV. Status: AUTHOR.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.

($\langle \cdots \rangle$ the expectation value in the wave function; here $\Delta$ is the spread, the root-mean-square deviation).

**Step 3: momentum is proportional to mass.** The momentum along the hidden direction is invisible to a four-dimensional observer; its energy is rest energy. For a mode of bulk mass $M$ (the mass function of the field; zero for the gauge fields and the graviton) in a flat hidden direction the dispersion relation of eight dimensions, seen at rest in four dimensions ($\vec{k} = 0$), gives the effective mass

$$E^2 = |\vec{k}|^2 + k_{x_0}^2 + M^2 \qquad \Rightarrow \qquad m_\mathrm{eff}^2 = \langle k_{x_0}^2 \rangle + M^2 \,:$$

**Symbols in this equation.**

- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_\mathrm{lat}\vec{n}_o$. Unit: GeV. Status: STANDARD.

- $k_{x_0}$: The component of the momentum along the hidden coordinate $x_0$, the momentum conjugate to $x_0$ (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the $x_0$ dependence. Unit: GeV. Status: AUTHOR.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_{\text{eff}}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $m_{\text{eff}}$: The effective mass of the author's argument: the mass that a particle receives from the $x_0$ dependence of its wave function, estimated by taking the momentum $k_{x_0}$ proportional to the mass and the kinetic energy proportional to the momentum squared over the mass. Unit: GeV. Status: AUTHOR.

without a bulk mass the hidden momentum is the mass.

**Step 4: the kinetic energy is proportional to the momentum squared over the mass.** For a hidden momentum small against the bulk mass the hidden kinetic energy adds to the rest energy:

$$m_{\text{eff}} \simeq M + E_{\text{kin}}, \qquad E_{\text{kin}} = \frac{\langle k_{x_0}^2 \rangle}{2M} \qquad \big(\langle k_{x_0}^2 \rangle \ll M^2\big).$$

**Symbols in this equation.**

- $m_{\text{eff}}$: The effective mass of the author's argument: the mass that a particle receives from the $x_0$ dependence of its wave function, estimated by taking the momentum $k_{x_0}$ proportional to the mass and the kinetic energy proportional to the momentum squared over the mass. Unit: GeV. Status: AUTHOR.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_{\text{eff}}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $E_{\text{kin}}$: The kinetic energy of the motion along $x_0$ in the author's argument, proportional to $k_{x_0}^2/m_{\text{eff}}$. Unit: GeV. Status: AUTHOR.
- $k_{x_0}$: The component of the momentum along the hidden coordinate $x_0$, the momentum conjugate to $x_0$ (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the $x_0$ dependence. Unit: GeV. Status: AUTHOR.

**Step 5: the bound.** Since $\langle k_{x_0}^2 \rangle \geq (\Delta k_{x_0})^2$, a localised wave function receives an effective mass

$$\langle k_{x_0}^2 \rangle \geq \big(\Delta k_{x_0}\big)^2 \geq \frac{\hbar^2}{4\big(\Delta x_0\big)^2} \qquad \Rightarrow \qquad m_{\text{eff}} \geq \frac{\hbar}{2\,\Delta x_0} \qquad (M = 0).$$

**Symbols in this equation.**

- $k_{x_0}$: The component of the momentum along the hidden coordinate $x_0$, the momentum conjugate to $x_0$ (the author's $(p)_{x_0}$); since $\hbar = 1$ it is also the wave number of the $x_0$ dependence. Unit: GeV. Status: AUTHOR.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\text{eff}}$ an addition to $N_{\text{eff}}$, $\Delta S_{\text{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $m_{\text{eff}}$: The effective mass of the author's argument: the mass that a particle receives from the $x_0$ dependence of its wave function, estimated by taking the momentum $k_{x_0}$ proportional to the mass and the kinetic energy proportional to the momentum squared over the mass. Unit: GeV. Status: AUTHOR.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_{\text{eff}}$ for dirac16complex. Unit: GeV. Status: DERIVED.

Only the proper hidden length $fy$ is free of the choice of chart (below), so the author's estimate of a mass is written with the root-mean-square proper distance $\Delta(fy)$ of the state from the fold, computed with the probability density $\varrho_\xi$ of its $x_0$ profile (in the flat measure; symmetric about the fold over the full period):

$$m_{\text{Heis}} = \frac{\hbar}{2\,\Delta(fy)}, \qquad \Delta(fy) = fL\,\Delta\xi, \qquad \big(\Delta\xi\big)^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \,\mathrm{d}\xi.$$

**Symbols in this equation.**

- $m_{\mathrm{Heis}}$: The author's uncertainty (Heisenberg) estimate of the mass of a state, $m_{\mathrm{Heis}} = \hbar/(2\Delta(fy))$, where $\Delta(fy) = fL\,\Delta\xi$ is the root-mean-square proper distance of the state from the fold ($\Delta\xi^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \,\mathrm{d}\xi$). It is the lower bound on the mass that the uncertainty relation would give if the hidden kinetic energy were the whole mass and the state were negligible at the tip. (The subscript stands for Heisenberg; UM-2 has no Higgs particle.) Unit: GeV. Value: 70.56 TeV (photon, gluons, graviton, $W^{\pm}$, $Z$); 53.80 TeV (quark-doublet zero mode); 93.08 TeV (lepton-doublet zero mode); 22.90 TeV (fold band of dirac16complex); 19.39 TeV (dark matter); at least $1/(2fL) = 11.05$ TeV for every profile on the cell. Status: DERIVED.
- $\hbar$: The reduced Planck constant, which sets the scale of the uncertainty relation; it equals 1 in the natural units of this document, and $\hbar c = 1.973269804 \times 10^{-16}$ GeV m converts an inverse mass into a length. Unit: GeV s (equal to 1 in natural units). Value: 1 (natural units). Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \leq y \leq 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\varrho_\xi$: The probability density in $\xi$ of the $x_0$ profile of a mode on one half of the period, normalised as $\int_{-1}^{0} \varrho_\xi \,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 / \int_{-1}^{0} \tilde{F}^2 \mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2} F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi / \int_{-1}^{0} \chi^\dagger\chi \,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.

**Step 6: the exceptions.** A wave function whose $x_0$ dependence costs no energy is massless even when it is localised: the graviton and the photon (and the gluons, whose colour symmetry is never broken), whose flat profile is an exact eigenfunction with the eigenvalue zero, protected by four-dimensional general covariance and gauge invariance; and the fold bands of dirac16complex, chiral zero modes that are massless for every mass profile. This is the author's list of exceptions ("graviton, photon and perhaps the quanta of the dirac16complex field"), made precise by the exact equations: the gluons belong to it, and of the quanta of dirac16complex the fold bands are massless while the bulk band (the dark matter) is massive. For these massless states the estimate $m_{\mathrm{Heis}}$ is NOT the mass (70.6 TeV for the photon): the argument is a heuristic, and the exact $x_0$ equations below decide.

**Proper versus coordinate localisation (DERIVED).** The photon zero mode, localised at the fold in proper length, has 61.9 % of its probability within $\pi/8$ of the tips in the coordinate $z = 6Hx_0$: localisation in the coordinate $x_0$ depends on the chart, because the proper hidden length $f|\cot z|\,\mathrm{d}x_0$ vanishes at the fold and diverges at the tips. A profile confined to a small proper distance from the fold ($\Delta\xi = 0.157$) extends over almost the whole coordinate range of $z$. Only the proper hidden length enters the masses; "$\Delta x_0$ small" must therefore be read as "$\Delta(fy)$ small".

## 12.2 The exact $x_0$ eigenvalue equations and what they give

Throughout this section the block is static ($f = f_*$), $\xi = y/L \in [-1, 0]$ on each half of a period, and the eigenvalues are dimensionless in units of the proper half cell: $\tilde{m} = m\,fL$ for bosons and $\tilde{\varepsilon} = \varepsilon\,fL$ for fermions.

### 12.2.1 Gauge fields and the graviton (DERIVED)

A four-dimensional component of a gauge field with the profile $F$ (gauge $A_y = 0$) has a reduced action per half whose stationary points are the eigenfunctions of the $x_0$ operator; the transverse-traceless graviton obeys the same operator in Einstein gravity:

$$\int_{-L}^{0}\mathrm{d}y\, f^4\mathrm{e}^{3Hy}\Big[\frac{(\partial_y F)^2}{f^2}-m^2F^2\Big]\ \text{stationary}: \qquad -\frac{1}{f^2}\,\mathrm{e}^{-3Hy}\,\partial_y\big(\mathrm{e}^{3Hy}\,\partial_y F\big)=m^2F,$$
$$\Leftrightarrow \qquad -\mathrm{e}^{-3HL\xi}\,\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\mathrm{e}^{3HL\xi}\,\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)=\tilde{m}^2F, \qquad \tilde{m}=m\,fL \tag{FE-50}$$

**Symbols in (FE-50).**

- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi=y/L$): of a gauge field (gauge $A_y=0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0=1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m}=m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta=0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2=30.1196$, $\tilde{m}_2^2=59.7284$, $\tilde{m}_3^2=109.0764$ ($HL=3$), i.e. $\tilde{m}_1=5.48813$, $\tilde{m}_2=7.72842$, $\tilde{m}_3=10.44397$. Status: DERIVED.

with the fold and tip conditions (the tip condition is the natural boundary condition of the tip mass term of IN-15)

$$\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=0}=0\ \ \text{(even)}\quad\text{or}\quad F(0)=0\ \ \text{(odd)}, \qquad \frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=-1}=\beta\,F(-1),$$
$$\beta=0\ \ \text{(photon, gluons, graviton)}, \qquad \beta=\beta_\mathrm{W}\ \ (W^\pm), \qquad \beta=\beta_\mathrm{Z}=\frac{\beta_\mathrm{W}}{\cos^2\theta_\mathrm{W}}\ \ (Z) \tag{FE-51}$$

**Symbols in (FE-51).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi=y/L$): of a gauge field (gauge $A_y=0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0=1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\beta$: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi=\beta F$ at $\xi=-1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- $\beta_\mathrm{W}$: The Robin parameter of $W^+$ and $W^-$ (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.

- $W_M^{\pm}$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV$^3$. Status: STANDARD.
- $\beta_{\mathrm{Z}}$: The Robin parameter of the $Z$, $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$. Unit: dimensionless. Value: 0.015343. Status: DERIVED.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.

With the rescaled profile the solutions are elementary and the eigenvalue conditions are

$$F = \mathrm{e}^{-3HL\xi/2}\,\tilde{F}, \qquad \varpi^2 = \tilde{m}^2 - \frac{9H^2L^2}{4}:$$
$$\text{even: } \Big[\varpi^2 + \frac{3HL}{2}\Big(\beta + \frac{3HL}{2}\Big)\Big]\sin\varpi - \beta\,\varpi\cos\varpi = 0, \qquad \text{odd: } \varpi\cos\varpi + \Big(\beta + \frac{3HL}{2}\Big)\sin\varpi = 0; \qquad \text{(FE-52)}$$
$$\beta = 0: \quad \tilde{m} = 0 \text{ (flat profile)}, \qquad \tilde{m}_n^2 = \frac{9H^2L^2}{4} + \varpi_n^2 \text{ (even)}, \qquad \varpi\cot\varpi = -\frac{3HL}{2} \text{ (odd)}$$

**Symbols in (FE-52).**

- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- $\varpi$, $\varpi_n$: The wave number of an oscillating $x_0$ profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); $\varpi_n$ that of the $n$-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- $\beta$: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.

($\varpi = \mathrm{i}|\varpi|$ below the gap $9H^2L^2/4$).

**Massless particles.** For $\beta = 0$ the flat profile \$F = \$ const is an exact eigenfunction with $\tilde{m} = 0$: the **graviton, the photon and the gluons are exactly massless**; their masslessness is protected by four-dimensional general covariance and gauge invariance. The excited levels are $\tilde{m}_n^2 = 9H^2L^2/4 + n^2\pi^2$ (even, $n \ge 1$) and the roots of $\varpi\cot\varpi = -3HL/2$ (odd); for $HL = 3$: $\tilde{m}_1^2 = 30.1196$, the first odd level 27.0890, the gap 20.25 (DERIVED, finite elements agree to $5 \times 10^{-10}$). The Kaluza-Klein scale is $m_{\mathrm{KK}} = \tilde{m}_1/(fL)$.

**$W^{\pm}$ and $Z$.** The tip term gives $\beta_{\mathrm{W}}$ for $W^+$ and $W^-$ (one value for both) and, by custodial symmetry, $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$, with $\theta_{\mathrm{W}}$ the angle of the gauge couplings. At small $\beta$

$$\tilde{m}^2 \simeq \beta\,\frac{3HL\,\mathrm{e}^{-3HL}}{1 - \mathrm{e}^{-3HL}} \qquad (\beta \ll 1),$$

**Symbols in this equation.**

- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- $\beta$: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

and $m_\mathrm{Z}/m_\mathrm{W} = 1/\cos\theta_\mathrm{W}$. Beyond leading order the bending of the profiles moves the mass ratio $m_\mathrm{W}^2/m_\mathrm{Z}^2 = \tilde{m}_\mathrm{W}^2/\tilde{m}_\mathrm{Z}^2$ away from $\cos^2\theta_\mathrm{W}$: the electroweak parameter exceeds 1. This tree-level custodial violation is what the measured $\rho_{\mathrm{EW},0} - 1 = 0.00038$ constrains, and the fit is (written out again)

$$\rho_\mathrm{EW} = \frac{\tilde{m}_\mathrm{W}^2}{\tilde{m}_\mathrm{Z}^2\cos^2\theta_\mathrm{W}} = \rho_{\mathrm{EW},0}, \qquad \cos^2\theta_\mathrm{W} = \frac{m_\mathrm{W}^2/m_\mathrm{Z}^2}{\rho_{\mathrm{EW},0}}, \qquad m_\mathrm{KK} = m_\mathrm{W}\,\frac{\tilde{m}_1}{\tilde{m}_\mathrm{W}}, \qquad m_\mathrm{Z} = m_\mathrm{W}\,\frac{\tilde{m}_\mathrm{Z}}{\tilde{m}_\mathrm{W}} \tag{FE-55}$$

**Symbols in (FE-55).**

- $\rho_\mathrm{EW}$: The electroweak parameter, $\rho_\mathrm{EW} = \tilde{m}_\mathrm{W}^2/(\tilde{m}_\mathrm{Z}^2\cos^2\theta_\mathrm{W})$. Unit: dimensionless. Status: DERIVED.
- $\tilde{m}_\mathrm{W}$: The dimensionless $x_0$ eigenvalue of the lowest $W^\pm$ mode ($\beta = \beta_\mathrm{W}$). Unit: dimensionless. Value: $\tilde{m}_\mathrm{W}^2 = 1.32149 \times 10^{-5}$. Status: DERIVED.
- $\tilde{m}_\mathrm{Z}$: The dimensionless $x_0$ eigenvalue of the lowest $Z$ mode ($\beta = \beta_\mathrm{Z}$). Unit: dimensionless. Value: $\tilde{m}_\mathrm{Z}^2 = 1.70150 \times 10^{-5}$. Status: DERIVED.
- $\theta_\mathrm{W}$: The weak mixing angle of the couplings, $\cos\theta_\mathrm{W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_\mathrm{W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_\mathrm{W} = 0.776368$. Status: DERIVED.
- $\rho_{\mathrm{EW},0}$: The measured value of the electroweak parameter (PDG fit). Unit: dimensionless. Value: 1.00038. Status: STANDARD.
- $m_\mathrm{W}$: The mass of $W^+$ and $W^-$. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $m_\mathrm{Z}$: The mass of the $Z$. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $m_n^\mathrm{KK}$, $m_\mathrm{KK}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.

With the measured masses (STANDARD: $m_\mathrm{W} = 80.3625$ GeV, $m_\mathrm{Z} = 91.1879$ GeV, PDG 2026) the angle of the couplings is $\cos^2\theta_\mathrm{W} = 0.776368$ (the on-shell value $m_\mathrm{W}^2/m_\mathrm{Z}^2$ is 0.776663), and the fit gives (DERIVED) $\beta_\mathrm{W} = 0.011912$, $\tilde{m}_\mathrm{W}^2 = 1.32149 \times 10^{-5}$, $m_\mathrm{KK} = 121.32$ TeV, with both masses reproduced ($m_\mathrm{Z} = 91.1879$ GeV). A positive tip term cannot make the $W$ heavier than the Dirichlet limit ($F(-1) = 0$, $\tilde{m}^2 = 0.010016$), i.e. $m_\mathrm{W} \le 0.01824\,m_\mathrm{KK}$. The angle of the couplings of this fit, $\sin^2\theta_\mathrm{W} = 0.223632$, differs from the $\overline{\mathrm{MS}}$ value 0.23122 used in the couplings to the fermions (the residual analysis: two values of one angle, of the size of the electroweak loop corrections that UM-2 does not compute; they change the total tree-level $Z$ width by 0.82 %, 0.83 % with the QCD factor).

### 12.2.2 Fermions (DERIVED)

The eight-dimensional Dirac equation with the canonical spin connection reduces EXACTLY (all eight blocks, general $a_4(x_4)$, $b_4(x_4)$, $y$-dependent mass function $M(y)$ and potential $v(y)$; DERIVED with the author's matrices) to the block Hamiltonian:

$$
\begin{aligned}
&\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\mathcal{V}\chi(y,x_4)\,:\\
&\gamma^M D_M\Psi - \big(M(y) - \mathrm{i}v(y)\gamma^{x_4}\big)\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\gamma^{x_4}\mathcal{V}\big[\partial_{x_4}\chi + \mathrm{i}h\chi\big],\\
&h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y), \qquad \mathrm{i}\,\partial_{x_4}\chi = h\chi;\\
&\text{in each block: } \gamma^{x_0} = \sigma_3, \qquad \mathcal{B} = \varsigma\varsigma_2, \qquad C = \varsigma_2\sigma_2, \qquad \Gamma_4 = -\varsigma\varsigma_2\sigma_3
\end{aligned}
\tag{FE-56}
$$

**Symbols in (FE-56).**

- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{V}$: The constant $16 \times 2$ matrix of the reduction ansatz that places the two-component spinor $\chi$ of one block into the 16 components of the eight-dimensional spinor. Unit: dimensionless. Status: DERIVED.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^0 \chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along $\gamma^{x_4}$): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $h$: The block Hamiltonian: the $2 \times 2$ differential operator of one block that generates the time evolution of $\chi$, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies $\varepsilon$. Unit: GeV. Status: DERIVED.
- $\varsigma$: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each $\pm1$, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$

(the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor $\chi$ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues $\mp 1$ define the halves $\Psi_A$ and $\Psi_B$, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- $\varsigma_2$: The second sign that labels the blocks ($\pm 1$). Unit: dimensionless. Status: DERIVED.
- $C$: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar{\Psi} = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

The factor $\mathrm{e}^{-\frac{3}{2}Hy}$ removes the $\frac{3}{2}(H/f)\gamma^{x_0}$ of the spin connection, and $\mathrm{e}^{a_4/2}b_4^{-1}$ removes its $\gamma^{x_4}$ part. For a constant bulk mass and $k = 0$ the general solution of $h\chi = 0$ is a combination of $(\mathrm{e}^{f\mu y}, 0)$ and $(0, \mathrm{e}^{-f\mu y})$; the boundary conditions of the Standard-Model fermions (written out again)

$$
\begin{aligned}
&P_{\mathrm{sel}} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-\mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2};\\
&\text{doublets } (Q,\ell): \quad P_{\mathrm{sel}}\big(1-\gamma^{x_0}\big)\psi\big|_{y=0} = \big(1-P_{\mathrm{sel}}\big)\big(1+\gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1-\gamma^{x_0}\big)\psi\big|_{y=-L} = 0;\\
&\text{singlets } (u_R, d_R, e_R, \nu_R): \quad P_{\mathrm{sel}}\big(1+\gamma^{x_0}\big)\psi\big|_{y=0} = \big(1-P_{\mathrm{sel}}\big)\big(1-\gamma^{x_0}\big)\psi\big|_{y=0} = 0, \qquad \big(1+\gamma^{x_0}\big)\psi\big|_{y=-L} = 0
\end{aligned}
\tag{FE-29}
$$

**Symbols in (FE-29).**

- $P_{\mathrm{sel}}$: The projector that selects the two blocks of the half $\mathcal{B} = -1$ in which the Standard-Model zero modes live, $P_{\mathrm{sel}} = \frac{1+\Gamma_4\gamma^{x_0}}{2}\,\frac{1-\mathrm{i}\gamma^{x_5}\gamma^{x_6}}{2}$. Unit: dimensionless. Status: INVENTED.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $Q_i,\ \bar{Q}_i$: The quark doublet of the generation $i$ (left-handed up and down quark of one generation, three colours) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\ell_i,\ \bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $\psi_i,\ \bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $u_R,\ u_{Ri},\ \bar{u}_{Ri}$: The up-type quark singlet: $u_R$ the multiplet, $u_{Ri}$ that of the generation $i$ ($u_R$, $c_R$, $t_R$), $\bar{u}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $d_R,\ d_{Ri},\ \bar{d}_{Ri}$: The down-type quark singlet: $d_R$ the multiplet, $d_{Ri}$ that of the generation $i$ ($d_R$, $s_R$, $b_R$), $\bar{d}_{Ri}$ its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.
- $e_R,\ e_{Ri}$: The right-handed charged-lepton singlet: $e_R$ the multiplet, $e_{Ri}$ that of the generation $i$ ($e_R$, $\mu_R$,

$\tau_R$). Unit: GeV$^{7/2}$. Status: ASSUMED.

- $\nu_R,\ \nu_{Ri}$: The right-handed neutrino (a gauge singlet): $\nu_R$ the multiplet, $\nu_{Ri}$ that of the generation $i$; its zero modes give Dirac neutrino masses. Unit: GeV$^{7/2}$. Status: ASSUMED.

select one component, and the zero modes are

$$\chi=\big(\mathrm{e}^{c\xi},\ 0\big)^T\ \ (\text{doublet},\ \Gamma_4=+1),\qquad \chi=\big(0,\ \mathrm{e}^{-c\xi}\big)^T\ \ (\text{singlet},\ \Gamma_4=-1),\qquad c=f\mu L,$$
$$h\big(\mathrm{e}^{c\xi},\ 0\big)^T=\varsigma\,\mathrm{e}^{-a_4}k\big(\mathrm{e}^{c\xi},\ 0\big)^T \tag{FE-57}$$

**Symbols in (FE-57).**

- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi=(\chi_1,\chi_2)^T=(a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y=1$. Unit: GeV$^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c=f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L>0$), $c_R$ of a singlet (localised at the fold for $c_R<0$). Unit: dimensionless. Value: doublets $c_L=3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4=\mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4=+1$ (left-handed), a singlet zero mode $\Gamma_4=-1$ (right-handed). Unit: dimensionless. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mu_{\psi i}$: The bulk mass of a Standard-Model fermion: $\mu_{\psi i}$ of the multiplet $\psi$ of the generation $i$; with the hidden direction it fixes the localisation $c=f\mu L$ of the zero mode. Unit: GeV. Value: $\mu_{\psi i}=c/(fL)$ with the localisation $c$ of the zero mode (doublets: $\mu_Q=5.2304\,H$, $\mu_\ell=9.2055\,H$). Status: ASSUMED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $h$: The block Hamiltonian: the $2\times2$ differential operator of one block that generates the time evolution of $\chi$, $\mathrm{i}\,\partial_{x_4}\chi=h\chi$; its eigenvalues are the energies $\varepsilon$. Unit: GeV. Status: DERIVED.
- $\varsigma$: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each $\pm1$, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $k$: The magnitude of the wave vector (the momentum, $\hbar=1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

the doublet zero mode is localised at the fold for $c_L>0$ and the singlet zero mode for $c_R<0$; at $k\neq0$ the zero modes are massless four-dimensional particles with the energy $\mathrm{e}^{-a_4}|k|$ (the physical momentum).

**Masses from the tip: one generation.** The tip Yukawa coupling turns the tip conditions into the self-adjoint mixing conditions (written out again)

$$\begin{aligned}&\text{one generation:}\quad b_L(-L)=\mathcal{K}\,b_R(-L),\qquad a_R(-L)=-\mathcal{K}\,a_L(-L),\qquad \chi=(a,\ \mathrm{i}b);\\ &\text{three generations:}\quad b_L(-L)=\mathcal{K}\,b_R(-L),\qquad a_R(-L)=-\mathcal{K}^\dagger a_L(-L),\qquad \mathcal{K}\in\{\mathcal{K}_\nu,\mathcal{K}_e,\mathcal{K}_u,\mathcal{K}_d\};\\ &\text{UM-2:}\quad \mathcal{K}_e=\mathcal{K}_u=\mathbf{1},\qquad \mathcal{K}_\nu=U,\qquad \mathcal{K}_d=V_{\mathrm{CKM}}\end{aligned} \tag{FE-30}$$

**Symbols in (FE-30).**

- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3 \times 3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i} \nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $V_{\mathrm{CKM}}$: The CKM matrix, the unitary $3 \times 3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

with $\mathcal{K}$ the dimensionless tip mixing. At $k = 0$ the doublet ($c_L > 0$) and singlet ($c_R < 0$) solutions that satisfy the fold conditions reach the tip with

$$\frac{\mathrm{d}a}{\mathrm{d}\xi} = c\,a - \tilde{\varepsilon}\,b, \qquad \frac{\mathrm{d}b}{\mathrm{d}\xi} = \tilde{\varepsilon}\,a - c\,b, \qquad \chi = (a,\ \mathrm{i}b), \qquad \tilde{\varepsilon} = \varepsilon f L, \qquad \upsilon_L = \sqrt{c_L^2 - \tilde{\varepsilon}^2}, \qquad \upsilon_R = \sqrt{c_R^2 - \tilde{\varepsilon}^2},$$
$$(a_L,\ b_L)\big|_{\xi=-1} = \Big(\cosh\upsilon_L - c_L \frac{\sinh\upsilon_L}{\upsilon_L}, -\tilde{\varepsilon}\,\frac{\sinh\upsilon_L}{\upsilon_L}\Big), \qquad (a_R,\ b_R)\big|_{\xi=-1} = \Big(\tilde{\varepsilon}\,\frac{\sinh\upsilon_R}{\upsilon_R}, \cosh\upsilon_R + c_R\frac{\sinh\upsilon_R}{\upsilon_R}\Big) \tag{FE-58}$$

**Symbols in (FE-58).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $a,\ b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.

- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k=0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon=(c^2-\tilde\varepsilon^2)^{1/2}$ of the doublet ($\upsilon_L$) and singlet ($\upsilon_R$) solutions at $k=0$. Unit: dimensionless. Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L=(1+\Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R=(1-\Gamma_4)/2$. Status: STANDARD.

so the four-dimensional mass is the lowest root of the exact $x_0$ eigenvalue equation (DERIVED; direct shooting of the coupled system agrees to $6\times10^{-10}$), whose first-order solution is the tip mixing times the tip amplitudes of the two chiral wave functions:

$$\begin{aligned}&\mathcal{D}(\tilde\varepsilon)=\tilde\varepsilon^2\,\frac{\sinh\upsilon_L}{\upsilon_L}\frac{\sinh\upsilon_R}{\upsilon_R}-\mathcal{K}^2\Big(\cosh\upsilon_L-c_L\frac{\sinh\upsilon_L}{\upsilon_L}\Big)\Big(\cosh\upsilon_R+c_R\frac{\sinh\upsilon_R}{\upsilon_R}\Big)=0,\\&\tilde\varepsilon\simeq\mathcal{K}\,\mathcal{Z}(c_L)\,\mathcal{Z}(-c_R),\qquad\mathcal{Z}(c)^2=\frac{c\,\mathrm{e}^{-c}}{\sinh c}\end{aligned}\tag{FE-59}$$

**Symbols in (FE-59).**

- $\mathcal{D}(\tilde\varepsilon)$: The mass function of one generation: its zeros are the dimensionless masses $\tilde\varepsilon$. Unit: dimensionless. Status: DERIVED.
- $\tilde\varepsilon$: The dimensionless energy (mass) of a fermion mode, $\tilde\varepsilon=\varepsilon fL$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon=(c^2-\tilde\varepsilon^2)^{1/2}$ of the doublet ($\upsilon_L$) and singlet ($\upsilon_R$) solutions at $k=0$. Unit: dimensionless. Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L=(1+\Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R=(1-\Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3\times3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e=\mathcal{K}_u=\mathbf{1}$, $\mathcal{K}_\nu=U$, $\mathcal{K}_d=V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c=f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L>0$), $c_R$ of a singlet (localised at the fold for $c_R<0$). Unit: dimensionless. Value: doublets $c_L=3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2=c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

(both zero modes are localised at the fold, so their tip amplitudes are exponentially small).

**Masses and mixing from the tip: three generations.** In each sector (neutrinos, charged leptons, up-type quarks, down-type quarks) the tip coupling is a $3\times3$ matrix $\mathcal{K}$ (the dimensionless form of $\mathcal{Y}_\nu$, $\mathcal{Y}_e$, $\mathcal{Y}_u$, $\mathcal{Y}_d$), and the conditions hold for the vectors of boundary values over the generations; they are self-adjoint for every complex $\mathcal{K}$ (DERIVED). With the doublet amplitudes $\boldsymbol{\alpha}$ and the singlet amplitudes $\boldsymbol{\beta}$ of the solutions above,

$$\Big[\mathrm{diag}\big(b_{L\alpha}\big)+\mathcal{K}\,\mathrm{diag}\Big(\frac{b_{Ri}}{a_{Ri}}\Big)\mathcal{K}^{\dagger}\,\mathrm{diag}\big(a_{L\alpha}\big)\Big]\boldsymbol{\alpha}=0;\qquad \det\big[\,\cdots\,\big]=\prod_{i=1}^{3}\frac{b_La_{Ri}+a_Lb_{Ri}}{a_{Ri}},$$
$$b_La_R+a_Lb_R=-\mathcal{D}(\tilde{\varepsilon})\big|_{\mathcal{K}=1},\qquad \boldsymbol{\alpha}^{(i)}=\mathcal{K}\,\mathbf{u}_i,\qquad \boldsymbol{\beta}^{(i)}\propto\mathbf{u}_i,\qquad U=\mathcal{K}_e^{\dagger}\mathcal{K}_\nu,\qquad V_{\mathrm{CKM}}=\mathcal{K}_u^{\dagger}\mathcal{K}_d \tag{FE-60}$$

**Symbols in (FE-60).**

- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $a$, $b$: The upper and lower components of a block spinor, $\chi=(a,ib)$, real at $k=0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L=(1+\Gamma_4)/2$. Status: STANDARD.
- $\alpha$: A flavour index running over $e,\mu,\tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3\times3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e=\mathcal{K}_u=\mathbf{1}$, $\mathcal{K}_\nu=U$, $\mathcal{K}_d=V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R=(1-\Gamma_4)/2$. Status: STANDARD.
- $i$: The generation index ($i=1,2,3$), which also labels the neutrino mass states $\nu_1,\nu_2,\nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $^{\dagger}$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\boldsymbol{\alpha}$, $\boldsymbol{\alpha}^{(i)}$: The vector of the three doublet amplitudes at the tip in the mass condition of three generations; $\boldsymbol{\alpha}^{(i)}$ its null vector for the mass state $i$ (the left-handed flavour content). Unit: dimensionless. Status: DERIVED.
- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon}=\varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\mathbf{u}_i$: The unit vector of the generation $i$ (the $i$-th column of the $3\times3$ identity). Unit: dimensionless. Status: STANDARD.
- $\boldsymbol{\beta}^{(i)}$: The vector of the three singlet amplitudes of the mass state $i$. Unit: dimensionless. Status: DERIVED.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha=\sum_i U_{\alpha i}\nu_i$ (rows $\alpha=e,\mu,\tau$, columns the mass states $i$); in UM-2 $U=\mathcal{K}_e^{\dagger}\mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12}=0.308$, $\sin^2\theta_{13}=0.02215$, $\sin^2\theta_{23}=0.470$, $\delta_{\mathrm{CP}}=212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $V_{\mathrm{CKM}}$: The CKM matrix, the unitary $3\times3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\mathrm{CKM}}=\mathcal{K}_u^{\dagger}\mathcal{K}_d$. Unit: dimensionless. Value: $s_{12}=0.22501$, $s_{13}=0.003732$, $s_{23}=0.04183$, $\delta=1.147$ (PDG). Status: STANDARD.

the masses are the roots of the determinant, and $\boldsymbol{\alpha}$ is the left-handed flavour content of each mass state. If the three doublets of a sector share one bulk mass and $\mathcal{K}$ is unitary, the determinant factorises: each factor is the one-generation condition with $\mathcal{K}=1$, so the masses are the one-generation roots, the left-handed rotation of the sector is its tip matrix (so that the mixing matrices are $U=\mathcal{K}_e^{\dagger}\mathcal{K}_\nu$ and $V_{\mathrm{CKM}}=\mathcal{K}_u^{\dagger}\mathcal{K}_d$), and the right-handed rotations are trivial (DERIVED with mpmath at 50 digits: the roots of the $3\times3$ determinant equal the one-generation roots, both refined in the same computation, to $3.0\times10^{-42}$ in all four sectors; the masses stored by the mass computation agree with them to $4.5\times10^{-11}$; the null vectors equal the columns of $\mathcal{K}$ to $1.1\times10^{-16}$; for a random complex non-unitary $\mathcal{K}$ the exact roots agree with the singular values of $\mathcal{Z}(c_L)\,\mathcal{K}\,\mathrm{diag}(\mathcal{Z}(-c_R))$ to $3.5\times10^{-5}$).

**The UM-2 choice (ASSUMED).** One doublet bulk mass per sector, common to the three generations ($c_L=f\mu_\ell L$ for the leptons, $c_L=f\mu_Q L$ for the quarks), fixed by the heaviest fermion of the sector with $c_R=-c_L$ (the tau and the top quark); unitary tip matrices $\mathcal{K}_e=\mathcal{K}_u=\mathbf{1}$, $\mathcal{K}_\nu=U$, $\mathcal{K}_d=V_{\mathrm{CKM}}$ (FE-30) in the basis in which the singlet bulk masses are diagonal. The targets are the measured masses (STANDARD, the PDG 2025 values: electron, muon, tau; up, down and strange quark in $\overline{\mathrm{MS}}$ at 2 GeV, charm and bottom at their own scale, the directly measured top mass, used as tree-level targets) and the neutrino masses (ASSUMED lightest mass and normal ordering, measured splittings; written out again)

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV} \qquad \text{(FE-100)}$$

**Symbols in (FE-100).**

- $m_1,\ m_2,\ m_3$: The masses of the neutrino mass states $\nu_1$, $\nu_2$, $\nu_3$ (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV ($m_1$ ASSUMED). Status: DERIVED.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\text{eff}}$ an addition to $N_{\text{eff}}$, $\Delta S_{\text{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\Delta m_{21}^2,\ \Delta m_{31}^2,\ \Delta m_{32}^2$: The differences of the squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Unit: GeV$^2$ (quoted in eV$^2$). Value: $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ eV$^2$, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ eV$^2$ (NuFIT 6.0). Status: STANDARD.

They require (DERIVED; exact roots; the first-order formula agrees to $1.1 \times 10^{-3}$ for the top quark and better than $3 \times 10^{-4}$ for every other fermion)

| fermion | mass | doublet $c_L$ | singlet $c_R$ | $\mu_{\psi i}/H$ (singlet) | singlet width $fL/(2\lvert c_R\rvert)$ | $\tilde{\varepsilon} = m\,fL$ |
|---|---|---|---|---|---|---|
| $\nu_1$ | 1.000 meV | 5.9529 | $-35.0452$ | $-54.194$ | $1.27 \times 10^{-22}$ m | $4.524 \times 10^{-17}$ |
| $\nu_2$ | 8.712 meV | 5.9529 | $-32.8481$ | $-50.796$ | $1.36 \times 10^{-22}$ m | $3.941 \times 10^{-16}$ |
| $\nu_3$ | 50.140 meV | 5.9529 | $-31.0702$ | $-48.047$ | $1.44 \times 10^{-22}$ m | $2.268 \times 10^{-15}$ |
| electron | 0.51100 MeV | 5.9529 | $-14.5539$ | $-22.506$ | $3.07 \times 10^{-22}$ m | $2.312 \times 10^{-8}$ |
| muon | 105.658 MeV | 5.9529 | $-8.9809$ | $-13.888$ | $4.97 \times 10^{-22}$ m | $4.779 \times 10^{-6}$ |
| tau | 1776.93 MeV | 5.9529 | $-5.9529$ | $-9.205$ | $7.50 \times 10^{-22}$ m | $8.038 \times 10^{-5}$ |
| up | 2.16 MeV | 3.3823 | $-15.4300$ | $-23.861$ | $2.89 \times 10^{-22}$ m | $9.771 \times 10^{-8}$ |
| charm | 1.2730 GeV | 3.3823 | $-8.7683$ | $-13.559$ | $5.09 \times 10^{-22}$ m | $5.758 \times 10^{-5}$ |
| top | 172.56 GeV | 3.3823 | $-3.3823$ | $-5.230$ | $1.32 \times 10^{-21}$ m | $7.806 \times 10^{-3}$ |
| down | 4.70 MeV | 3.3823 | $-14.6258$ | $-22.617$ | $3.05 \times 10^{-22}$ m | $2.126 \times 10^{-7}$ |
| strange | 93.5 MeV | 3.3823 | $-11.5158$ | $-17.808$ | $3.88 \times 10^{-22}$ m | $4.230 \times 10^{-6}$ |
| bottom | 4.183 GeV | 3.3823 | $-7.5005$ | $-11.599$ | $5.95 \times 10^{-22}$ m | $1.892 \times 10^{-4}$ |

The doublet bulk masses are $\mu_\ell = 9.2055\,H$ (width $7.50 \times 10^{-22}$ m) and $\mu_Q = 5.2304\,H$ (width $1.32 \times 10^{-21}$ m). Order-one bulk masses give the whole hierarchy, because the tip amplitudes depend exponentially on $c$; within a sector the lighter fermion has the sharper singlet localisation. The neutrinos are Dirac fermions; their right-handed partners $\nu_{Ri}$ are gauge singlets that are never thermalised. (Version 1 gave every fermion its own $c_L = -c_R$; with three generations that rule would give the members of one doublet different bulk masses, against the single bulk mass of a multiplet, and flavour-changing Kaluza-Klein couplings.)

**Mixing matrices (STANDARD parametrisation, measured values).** Both tip matrices are written in the standard form

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}\mathrm{e}^{-\mathrm{i}\delta_{\mathrm{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{23}c_{13} \end{pmatrix}, \qquad J_{\mathrm{CP}} = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta_{\mathrm{CP}} \tag{FE-61}$$

**Symbols in (FE-61).**

- $U$: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- $c_{12}$, $c_{13}$, $c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $s_{12}$, $s_{13}$, $s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\delta_{\mathrm{CP}}$: The CP-violating phase of a mixing matrix. Unit: dimensionless. Value: 212° (PMNS), 1.147 rad (CKM). Status: STANDARD.
- $J_{\mathrm{CP}}$: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: $-0.01777$ (PMNS), $3.117 \times 10^{-5}$ (CKM). Status: STANDARD.

with, for the PMNS matrix (NuFIT 6.0, normal ordering), $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\mathrm{CP}} = 212°$ ($\theta_{12} = 33.71°$, $\theta_{23} = 43.28°$, $\theta_{13} = 8.56°$; $J_{\mathrm{CP}} = -0.01777$), and for the CKM matrix (PDG global fit) $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta_{\mathrm{CP}} = 1.147$ ($J_{\mathrm{CP}} = 3.117 \times 10^{-5}$). Both are unitary to $1.1 \times 10^{-16}$ (DERIVED). The magnitudes are

$$|U| = \begin{pmatrix} 0.8226 & 0.5488 & 0.1488 \\ 0.3351 & 0.6543 & 0.6779 \\ 0.4594 & 0.5203 & 0.7199 \end{pmatrix}, \qquad |V_{\mathrm{CKM}}| = \begin{pmatrix} 0.97435 & 0.22501 & 0.00373 \\ 0.22488 & 0.97349 & 0.04183 \\ 0.00858 & 0.04111 & 0.99912 \end{pmatrix}.$$

**Symbols in this equation.**

- $U$: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- $V_{\mathrm{CKM}}$: The CKM matrix, the unitary $3 \times 3$ mixing matrix of the quarks (the same parametrisation as $U$); in UM-2 $V_{\mathrm{CKM}} = \mathcal{K}_u^\dagger \mathcal{K}_d$. Unit: dimensionless. Value: $s_{12} = 0.22501$, $s_{13} = 0.003732$, $s_{23} = 0.04183$, $\delta = 1.147$ (PDG). Status: STANDARD.

**No flavour-changing neutral currents at tree level (DERIVED).** Because the doublets of a sector share one bulk mass and the right-handed rotations are trivial, the couplings of the Kaluza-Klein gauge modes are flavour diagonal in the mass basis to $10^{-17}$ (the section on the interactions).

### 12.2.3 The quanta of dirac16complex (DERIVED)

$\Psi$ has the same block Hamiltonian with $M = m_\Psi$ and the boundary conditions written out again:

$$\left(1 - \gamma^{x_0}\right)\Psi\big|_{y=0} = 0 \;\; \text{(even)} \quad \text{or} \quad \left(1 + \gamma^{x_0}\right)\Psi\big|_{y=0} = 0 \;\; \text{(odd)}, \qquad \left(1 - \gamma^{x_0}\right)\Psi\big|_{y=-L} = 0 \tag{FE-28}$$

**Symbols in (FE-28).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M \gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: GeV$^{7/2}$. Status: RECORD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

both parities being modes of the full period. With $c_\Psi = f\, m_\Psi L$ the exact $k = 0$ spectrum is

$$\text{even: } \chi = \left(\mathrm{e}^{c_\Psi \xi},\ 0\right)^T,\ \varepsilon = 0 \text{ (all eight blocks)}, \qquad \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi_n}{fL}\right)^2},\ \varpi_n = \pi,\ 2\pi,\ \ldots;$$
$$\text{odd: } \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi}{fL}\right)^2}, \quad \tan\varpi = -\frac{\varpi}{c_\Psi}; \qquad c_\Psi = f\, m_\Psi L; \qquad \text{fold band: } \varepsilon = \mathrm{e}^{-a_4}|k| \tag{FE-62}$$

**Symbols in (FE-62).**

- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: GeV$^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $c_\Psi$: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\varpi$, $\varpi_n$: The wave number of an oscillating $x_0$ profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); $\varpi_n$ that of the $n$-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

The **fold band** $\varepsilon = \mathrm{e}^{-a_4}|k|$ is massless: 8 states per 3-momentum, 4 in $\Psi_A$ (two Weyl species with $\Gamma_4 = +1$) and 4 in $\Psi_B$ (two Weyl species with $\Gamma_4 = -1$). The **bulk band** is massive; at the stabilised block ($f_* = 0.215556$, $c_\Psi = 0.646667$) its lowest level is the odd level $\varepsilon_{\mathrm{DM}} = 3.102204\, m_\Psi$ ($\varpi = 1.89901$, a hidden wave number $\varpi/(fL) = 2.93661\, m_\Psi$), and the lowest even bulk level is $4.959981\, m_\Psi$. The odd level is the dark-matter candidate (the section on the dark sector).

## 12.3 What the uncertainty relation does and does not bound (DERIVED)

For a boson the eigenvalue problem is (written out again)

$$\int_{-L}^{0}\mathrm{d}y\,f^4\mathrm{e}^{3Hy}\Big[\frac{(\partial_y F)^2}{f^2}-m^2F^2\Big]\ \text{stationary}: \qquad -\frac{1}{f^2}\,\mathrm{e}^{-3Hy}\,\partial_y\big(\mathrm{e}^{3Hy}\,\partial_y F\big)=m^2F,$$
$$\Leftrightarrow \qquad -\mathrm{e}^{-3HL\xi}\,\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\mathrm{e}^{3HL\xi}\,\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)=\tilde m^2F, \qquad \tilde m=m\,fL \tag{FE-50}$$

**Symbols in (FE-50).**

- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f=\mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi=y/L$): of a gauge field (gauge $A_y=0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0=1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\tilde m$, $\tilde m_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde m=m\,fL$; $\tilde m_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta=0$ ($\tilde m_1$ the first). Unit: dimensionless. Value: $\tilde m_1^2=30.1196$, $\tilde m_2^2=59.7284$, $\tilde m_3^2=109.0764$ ($HL=3$), i.e. $\tilde m_1=5.48813$, $\tilde m_2=7.72842$, $\tilde m_3=10.44397$. Status: DERIVED.

$$\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=0}=0\ \ \text{(even)}\quad\text{or}\quad F(0)=0\ \ \text{(odd)}, \qquad \frac{\mathrm{d}F}{\mathrm{d}\xi}\Big|_{\xi=-1}=\beta\,F(-1),$$
$$\beta=0\ \ \text{(photon, gluons, graviton)}, \qquad \beta=\beta_\mathrm{W}\ \ (W^\pm), \qquad \beta=\beta_\mathrm{Z}=\frac{\beta_\mathrm{W}}{\cos^2\theta_\mathrm{W}}\ \ (Z) \tag{FE-51}$$

**Symbols in (FE-51).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi=y/L$): of a gauge field (gauge $A_y=0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0=1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi=y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\beta$: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi=\beta F$ at $\xi=-1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.
- $\beta_\mathrm{W}$: The Robin parameter of $W^+$ and $W^-$ (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.

- $W_M^{\pm}$, $W^+$, $W^-$: The charged weak bosons, canonically normalised: $W_M^{\pm} = (W_M^1 \mp \mathrm{i}W_M^2)/(\sqrt{2}\,\hat{g}_2)$; $W^+$ carries the electric charge $+e$ and $W^-$ the charge $-e$ (they are each other's antiparticles and have one mass). Unit: GeV$^3$. Status: STANDARD.
- $\beta_{\mathrm{Z}}$: The Robin parameter of the $Z$, $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$. Unit: dimensionless. Value: 0.015343. Status: DERIVED.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $Z_M$: The $Z$ boson field, canonically normalised, $Z_M = (W_M^3 - B_M)/\hat{g}_Z$. Unit: GeV$^3$. Status: STANDARD.

and the rescaled profile $\tilde{F} = \mathrm{e}^{3HL\xi/2}F$ obeys the exact identity

$$\mathrm{e}^{3HL\xi}\Big(\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)^2 = \Big(\frac{\mathrm{d}\tilde{F}}{\mathrm{d}\xi}\Big)^2 + \frac{9H^2L^2}{4}\,\tilde{F}^2 - \frac{3HL}{2}\,\frac{\mathrm{d}\big(\tilde{F}^2\big)}{\mathrm{d}\xi},$$

**Symbols in this equation.**

- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.

which turns the Rayleigh quotient of FE-50 with the conditions FE-51 into the decomposition of every bosonic eigenvalue into a hidden kinetic energy, a warp term, a fold term and a tip term:

$$\tilde{m}^2 = \langle k_\xi^2\rangle + \frac{9H^2L^2}{4} - \frac{3HL}{2}\,\varrho_\xi(0) + \Big(\frac{3HL}{2} + \beta\Big)\varrho_\xi(-1), \qquad \langle k_\xi^2\rangle = \frac{\int_{-1}^0 \big(\mathrm{d}\tilde{F}/\mathrm{d}\xi\big)^2\mathrm{d}\xi}{\int_{-1}^0 \tilde{F}^2\mathrm{d}\xi}, \qquad \varrho_\xi = \frac{\tilde{F}^2}{\int_{-1}^0 \tilde{F}^2\mathrm{d}\xi}.$$

**Symbols in this equation.**

- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- $k_\xi$: The wave number conjugate to $\xi$, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^0(\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2\mathrm{d}\xi/\int_{-1}^0\tilde{F}^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^0(\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi/\int_{-1}^0\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\varrho_\xi$: The probability density in $\xi$ of the $x_0$ profile of a mode on one half of the period, normalised as $\int_{-1}^0\varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2/\int_{-1}^0\tilde{F}^2\mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi/\int_{-1}^0\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- $\beta$: The Robin parameter of the tip condition of a boson profile, $\mathrm{d}F/\mathrm{d}\xi = \beta F$ at $\xi = -1$ (zero for the massless gauge fields and the graviton). Unit: dimensionless. Status: DERIVED.

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\tilde{F}$: The rescaled boson profile, $F = \mathrm{e}^{-3HL\xi/2}\tilde{F}$. Unit: dimensionless. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.

For a fermion the eigenvalue problem is the block Hamiltonian (written out again)

$$
\begin{aligned}
&\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\mathcal{V}\chi(y,x_4)\,:\\
&\gamma^M D_M\Psi - \big(M(y) - \mathrm{i}v(y)\gamma^{x_4}\big)\Psi = \mathrm{e}^{\mathrm{i}kx_1}\,\mathrm{e}^{-\frac{3}{2}Hy}\,\mathrm{e}^{a_4/2}\,b_4^{-1}\,\gamma^{x_4}\mathcal{V}\big[\partial_{x_4}\chi + \mathrm{i}h\chi\big],\\
&h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y), \qquad \mathrm{i}\,\partial_{x_4}\chi = h\chi;\\
&\text{in each block: } \gamma^{x_0} = \sigma_3, \qquad \mathcal{B} = \varsigma\varsigma_2, \qquad C = \varsigma_2\sigma_2, \qquad \Gamma_4 = -\varsigma\varsigma_2\sigma_3
\end{aligned}
\tag{FE-56}
$$

**Symbols in (FE-56).**

- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $b_4$: The author's metric function of the time $x_4$ in the hidden and extra-time components, $b_4 > 0$. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{V}$: The constant $16 \times 2$ matrix of the reduction ansatz that places the two-component spinor $\chi$ of one block into the 16 components of the eight-dimensional spinor. Unit: dimensionless. Status: DERIVED.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $D_M$, $D_\mu$: The covariant derivative: on a spinor $D_M = \partial_M + \Omega_M$ (spin connection), on a Standard-Model fermion with the gauge fields added; on the sigma field $D_\mu\Sigma$ with the $W$ and $B$ fields. Unit: 1/length. Status: STANDARD.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along $\gamma^{x_4}$): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$

with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.

- $h$: The block Hamiltonian: the $2\times 2$ differential operator of one block that generates the time evolution of $\chi$, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies $\varepsilon$. Unit: GeV. Status: DERIVED.
- $\varsigma$: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each $\pm 1$, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor $\chi$ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{B}$: The author's operator $\mathcal{B} = -\mathrm{i}C\gamma^{x_4}$; its eigenvalues $\mp 1$ define the halves $\Psi_A$ and $\Psi_B$, and it commutes with every interaction of the good sector. Unit: dimensionless. Status: AUTHOR.
- $\varsigma_2$: The second sign that labels the blocks ($\pm 1$). Unit: dimensionless. Status: DERIVED.
- $C$: The matrix of the Dirac adjoint, $C = \gamma^{x_0}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}$ ($\bar\Psi = \Psi^\dagger C$); in each block of the reduction $C = \varsigma_2\sigma_2$. Unit: dimensionless. Status: AUTHOR.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

and for a fermion of localisation $c$ the same reasoning with $h$ at $k = 0$ gives

$$\tilde\varepsilon^2 = \langle k_\xi^2\rangle + c^2 - c\,\frac{\left[\chi^\dagger\sigma_3\chi\right]_{-1}^{0}}{\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi}, \qquad \langle k_\xi^2\rangle = \frac{\int_{-1}^{0}\left(\mathrm{d}\chi/\mathrm{d}\xi\right)^\dagger\left(\mathrm{d}\chi/\mathrm{d}\xi\right)\mathrm{d}\xi}{\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi}, \qquad \varrho_\xi = \frac{\chi^\dagger\chi}{\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi}.$$

**Symbols in this equation.**

- $\tilde\varepsilon$: The dimensionless energy (mass) of a fermion mode, $\tilde\varepsilon = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $k_\xi$: The wave number conjugate to $\xi$, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^{0}(\mathrm{d}\tilde F/\mathrm{d}\xi)^2\mathrm{d}\xi\big/\int_{-1}^{0}\tilde F^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde F$) and $\int_{-1}^{0}(\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi\big/\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor $\chi$ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\varrho_\xi$: The probability density in $\xi$ of the $x_0$ profile of a mode on one half of the period, normalised as $\int_{-1}^{0}\varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde F^2\big/\int_{-1}^{0}\tilde F^2\mathrm{d}\xi$ for a boson ($\tilde F = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi\big/\int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.

The uncertainty relation bounds only the hidden kinetic energy $\langle k_\xi^2\rangle$. On the cell, which ends at the tips, it reads

$$\sqrt{\langle k_\xi^2\rangle}\,\Delta\xi \geq \frac{1-\varrho_\xi(-1)}{2}, \qquad (\Delta\xi)^2 = \int_{-1}^{0} \xi^2 \varrho_\xi \,\mathrm{d}\xi,$$

**Symbols in this equation.**

- $k_\xi$: The wave number conjugate to $\xi$, i.e. the hidden momentum in units of $1/(fL)$. Its mean square in a state is the hidden kinetic energy in units of $1/(fL)^2$: $\langle k_\xi^2\rangle = \int_{-1}^{0}(\mathrm{d}\tilde{F}/\mathrm{d}\xi)^2\mathrm{d}\xi / \int_{-1}^{0}\tilde{F}^2\mathrm{d}\xi$ for a boson (flat-measure profile $\tilde{F}$) and $\int_{-1}^{0}(\mathrm{d}\chi/\mathrm{d}\xi)^\dagger(\mathrm{d}\chi/\mathrm{d}\xi)\,\mathrm{d}\xi / \int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. Unit: dimensionless. Status: DERIVED.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $\varrho_\xi$: The probability density in $\xi$ of the $x_0$ profile of a mode on one half of the period, normalised as $\int_{-1}^{0}\varrho_\xi\,\mathrm{d}\xi = 1$: $\varrho_\xi = \tilde{F}^2 / \int_{-1}^{0}\tilde{F}^2\mathrm{d}\xi$ for a boson ($\tilde{F} = \mathrm{e}^{3HL\xi/2}F$, the flat measure) and $\varrho_\xi = \chi^\dagger\chi / \int_{-1}^{0}\chi^\dagger\chi\,\mathrm{d}\xi$ for a fermion. $\varrho_\xi(0)$ is its value at the fold, $\varrho_\xi(-1)$ its value at the tip; over the full period the density is symmetric about the fold. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.

since $1 = \int\varrho_\xi\,\mathrm{d}\xi = \varrho_\xi(-1) - \int\xi\,(\mathrm{d}\varrho_\xi/\mathrm{d}\xi)\,\mathrm{d}\xi$ (integration by parts; the fold end $\xi = 0$ contributes nothing) and the Cauchy-Schwarz inequality bounds $|\int\xi\,(\mathrm{d}\varrho_\xi/\mathrm{d}\xi)\,\mathrm{d}\xi|$ by $2\Delta\xi\sqrt{\langle k_\xi^2\rangle}$; $\Delta(fy) = fL\,\Delta\xi$ is the root-mean-square proper distance from the fold. The familiar $\langle k_\xi^2\rangle \geq 1/(4\Delta\xi^2)$ is guaranteed only for profiles negligible at the tip (it can hold otherwise: the first even Kaluza-Klein mode, with $\varrho_\xi(-1) = 0.655$, has $\langle k_\xi^2\rangle\Delta\xi^2 = 4.59$); the photon zero mode has 0.497, the broad fold zero mode of $\Psi$ ($\varrho_\xi(-1) = 0.489$) only 0.097. The relation with the tip term is a theorem; it is also checked numerically for each of the 21 computed states (the zero mode of the photon, gluons and graviton, $W^\pm$, $Z$, the first even and odd Kaluza-Klein modes, the two doublet zero modes, the twelve singlet zero modes of the fermions, the fold zero mode of $\Psi$ and the dark matter; DERIVED). Three classes follow:

- **massless**: profiles flat relative to the geometry (\$F = \$ const or $h\chi = 0$): the fold term cancels the hidden kinetic energy and the warp exactly; the graviton, the photon, the gluons and the fold bands of $\Psi$. They are localised at the fold ($\Delta(fy) = 1.40\times10^{-21}$ m for the photon) and yet massless;
- **heavy**: the $x_0$ dependence fills the cell. For the gauge fields and the graviton every odd mode and every excitation has $m \geq 3H/(2f_*) = 99.5$ TeV (the gap theorem of the boson operator: one even level below the gap for every $\beta \geq 0$, no odd level); the excitations of $\Psi$ start at its lowest odd level, $3.10\,m_\Psi = 44.35$ TeV (the dark matter), since the Dirac operator has no warp gap. For the computed states of this class the author's estimate $m_{\mathrm{Heis}}$ lies below the mass (by factors 2.3 to 7.5, the dark matter included): the uncertainty argument is the reason of these masses (observed for these states, not derived for all);
- **light**: the $x_0$ dependence that costs mass sits at the tip, where the probability is tiny: $W^\pm$, $Z$ and the twelve fermions. Their masses lie $3\times10^{-3}$ (the top quark) to $10^{-17}$ (the lightest neutrino) below $m_{\mathrm{Heis}}$ (that of the broader of the fermion's two zero modes; $10^{-3}$ for $W^\pm$), and a sharper localisation of either zero mode gives a LIGHTER fermion ($\mathrm{d}\ln m/\mathrm{d}\ln c$ from $-2.9$ to $-34.5$): the reverse of the uncertainty intuition. These masses come from the tip of the cell, where electroweak symmetry is broken, and from the exponentially small overlap of the chiral zero modes there.

**Verdict.** Every four-dimensional mass of UM-2 is an $x_0$ eigenvalue of a wave function periodic in $x_0$ with the period $\pi/(6H)$, and the massless particles are exactly those protected by symmetry (the author's mechanism holds in this precise sense). The uncertainty relation bounds the hidden kinetic energy, not the mass: it is the reason of the heavy masses (the Kaluza-Klein modes, the odd modes, the dark matter), while the light masses ($W^\pm$, $Z$ and all twelve fermions) need two further ingredients that do not follow from it, the tip of the cell and the exponentially localised chiral zero modes, whose overlap at the tip produces the hierarchy of the fermion masses. The mixing matrices are tip matrices and are not explained by the mechanism.

# 13 Dimensional reduction and the four-dimensional constants

The good sector (no dependence on $x_5$, $x_6$, $x_7$) is kept (ASSUMED): a mode with the wave number $k_\mathrm{t}$ along the extra times has the squared energy

$$\varepsilon^2 = M^2 + k^2 - \frac{k_\mathrm{t}^2}{f^2}$$

**Symbols in this equation.**

- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $M(y)$: The mass function of the block Hamiltonian (the scalar potential): $M = \mu_{\psi i}$ for a Standard-Model fermion, $M = M_\mathrm{eff}$ for dirac16complex. Unit: GeV. Status: DERIVED.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $k_\mathrm{t}$: The wave number (momentum) of a mode along the compact extra times $x_5, x_6, x_7$; UM-2 keeps the good sector $k_\mathrm{t} = 0$, because a mode with $k_\mathrm{t} > fM$ would be tachyonic. Unit: GeV (1/length). Value: 0 in the good sector. Status: ASSUMED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

and is tachyonic for $k_\mathrm{t} > fM$ (OPEN). The extra times are compact with the coordinate volume $v_\mathrm{t}$ (ASSUMED). Integrating over one period of $x_0$ (both halves) and the extra times, the four-dimensional theory on the static block has (DERIVED)

$$\begin{aligned} &G^{(4)}_{\mu\nu} + \Lambda_4\, g^{(4)}_{\mu\nu} = 8\pi G_\mathrm{N}\, \mathcal{T}^{(4)}_{\mu\nu}, \qquad 8\pi G_\mathrm{N} = \frac{\kappa}{\mathcal{G}(K_*)\, V_\perp}, \qquad V_\perp = \frac{2 f^4 v_\mathrm{t}\bigl(1 - \mathrm{e}^{-3HL}\bigr)}{3H}, \\ &\mathcal{T}^{(4)}_{\mu\nu} = \int \mathrm{d}x_0\, \mathrm{d}x_5\, \mathrm{d}x_6\, \mathrm{d}x_7\, \sqrt{|g_\perp|}\, \mathcal{T}_{\mu\nu}, \qquad g_3^2 = \frac{\hat g_3^2}{V_\perp}, \qquad g_2^2 = \frac{\hat g_2^2}{V_\perp}, \qquad g_1^2 = \frac{\hat g_1^2}{V_\perp} \end{aligned} \tag{FE-49}$$

**Symbols in (FE-49).**

- $G^{(4)}_{\mu\nu}$: The Einstein tensor of the four-dimensional metric, $R^{(4)}_{\mu\nu} - \frac{1}{2} g^{(4)}_{\mu\nu} R^{(4)}$. Unit: 1/length$^2$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- $\Lambda_4$: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_\mathrm{stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: GeV$^2$. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_\mathrm{N}$: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: GeV$^{-2}$. Value: $6.709 \times 10^{-39}$ GeV$^{-2}$ ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{T}^{(4)}_{\mu\nu}$: The four-dimensional energy-momentum tensor: $\mathcal{T}_{\mu\nu}$ integrated over the block (both halves of a period of $x_0$ and the extra times). Unit: GeV$^4$. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV$^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV$^{-2}$. Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.

- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $V_\perp$: The proper volume of the block per unit four-volume (both halves of a period of $x_0$ and the extra times), $V_\perp = 2f^4 v_\mathrm{t}(1 - \mathrm{e}^{-3HL})/(3H)$. Unit: $\mathrm{GeV}^{-4}$. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_\mathrm{t}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $g_\perp$: The metric of the four transverse directions $x_0, x_5, x_6, x_7$ (the block); $|g_\perp|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $g_3$: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $g_1$: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.

with the reduced Planck mass

$$\bar{M}_\mathrm{Pl}^2 = \frac{\mathcal{G}(K_*)\, V_\perp}{\kappa} = \frac{1}{8\pi G_\mathrm{N}}.$$

**Symbols in this equation.**

- $\bar{M}_\mathrm{Pl}$: The reduced Planck mass, $\bar{M}_\mathrm{Pl} = (8\pi G_\mathrm{N})^{-1/2} = M_\mathrm{Pl}/\sqrt{8\pi}$. Unit: GeV. Value: $2.435 \times 10^{18}$ GeV. Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $V_\perp$: The proper volume of the block per unit four-volume (both halves of a period of $x_0$ and the extra times), $V_\perp = 2f^4 v_\mathrm{t}(1 - \mathrm{e}^{-3HL})/(3H)$. Unit: $\mathrm{GeV}^{-4}$. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ $\mathrm{GeV}^{-2}$. Status: AUTHOR.

- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\mathrm{N}}$: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.

Every four-dimensional constant depends on the block (DERIVED): $\alpha_{\mathrm{em}}$ and the gauge couplings $\propto f^{-4}$; Newton's constant $G_{\mathrm{N}} \propto f^{+2.05}$, because the Lovelock factor varies too,

$$\frac{\mathrm{d}\ln G_{\mathrm{N}}}{\mathrm{d}\ln f} = -4 + 2\,\frac{\mathrm{d}\ln \mathcal{G}}{\mathrm{d}\ln K}\Big|_{K=K_*} = 2.05;$$

**Symbols in this equation.**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $G_{\mathrm{N}}$: Newton's constant of four-dimensional gravity, $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_{\mathrm{Pl}} = G_{\mathrm{N}}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.

the Kaluza-Klein masses $\propto f^{-1}$, $m_{\mathrm{W}} \propto f^{-0.501}$, $M_{\mathrm{DM}} \propto f^{-0.772}$, and the fermion masses $m = \tilde{\varepsilon}/(fL)$, from the first-order root of the mass condition (written out again)

$$\begin{aligned} &\mathcal{D}(\tilde{\varepsilon}) = \tilde{\varepsilon}^2\,\frac{\sinh \upsilon_L}{\upsilon_L}\,\frac{\sinh \upsilon_R}{\upsilon_R} - \mathcal{K}^2\Big(\cosh \upsilon_L - c_L\frac{\sinh \upsilon_L}{\upsilon_L}\Big)\Big(\cosh \upsilon_R + c_R\frac{\sinh \upsilon_R}{\upsilon_R}\Big) = 0, \\ &\tilde{\varepsilon} \simeq \mathcal{K}\,\mathcal{Z}(c_L)\,\mathcal{Z}(-c_R), \qquad \mathcal{Z}(c)^2 = \frac{c\,\mathrm{e}^{-c}}{\sinh c} \end{aligned} \tag{FE-59}$$

**Symbols in (FE-59).**

- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L$, $\upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet ($\upsilon_L$) and singlet ($\upsilon_R$) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3 \times 3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.

- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

with localisations $c = f\mu L$ proportional to $f$:

$$\frac{\mathrm{d}\ln m}{\mathrm{d}\ln f} = -1 + \frac{1}{2}\Big[\frac{\mathrm{d}\ln \mathcal{Z}(c_L)^2}{\mathrm{d}\ln c_L} + \frac{\mathrm{d}\ln \mathcal{Z}(-c_R)^2}{\mathrm{d}\ln(-c_R)}\Big] \tag{FE-85}$$

**Symbols in (FE-85).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.

$-6.8$ (top quark) to $-41.0$ (lightest neutrino), with $-20.5$ for the electron. The masses are constant only because the block is stabilised. A continuing deflation today is excluded (ESTIMATE, bounds from memory): since $\mathrm{d}\ln X/\mathrm{d}x_4 = (\mathrm{d}\ln X/\mathrm{d}\ln f)\,\mathcal{H}_f$ for every quantity $X$, the atomic-clock bound $|\mathrm{d}\ln \alpha_{\mathrm{em}}/\mathrm{d}x_4| < 10^{-17}$ per year requires $|\mathcal{H}_f| < 2.5\times10^{-18}$ per year $= 3.6\times10^{-8}\,\mathcal{H}_0$, and lunar ranging ($|\mathrm{d}\ln G_{\mathrm{N}}/\mathrm{d}x_4| < 10^{-13}$ per year) requires $|\mathcal{H}_f| < 7.1\times10^{-4}\,\mathcal{H}_0$; a block deflating at the Hubble rate would change the electron mass by $1.4\times10^{-9}$ per year. The extra times are irrelevant at low energies not because they keep deflating but because they are small (the proper half cell $f_* L = 8.9\times10^{-21}$ m, Kaluza-Klein masses above 100 TeV) and frozen.

# 14 Gravity: the background, the stabilised block, the deflating extra times and four-dimensional gravity

## 14.1 The background field equations and their solutions (DERIVED)

### 14.1.1 The field equations of the order-0 metric

The gravitational field equations are the Einstein-Lovelock equations with the source of all fields (written out again):

$$\begin{aligned}
&\sum_{p=1}^{3} \alpha_{(p)}\, \mathcal{E}_{(p)}{}^{M}{}_{N} + \Lambda\,\delta^{M}_{N} = \kappa\,\mathcal{T}^{M}{}_{N},\\
&\mathcal{E}_{(p)}{}^{M}{}_{N} = -\frac{1}{2^{p+1}}\,\delta^{M M_1 N_1 \cdots M_p N_p}_{N P_1 Q_1 \cdots P_p Q_p}\, R^{P_1 Q_1}{}_{M_1 N_1} \cdots R^{P_p Q_p}{}_{M_p N_p}
\end{aligned} \tag{FE-1}$$

**Symbols in (FE-1).**

- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order $p$, $\alpha_{(p)} = w_{(p)} H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: $\mathrm{GeV}^{2-2p}$. Status: AUTHOR.

- $\mathcal{E}_{(p)}{}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^{M}{}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: GeV$^2$. Value: $-10H^2$. Status: AUTHOR.
- $\delta$: The Kronecker delta: with two indices $\delta^{M}_{N} = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta^{M_1\cdots M_n}_{N_1\cdots N_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta^{M_a}_{N_b}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV$^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ GeV$^{-2}$. Status: AUTHOR.
- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV$^8$. Status: STANDARD.
- $P$: A coordinate index of eight dimensions (also $P_1, P_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $Q$: A coordinate index of eight dimensions (also $Q_1, Q_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^{P}{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: 1/length$^2$. Status: STANDARD.

$$\Delta S_{\mathrm{m}} = -\frac{1}{2}\int \mathrm{d}^8X\,\sqrt{|g|}\,\mathcal{T}_{MN}\,\Delta g^{MN}, \qquad \rho = -\mathcal{T}^{x_4}{}_{x_4}, \qquad p_M = \mathcal{T}^{M}{}_{M} \quad (M \neq x_4, \text{ no sum}),$$
$$\mathcal{T}^{M}{}_{N} = \mathrm{diag}\big(p_0,\ p_3,\ p_3,\ p_3,\ -\rho,\ p_5,\ p_5,\ p_5\big) \quad \text{(a diagonal source)}, \qquad \bar{p} = \frac{p_0 + 3p_5}{4} \tag{FE-2}$$

**Symbols in (FE-2).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $S_{\mathrm{m}}$: The matter action: all terms of the action except $S_{\mathrm{EL}}$ (the stabiliser included). Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1\cdots\mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- $g_{MN}$: The metric tensor of the eight-dimensional spacetime (signature $+,+,+,+,-,-,-,-$ in the order $x_0, \ldots, x_7$): $g_{MN}\,\mathrm{d}X^M\mathrm{d}X^N$ is the squared interval; $g^{MN}$ is its inverse, $g^{M}{}_{N} = \delta^{M}_{N}$, and $|g|$ the absolute value of its determinant, so that $\sqrt{|g|}\,\mathrm{d}^8X$ is the proper volume element. Unit: dimensionless. Status: AUTHOR.
- $\mathcal{T}^{M}{}_{N}$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: GeV$^8$. Status: STANDARD.
- $M$: A coordinate index of eight dimensions (also with a numbered copy $M_1, M_2, \ldots$): it runs over $x_0, x_1, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $N$: A coordinate index of eight dimensions (also $N_1, N_2, \ldots$), running over $x_0, \ldots, x_7$; a repeated index is summed. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$

is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.

- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

For the order-0 metric the curvature is pair-diagonal, with the sectional curvatures (written out again)

$$
\begin{aligned}
&R^{x_1x_2}{}_{x_1x_2} = \mathcal{H}^2, \qquad R^{x_1x_4}{}_{x_1x_4} = \mathcal{H}^2 + \dot{\mathcal{H}}, \qquad R^{x_1x_0}{}_{x_1x_0} = \mathcal{H}\mathcal{H}_f, \qquad R^{x_4x_0}{}_{x_4x_0} = \mathcal{H}_f^2 + \dot{\mathcal{H}}_f,\\
&R^{x_0x_5}{}_{x_0x_5} = \mathcal{H}_f^2 - H^2K;
\end{aligned}
\tag{FE-33}
$$

the same for every pair of the same kind; all other $R^{ab}{}_{cd}$ with $\{a,b\} \neq \{c,d\}$ vanish

**Symbols in (FE-33).**

- $R^{MN}{}_{PQ}$, $R_{MN}$, $R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor $R_{MN} = R^P{}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; $R^{ab}{}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a,b)$. Unit: $1/\mathrm{length}^2$. Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $x_2$: The second coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.

and for a pair-diagonal curvature the Lovelock densities and tensors are sums over disjoint pairs (written out again):

$$
\begin{aligned}
&\mathcal{L}_{(1)} = 2\sum_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(2)} = 8\sum_{\text{2 disjoint pairs}}\ \prod_{\text{pairs}} R^{ab}{}_{ab}, \qquad \mathcal{L}_{(3)} = 48\sum_{\text{3 disjoint pairs}}\ \prod_{\text{pairs}} R^{ab}{}_{ab},\\
&\mathcal{E}_{(1)}{}^c{}_c = -\sum_{\text{pairs avoiding } c} R^{ab}{}_{ab}, \qquad \mathcal{E}_{(2)}{}^c{}_c = -4\sum_{\text{2 disjoint pairs avoiding } c}\ \prod_{\text{pairs}} R^{ab}{}_{ab},\\
&\mathcal{E}_{(3)}{}^c{}_c = -24\sum_{\text{3 disjoint pairs avoiding } c}\ \prod_{\text{pairs}} R^{ab}{}_{ab}
\end{aligned}
\tag{FE-34}
$$

**Symbols in (FE-34).**

- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\mathrm{length}^{2p}$. Status: STANDARD.

- ${R^{MN}}_{PQ},\ {R_{MN}},\ R$: The curvature of the metric $g_{MN}$: with four indices the Riemann tensor, with two indices the Ricci tensor ${R_{MN}} = {R^{P}}_{MPN}$, without indices the Ricci (scalar) curvature $R = g^{MN}R_{MN}$; ${R^{ab}}_{ab}$ (no sum) is the sectional curvature of the coordinate plane $(a, b)$. Unit: 1/length$^2$. Status: STANDARD.
- $a$: A frame (orthonormal) index running over $0, 1, \ldots, 7$, aligned with $x_0, \ldots, x_7$; raised and lowered with $\eta_{ab}$; a repeated index is summed (except in a sectional curvature ${R^{ab}}_{ab}$). Status: STANDARD.
- $b$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.
- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^{M}}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $c$: A frame index running over $0, \ldots, 7$ (as $a$). Status: STANDARD.

The three Lovelock tensors of the order-0 metric are therefore (exact, sympy, four routes)

$$
\begin{aligned}
{\mathcal{E}_{(1)}}^{x_4}{}_{x_4} &= -3\,\mathcal{H}^2 - 12\,\mathcal{H}\mathcal{H}_f - 6\,\mathcal{H}_f^2 + 6\,KH^2 \\
{\mathcal{E}_{(1)}}^{x_1}{}_{x_1} &= -3\,\mathcal{H}^2 - 8\,\mathcal{H}\mathcal{H}_f - 10\,\mathcal{H}_f^2 - 2\,\dot{\mathcal{H}} - 4\,\dot{\mathcal{H}}_f + 6\,KH^2 \\
{\mathcal{E}_{(1)}}^{x_0}{}_{x_0} &= -6\,\mathcal{H}^2 - 9\,\mathcal{H}\mathcal{H}_f - 6\,\mathcal{H}_f^2 - 3\,\dot{\mathcal{H}} - 3\,\dot{\mathcal{H}}_f + 3\,KH^2
\end{aligned}
\tag{FE-35}
$$

**Symbols in (FE-35).**

- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^{M}}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

$$
\begin{aligned}
{\mathcal{E}_{(2)}}^{x_4}{}_{x_4} &= -48\,\mathcal{H}^3\mathcal{H}_f - 216\,\mathcal{H}^2\mathcal{H}_f^2 - 144\,\mathcal{H}\mathcal{H}_f^3 - 12\,\mathcal{H}_f^4 + 72\,K\mathcal{H}^2H^2 \\
&\quad + 144\,K\mathcal{H}\mathcal{H}_fH^2 + 24\,K\mathcal{H}_f^2H^2 - 12\,K^2H^4 \\
{\mathcal{E}_{(2)}}^{x_1}{}_{x_1} &= -32\,\mathcal{H}^3\mathcal{H}_f - 136\,\mathcal{H}^2\mathcal{H}_f^2 - 192\,\mathcal{H}\mathcal{H}_f^3 - 60\,\mathcal{H}_f^4 \\
&\quad - 32\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f - 48\,\dot{\mathcal{H}}\mathcal{H}_f^2 - 16\,\dot{\mathcal{H}}_f\mathcal{H}^2 - 96\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f \\
&\quad - 48\,\dot{\mathcal{H}}_f\mathcal{H}_f^2 + 72\,K\mathcal{H}^2H^2 + 96\,K\mathcal{H}\mathcal{H}_fH^2 + 72\,K\mathcal{H}_f^2H^2 + 48\,\dot{\mathcal{H}}KH^2 \\
&\quad + 48\,\dot{\mathcal{H}}_fKH^2 - 12\,K^2H^4 \\
{\mathcal{E}_{(2)}}^{x_0}{}_{x_0} &= -12\,\mathcal{H}^4 - 108\,\mathcal{H}^3\mathcal{H}_f - 180\,\mathcal{H}^2\mathcal{H}_f^2 - 108\,\mathcal{H}\mathcal{H}_f^3 - 12\,\mathcal{H}_f^4 \\
&\quad - 12\,\dot{\mathcal{H}}\mathcal{H}^2 - 72\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f - 36\,\dot{\mathcal{H}}\mathcal{H}_f^2 - 36\,\dot{\mathcal{H}}_f\mathcal{H}^2 \\
&\quad - 72\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f - 12\,\dot{\mathcal{H}}_f\mathcal{H}_f^2 + 72\,K\mathcal{H}^2H^2 + 36\,K\mathcal{H}\mathcal{H}_fH^2 + 12\,K\mathcal{H}_f^2H^2 \\
&\quad + 36\,\dot{\mathcal{H}}KH^2 + 12\,\dot{\mathcal{H}}_fKH^2
\end{aligned}
\tag{FE-36}
$$

**Symbols in (FE-36).**

- ${\mathcal{E}_{(p)}}^{M}{}_{N}$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor ${R^{M}}_{N} - \frac{1}{2}\delta^{M}_{N}R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

$$
\begin{aligned}
{\mathcal{E}_{(3)}}^{x_4}{}_{x_4} =\ & -1440\,\mathcal{H}^3\mathcal{H}_f^3 - 1080\,\mathcal{H}^2\mathcal{H}_f^4 + 864\,K\mathcal{H}^3\mathcal{H}_f H^2 + 1296\,K\mathcal{H}^2\mathcal{H}_f^2 H^2 \\
& - 216\,K^2\mathcal{H}^2 H^4 \\
{\mathcal{E}_{(3)}}^{x_1}{}_{x_1} =\ & -576\,\mathcal{H}^3\mathcal{H}_f^3 - 1368\,\mathcal{H}^2\mathcal{H}_f^4 - 576\,\mathcal{H}\mathcal{H}_f^5 - 576\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f^3 \\
& - 144\,\dot{\mathcal{H}}\mathcal{H}_f^4 - 864\,\dot{\mathcal{H}}_f\mathcal{H}^2\mathcal{H}_f^2 - 576\,\dot{\mathcal{H}}_f\mathcal{H}\mathcal{H}_f^3 + 576\,K\mathcal{H}^3\mathcal{H}_f H^2 \\
& + 1008\,K\mathcal{H}^2\mathcal{H}_f^2 H^2 + 576\,K\mathcal{H}\mathcal{H}_f^3 H^2 + 576\,\dot{\mathcal{H}}K\mathcal{H}\mathcal{H}_f H^2 \\
& + 288\,\dot{\mathcal{H}}K\mathcal{H}_f^2 H^2 + 288\,\dot{\mathcal{H}}_f K\mathcal{H}^2 H^2 + 576\,\dot{\mathcal{H}}_f K\mathcal{H}\mathcal{H}_f H^2 - 216\,K^2\mathcal{H}^2 H^4 \\
& - 144\,\dot{\mathcal{H}}K^2 H^4 \\
{\mathcal{E}_{(3)}}^{x_0}{}_{x_0} =\ & -648\,\mathcal{H}^4\mathcal{H}_f^2 - 1224\,\mathcal{H}^3\mathcal{H}_f^3 - 648\,\mathcal{H}^2\mathcal{H}_f^4 - 648\,\dot{\mathcal{H}}\mathcal{H}^2\mathcal{H}_f^2 \\
& - 432\,\dot{\mathcal{H}}\mathcal{H}\mathcal{H}_f^3 - 432\,\dot{\mathcal{H}}_f\mathcal{H}^3\mathcal{H}_f - 648\,\dot{\mathcal{H}}_f\mathcal{H}^2\mathcal{H}_f^2 + 216\,K\mathcal{H}^4 H^2 \\
& + 648\,K\mathcal{H}^3\mathcal{H}_f H^2 + 216\,K\mathcal{H}^2\mathcal{H}_f^2 H^2 + 216\,\dot{\mathcal{H}}K\mathcal{H}^2 H^2 + 432\,\dot{\mathcal{H}}K\mathcal{H}\mathcal{H}_f H^2 \\
& + 216\,\dot{\mathcal{H}}_f K\mathcal{H}^2 H^2
\end{aligned}
\tag{FE-37}
$$

**Symbols in (FE-37).**

- ${\mathcal{E}_{(p)}}^M{}_N$: The Lovelock tensor of order $p$, the metric variation of $\sqrt{|g|}\mathcal{L}_{(p)}$: $\mathcal{E}_{(1)}$ is the Einstein tensor $R^M{}_N - \frac{1}{2}\delta^M_N R$; every $\mathcal{E}_{(p)}$ is divergence free. Unit: 1/length$^{2p}$. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_\mathrm{Pl}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.

Every off-diagonal component of every Lovelock tensor vanishes and the hidden and extra-time components coincide, so the source must be diagonal, independent of $x_0$ and block-isotropic ($p_0 = p_5$ in FE-2). The three independent components are the $x_4$ constraint (first order in time derivatives), the 3-space component and the block component; the last two are linear in $(\dot{\mathcal{H}}, \dot{\mathcal{H}}_f)$. With the author's weights and $\Lambda = -10H^2$ they are (DERIVED, sympy):

$$
\begin{aligned}
x_4: \quad & -\frac{5\,\mathcal{H}^3\mathcal{H}_f^3}{4H^4}-\frac{15\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4}+\frac{3\,K\mathcal{H}^3\mathcal{H}_f}{4H^2}\\
& +\frac{9\,K\mathcal{H}^2\mathcal{H}_f^2}{8H^2}-\frac{3\,\mathcal{H}^3\mathcal{H}_f}{2H^2}-\frac{27\,\mathcal{H}^2\mathcal{H}_f^2}{4H^2}\\
& -\frac{9\,\mathcal{H}\mathcal{H}_f^3}{2H^2}-\frac{3\,\mathcal{H}_f^4}{8H^2}-\frac{3\,K^2\mathcal{H}^2}{16}+\frac{9\,K\mathcal{H}^2}{4}\\
& +\frac{9\,K\mathcal{H}\mathcal{H}_f}{2}+\frac{3\,K\mathcal{H}_f^2}{4}-3\,\mathcal{H}^2-12\,\mathcal{H}\mathcal{H}_f-6\,\mathcal{H}_f^2-\frac{3\,K^2H^2}{8}+6\,KH^2-10\,H^2\\
& =-\kappa\rho
\end{aligned}
\tag{FE-38}
$$

**Symbols in (FE-38).**

- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f=\dot{f}/f=\dot{b}_4/(2b_4)-\dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $K$: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp=8\pi G_{\mathrm{N}}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\rho$: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.

$$
\begin{aligned}
x_1: \quad & \Big[-\frac{\mathcal{H}\mathcal{H}_f^3}{2H^4}-\frac{\mathcal{H}_f^4}{8H^4}+\frac{K\mathcal{H}\mathcal{H}_f}{2H^2}+\frac{K\mathcal{H}_f^2}{4H^2}\\
& -\frac{\mathcal{H}\mathcal{H}_f}{H^2}-\frac{3\,\mathcal{H}_f^2}{2H^2}-\frac{K^2}{8}+\frac{3\,K}{2}-2\Big]\dot{\mathcal{H}}\\
& +\Big[-\frac{3\,\mathcal{H}^2\mathcal{H}_f^2}{4H^4}-\frac{\mathcal{H}\mathcal{H}_f^3}{2H^4}+\frac{K\mathcal{H}^2}{4H^2}\\
& +\frac{K\mathcal{H}\mathcal{H}_f}{2H^2}-\frac{\mathcal{H}^2}{2H^2}-\frac{3\,\mathcal{H}\mathcal{H}_f}{H^2}-\frac{3\,\mathcal{H}_f^2}{2H^2}+\frac{3\,K}{2}-4\Big]\dot{\mathcal{H}}_f\\
& -\frac{\mathcal{H}^3\mathcal{H}_f^3}{2H^4}-\frac{19\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4}-\frac{\mathcal{H}\mathcal{H}_f^5}{2H^4}+\frac{K\mathcal{H}^3\mathcal{H}_f}{2H^2}\\
& +\frac{7\,K\mathcal{H}^2\mathcal{H}_f^2}{8H^2}+\frac{K\mathcal{H}\mathcal{H}_f^3}{2H^2}-\frac{\mathcal{H}^3\mathcal{H}_f}{H^2}-\frac{17\,\mathcal{H}^2\mathcal{H}_f^2}{4H^2}\\
& -\frac{6\,\mathcal{H}\mathcal{H}_f^3}{H^2}-\frac{15\,\mathcal{H}_f^4}{8H^2}-\frac{3\,K^2\mathcal{H}^2}{16}+\frac{9\,K\mathcal{H}^2}{4}\\
& +3\,K\mathcal{H}\mathcal{H}_f+\frac{9\,K\mathcal{H}_f^2}{4}-3\,\mathcal{H}^2-8\,\mathcal{H}\mathcal{H}_f-10\,\mathcal{H}_f^2-\frac{3\,K^2H^2}{8}+6\,KH^2-10\,H^2\\
& =\kappa p_3
\end{aligned}
\tag{FE-39}
$$

**Symbols in (FE-39).**

- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f=\dot{f}/f=\dot{b}_4/(2b_4)-\dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.

- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV$^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV$^{-2}$. Status: AUTHOR.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: GeV$^8$. Status: STANDARD.

$$
\begin{aligned}
x_0:\ &\Big[-\frac{9\,\mathcal{H}^2\mathcal{H}_f^2}{16H^4}-\frac{3\,\mathcal{H}\mathcal{H}_f^3}{8H^4}+\frac{3\,K\mathcal{H}^2}{16H^2}+\frac{3\,K\mathcal{H}\mathcal{H}_f}{8H^2}\\
&-\frac{3\,\mathcal{H}^2}{8H^2}-\frac{9\,\mathcal{H}\mathcal{H}_f}{4H^2}-\frac{9\,\mathcal{H}_f^2}{8H^2}+\frac{9\,K}{8}-3\Big]\dot{\mathcal{H}}\\
&+\Big[-\frac{3\,\mathcal{H}^3\mathcal{H}_f}{8H^4}-\frac{9\,\mathcal{H}^2\mathcal{H}_f^2}{16H^4}+\frac{3\,K\mathcal{H}^2}{16H^2}-\frac{9\,\mathcal{H}^2}{8H^2}\\
&-\frac{9\,\mathcal{H}\mathcal{H}_f}{4H^2}-\frac{3\,\mathcal{H}_f^2}{8H^2}+\frac{3\,K}{8}-3\Big]\dot{\mathcal{H}}_f\\
&-\frac{9\,\mathcal{H}^4\mathcal{H}_f^2}{16H^4}-\frac{17\,\mathcal{H}^3\mathcal{H}_f^3}{16H^4}-\frac{9\,\mathcal{H}^2\mathcal{H}_f^4}{16H^4}+\frac{3\,K\mathcal{H}^4}{16H^2}\\
&+\frac{9\,K\mathcal{H}^3\mathcal{H}_f}{16H^2}+\frac{3\,K\mathcal{H}^2\mathcal{H}_f^2}{16H^2}-\frac{3\,\mathcal{H}^4}{8H^2}-\frac{27\,\mathcal{H}^3\mathcal{H}_f}{8H^2}\\
&-\frac{45\,\mathcal{H}^2\mathcal{H}_f^2}{8H^2}-\frac{27\,\mathcal{H}\mathcal{H}_f^3}{8H^2}-\frac{3\,\mathcal{H}_f^4}{8H^2}\\
&+\frac{9\,K\mathcal{H}^2}{4}+\frac{9\,K\mathcal{H}\mathcal{H}_f}{8}+\frac{3\,K\mathcal{H}_f^2}{8}-6\,\mathcal{H}^2-9\,\mathcal{H}\mathcal{H}_f\\
&-6\,\mathcal{H}_f^2+3\,KH^2-10\,H^2\\
&=\kappa p_0;\qquad x_5,x_6,x_7:\ \text{the same left side}=\kappa p_5;\\
&\text{block-isotropic order-0 metric: the same left side}=\kappa\bar{p},\qquad \bar{p}=\tfrac{1}{4}\big(p_0+3p_5\big)
\end{aligned}
\tag{FE-40}
$$

**Symbols in (FE-40).**

- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot{f}/f = \dot{b}_4/(2b_4) - \dot{a}_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: GeV$^2$. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: GeV$^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\rm N}\mathcal{G}(K_*) = 2.328\times10^{-36}$ GeV$^{-2}$. Status: AUTHOR.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: GeV$^8$. Status: STANDARD.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: GeV$^8$. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: GeV$^8$ (or GeV$^4$ for a four-dimensional density). Status: DERIVED.

and the source is conserved:

$$\dot\rho = -3\mathcal{H}(\rho + p_3) - \mathcal{H}_f(\rho + p_0) - 3\mathcal{H}_f(\rho + p_5) = -3\mathcal{H}(\rho + p_3) - 4\mathcal{H}_f(\rho + \bar p) \tag{FE-41}$$

**Symbols in (FE-41).**

- $\dot\rho$: The time derivative $\mathrm{d}\rho/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^9$. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot f/f = \dot b_4/(2b_4) - \dot a_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\bar p$: The hidden pressure, the pressure averaged over the block, $\bar p = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar p = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

A source with $p_0 \neq p_5$ (the Kohn-Sham states of $\Psi$ and the matter at the fold have it; the dark matter has $p_0/\rho = 0.772$, $p_5 = 0$) acts on the common block scale $f$ through the block-averaged pressure $\bar p = (p_0 + 3p_5)/4$ (the variation of the action with respect to $f$), while $p_0 - p_5$ drives a block-shape mode that the order-0 metric does not contain (OPEN). The residual analysis quantifies it: the shape source is about 4 times the kept $\bar p$ for non-relativistic dark matter (0.9 for the self-interacting test states), with an absolute size suppressed like that of $\bar p$ ($\kappa p/\kappa\rho_{\mathrm{stab}} \leq 3\times10^{-32}$ at the freeze-out), if the shape mode is stable (OPEN); for matter its size relative to the stabiliser is of order $(\mathcal{H}/H)^2$, $10^{-92}$ today.

To first order about the static block with block-averaged matter sources ($\mathcal{H}^2$ counts as first order: the $x_4$ constraint gives $3\mathcal{G}(K_*)\mathcal{H}^2 = \kappa\Delta\rho$, because the $K$ derivative of the static balance vanishes at $K_*$) the equations give (DERIVED)

$$\begin{aligned}
\Delta\dot{\mathcal{H}} &= \Xi_{\mathcal{H}K}\,\Delta K + \Xi_{\mathcal{H}3}\,\kappa\Delta p_3 + \Xi_{\mathcal{H}p}\,\kappa\Delta\bar p + \Xi_{\mathcal{H}\rho}\,\kappa\Delta\rho, \qquad 3\mathcal{G}(K_*)\,\mathcal{H}^2 = \kappa\Delta\rho,\\
\Delta\dot{\mathcal{H}}_f &= \Xi_{fK}\,\Delta K + \Xi_{f3}\,\kappa\Delta p_3 + \Xi_{fp}\,\kappa\Delta\bar p + \Xi_{f\rho}\,\kappa\Delta\rho,\\
\Delta\ddot K + \omega_{\mathrm{r}}^2\,\Delta K &= -2K_*\big(\Xi_{f3}\,\kappa\Delta p_3 + \Xi_{fp}\,\kappa\Delta\bar p + \Xi_{f\rho}\,\kappa\Delta\rho\big)
\end{aligned} \tag{FE-45}$$

**Symbols in (FE-45).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\Xi_{\mathcal{H}K}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about $K_*$). Unit: $\mathrm{GeV}^2$. Value: $6(238 - 19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\Xi_{\mathcal{H}3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $2283/4498 - 162\sqrt{51}/2249 = -0.006853$. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\Xi_{\mathcal{H}p}$: The coefficient of $\kappa\Delta\bar p$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $(28\sqrt{51} - 114)/2249 = 0.038221$. Status: DERIVED.

- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.
- $\Xi_{\mathcal{H}\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $135021/265382 - 10308\sqrt{51}/132691 = -0.045997$. Status: DERIVED.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\Xi_{fK}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: $\mathrm{GeV}^2$. Value: $(4063 - 485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- $\Xi_{f3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $21\sqrt{51}/2249 - 171/4498 = 0.028666$. Status: DERIVED.
- $\Xi_{fp}$: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $239\sqrt{51}/6747 - 485/2249 = 0.037320$. Status: DERIVED.
- $\Xi_{f\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $57/4498 - 7\sqrt{51}/2249 = -0.009555$. Status: DERIVED.
- $\ddot{K}$: The second time derivative $\mathrm{d}^2K/\mathrm{d}x_4^2$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\omega_{\mathrm{r}}$: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.

with the coefficients (exact, sympy; numerically $0.272956H^2$, $0.266522H^2$, $-0.006853$, $0.038221$, $-0.045997$, $0.028666$, $0.037320$, $-0.009555$)

$$\Xi_{\mathcal{H}K} = \frac{6\big(238 - 19\sqrt{51}\big)}{2249}H^2, \qquad \Xi_{fK} = \frac{4063 - 485\sqrt{51}}{2249}H^2, \qquad \Xi_{\mathcal{H}3} = \frac{2283}{4498} - \frac{162\sqrt{51}}{2249},$$
$$\Xi_{\mathcal{H}p} = \frac{28\sqrt{51} - 114}{2249}, \qquad \Xi_{\mathcal{H}\rho} = \frac{135021}{265382} - \frac{10308\sqrt{51}}{132691}, \qquad \Xi_{f3} = \frac{21\sqrt{51}}{2249} - \frac{171}{4498}, \tag{FE-46}$$
$$\Xi_{fp} = \frac{239\sqrt{51}}{6747} - \frac{485}{2249}, \qquad \Xi_{f\rho} = \frac{57}{4498} - \frac{7\sqrt{51}}{2249}$$

**Symbols in (FE-46).**

- $\Xi_{\mathcal{H}K}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about $K_*$). Unit: $\mathrm{GeV}^2$. Value: $6(238 - 19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\Xi_{fK}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: $\mathrm{GeV}^2$. Value: $(4063 - 485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- $\Xi_{\mathcal{H}3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $2283/4498 - 162\sqrt{51}/2249 = -0.006853$. Status: DERIVED.
- $\Xi_{\mathcal{H}p}$: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $(28\sqrt{51} - 114)/2249 = 0.038221$. Status: DERIVED.
- $\Xi_{\mathcal{H}\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}$. Unit: dimensionless. Value: $135021/265382 - 10308\sqrt{51}/132691 = -0.045997$. Status: DERIVED.
- $\Xi_{f3}$: The coefficient of $\kappa\Delta p_3$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $21\sqrt{51}/2249 - 171/4498 = 0.028666$. Status: DERIVED.
- $\Xi_{fp}$: The coefficient of $\kappa\Delta\bar{p}$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $239\sqrt{51}/6747 - 485/2249 = 0.037320$. Status: DERIVED.

- $\Xi_{f\rho}$: The coefficient of $\kappa\Delta\rho$ in the equation for $\Delta\dot{\mathcal{H}}_f$. Unit: dimensionless. Value: $57/4498 - 7\sqrt{51}/2249 = -0.009555$. Status: DERIVED.

In the adiabatic response (the second time derivative of $\Delta K$ neglected) the block shifts and 3-space obeys exactly the four-dimensional Friedmann acceleration equation (with $c_\rho = 0.035852$, $c_3 = -0.107556$, $c_p = -0.140028$):

$$\Delta K = \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*}, \qquad c_3 = -3c_\rho, \qquad \Delta\dot{\mathcal{H}} = -\frac{\kappa(\Delta\rho + \Delta p_3)}{2\mathcal{G}(K_*)} \tag{FE-47}$$

**Symbols in (FE-47).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $c_\rho$: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response $\Delta K$. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $c_3$: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $c_p$: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.

with $8\pi G_{\mathrm{N}} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Radiation ($p_3 = \rho/3$) drops out of the block response exactly ($c_3 = -3c_\rho$).

### 14.1.2 The static block and attractive gravity with the author's weights (DERIVED)

For a static block ($\mathcal{H}_f = 0$, $K$ constant) and $\dot{\mathcal{H}} = 0$ the components are

$$
\begin{aligned}
&x_4:\ -3\mathcal{G}(K)\,\mathcal{H}^2+6H^2K\big(1-2w_{(2)}K\big)+\Lambda+\kappa\rho=0, \qquad x_1:\ \text{the same with } -\kappa p_3 \text{ in place of } \kappa\rho,\\
&x_0:\ 6\big(12w_{(2)}K-1\big)\mathcal{H}^2-\frac{12\big(w_{(2)}-18w_{(3)}K\big)}{H^2}\,\mathcal{H}^4+3H^2K+\Lambda-\kappa p_0=0,\\
&x_5,x_6,x_7:\ \text{the same with } \kappa p_5 \text{ in place of } \kappa p_0; \qquad \text{block-isotropic order-0 metric: the same with } \kappa\bar{p},\\
&\mathcal{G}(K)=1-24w_{(2)}K+72w_{(3)}K^2=\frac{K^2-12K+16}{16}
\end{aligned}
\tag{FE-42}
$$

**Symbols in (FE-42).**

- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K)=1-24w_{(2)}K+72w_{(3)}K^2=(K^2-12K+16)/16$; four-dimensional gravity is attractive for $\mathcal{G}>0$. Unit: dimensionless. Value: $\mathcal{G}(K_*)=20/3+\sqrt{51}=13.8081$. Status: DERIVED.
- $K$: The block variable, $K=1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)}=1$, $w_{(2)}=1/32$, $w_{(3)}=1/1152$. Unit: dimensionless. Value: $(1,1/32,1/1152)$. Status: AUTHOR.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda=-10H^2$. Unit: $\mathrm{GeV}^2$. Value: $-10H^2$. Status: AUTHOR.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa=8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp=8\pi G_{\mathrm{N}}\mathcal{G}(K_*)=2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\rho$: An energy density: in eight dimensions $\rho=-\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $x_1$: The first coordinate of ordinary 3-space. Unit: length. Status: AUTHOR.
- $p_3$: The pressure of ordinary 3-space, $p_3=\mathcal{T}^{x_1}{}_{x_1}=\mathcal{T}^{x_2}{}_{x_2}=\mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0=\mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $p_5$: The pressure along the extra times, $p_5=\mathcal{T}^{x_5}{}_{x_5}=\mathcal{T}^{x_6}{}_{x_6}=\mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p}=(p_0+3p_5)/4$; for a species at fixed temperature $\bar{p}=\frac{1}{4}\,\partial p/\partial\ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

$\mathcal{G}(K)$ is the coefficient of the four-dimensional Einstein-Hilbert term (FE-48 below): four-dimensional gravity is attractive for $\mathcal{G}>0$. Without sources the only static block with $\mathcal{H}=0$ is $K=1/(4w_{(2)})=8$, which needs the cosmological constant $-24H^2$ and has $\mathcal{G}=-1$: with the author's weights the sourceless block has REPULSIVE gravity, and the author's $\Lambda=-10H^2$ is not the value it needs. For the author's weights $\mathcal{G}(K)=(K^2-12K+16)/16$, positive only for $K<6-2\sqrt{5}$ or $K>6+2\sqrt{5}=10.47$. With the block stabiliser ($p_3=-\rho_{\mathrm{stab}}$, $p_0=p_5=\rho_{\mathrm{stab}}$) and $\mathcal{H}=0$ the two conditions give the stabilised block:

$$
\begin{aligned}
&\kappa\rho_{\mathrm{stab}}=10H^2-6H^2K+\tfrac{3}{8}H^2K^2\ \ (x_4), \qquad \kappa p_0=\kappa p_5=\kappa\rho_{\mathrm{stab}}=3H^2K-10H^2\ \ (x_0;\ x_5,x_6,x_7),\\
&\Rightarrow\ K_\pm=12\pm\tfrac{4}{3}\sqrt{51}, \qquad K_*=12+\tfrac{4}{3}\sqrt{51}=21.5219, \qquad f_*=K_*^{-1/2}=0.215556,\\
&\mathcal{G}(K_*)=\tfrac{20}{3}+\sqrt{51}=13.8081, \qquad \kappa\rho_{\mathrm{stab}}(f_*)=\big(26+4\sqrt{51}\big)H^2
\end{aligned}
\tag{FE-43}
$$

**Symbols in (FE-43).**

- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328\times10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\rho_\mathrm{stab}$: The energy density of the block stabiliser, a function of the block scale, $\rho_\mathrm{stab}\propto f^{-8}$; its value at the static block is $\rho_\mathrm{stab}(f_*)$. Unit: $\mathrm{GeV}^8$. Value: $\kappa\rho_\mathrm{stab}(f_*) = (26+4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $K_\pm$: The two static blocks with the stabiliser, $K_\pm = 12\pm\frac{4}{3}\sqrt{51}$. Unit: dimensionless. Value: 21.5219 and 2.4781. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12+\frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $f_*$: The block scale of the stabilised (static) block, $f_* = K_*^{-1/2}$. Unit: dimensionless. Value: 0.215556. Status: DERIVED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1-24w_{(2)}K+72w_{(3)}K^2 = (K^2-12K+16)/16$; four-dimensional gravity is attractive for $\mathcal{G}>0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3+\sqrt{51} = 13.8081$. Status: DERIVED.

The upper root is the UM-2 vacuum, with attractive four-dimensional gravity and the author's weights and $\Lambda$ unchanged; the lower root $K_- = 2.4781$ has $\mathcal{G} = -0.4748$.

### 14.1.3 Stability: the radion (DERIVED)

Linearising the 3-space and block components (solved for $\dot{\mathcal{H}}$ and $\dot{\mathcal{H}}_f$) with $\dot K = -2K\mathcal{H}_f$ and $\rho_\mathrm{stab}\propto K^4$ about $K_*$:

$$\begin{aligned}&\Delta\dot{\mathcal{H}} = \Xi_{\mathcal{H}K}\,\Delta K, \qquad \Delta\dot{\mathcal{H}}_f = \Xi_{fK}\,\Delta K, \qquad \Delta\dot K = -2K_*\,\Delta\mathcal{H}_f,\\ &\omega_\mathrm{r}^2 = 2K_*\,\Xi_{fK} = \Big(\frac{31552}{2249}-\frac{2416\sqrt{51}}{6747}\Big)H^2 = (3.38705\,H)^2\end{aligned} \tag{FE-44}$$

**Symbols in (FE-44).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2-m_1^2$ a difference of squared masses, $\Delta N_\mathrm{eff}$ an addition to $N_\mathrm{eff}$, $\Delta S_\mathrm{m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$). Status: STANDARD.
- $\dot{\mathcal{H}}$: The time derivative $\mathrm{d}\mathcal{H}/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\Xi_{\mathcal{H}K}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}$ (linearised block about $K_*$). Unit: $\mathrm{GeV}^2$. Value: $6(238-19\sqrt{51})H^2/2249 = 0.272956\,H^2$. Status: DERIVED.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $\dot{\mathcal{H}}_f$: The time derivative $\mathrm{d}\mathcal{H}_f/\mathrm{d}x_4$. Unit: $\mathrm{GeV}^2$. Status: DERIVED.
- $\Xi_{fK}$: The coefficient of $\Delta K$ in the first-order equation for $\Delta\dot{\mathcal{H}}_f$. Unit: $\mathrm{GeV}^2$. Value: $(4063-485\sqrt{51})H^2/2249 = 0.266522\,H^2$. Status: DERIVED.
- $\dot K$: The time derivative $\mathrm{d}K/\mathrm{d}x_4$. Unit: GeV. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12+\frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $\mathcal{H}_f$: The expansion rate of the block, $\mathcal{H}_f = \dot f/f = \dot b_4/(2b_4)-\dot a_4$ (negative when the extra times and $x_0$ deflate). Unit: GeV. Status: DERIVED.

- $\omega_{\rm r}$: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.

with the eigenvalues 0 and $\pm{\rm i}\,\omega_{\rm r}$: the block is stable and oscillates; the zero eigenvalue is the direction of a small four-dimensional cosmological term. The radion (the quantum of block-size oscillations) has the mass $\omega_{\rm r} = 3.38705H = 48.42$ TeV. Stability holds for each of the stabiliser laws whose hidden pressure is $\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, 1, $\frac{5}{4}$, $\frac{3}{2}$, 2 or 3 times their energy density (radion masses $3.33H$ to $4.10H$).

### 14.1.4 The deflating extra times (DERIVED, full nonlinear equations)

The full nonlinear equations FE-38 to FE-40 of this section (all three components, the $x_4$ constraint preserved to $6\times10^{-10}$) were integrated from an expanding 3-space ($\mathcal{H}$ from the constraint) and a block LARGER than the stabilised one ($K < K_*$ initially, i.e. $f > f_*$) that deflates at $\mathcal{H}_f = -H/2$, $-H/3$, $-H/10$, for initial $K$ from 1.5 to 20. Every start with $K$ above the separatrix 2.6983, 2.5991, 2.5301 (for the three initial rates; just above the unstable static block $K_- = 2.4781$, whose stable manifold the separatrix is) ends at $K_*$: $x_5$, $x_6$, $x_7$ and $x_0$ deflate (at rates up to $|\mathcal{H}_f| = 2.63H$ in the stored runs) by up to the factor $(K_*/2.6983)^{1/2} = 2.917$, reach $K_*$ for the first time within $x_4 < 1/H$, and then oscillate about $K_*$ with the radion frequency ($3.3868H$ measured, $3.3870H$ linear; the relative difference $7.9\times10^{-5}$ is the anharmonicity of a finite oscillation, residual analysis), while 3-space keeps expanding; the oscillation energy redshifts like dust ($\mathcal{H}x_4 = 0.667$ to 0.670 late, 2/3 for dust). Starts below the separatrix (still larger blocks) become singular within $x_4 = 0.2/H$ to $1.4/H$ or admit no expanding initial data. The extra times therefore deflate exponentially only for a short primordial era and are then frozen, which is what the constancy of the masses requires (the previous section).

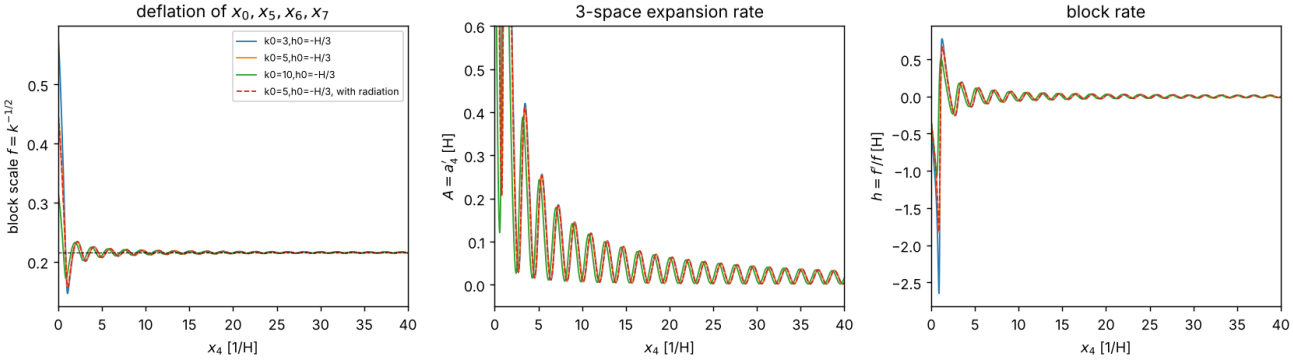



Figure 1: Deflation of the extra times into the stabilised block (full nonlinear Einstein-Lovelock equations with the author's weights).

## 14.2 The four-dimensional effective gravity (DERIVED)

On a product of a four-dimensional metric with the static block (constant curvature $-H^2K$), the Lovelock densities factorise; the four-dimensional quadratic density is topological and the cubic one vanishes in four dimensions, so the reduced Lagrangian (gravity plus the stabiliser, in units of $1/(2\kappa)$) is

$$
\begin{aligned}
&\mathcal{L}_{(1)} = \mathcal{L}^{(4)}_{(1)} + \mathcal{L}^{(\rm int)}_{(1)}, \qquad \mathcal{L}_{(2)} = \mathcal{L}^{(4)}_{(2)} + 2\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\rm int)}_{(1)} + \mathcal{L}^{(\rm int)}_{(2)}, \qquad \mathcal{L}_{(3)} = 3\mathcal{L}^{(4)}_{(2)}\mathcal{L}^{(\rm int)}_{(1)} + 3\mathcal{L}^{(4)}_{(1)}\mathcal{L}^{(\rm int)}_{(2)},\\
&\mathcal{L}^{(4)}_{(1)} = R^{(4)}, \qquad \mathcal{L}^{(4)}_{(3)} = 0, \qquad \mathcal{L}^{(\rm int)}_{(1)} = -12H^2K, \qquad \mathcal{L}^{(\rm int)}_{(2)} = 24H^4K^2, \qquad \mathcal{L}^{(\rm int)}_{(3)} = 0,\\
&\sum_{p=1}^{3}\alpha_{(p)}\,\mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_{\rm stab} = \mathcal{G}(K)\,R^{(4)} + (\text{topological}) - 2\mathcal{G}(K)\,\Lambda_4,\\
&\Lambda_4 = \frac{6H^2K\big(1-2w_{(2)}K\big) + \Lambda + \kappa\rho_{\rm stab}}{\mathcal{G}(K)}
\end{aligned}
\qquad \text{(FE-48)}
$$

**Symbols in (FE-48).**

- $\mathcal{L}_{(p)}$: The Lovelock density of order $p$ of the metric (a scalar built from $p$ Riemann tensors): $\mathcal{L}_{(1)} = R$ (Einstein-Hilbert), $\mathcal{L}_{(2)}$ the Gauss-Bonnet density, $\mathcal{L}_{(3)}$ the cubic Lovelock density. Unit: $1/\text{length}^{2p}$. Status: STANDARD.
- $\mathcal{L}^{(4)}_{(r)}$: The Lovelock density of order $r$ of the four-dimensional metric $g^{(4)}_{\mu\nu}$ ($\mathcal{L}^{(4)}_{(1)} = R^{(4)}$; the order 2 is topological and the order 3 vanishes in four dimensions). Unit: $1/\text{length}^{2r}$. Status: STANDARD.
- $\mathcal{L}^{(\text{int})}_{(r)}$: The Lovelock density of order $r$ of the static block (a four-dimensional space of constant curvature $-H^2K$): 1, $-12H^2K$, $24H^4K^2$, 0 for $r = 0, 1, 2, 3$. Unit: $1/\text{length}^{2r}$. Status: DERIVED.
- $R^{(4)}$: The Ricci scalar of the four-dimensional metric $g^{(4)}_{\mu\nu}$. Unit: $1/\text{length}^2$. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\text{W}$ and $\rho_{\text{EW},0}$). Status: AUTHOR.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $p$: The Lovelock order $p = 1, 2, 3$ (Einstein, Gauss-Bonnet, cubic). Status: STANDARD.
- $\alpha_{(p)}$: The Lovelock coupling of order $p$, $\alpha_{(p)} = w_{(p)}H^{2-2p}$ ($\alpha_{(1)} = 1$). Unit: $\text{GeV}^{2-2p}$. Status: AUTHOR.
- $\Lambda$: The eight-dimensional cosmological constant, $\Lambda = -10H^2$. Unit: $\text{GeV}^2$. Value: $-10H^2$. Status: AUTHOR.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\text{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\text{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}$ $\text{GeV}^{-2}$. Status: AUTHOR.
- $\rho_\text{stab}$: The energy density of the block stabiliser, a function of the block scale, $\rho_\text{stab} \propto f^{-8}$; its value at the static block is $\rho_\text{stab}(f_*)$. Unit: $\text{GeV}^8$. Value: $\kappa\rho_\text{stab}(f_*) = (26 + 4\sqrt{51})H^2 = 54.566\,H^2$. Status: ASSUMED.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $\Lambda_4$: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_\text{stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: $\text{GeV}^2$. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $w_{(p)}$: The author's Lovelock weights, $w_{(1)} = 1$, $w_{(2)} = 1/32$, $w_{(3)} = 1/1152$. Unit: dimensionless. Value: $(1, 1/32, 1/1152)$. Status: AUTHOR.

and the four-dimensional gravity of UM-2 is EXACTLY general relativity (written out again):

$$
\begin{aligned}
&G^{(4)}_{\mu\nu} + \Lambda_4\, g^{(4)}_{\mu\nu} = 8\pi G_\text{N}\,\mathcal{T}^{(4)}_{\mu\nu}, \qquad 8\pi G_\text{N} = \frac{\kappa}{\mathcal{G}(K_*)\,V_\perp}, \qquad V_\perp = \frac{2f^4 v_\text{t}\big(1 - \text{e}^{-3HL}\big)}{3H}, \\
&\mathcal{T}^{(4)}_{\mu\nu} = \int \text{d}x_0\,\text{d}x_5\,\text{d}x_6\,\text{d}x_7\,\sqrt{|g_\perp|}\,\mathcal{T}_{\mu\nu}, \qquad g_3^2 = \frac{\hat{g}_3^2}{V_\perp}, \qquad g_2^2 = \frac{\hat{g}_2^2}{V_\perp}, \qquad g_1^2 = \frac{\hat{g}_1^2}{V_\perp}
\end{aligned}
\tag{FE-49}
$$

**Symbols in (FE-49).**

- $G^{(4)}_{\mu\nu}$: The Einstein tensor of the four-dimensional metric, $R^{(4)}_{\mu\nu} - \frac{1}{2}g^{(4)}_{\mu\nu}R^{(4)}$. Unit: $1/\text{length}^2$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- $\Lambda_4$: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_\text{stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: $\text{GeV}^2$. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_\text{N}$: Newton's constant of four-dimensional gravity, $8\pi G_\text{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\text{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\text{GeV}^{-2}$ ($M_\text{Pl} = G_\text{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{T}^{(4)}_{\mu\nu}$: The four-dimensional energy-momentum tensor: $\mathcal{T}_{\mu\nu}$ integrated over the block (both halves of a period of $x_0$ and the extra times). Unit: $\text{GeV}^4$. Status: DERIVED.

- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_\mathrm{N}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $V_\perp$: The proper volume of the block per unit four-volume (both halves of a period of $x_0$ and the extra times), $V_\perp = 2f^4 v_\mathrm{t}(1 - \mathrm{e}^{-3HL})/(3H)$. Unit: $\mathrm{GeV}^{-4}$. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_\mathrm{t}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z = 6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $x_5$: The first of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_6$: The second of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $x_7$: The third of the three extra time-like coordinates. Unit: length. Status: AUTHOR.
- $g_\perp$: The metric of the four transverse directions $x_0, x_5, x_6, x_7$ (the block); $|g_\perp|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $\mathcal{T}^M{}_N$: The energy-momentum tensor of all matter (every field except gravity, the stabiliser included), the source of gravity; defined by the change of the matter action under a change of the metric. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $g_3$: The four-dimensional strong coupling ($g_3^2 = 4\pi\alpha_3$). Unit: dimensionless. Value: $g_3^2 = 0.7786$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_3$: The eight-dimensional SU(3) coupling (it multiplies the gluon kinetic term as $1/\hat{g}_3^2$); $g_3^2 = \hat{g}_3^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_2$: The eight-dimensional SU(2) coupling; $g_2^2 = \hat{g}_2^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.
- $g_1$: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $\hat{g}_1$: The eight-dimensional $\mathrm{U}(1)_Y$ coupling; $g_1^2 = \hat{g}_1^2/V_\perp$. Unit: $\mathrm{GeV}^{-2}$. Status: ASSUMED.

The cosmological term $\Lambda_4$ vanishes at the static point by construction. The observed dark energy is a small residual of this balance, $\Lambda_4/H^2 \sim (\mathcal{H}_0/H)^2 = 1.0 \times 10^{-92}$: tuned (the cosmological-constant problem, OPEN).

# 15 dirac16complex in the cell: the Kohn-Sham description and the coupled solution

$\Psi$ is quantised canonically with $x_4$ as the time; the anticommutator of the field and its adjoint on a slice carries the indefinite operator $\mathcal{B}$, which is handled in the good sector by counting the positive branch of every block as particles (the record's convention, ASSUMED). The interacting many-body state is described by Kohn-Sham theory, coupled to gravity and to all Standard-Model fields: orbitals of the block Hamiltonian with the mean fields of the self-interaction and of the three portals (the I13 term acts on the lower component only),

$$h\,\chi_o = \varepsilon_o\,\chi_o, \qquad h = \varsigma\Big[-\frac{\mathrm{i}}{f}\sigma_1\partial_y + M_{\rm eff}(y)\,\sigma_2 + \mathrm{e}^{-a_4}k\,\sigma_3\Big] + v(y) + \frac{1-\sigma_3}{2}\,v_{\rm b}(y) \tag{FE-63}$$

**Symbols in (FE-63).**

- $h$: The block Hamiltonian: the $2\times 2$ differential operator of one block that generates the time evolution of $\chi$, $\mathrm{i}\,\partial_{x_4}\chi = h\chi$; its eigenvalues are the energies $\varepsilon$. Unit: GeV. Status: DERIVED.
- $\chi,\ \chi_o,\ \chi_1,\ \chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1,\chi_2)^T = (a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0}\chi_o^\dagger\chi_o\,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $\varepsilon,\ \varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k=0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $\varsigma$: The first of the three signs $(\varsigma,\varsigma_2,\varsigma_3)$, each $\pm1$, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_1,\ \sigma_2,\ \sigma_3$: The Pauli matrices acting on the two components of a block spinor $\chi$ ($\sigma_3 = \mathrm{diag}(1,-1)$). Unit: dimensionless. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $y$: The label $y$ in $\partial_y$: the derivative with respect to the hidden coordinate $y$. Status: STANDARD.
- $M_{\rm eff}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: $m_\Psi$ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $k$: The magnitude of the wave vector (the momentum, $\hbar=1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along $\gamma^{x_4}$): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $v_{\rm b}(y)$: The mean field of I13 (baryon densities), acting on the lower component of $\chi$ only. Unit: GeV. Status: DERIVED.

with the mean fields

$$M_{\rm eff}(y) = m_\Psi + \frac{15}{16}\,\lambda\,n^{\rm sc}(y) + \frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}, \qquad v(y) = -\frac{\lambda}{16}\,n(y) + \frac{c_{\rm V}}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi} q_\psi\,n_{\psi i}(y),$$
$$v_{\rm b}(y) = \frac{c_{\rm A}}{\Lambda_\star^6}\sum_{i=1}^{3}\sum_{\psi} q_\psi^B\,n_{\psi i}(y) \tag{FE-64}$$

**Symbols in (FE-64).**

- $M_{\rm eff}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: $m_\Psi$ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.

- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7\times 10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times 10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $n^{\mathrm{sc}}$, $n_o^{\mathrm{sc}}$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and $n_o^{\mathrm{sc}}$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $\mathcal{F}$: The gauge-kinetic operator of the portal I12, $\mathcal{F} = c_{\mathrm{G}}G^A_{MN}G^{AMN} + c_{\mathrm{W}}W^I_{MN}W^{IMN} + c_{\mathrm{B}}B_{MN}B^{MN}$; $\langle\mathcal{F}\rangle$ its mean value in an environment. Unit: $\mathrm{GeV}^4$. Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along $\gamma^{x_4}$): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $c_{\mathrm{V}}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_{\mathrm{V}} = 4.38\times 10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $v_{\mathrm{b}}(y)$: The mean field of I13 (baryon densities), acting on the lower component of $\chi$ only. Unit: GeV. Status: DERIVED.
- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278\times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic). Status: INVENTED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.

($n_{\psi i}$ the proper eight-dimensional densities of the Standard-Model fermions of the three generations, $\langle\mathcal{F}\rangle$ the condensate of the gauge invariant of I12; in the QCD vacuum the gluon condensate, a uniform mass shift), the densities, momentum shells, cell volume and occupations

$$
\begin{aligned}
&n = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o, \qquad n^{\mathrm{sc}} = \sum_o \tfrac{1}{2}\, d_o\, \phi_o\, n_o^{\mathrm{sc}}, \qquad n_o = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\,\chi_o^\dagger\chi_o,\\
&n_o^{\mathrm{sc}} = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\,\chi_o^\dagger\varsigma\sigma_2\chi_o, \qquad n_o^{\mathrm{t}} = \frac{\mathrm{e}^{-3Hy}}{\mathrm{Vol}_7}\,\chi_o^\dagger\varsigma\sigma_3\chi_o, \qquad d_o = 4\, r_3\big(|\vec n_o|^2\big), \qquad \vec k_o = k_{\mathrm{lat}}\,\vec n_o,\\
&\mathrm{Vol}_7 = \Big(\frac{2\pi}{k_{\mathrm{lat}}}\Big)^3 v_{\mathrm{t}}\, f^4, \qquad \phi_o = \frac{1}{\mathrm{e}^{(\varepsilon_o-\mu_{\mathrm{chem}})/T}+1} \quad \text{(Mermin; or 0 and 1 for the aufbau)}
\end{aligned}
\tag{FE-65}
$$

**Symbols in (FE-65).**

- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $d_o$, $d_s$: A degeneracy: $d_o = 4r_3(|\vec n_o|^2)$ the number of states of the momentum shell of the orbital $o$; $d_s$ the number of internal states (spin, colour, particle and antiparticle) of the species $s$. Unit: dimensionless. Status: STANDARD.

- $\phi_o$: The occupation of the Kohn-Sham orbital $o$ (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- $n^{\mathrm{sc}}$, $n_o^{\mathrm{sc}}$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and $n_o^{\mathrm{sc}}$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $\mathrm{Vol}_7$: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: $\mathrm{GeV}^{-6}$. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\varsigma$: The first of the three signs $(\varsigma, \varsigma_2, \varsigma_3)$, each $\pm 1$, that label the eight blocks of the reduction. Unit: dimensionless. Status: DERIVED.
- $\sigma_1$, $\sigma_2$, $\sigma_3$: The Pauli matrices acting on the two components of a block spinor $\chi$ ($\sigma_3 = \mathrm{diag}(1, -1)$). Unit: dimensionless. Status: STANDARD.
- $n_o^{\mathrm{t}}$: The density of the orbital $o$ along the momentum term ($\chi_o^\dagger \varsigma \sigma_3 \chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $r_3$: The number of representations of an integer as an ordered sum of three squares of integers (signs counted). Unit: dimensionless. Status: STANDARD.
- $\vec{n}_o$: The integer lattice vector of the momentum shell of the orbital $o$ ($\vec{n} \in \mathbb{Z}^3$). Unit: dimensionless. Status: RECORD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $k_{\mathrm{lat}}$: The spacing of the momentum lattice of the Kohn-Sham cell. Unit: GeV. Value: $0.25\,H$. Status: RECORD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $v_{\mathrm{t}}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $\mu_{\mathrm{chem}}$: The chemical potential of the Mermin occupations. Unit: GeV. Status: RECORD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

($k_{\mathrm{lat}} = 0.25H$ in the computations; the proper seven-volume of a cell, with its hidden extent, is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$; the occupations are those of the aufbau or of the Mermin functional), the Kohn-Sham energy

$$E_{\mathrm{KS}} = \sum_o d_o\, \phi_o\, \varepsilon_o - 2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\, e_{\mathrm{int}}\,\mathrm{d}y, \qquad e_{\mathrm{int}} = \frac{15}{32}\,\lambda\,(n^{\mathrm{sc}})^2 - \frac{1}{32}\,\lambda\, n^2 \tag{FE-66}$$

**Symbols in (FE-66).**

- $E_{\mathrm{KS}}$: The Kohn-Sham energy of a cell. Unit: GeV. Status: RECORD.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.

- $d_o$, $d_s$: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital $o$; $d_s$ the number of internal states (spin, colour, particle and antiparticle) of the species $s$. Unit: dimensionless. Status: STANDARD.
- $\phi_o$: The occupation of the Kohn-Sham orbital $o$ (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $\mathrm{Vol}_7$: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - \mathrm{e}^{-3HL})/(3H)$). Unit: $\mathrm{GeV}^{-6}$. Value: 34.27 in units $H = 1$, $v_{\mathrm{t}} = 1$, $f = f_*$. Status: RECORD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $e_{\mathrm{int}}$: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: $\mathrm{GeV}^8$. Status: RECORD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar{\Psi}\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $n^{\mathrm{sc}}$, $n^{\mathrm{sc}}_o$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and $n^{\mathrm{sc}}_o$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.

and the energy-momentum tensor of a Kohn-Sham state, the source of $\Psi$ in the gravitational equations (its block-averaged pressure $\bar{p} = (p_0 + 3p_5)/4$ pushes on the block):

$$\begin{aligned} \rho &= \sum_o \tfrac{1}{2}\, d_o \phi_o\, \varepsilon_o\, n_o - e_{\mathrm{int}}, \qquad p_3 = \sum_o \tfrac{1}{2}\, d_o \phi_o\, \frac{\mathrm{e}^{-a_4|\vec{k}_o|}}{3}\, n^{\mathrm{t}}_o + e_{\mathrm{int}}, \qquad p_5 = e_{\mathrm{int}}, \\ p_0 &= \sum_o \tfrac{1}{2}\, d_o \phi_o \Big[ (\varepsilon_o - v)\, n_o - M_{\mathrm{eff}}\, n^{\mathrm{sc}}_o - \mathrm{e}^{-a_4|\vec{k}_o|}\, n^{\mathrm{t}}_o \Big] + e_{\mathrm{int}} \end{aligned} \tag{FE-67}$$

**Symbols in (FE-67).**

- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $d_o$, $d_s$: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital $o$; $d_s$ the number of internal states (spin, colour, particle and antiparticle) of the species $s$. Unit: dimensionless. Status: STANDARD.
- $\phi_o$: The occupation of the Kohn-Sham orbital $o$ (Fermi-Dirac for the Mermin functional, 0 or 1 for the aufbau). Unit: dimensionless. Status: RECORD.
- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.

- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $e_{\mathrm{int}}$: The interaction energy density of the contact term in the Kohn-Sham energy and energy-momentum tensor. Unit: $\mathrm{GeV}^8$. Status: RECORD.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $n_o^{\mathrm{t}}$: The density of the orbital $o$ along the momentum term ($\chi_o^\dagger \varsigma\sigma_3\chi_o$ times the weight $\mathrm{e}^{-3Hy}/\mathrm{Vol}_7$); it carries the 3-space pressure. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $v(y)$: The vector potential of the block Hamiltonian (the mean field along $\gamma^{x_4}$): for dirac16complex the contact term and the mean field of I11; zero for a Standard-Model fermion. Unit: GeV. Status: DERIVED.
- $M_{\mathrm{eff}}(y)$: The effective (scalar) mass field of the Kohn-Sham Hamiltonian: $m_\Psi$ plus the mean fields of the contact term and of I12. Unit: GeV. Status: DERIVED.
- $n^{\mathrm{sc}}$, $n_o^{\mathrm{sc}}$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar{\Psi}\Psi$, per proper 7-volume) and $n_o^{\mathrm{sc}}$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.

Every self-consistent state satisfies the energy identity and the $y$-conservation law (DERIVED, to $4\times10^{-15}$ and $9\times10^{-8}$ with Richardson extrapolation)

$$E_{\mathrm{KS}} = 2\,\mathrm{Vol}_7 \int_{-L}^{0} \mathrm{e}^{3Hy}\rho\,\mathrm{d}y, \qquad p_0\big|_{y=0} - \mathrm{e}^{-3HL}\,p_0\big|_{y=-L} = 3H\int_{-L}^{0}\mathrm{e}^{3Hy}p_5\,\mathrm{d}y \tag{FE-68}$$

**Symbols in (FE-68).**

- $E_{\mathrm{KS}}$: The Kohn-Sham energy of a cell. Unit: GeV. Status: RECORD.
- $\mathrm{Vol}_7$: The normalisation volume of a Kohn-Sham cell, $\mathrm{Vol}_7 = (2\pi/k_{\mathrm{lat}})^3 v_{\mathrm{t}} f^4$ (the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1-\mathrm{e}^{-3HL})/(3H)$). Unit: $\mathrm{GeV}^{-6}$. Value: 34.27 in units $H=1$, $v_{\mathrm{t}}=1$, $f=f_*$. Status: RECORD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le 0$. Unit: length. Status: DERIVED.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $p_0$: The pressure along the hidden direction $x_0$, $p_0 = \mathcal{T}^{x_0}{}_{x_0}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $p_5$: The pressure along the extra times, $p_5 = \mathcal{T}^{x_5}{}_{x_5} = \mathcal{T}^{x_6}{}_{x_6} = \mathcal{T}^{x_7}{}_{x_7}$. Unit: $\mathrm{GeV}^8$. Status: STANDARD.

and the orbitals obey the boundary conditions of dirac16complex (written out again)

$$\big(1-\gamma^{x_0}\big)\Psi\big|_{y=0}=0\ \ \text{(even)}\quad\text{or}\quad\big(1+\gamma^{x_0}\big)\Psi\big|_{y=0}=0\ \ \text{(odd)},\qquad\big(1-\gamma^{x_0}\big)\Psi\big|_{y=-L}=0 \tag{FE-28}$$

**Symbols in (FE-28).**

- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M=\mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M=g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $x_0$: The hidden space-like coordinate (listed first); the metric depends on it through $z=6Hx_0$, periodically with the period $\pi/(6H)$. Unit: length. Status: AUTHOR.
- $\Psi$, $\bar\Psi$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar\Psi=\Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $y$: The hidden coordinate of the $y$ chart, $y=\ln|\sin z|/(6H)$, with $y=0$ at the fold ($z=\pi/2$) and $y=-L$ at the regular tip; each half of a period of $x_0$ is the interval $-L\le y\le0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.

in their block form:

$$\chi=(\chi_1,\ \chi_2)^T:\qquad\chi_2(0)=0\ \ \text{(even)}\quad\text{or}\quad\chi_1(0)=0\ \ \text{(odd)},\qquad\chi_2(-L)=0 \tag{FE-69}$$

**Symbols in (FE-69).**

- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi=(\chi_1,\chi_2)^T=(a,\mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^0\chi_o^\dagger\chi_o\,\mathrm{d}y=1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.

The Kohn-Sham solver reproduces the record's reference states to $10^{-13}$; the dark matter at rest has $p_3=0$ and $p_0/\rho=0.772034=-\mathrm{d}\ln\varepsilon_{\mathrm{DM}}/\mathrm{d}\ln f$ (two independent computations agree to $10^{-9}$).

## 15.1 The coupled solution (DERIVED)

The coupled equations are solved by iterating the Kohn-Sham equations of $\Psi$ with the mean fields of the Standard-Model fields and the response of the block until self-consistency (16 checks). The four-dimensional couplings of a state of $\Psi$ to the Standard-Model fermions are the overlaps of the portals on the zero modes (written out again):

$$\begin{aligned}
&\mathcal{L}_4=\big(G_{\mathrm{V}}\,q_f\,\tilde I_{of}+G_{\mathrm{A}}\,q_f^B\,\tilde I_{of}^{(b)}\big)\big(\mathrm{i}\bar\Psi_o\gamma^\mu\Psi_o\big)\big(\mathrm{i}\bar\psi_f\gamma_\mu\psi_f\big),\\
&\tilde I_{of}=\frac{1}{H}\int_{-L}^0\big(a_o^2+b_o^2\big)\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y,\qquad\tilde I_{of}^{(b)}=\frac{1}{H}\int_{-L}^0 b_o^2\,\varrho_f\,\mathrm{e}^{-3Hy}\,\mathrm{d}y,\qquad G_{\mathrm{V}}=\frac{c_{\mathrm{V}}H}{2f^4v_{\mathrm{t}}\Lambda_\star^6},\qquad G_{\mathrm{A}}=\frac{c_{\mathrm{A}}H}{2f^4v_{\mathrm{t}}\Lambda_\star^6},\\
&n_{\psi i}(y)=\frac{n_f\,\varrho_f(y)\,\mathrm{e}^{-3Hy}}{2f^4v_{\mathrm{t}}},\qquad\varrho(y;c)=\frac{2c}{L}\,\frac{\mathrm{e}^{2cy/L}}{1-\mathrm{e}^{-2c}}
\end{aligned} \tag{FE-70}$$

**Symbols in (FE-70).**

- $\mathcal{L}_4$: The four-dimensional contact Lagrangian of I11 and I13 between a state $o$ of dirac16complex and a Standard-Model fermion component $f$. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}}=c_{\mathrm{V}}H/(2f^4v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}}=0.001$). Status: DERIVED.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q=q_{u_R}=q_{d_R}=\frac13$, $q_\ell=q_{e_R}=q_{\nu_R}=-1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.

- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_\mathrm{c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $G_\mathrm{A}$: The four-dimensional strength of the portal I13 on the zero modes, $G_\mathrm{A} = c_\mathrm{A} H/(2f^4 v_\mathrm{t}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^{0} b_o^2\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\Psi$, $\bar{\Psi}$: dirac16complex: the 16-component complex spinor field of eight dimensions of the record (the author's field), the source of the dark matter (its lowest odd level) and of the massless fold bands; $\bar{\Psi} = \Psi^\dagger C$ is its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: RECORD.
- $\gamma^a$, $\gamma^M$, $\gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16\times16$ matrices of the Clifford algebra of signature $(4,4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $a$, $b$: The upper and lower components of a block spinor, $\chi = (a, \mathrm{i}b)$, real at $k = 0$: $a_L, b_L$ of a doublet and $a_R, b_R$ of a singlet solution ($a_{Ri}$, $b_{Ri}$ of the generation $i$, $b_{L\alpha}$, $a_{L\alpha}$ of the doublet in the flavour basis), $a_o, b_o$ of the orbital $o$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- $\varrho_f(y)$, $\varrho(y;c)$: The $x_0$ profile of a zero mode as a probability density on one half, $\int_{-L}^{0}\varrho\,\mathrm{d}y = 1$: $\varrho_f$ that of the fermion component $f$, $\varrho(y;c)$ its form for the localisation $c$. Unit: 1/length (GeV). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $c_\mathrm{V}$: The coefficient of the vector portal I11. Unit: dimensionless. Value: benchmark through $G_\mathrm{V} = 4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$. Status: INVENTED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_\mathrm{t}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_\mathrm{t} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.

- $c_{\mathrm{A}}$: The coefficient of the $x_0$-chiral vector portal I13. Unit: dimensionless. Value: fixed through $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ GeV$^{-2}$ (dark-matter relic). Status: INVENTED.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: GeV$^7$ (eight dimensions), GeV$^3$ (four dimensions). Status: STANDARD.
- $\psi$: The label of a Standard-Model multiplet: $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ (quark doublet, up and down quark singlets, lepton doublet, charged-lepton singlet, right-handed neutrino); a sum over $\psi$ runs over the six. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.

| zero mode $f$ | $\tilde{I}_{of}$, dark matter | $\tilde{I}^{(b)}_{of}$, dark matter | $\tilde{I}_{of}$, fold band |
|---|---|---|---|
| quark doublet $Q$ | 8.272 | 2.613 | 6.685 |
| lepton doublet $\ell$ | 1.252 | 0.913 | 1.659 |
| up $u_R$ | 0.385 | 0.379 | 0.792 |
| charm $u_R$ | 0.606 | 0.557 | 1.060 |
| top $u_R$ | 8.272 | 2.613 | 6.685 |
| down $d_R$ | 0.396 | 0.389 | 0.807 |
| strange $d_R$ | 0.462 | 0.446 | 0.893 |
| bottom $d_R$ | 0.762 | 0.661 | 1.221 |
| electron $e_R$ | 0.397 | 0.390 | 0.808 |
| muon $e_R$ | 0.588 | 0.544 | 1.041 |
| tau $e_R$ | 1.252 | 0.913 | 1.659 |
| right-handed neutrinos $\nu_{Ri}$ | 0.302-0.308 | 0.301-0.308 | 0.668-0.678 |

($\nu_{Ri}$: the three right-handed neutrinos; the top and tau singlets share the profile of their doublets, $c_R = -c_L$; $\tilde{I}^{(b)}_{of} = 0$ for the fold band.) Each overlap equals the derivative of the self-consistent Kohn-Sham level with respect to the amplitude of the corresponding mean field to $3 \times 10^{-10}$ (I11) and $2 \times 10^{-10}$ (I13; Richardson in the amplitude); in the interacting state (eight fold zero modes, eight dark-matter quanta, $\lambda_{\mathrm{test}}$) the response is screened by only $3.5 \times 10^{-4}$. The fold band stays massless under every I11 mean field (its first-order change of the gap per unit amplitude, $7.9 \times 10^{-11}$ for the electron density, $9.3 \times 10^{-11}$ for the neutrino density and $3.0 \times 10^{-9}$ for the up-quark density, is rounding; zero in exact arithmetic; the dispersion stays linear), under the I12 gluon condensate ($1.4 \times 10^{-10}$, rounding) and under I13 (computed shifts at most $1.2 \times 10^{-13}$, rounding; zero in exact arithmetic).

**The Standard-Model fields acting on $\Psi$** in ten environments. To first order in the mean fields (Hellmann-Feynman; universal I12 coefficients) a Kohn-Sham level moves by

$$\Delta\varepsilon_o = \sum_f \left(G_{\mathrm{V}}\, q_f\, \tilde{I}_{of} + G_{\mathrm{A}}\, q_f^B\, \tilde{I}^{(b)}_{of}\right) n_f + \mathcal{O}_o\, \frac{c_{\mathrm{I12}}}{\Lambda_\star^3} \Big[\big\langle G^A_{\mu\nu} G^{A\mu\nu}\big\rangle + \big\langle W^I_{\mu\nu} W^{I\mu\nu}\big\rangle + \big\langle B_{\mu\nu} B^{\mu\nu}\big\rangle\Big], \qquad \mathcal{O}_o = \frac{\partial\varepsilon_o}{\partial m_\Psi} \quad \text{(FE-72)}$$

**Symbols in (FE-72).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$

an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.

- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38 \times 10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- $q_\psi$, $q_f$: The $B - L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2 + b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $\mathcal{O}_o$, $\mathcal{O}_{\mathrm{DM}}$: The scalar overlap of a state of dirac16complex, $\mathcal{O}_o = \partial\varepsilon_o/\partial m_\Psi$ (Hellmann-Feynman: the integrated scalar density of the orbital); $\mathcal{O}_{\mathrm{DM}}$ that of the dark matter. Unit: dimensionless. Value: $\mathcal{O}_{\mathrm{DM}} = 0.7072$. Status: DERIVED.
- $c_{\mathrm{I12}}$: The common coefficient of the three terms of I12 when they are universal ($c_{\mathrm{G}} = c_{\mathrm{W}} = c_{\mathrm{B}} = c_{\mathrm{I12}}$). Unit: dimensionless. Value: $c_{\mathrm{I12}}/\Lambda_\star^3 = 1.106 \times 10^{-14}$ $\mathrm{GeV}^{-3}$ (unitarity-safe benchmark). Status: INVENTED.
- $\Lambda_\star$: The scale of the portal operators I11, I12, I13. Unit: GeV. Value: only the combinations $c/\Lambda_\star^n$ enter. Status: INVENTED.
- $G_M^A$, $G_M$, $G_{MN}$, $G_{MN}^A$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G_M^A$ ($A = 1, \ldots, 8$), the matrix $G_M = G_M^A t^A$, the field strength $G_{MN}$ (matrix) and its components $G_{MN}^A$; $G_{\mu\nu}^A$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1, \ldots, 8$; a repeated index is summed. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- $W_M^I$, $W_M$, $W_{MN}$, $W_{MN}^I$: The SU(2) gauge field (geometric normalisation): the components $W_M^I$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W_M^I \tau^I/2$, the field strength $W_{MN}$ and its components $W_{MN}^I$; with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.

- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.

and with unpolarised fermions (each chirality carries half of the density) and the net quark densities of protons and neutrons the dark-matter level moves by

$$\Delta\varepsilon_{\mathrm{I11}} = G_{\mathrm{V}}\Big[\frac{1}{3}\,n_{\mathrm{u}}\,\frac{\tilde{I}_Q+\tilde{I}_{u_R}}{2}+\frac{1}{3}\,n_{\mathrm{d}}\,\frac{\tilde{I}_Q+\tilde{I}_{d_R}}{2}-n_{\mathrm{e}}\,\frac{\tilde{I}_\ell+\tilde{I}_{e_R}}{2}\Big],$$
$$\Delta\varepsilon_{\mathrm{I13}} = G_{\mathrm{A}}\Big[\frac{1}{3}\,n_{\mathrm{u}}\,\frac{\tilde{I}^{(b)}_Q+\tilde{I}^{(b)}_{u_R}}{2}+\frac{1}{3}\,n_{\mathrm{d}}\,\frac{\tilde{I}^{(b)}_Q+\tilde{I}^{(b)}_{d_R}}{2}\Big],\qquad n_{\mathrm{u}}=2n_{\mathrm{p}}+n_{\mathrm{n}},\qquad n_{\mathrm{d}}=n_{\mathrm{p}}+2n_{\mathrm{n}} \tag{FE-73}$$

**Symbols in (FE-73).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$). Status: STANDARD.
- $\Delta\varepsilon_{\mathrm{I11}}$, $\Delta\varepsilon_{\mathrm{I13}}$: The shifts of the dark-matter level by the mean fields of I11 and of I13 in matter. Unit: GeV. Value: at most $8.2\times10^{-10}$ GeV (I13, at the freeze-out). Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}}H/(2f^4v_{\mathrm{t}}\Lambda^6_\star)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}}=0.001$). Status: DERIVED.
- $n_{\mathrm{u}}$, $n_{\mathrm{d}}$: The number densities of up and down quarks bound in nucleons, $n_{\mathrm{u}}=2n_{\mathrm{p}}+n_{\mathrm{n}}$, $n_{\mathrm{d}}=n_{\mathrm{p}}+2n_{\mathrm{n}}$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q=8.2722$, $\tilde{I}_\ell=1.2524$, $\tilde{I}_{u_R}=0.3854$, $\tilde{I}_{d_R}=0.3961$, $\tilde{I}_{e_R}=0.3972$. Status: DERIVED.
- $n_{\mathrm{e}}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4v_{\mathrm{t}}\Lambda^6_\star)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_Q$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_Q=2.6135$. Status: DERIVED.
- $n_{\mathrm{p}}$: The number density of protons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_{\mathrm{n}}$: The number density of neutrons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.

The gluon condensate follows from the trace anomaly of QCD,

$$\big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(T) = \big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(0) - \frac{32\pi^2}{b_3}\,\Theta_{\mathrm{QCD}},\qquad \big\langle G^A_{\mu\nu}G^{A\mu\nu}\big\rangle(0) = 4\pi^2\times0.012\ \mathrm{GeV}^4,\qquad b_3 = 11-\frac{2}{3}\,N_{\mathrm{f}} \tag{FE-74}$$

**Symbols in (FE-74).**

- $G^A_M$, $G_M$, $G_{MN}$, $G^A_{MN}$: The gluon field (SU(3)), in the geometric normalisation of the action (couplings in front of the kinetic terms): the components $G^A_M$ ($A=1,\ldots,8$), the matrix $G_M = G^A_M t^A$, the field strength $G_{MN}$ (matrix) and its components $G^A_{MN}$; $G^A_{\mu\nu}$ are its components along the observable directions (the zero mode, flat in $x_0$). Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $A$: A colour (adjoint SU(3)) index running over $1,\ldots,8$; a repeated index is summed. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1,x_2,x_3,x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1,\ldots,x_4$ (as $\mu$). Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $b_3,\ b_2,\ b_1,\ b_{\mathrm{G}}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\mathrm{ren}} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\mathrm{f}}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\mathrm{G}}$ that of the group G (not related to the metric function $b_4$). Unit: dimensionless. Status: STANDARD.
- $\Theta_{\mathrm{QCD}},\ \Theta_{\mathrm{G}},\ \Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\mathrm{QCD}}$ of the quark-gluon plasma or hadron gas, $\Theta_{\mathrm{G}}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_{\mathrm{W}}$ of $W^{\pm}$, $\Theta_{\mathrm{Z}}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $N_{\mathrm{f}}$: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.

with the QCD interaction measure (a lattice-like shape below 1 GeV, an ESTIMATE, and leading-order perturbation theory above, with $\alpha_3$ at the scale $2\pi T$)

$$\frac{\Theta_{\mathrm{QCD}}}{T^4} = 4\exp\Big[-\frac{\ln^2(T/0.2\,\mathrm{GeV})}{0.5}\Big]\ \ (T<1\,\mathrm{GeV}), \qquad \frac{\Theta_{\mathrm{QCD}}}{T^4} = \frac{b_3}{3}\Big(1+\frac{5N_{\mathrm{f}}}{12}\Big)\alpha_3^2\ \ (T>1\,\mathrm{GeV}),$$
$$p_{\mathrm{QCD}} = \frac{8\pi^2}{45}T^4\Big[1+\frac{21N_{\mathrm{f}}}{32}-\frac{15}{4}\Big(1+\frac{5N_{\mathrm{f}}}{12}\Big)\frac{\alpha_3}{\pi}\Big], \qquad \frac{\mathrm{d}\alpha_3}{\mathrm{d}\ln T} = -\frac{b_3}{2\pi}\,\alpha_3^2, \qquad \Theta_{\mathrm{QCD}} = T^5\,\frac{\mathrm{d}}{\mathrm{d}T}\Big(\frac{p_{\mathrm{QCD}}}{T^4}\Big) \tag{FE-75}$$

**Symbols in (FE-75).**

- $\Theta_{\mathrm{QCD}},\ \Theta_{\mathrm{G}},\ \Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\mathrm{QCD}}$ of the quark-gluon plasma or hadron gas, $\Theta_{\mathrm{G}}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_{\mathrm{W}}$ of $W^{\pm}$, $\Theta_{\mathrm{Z}}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $b_3,\ b_2,\ b_1,\ b_{\mathrm{G}}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\mathrm{ren}} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\mathrm{f}}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\mathrm{G}}$ that of the group G (not related to the metric function $b_4$). Unit: dimensionless. Status: STANDARD.
- $N_{\mathrm{f}}$: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.
- $\alpha_3$: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- $p_{\mathrm{QCD}}$: The pressure of the quark-gluon plasma (leading orders). Unit: $\mathrm{GeV}^4$. Status: ESTIMATE.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.

For the massive vector bosons the free part of the thermal condensate follows from the on-shell Proca Lagrangian, and above $m_{\mathrm{W}}$ the interaction pressure of the electroweak plasma dominates (Feynman-Hellmann; for QCD it reproduces the trace-anomaly condensate exactly, DERIVED to $10^{-16}$; for $SU(2)$ $d_{\mathrm{adj}} = 3$, $C_{\mathrm{adj}} = 2$ and twelve Weyl doublets, $T_{\mathrm{F}}N_{\mathrm{f}} = 3$; for $U(1)_Y$ $d_{\mathrm{adj}} = 1$, $C_{\mathrm{adj}} = 0$ and $\frac{5}{2}T_{\mathrm{F}}N_{\mathrm{f}} \to \frac{25}{2}$; no scalar term, ESTIMATE; below $m_{\mathrm{W}}$ the order-$\alpha$ part is multiplied by the thermal suppression of a massive vector, ESTIMATE):

$$\big\langle F^V_{\mu\nu}F^{V\,\mu\nu}\big\rangle(T) = -2\,\Theta_V\ \ (\text{for } V = W^{\pm}, Z), \qquad \big\langle F^2_{\mathrm{geo}}\big\rangle = -4\,\frac{\partial p}{\partial\big(1/g_{\mathrm{G}}^2\big)} = 4g_{\mathrm{G}}^2\,p_{\alpha,\mathrm{G}},$$
$$p_{\alpha,\mathrm{G}} = -\frac{g_{\mathrm{G}}^2T^4}{144}\,d_{\mathrm{adj}}\Big[C_{\mathrm{adj}} + \frac{5}{2}\,T_{\mathrm{F}}N_{\mathrm{f}}\Big] \tag{FE-76}$$

**Symbols in (FE-76).**

- $F^V_{\mu\nu}$: The field strength of the canonically normalised massive vector $V = W^{\pm}$ or $Z$, $F^V_{\mu\nu} = \partial_\mu V_\nu - \partial_\nu V_\mu$. Unit: $\mathrm{GeV}^2$. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.

- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $\Theta_{\mathrm{QCD}}$, $\Theta_{\mathrm{G}}$, $\Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\mathrm{QCD}}$ of the quark-gluon plasma or hadron gas, $\Theta_{\mathrm{G}}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_{\mathrm{W}}$ of $W^{\pm}$, $\Theta_{\mathrm{Z}}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $F_{\mathrm{geo}}$: The field strength of the gauge group G in geometric normalisation; $\langle F^2_{\mathrm{geo}}\rangle$ is its condensate ($G^A_{\mu\nu}G^{A\mu\nu}$, $W^I_{\mu\nu}W^{I\mu\nu}$ or $B_{\mu\nu}B^{\mu\nu}$). Unit: $\mathrm{GeV}^2$. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $g_{\mathrm{G}}$: The four-dimensional coupling of the gauge group G (one of $g_3$, $g_2$, $g_1$). Unit: dimensionless. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-$\alpha$ (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $d_{\mathrm{adj}}$: The dimension of the adjoint representation of the gauge group (8, 3, 1). Unit: dimensionless. Status: STANDARD.
- $C_{\mathrm{adj}}$: The quadratic Casimir of the adjoint representation (3, 2, 0). Unit: dimensionless. Status: STANDARD.
- $T_{\mathrm{F}}$: The Dynkin index of the fermion representation ($\frac{1}{2}$ per Dirac fundamental; for $\mathrm{U}(1)_Y$ the sum of the squared hypercharges). Unit: dimensionless. Status: STANDARD.
- $N_{\mathrm{f}}$: The number of active quark flavours (quarks lighter than the scale). Unit: dimensionless. Status: STANDARD.

with the couplings taken at $\mu_{\mathrm{ren}} = 2\pi T$, one loop from the $\overline{\mathrm{MS}}$ values at $m_{\mathrm{Z}}$ without a Higgs particle ($\alpha_{\mathrm{em}} = 1/127.95$, $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$),

$$\frac{1}{\alpha_2(\mu_{\mathrm{ren}})} = \frac{\sin^2\theta_{\overline{\mathrm{MS}}}}{\alpha_{\mathrm{em}}} + \frac{b_2}{2\pi}\ln\frac{\mu_{\mathrm{ren}}}{m_{\mathrm{Z}}}, \qquad \frac{1}{\alpha_1(\mu_{\mathrm{ren}})} = \frac{1-\sin^2\theta_{\overline{\mathrm{MS}}}}{\alpha_{\mathrm{em}}} + \frac{b_1}{2\pi}\ln\frac{\mu_{\mathrm{ren}}}{m_{\mathrm{Z}}}, \qquad b_2 = \frac{10}{3}, \quad b_1 = -\frac{20}{3} \tag{FE-77}$$

**Symbols in (FE-77).**

- $\alpha_2$: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- $\mu_{\mathrm{ren}}$: The renormalisation scale of a running coupling (taken as $2\pi T$ for the condensates of a plasma). Unit: GeV. Status: STANDARD.
- $\theta_{\overline{\mathrm{MS}}}$: The weak mixing angle in the $\overline{\mathrm{MS}}$ scheme at $m_{\mathrm{Z}}$. Unit: dimensionless. Value: $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$. Status: STANDARD.
- $\alpha_{\mathrm{em}}$: The electromagnetic fine-structure constant ($\overline{\mathrm{MS}}$, at $m_{\mathrm{Z}}$). Unit: dimensionless. Value: 1/127.95. Status: STANDARD.
- $b_3$, $b_2$, $b_1$, $b_{\mathrm{G}}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\mathrm{ren}} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\mathrm{f}}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\mathrm{G}}$ that of the group G (not related to the metric function $b_4$). Unit: dimensionless. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $m_{\mathrm{Z}}$: The mass of the $Z$. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $\alpha_1$: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.

Hence the electroweak condensates of the hot plasma

$$\langle W^I_{\mu\nu}W^{I\mu\nu}\rangle = g_2^2\Big[2\big(-2\Theta_{\mathrm{W}}\big)+\cos^2\theta_{\mathrm{W}}\big(-2\Theta_{\mathrm{Z}}\big)\Big]+4g_2^2\,p_{\alpha,2}, \qquad \langle B_{\mu\nu}B^{\mu\nu}\rangle = g_1^2\sin^2\theta_{\mathrm{W}}\big(-2\Theta_{\mathrm{Z}}\big)+4g_1^2\,p_{\alpha,1} \tag{FE-78}$$

**Symbols in (FE-78).**

- $W^I_M$, $W_M$, $W_{MN}$, $W^I_{MN}$: The SU(2) gauge field (geometric normalisation): the components $W^I_M$ ($I = 1, 2, 3$; $W^3$ the third component), the matrix $W_M = W^I_M\tau^I/2$, the field strength $W_{MN}$ and its components $W^I_{MN}$;

with Greek indices the components along the observable directions. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.

- $I$: A weak-isospin (adjoint SU(2)) index running over $1, 2, 3$; a repeated index is summed. Status: STANDARD.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_{\mathrm{DM}}$. Status: DERIVED.
- $\Theta_{\mathrm{QCD}}$, $\Theta_{\mathrm{G}}$, $\Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\mathrm{QCD}}$ of the quark-gluon plasma or hadron gas, $\Theta_{\mathrm{G}}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_{\mathrm{W}}$ of $W^{\pm}$, $\Theta_{\mathrm{Z}}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\theta_{\mathrm{W}}$: The weak mixing angle of the couplings, $\cos\theta_{\mathrm{W}} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\mathrm{W}} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_{\mathrm{W}} = 0.776368$. Status: DERIVED.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-$\alpha$ (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $B_M$, $B_{MN}$: The hypercharge gauge field $\mathrm{U}(1)_Y$ (geometric normalisation) and its field strength. Unit: GeV (field), $\mathrm{GeV}^2$ (field strength). Status: ASSUMED.
- $g_1$: The four-dimensional hypercharge coupling ($g_1^2 = 4\pi\alpha_1$, Standard-Model normalisation without the GUT factor). Unit: dimensionless. Value: $g_1^2 = 0.1380$ at $2M_{\mathrm{DM}}$. Status: DERIVED.

At the freeze-out $\langle G^A_{\mu\nu} G^{A\mu\nu}\rangle = -8.1\times10^{12}$ $\mathrm{GeV}^4$, $\langle W^I_{\mu\nu} W^{I\mu\nu}\rangle = -5.0\times10^{11}$ $\mathrm{GeV}^4$ (almost all from the interaction part) and $\langle B_{\mu\nu} B^{\mu\nu}\rangle = -2.6\times10^{10}$ $\mathrm{GeV}^4$ (geometric normalisation). The densities of the hot environments are

$$
\begin{aligned}
&n_{\mathrm{p}} + n_{\mathrm{n}} = \eta_{\mathrm{b}}\, n_\gamma, \qquad n_\gamma = \frac{2\zeta(3)}{\pi^2}\, T^3, \qquad \frac{n_{\mathrm{n}}}{n_{\mathrm{p}}} = \mathrm{e}^{-(m_{\mathrm{n}}-m_{\mathrm{p}})/T}, \qquad n_{\mathrm{e}} = n_{\mathrm{p}},\\
&n_{\mathrm{DM,tot}} = 2\big(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}\big) = 2\cdot 4\left(\frac{M_{\mathrm{DM}}T}{2\pi}\right)^{3/2} \mathrm{e}^{-M_{\mathrm{DM}}/T} \quad \left(\frac{M_{\mathrm{DM}}}{T} < 40\right)
\end{aligned}
\tag{FE-86}
$$

**Symbols in (FE-86).**

- $n_{\mathrm{p}}$: The number density of protons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_{\mathrm{n}}$: The number density of neutrons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\eta_{\mathrm{b}}$: The ratio of the baryon density to the photon density. Unit: dimensionless. Value: $6.1\times10^{-10}$. Status: STANDARD.
- $n_\gamma$: The number density of photons. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\zeta$: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers $n$. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $m_{\mathrm{p}}$, $m_{\mathrm{n}}$: The masses of the proton and the neutron. Unit: GeV. Value: 0.938272 GeV, 0.939565 GeV. Status: STANDARD.
- $n_{\mathrm{e}}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_{\mathrm{DM,tot}}$: The total number density of dark-matter quanta: both species (the lowest odd levels of $\Psi_A$ and $\Psi_B$), particles and antiparticles, $n_{\mathrm{DM,tot}} = 2(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}})$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_{\mathrm{DM}}$, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

($\eta_{\mathrm{b}} = 6.1\times10^{-10}$, $m_{\mathrm{n}} - m_{\mathrm{p}} = 1.293$ MeV; the dark matter in thermal equilibrium for both species while $M_{\mathrm{DM}}/T < 40$, its relic density afterwards). The cold environments are given by their matter: the cosmic mean today (the Planck baryon density with the helium fraction 0.2457 and $\Omega_{\mathrm{DM}}h^2_{100} = 0.120$); the Galaxy at the Sun (one proton and one electron per $\mathrm{cm}^3$, dark matter 0.4 $\mathrm{GeV/cm}^3$); the Earth (mean density 5.51 $\mathrm{g/cm}^3$ with 48 % protons); the

centre of the Sun (150 g/cm$^3$, 66 % protons, 32 % neutrons); a white dwarf ($10^6$ g/cm$^3$ of carbon and oxygen, equal numbers of protons and neutrons); a neutron-star core (0.3 fm$^{-3}$ with 10 % protons, as many electrons as protons); the local dark-matter density 0.4 GeV/cm$^3$ in all bound objects. Results (DERIVED):

| environment | $\Delta\varepsilon_{\rm DM}/M_{\rm DM}$ | $\Delta \ln f$ |
|---|---|---|
| cosmic mean today; the Galaxy at the Sun; the Earth; the centre of the Sun; a white dwarf | $8.4\times10^{-20}$ (the QCD vacuum condensate) | $7\times10^{-94}$ to $2\times10^{-57}$ |
| neutron-star core (0.3 fm$^{-3}$) | $1.0\times10^{-16}$ | $9\times10^{-49}$ |
| nucleosynthesis ($T=1$ MeV) | $8.4\times10^{-20}$ | $7\times10^{-60}$ |
| QCD crossover ($T=200$ MeV) | $4.4\times10^{-20}$ | $3\times10^{-48}$ |
| electroweak era ($T=100$ GeV) | $-5.5\times10^{-11}$ | $7\times10^{-39}$ |
| dark-matter freeze-out ($T=1.42$ TeV, $x_{\rm f}=31.23$) | $-1.5\times10^{-6}$ | $1.1\times10^{-34}$ |

**$\Psi$ acting on the Standard Model.** Through I12 the scalar density of the dark matter shifts the gauge couplings, and through the kinetic function of the gauge fields along $x_0$ it changes the $x_0$ eigenvalue of the $W$:

$$\frac{\Delta\alpha_3}{\alpha_3}=-4G_{\rm g}\,n_{\rm DM,tot},\qquad \frac{\Delta\alpha_2}{\alpha_2}=-4G_{\rm W}\,n_{\rm DM,tot},\qquad \frac{\Delta\alpha_1}{\alpha_1}=-4G_{\rm B}\,n_{\rm DM,tot},$$
$$-\frac{1}{\Upsilon\,{\rm e}^{3HL\xi}}\frac{\mathrm{d}}{\mathrm{d}\xi}\Big(\Upsilon\,{\rm e}^{3HL\xi}\frac{\mathrm{d}F}{\mathrm{d}\xi}\Big)=\tilde m^2F,\qquad \Upsilon(\xi)=1+A_{\rm mf}\,n_{\rm DM}^{\rm sc},\qquad n_{\rm DM}^{\rm sc}=2a_{\rm DM}b_{\rm DM},\tag{FE-79}$$
$$\frac{\mathrm{d}F}{\mathrm{d}\xi}(0)=0,\qquad \Upsilon(-1)\,\frac{\mathrm{d}F}{\mathrm{d}\xi}(-1)=\beta_{\rm W}F(-1)$$

**Symbols in (FE-79).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2=m_2^2-m_1^2$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$). Status: STANDARD.
- $\alpha_3$: The strong fine-structure constant, $\alpha_3=g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\rm Z})=0.1180$; $\alpha_3(m_{\rm KK})=0.0606$. Status: STANDARD.
- $G_{\rm g}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: GeV$^{-3}$. Value: $6.09\times10^{-15}$ GeV$^{-3}$ (benchmark). Status: DERIVED.
- $n_{\rm DM,tot}$: The total number density of dark-matter quanta: both species (the lowest odd levels of $\Psi_A$ and $\Psi_B$), particles and antiparticles, $n_{\rm DM,tot}=2(n_{\rm DM}+\bar n_{\rm DM})$. Unit: GeV$^3$. Status: STANDARD.
- $\alpha_2$: The SU(2) fine-structure constant, $\alpha_2=g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- $G_{\rm W}$: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: GeV$^{-3}$. Value: $2.96\times10^{-15}$ GeV$^{-3}$. Status: DERIVED.
- $\alpha_1$: The hypercharge fine-structure constant, $\alpha_1=g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- $G_{\rm B}$: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: GeV$^{-3}$. Value: $1.08\times10^{-15}$ GeV$^{-3}$. Status: DERIVED.
- $\Upsilon(\xi)$: The factor by which the scalar density of the dark matter changes the gauge kinetic term along $x_0$ (I12), $\Upsilon=1+A_{\rm mf}n_{\rm DM}^{\rm sc}$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; e$^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda=-10H^2$ and the mass parameter $m_\Psi=H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\rm W}$ and $\rho_{\rm EW,0}$). Status: AUTHOR.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y=-L$; $HL=3$. Unit: length. Value: $L=3/H$ ($fL=8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.

- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $F(\xi)$, $F_n$: The $x_0$ profile of a boson mode on one half of the period (a function of $y$ or of $\xi = y/L$): of a gauge field (gauge $A_y = 0$) or of the transverse-traceless graviton; $F_n$ is the profile of the $n$-th Kaluza-Klein level of the graviton tower ($F_0 = 1$ the flat zero mode). Unit: dimensionless. Status: DERIVED.
- $\tilde{m}$, $\tilde{m}_n$: The dimensionless $x_0$ eigenvalue of a boson mode, $\tilde{m} = m\,fL$; $\tilde{m}_n$ the $n$-th even Kaluza-Klein level of the tower with $\beta = 0$ ($\tilde{m}_1$ the first). Unit: dimensionless. Value: $\tilde{m}_1^2 = 30.1196$, $\tilde{m}_2^2 = 59.7284$, $\tilde{m}_3^2 = 109.0764$ ($HL = 3$), i.e. $\tilde{m}_1 = 5.48813$, $\tilde{m}_2 = 7.72842$, $\tilde{m}_3 = 10.44397$. Status: DERIVED.
- $A_{\mathrm{mf}}$: The amplitude of the I12 mean field in the $x_0$ problem of the gauge fields (a bookkeeping parameter: the result is linear in it). Unit: $\mathrm{GeV}^{-1}$ (with the orbital normalisation). Status: DERIVED.
- $n_{\mathrm{DM}}^{\mathrm{sc}}$: The scalar density of the dark-matter orbital, $n_{\mathrm{DM}}^{\mathrm{sc}} = 2a_{\mathrm{DM}}b_{\mathrm{DM}}$. Unit: GeV (normalised orbital). Status: DERIVED.
- $a_{\mathrm{DM}}$, $b_{\mathrm{DM}}$: The upper and lower components of the dark-matter orbital (block spinor $(a, \mathrm{i}b)$, normalised on one half). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- $\beta_{\mathrm{W}}$: The Robin parameter of $W^+$ and $W^-$ (set by the tip mass term of the sigma field). Unit: dimensionless. Value: 0.011912 (fitted). Status: DERIVED.

The coupling shifts are at most $1.7 \times 10^{-16}$ (at the freeze-out), $10^{-36}$ from nucleosynthesis on and $2 \times 10^{-60}$ today in the Galaxy; the eigenvalue of the $W$ changes by $\mathrm{d}\ln\tilde{m}_{\mathrm{W}}^2/\mathrm{d}A_{\mathrm{mf}} = -0.11125$ per unit amplitude (first-order perturbation theory, confirmed by shooting to $3 \times 10^{-11}$; version 3 quoted $-0.1126$ from finite elements, limited by rounding), while the photon and the gluons stay exactly massless (their zero mode has no slope, so the shift vanishes identically); at the physical amplitude of the freeze-out the shift of $\tilde{m}_{\mathrm{W}}^2$ is $1.9 \times 10^{-17}$.

**Gravity closes the loop.** The block responds to the block-averaged sources of all species with the adiabatic response of the gravity section:

$$\begin{aligned} \Delta K &= \frac{c_\rho\,\kappa\Delta\rho + c_3\,\kappa\Delta p_3 + c_p\,\kappa\Delta\bar{p}}{H^2}, \qquad \Delta\ln f = -\frac{\Delta K}{2K_*}, \\ \Delta\ln m_s &= \frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f}\,\Delta\ln f, \qquad \kappa\Delta\rho = 8\pi G_{\mathrm{N}}\,\mathcal{G}(K_*)\,\rho_4 \end{aligned} \tag{FE-80}$$

**Symbols in (FE-80).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $K$: The block variable, $K = 1/f^2$. Unit: dimensionless. Status: DERIVED.
- $c_\rho$: The coefficient of $\kappa\Delta\rho$ in the adiabatic block response $\Delta K$. Unit: dimensionless. Value: 0.0358521. Status: DERIVED.
- $\kappa$: The eight-dimensional gravitational coupling, $\kappa = 8\pi G_8$. Unit: $\mathrm{GeV}^{-6}$. Value: $\kappa/V_\perp = 8\pi G_{\mathrm{N}}\mathcal{G}(K_*) = 2.328 \times 10^{-36}\ \mathrm{GeV}^{-2}$. Status: AUTHOR.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $c_3$: The coefficient of $\kappa\Delta p_3$ in the adiabatic block response, $c_3 = -3c_\rho$ (so radiation drops out). Unit: dimensionless. Value: -0.107556. Status: DERIVED.
- $p_3$: The pressure of ordinary 3-space, $p_3 = \mathcal{T}^{x_1}{}_{x_1} = \mathcal{T}^{x_2}{}_{x_2} = \mathcal{T}^{x_3}{}_{x_3}$ for an isotropic source. Unit: $\mathrm{GeV}^8$. Status: STANDARD.
- $c_p$: The coefficient of $\kappa\Delta\bar{p}$ in the adiabatic block response. Unit: dimensionless. Value: -0.140028. Status: DERIVED.
- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial\ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.

- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_\mathrm{W}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $K_*$: The block variable of the stabilised block, $K_* = 12 + \frac{4}{3}\sqrt{51}$ (the upper root $K_+$). Unit: dimensionless. Value: 21.5219. Status: DERIVED.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $s$: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, $Z$, quarks, leptons, nucleons, dark matter); a sum over $s$ runs over the species present. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_\mathrm{N}$: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- $\mathcal{G}(K)$: The coefficient of the four-dimensional Einstein-Hilbert term on a static block, $\mathcal{G}(K) = 1 - 24w_{(2)}K + 72w_{(3)}K^2 = (K^2 - 12K + 16)/16$; four-dimensional gravity is attractive for $\mathcal{G} > 0$. Unit: dimensionless. Value: $\mathcal{G}(K_*) = 20/3 + \sqrt{51} = 13.8081$. Status: DERIVED.
- $\rho_4$: A four-dimensional energy density (per 3-volume); its block average is $\kappa\Delta\rho = 8\pi G_\mathrm{N}\mathcal{G}(K_*)\rho_4$. Unit: $\mathrm{GeV}^4$. Status: DERIVED.

in which radiation drops out exactly ($c_3 = -3c_\rho$), so only the trace $\rho - 3p_3$ and the hidden pressure act. The hidden pressure is the response of the pressure to the block scale at fixed temperature, $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$:

$$\bar{p} = -\frac{1}{4}\sum_s \left(\rho_s - 3p_s\right)\frac{\mathrm{d}\ln m_s}{\mathrm{d}\ln f} - \frac{1}{4}\,\Theta_\mathrm{QCD}\,\frac{\mathrm{d}\ln \Lambda_\mathrm{QCD}^{(N_\mathrm{f})}}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,2}\,\frac{\mathrm{d}\ln\alpha_2}{\mathrm{d}\ln f} + \frac{1}{4}\,p_{\alpha,1}\,\frac{\mathrm{d}\ln\alpha_1}{\mathrm{d}\ln f} \qquad \text{(FE-81)}$$

**Symbols in (FE-81).**

- $\bar{p}$: The hidden pressure, the pressure averaged over the block, $\bar{p} = (p_0 + 3p_5)/4$; for a species at fixed temperature $\bar{p} = \frac{1}{4}\,\partial p/\partial \ln f$. Unit: $\mathrm{GeV}^8$ (or $\mathrm{GeV}^4$ for a four-dimensional density). Status: DERIVED.
- $s$: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, $Z$, quarks, leptons, nucleons, dark matter); a sum over $s$ runs over the species present. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\Theta_\mathrm{QCD}$, $\Theta_\mathrm{G}$, $\Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_\mathrm{QCD}$ of the quark-gluon plasma or hadron gas, $\Theta_\mathrm{G}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_\mathrm{W}$ of $W^\pm$, $\Theta_\mathrm{Z}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\Lambda_\mathrm{QCD}^{(N_\mathrm{f})}$: The QCD scale with $N_\mathrm{f}$ active flavours ($\Lambda_\mathrm{QCD}^{(6)}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.

- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-$\alpha$ (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\alpha_2$: The SU(2) fine-structure constant, $\alpha_2 = g_2^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.
- $\alpha_1$: The hypercharge fine-structure constant, $\alpha_1 = g_1^2/(4\pi)$. Unit: dimensionless. Status: STANDARD.

(the plasma species as ideal gases with their $x_0$ masses, the nucleons and electrons of the cold environments as degenerate Fermi gases, the QCD plasma through its scale, the electroweak plasma through its couplings), with the block dependence of a gauge coupling and the trace of an electroweak plasma (the bulk couplings scale as $f^{-4}$ and $m_{\mathrm{KK}}$ as $1/f$; leading-order trace anomaly)

$$\frac{\mathrm{d}\ln\alpha_{\mathrm{G}}(\mu_{\mathrm{ren}})}{\mathrm{d}\ln f} = -\alpha_{\mathrm{G}}(\mu_{\mathrm{ren}})\Big[\frac{4}{\alpha_{\mathrm{G}}(m_{\mathrm{KK}})} + \frac{b_{\mathrm{G}}}{2\pi}\Big], \qquad \Theta_{\mathrm{G}} = -\frac{b_{\mathrm{G}}\,\alpha_{\mathrm{G}}}{2\pi}\,p_{\alpha,\mathrm{G}} \tag{FE-82}$$

**Symbols in (FE-82).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\alpha_{\mathrm{G}}$: The fine-structure constant of the gauge group G (one of $\alpha_3$, $\alpha_2$, $\alpha_1$). Unit: dimensionless. Status: STANDARD.
- $\mu_{\mathrm{ren}}$: The renormalisation scale of a running coupling (taken as $2\pi T$ for the condensates of a plasma). Unit: GeV. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $m_n^{\mathrm{KK}}$, $m_{\mathrm{KK}}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $b_3$, $b_2$, $b_1$, $b_{\mathrm{G}}$: The one-loop coefficients of the running couplings, $\mathrm{d}\alpha/\mathrm{d}\ln\mu_{\mathrm{ren}} = -b\,\alpha^2/(2\pi)$: $b_3 = 11 - \frac{2}{3}N_{\mathrm{f}}$ (SU(3)), $b_2 = \frac{10}{3}$ (SU(2) without a Higgs particle), $b_1 = -\frac{20}{3}$ ($\mathrm{U}(1)_Y$), $b_{\mathrm{G}}$ that of the group G (not related to the metric function $b_4$). Unit: dimensionless. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Theta_{\mathrm{QCD}}$, $\Theta_{\mathrm{G}}$, $\Theta_V$: The trace $\rho - 3p$ of the energy-momentum tensor (interaction measure) of a plasma component: $\Theta_{\mathrm{QCD}}$ of the quark-gluon plasma or hadron gas, $\Theta_{\mathrm{G}}$ of the gauge plasma of the group G, $\Theta_V$ of the thermal gas of the massive vector $V$ ($\Theta_{\mathrm{W}}$ of $W^{\pm}$, $\Theta_{\mathrm{Z}}$ of $Z$). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $p_{\alpha,\mathrm{G}}$, $p_{\alpha,2}$, $p_{\alpha,1}$: The order-$\alpha$ (interaction) part of the pressure of the gauge plasma of the group G; $p_{\alpha,2}$ of SU(2), $p_{\alpha,1}$ of $\mathrm{U}(1)_Y$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.

The nucleon pushes through the QCD scale (one loop, ESTIMATE; $\sigma_{\pi N} = 45$ MeV and $\sigma_{\mathrm{s}} = 40$ MeV from memory):

$$\frac{1}{\alpha_3(m_{\mathrm{KK}})} = \frac{1}{\alpha_3(m_{\mathrm{Z}})} + \frac{7}{2\pi}\ln\frac{m_{\mathrm{KK}}}{m_{\mathrm{Z}}}, \qquad \Lambda_{\mathrm{QCD}}^{(6)} = m_{\mathrm{KK}}\,\mathrm{e}^{-2\pi/(7\alpha_3(m_{\mathrm{KK}}))}, \qquad \Lambda_{\mathrm{QCD}}^{(3)} = \big(\Lambda_{\mathrm{QCD}}^{(6)}\big)^{21/27}\big(m_{\mathrm{c}}m_{\mathrm{b}}m_{\mathrm{t}}\big)^{2/27},$$
$$\frac{\mathrm{d}\ln m_{\mathrm{p}}}{\mathrm{d}\ln f} = \Big(1 - \frac{\sigma_{\pi N} + \sigma_{\mathrm{s}}}{m_{\mathrm{p}}}\Big)\frac{\mathrm{d}\ln\Lambda_{\mathrm{QCD}}^{(3)}}{\mathrm{d}\ln f} + \frac{\sigma_{\pi N}}{2m_{\mathrm{p}}}\Big(\frac{\mathrm{d}\ln m_{\mathrm{u}}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_{\mathrm{d}}}{\mathrm{d}\ln f}\Big) + \frac{\sigma_{\mathrm{s}}}{m_{\mathrm{p}}}\,\frac{\mathrm{d}\ln m_{\mathrm{s}}}{\mathrm{d}\ln f} \tag{FE-83}$$

**Symbols in (FE-83).**

- $\alpha_3$: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_{\mathrm{Z}}) = 0.1180$; $\alpha_3(m_{\mathrm{KK}}) = 0.0606$. Status: STANDARD.
- $m_n^{\mathrm{KK}}$, $m_{\mathrm{KK}}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^{\mathrm{KK}} = \tilde{m}_n/(fL)$; $m_{\mathrm{KK}} = m_1^{\mathrm{KK}}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_{\mathrm{KK}} = 121.32$ TeV. Status: DERIVED.
- $m_{\mathrm{Z}}$: The mass of the $Z$. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\Lambda_{\mathrm{QCD}}^{(N_{\mathrm{f}})}$: The QCD scale with $N_{\mathrm{f}}$ active flavours ($\Lambda_{\mathrm{QCD}}^{(6)}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $m_\mathrm{u}$, $m_\mathrm{d}$, $m_\mathrm{s}$, $m_\mathrm{c}$, $m_\mathrm{b}$, $m_\mathrm{t}$: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the $x_0$ masses, fitted to the PDG values). Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $m_\mathrm{p}$, $m_\mathrm{n}$: The masses of the proton and the neutron. Unit: GeV. Value: 0.938272 GeV, 0.939565 GeV. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\sigma_{\pi N}$: The pion-nucleon sigma term (the contribution of the up and down quark masses to the nucleon mass). Unit: GeV. Value: 45 MeV. Status: STANDARD.
- $\sigma_\mathrm{s}$: The strange sigma term of the nucleon. Unit: GeV. Value: 40 MeV. Status: STANDARD.

$$\begin{aligned}
&\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} = -1 - \frac{8\pi}{7\alpha_3(m_\mathrm{KK})}, \qquad \frac{\mathrm{d}\ln\Lambda^{(5)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{23}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{23}\frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f},\\
&\frac{\mathrm{d}\ln\Lambda^{(4)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{25}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{25}\Big(\frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{b}}{\mathrm{d}\ln f}\Big),\\
&\frac{\mathrm{d}\ln\Lambda^{(3)}_\mathrm{QCD}}{\mathrm{d}\ln f} = \frac{21}{27}\frac{\mathrm{d}\ln\Lambda^{(6)}_\mathrm{QCD}}{\mathrm{d}\ln f} + \frac{2}{27}\Big(\frac{\mathrm{d}\ln m_\mathrm{c}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{b}}{\mathrm{d}\ln f} + \frac{\mathrm{d}\ln m_\mathrm{t}}{\mathrm{d}\ln f}\Big)
\end{aligned} \tag{FE-84}$$

**Symbols in (FE-84).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\Lambda^{(N_\mathrm{f})}_\mathrm{QCD}$: The QCD scale with $N_\mathrm{f}$ active flavours ($\Lambda^{(6)}_\mathrm{QCD}$, $^{(5)}$, $^{(4)}$, $^{(3)}$), defined by the one-loop running. Unit: GeV. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\alpha_3$: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_\mathrm{Z}) = 0.1180$; $\alpha_3(m_\mathrm{KK}) = 0.0606$. Status: STANDARD.
- $m_n^\mathrm{KK}$, $m_\mathrm{KK}$: The mass of the $n$-th Kaluza-Klein graviton (and gauge) level, $m_n^\mathrm{KK} = \tilde{m}_n/(fL)$; $m_\mathrm{KK} = m_1^\mathrm{KK}$ is the Kaluza-Klein scale. Unit: GeV. Value: $m_\mathrm{KK} = 121.32$ TeV. Status: DERIVED.
- $m_\mathrm{u}$, $m_\mathrm{d}$, $m_\mathrm{s}$, $m_\mathrm{c}$, $m_\mathrm{b}$, $m_\mathrm{t}$: The masses of the up, down, strange, charm, bottom and top quarks (the upright subscript names the quark). Unit: GeV. Value: 2.16 MeV, 4.70 MeV, 93.5 MeV, 1.273 GeV, 4.183 GeV, 172.56 GeV (the $x_0$ masses, fitted to the PDG values). Status: DERIVED.

with the quark sensitivities of the block dependence of a fermion mass (written out again)

$$\frac{\mathrm{d}\ln m}{\mathrm{d}\ln f} = -1 + \frac{1}{2}\Big[\frac{\mathrm{d}\ln \mathcal{Z}(c_L)^2}{\mathrm{d}\ln c_L} + \frac{\mathrm{d}\ln \mathcal{Z}(-c_R)^2}{\mathrm{d}\ln(-c_R)}\Big] \tag{FE-85}$$

**Symbols in (FE-85).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-\alpha_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.

- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.

$\mathrm{d}\ln m/\mathrm{d}\ln f = -18.82$, $-18.01$, $-14.90$, $-12.15$, $-10.89$, $-6.77$ for the up, down, strange, charm, bottom and top quarks, so that $\alpha_3(m_{\mathrm{KK}}) = 0.0606$, $\mathrm{d}\ln\Lambda^{(6)}_{\mathrm{QCD}}/\mathrm{d}\ln f = -60.2$, $\mathrm{d}\ln\Lambda^{(3)}_{\mathrm{QCD}}/\mathrm{d}\ln f = -49.0$ and $\mathrm{d}\ln m_{\mathrm{p}}/\mathrm{d}\ln f = -46.1$: the nucleon mass is the most block-sensitive mass of UM-2, more than the lightest neutrino ($-41.0$). The result: $\Delta\ln f \le 1.1\times10^{-34}$, at the freeze-out, where the hot plasma has $\bar{p} = 3.1\times10^{12}$ GeV$^4$ (mostly the QCD interaction measure with $\mathrm{d}\ln\Lambda_{\mathrm{QCD}}/\mathrm{d}\ln f = -60$) and $\rho - 3p_3 = 2.2\times10^{11}$ GeV$^4$; at most $9\times10^{-49}$ from nucleosynthesis to today; the gain of the loop mass → source → block → mass is at most $1.1\times10^{-33}$, so the iteration converges at once. (The plasma energy density at the freeze-out in these block sources, $1.4161\times10^{14}$ GeV$^4$, exceeds that of the ideal-gas thermal history by 0.13 %, by design: the block sources contain the trace anomaly of the interacting plasma; residual analysis.)

**Domain of validity.** The I12 benchmark changes the strong coupling by 10 % through the thermal dark-matter density only at $T = 27$ TeV ($M_{\mathrm{DM}}/T = 1.6$): the coupled effective description holds through the freeze-out and everything later.

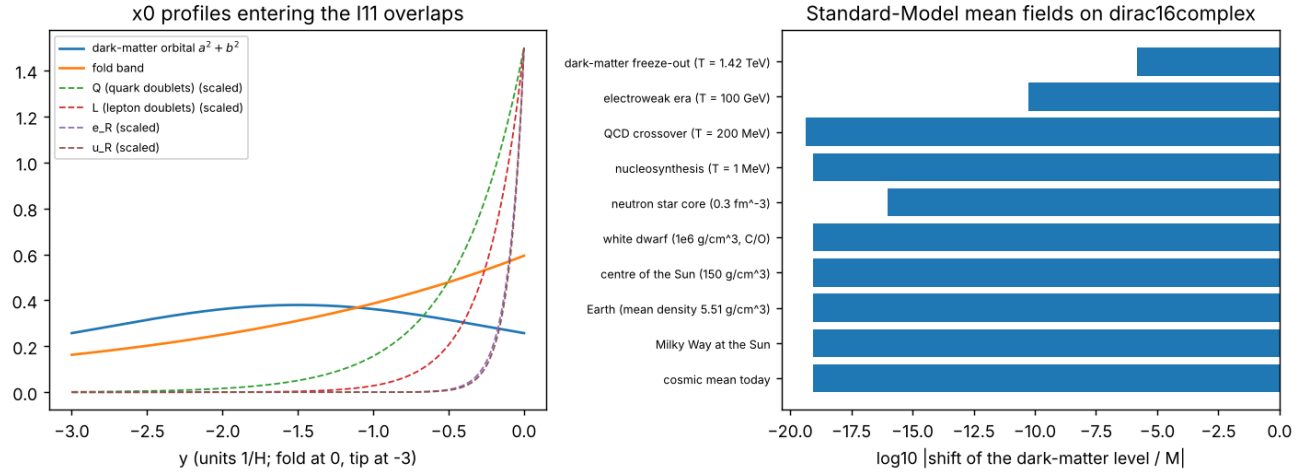



Figure 2: The coupled solution. Left: the $x_0$ profiles that enter the overlaps of FE-70 against $Hy$ (fold at 0, tip at $-3$): the dark-matter orbital $a^2 + b^2$, the fold band and, scaled, the zero modes of the quark doublets, the lepton doublets (labelled L in the figure) and two singlets. Right: the shift of the dark-matter level by the Standard-Model mean fields in the ten environments, the decimal logarithm of its ratio to $M_{\mathrm{DM}}$ (labelled M in the figure).

# 16 Three neutrino flavours: masses, mixing and flavour conversion (DERIVED)

## 16.1 Masses and mixing

The neutrinos are Dirac fermions whose masses are $x_0$ eigenvalues: the lepton doublet zero mode $c_L = 5.9529$ mixed at the tip with the singlet zero modes $c_R = -35.0452$, $-32.8481$, $-31.0702$, through the exact mass condition (written out again; $\mathcal{K} = 1$ for each mass state, because $\mathcal{K}_\nu = U$ is unitary and the three doublets share one bulk

mass)

$$\mathcal{D}(\tilde{\varepsilon}) = \tilde{\varepsilon}^2 \frac{\sinh \upsilon_L}{\upsilon_L} \frac{\sinh \upsilon_R}{\upsilon_R} - \mathcal{K}^2 \Big( \cosh \upsilon_L - c_L \frac{\sinh \upsilon_L}{\upsilon_L} \Big) \Big( \cosh \upsilon_R + c_R \frac{\sinh \upsilon_R}{\upsilon_R} \Big) = 0,$$
$$\tilde{\varepsilon} \simeq \mathcal{K}\, \mathcal{Z}(c_L)\, \mathcal{Z}(-c_R), \qquad \mathcal{Z}(c)^2 = \frac{c\, \mathrm{e}^{-c}}{\sinh c} \tag{FE-59}$$

**Symbols in (FE-59).**

- $\mathcal{D}(\tilde{\varepsilon})$: The mass function of one generation: its zeros are the dimensionless masses $\tilde{\varepsilon}$. Unit: dimensionless. Status: DERIVED.
- $\tilde{\varepsilon}$: The dimensionless energy (mass) of a fermion mode, $\tilde{\varepsilon} = \varepsilon f L$. Unit: dimensionless. Status: DERIVED.
- $\upsilon_L,\ \upsilon_R$: The auxiliary roots $\upsilon = (c^2 - \tilde{\varepsilon}^2)^{1/2}$ of the doublet ($\upsilon_L$) and singlet ($\upsilon_R$) solutions at $k = 0$. Unit: dimensionless. Status: DERIVED.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $\mathcal{K}$: The tip mixing matrix of a sector in the boundary conditions of the fermion zero modes (a number $\mathcal{K}$ for one generation, a $3 \times 3$ matrix for three): $\mathcal{K}_\nu$, $\mathcal{K}_e$, $\mathcal{K}_u$, $\mathcal{K}_d$ of the neutrino, charged-lepton, up-quark and down-quark sectors; UM-2 takes $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$. Unit: dimensionless. Status: ASSUMED.
- $c,\ c_L,\ c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2 = c\, \mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.

with $\tilde{\varepsilon} = m_i f L$. The lightest mass is not measured; UM-2 takes (ASSUMED) the normal ordering, the best fit of NuFIT 6.0 (the inverted ordering is disfavoured by $\Delta\chi^2 = 6.1$), and

$$m_1 = 1.000 \text{ meV}, \qquad m_2 = \sqrt{m_1^2 + \Delta m_{21}^2} = 8.712 \text{ meV}, \qquad m_3 = \sqrt{m_1^2 + \Delta m_{31}^2} = 50.140 \text{ meV} \tag{FE-100}$$

**Symbols in (FE-100).**

- $m_1,\ m_2,\ m_3$: The masses of the neutrino mass states $\nu_1$, $\nu_2$, $\nu_3$ (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV ($m_1$ ASSUMED). Status: DERIVED.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\Delta m_{21}^2,\ \Delta m_{31}^2,\ \Delta m_{32}^2$: The differences of the squared neutrino masses, $\Delta m_{ij}^2 = m_i^2 - m_j^2$. Unit: $\mathrm{GeV}^2$ (quoted in $\mathrm{eV}^2$). Value: $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ $\mathrm{eV}^2$, $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ $\mathrm{eV}^2$ (NuFIT 6.0). Status: STANDARD.

with $\Delta m_{21}^2 = 7.49 \times 10^{-5}$ $\mathrm{eV}^2$ and $\Delta m_{31}^2 = 2.513 \times 10^{-3}$ $\mathrm{eV}^2$ (NuFIT 6.0). The sum $\sum_i m_i = 0.0599$ eV lies above the minimum 0.0588 eV of the normal ordering and below the DESI DR2 bound 0.064 eV (95 %, ΛCDM), which allows $m_1 < 4.1$ meV. The mixing is the neutrino tip matrix, $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu = \mathcal{K}_\nu$, in the standard parametrisation (written out again)

$$U = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}\mathrm{e}^{-\mathrm{i}\delta_{\mathrm{CP}}} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{12}c_{23} - s_{12}s_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & -c_{12}s_{23} - s_{12}c_{23}s_{13}\mathrm{e}^{\mathrm{i}\delta_{\mathrm{CP}}} & c_{23}c_{13} \end{pmatrix}, \qquad J_{\mathrm{CP}} = c_{12}s_{12}c_{23}s_{23}c_{13}^2 s_{13}\sin\delta_{\mathrm{CP}}$$

(FE-61)

**Symbols in (FE-61).**

- $U$: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- $c_{12},\ c_{13},\ c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $s_{12},\ s_{13},\ s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\delta_{\mathrm{CP}}$: The CP-violating phase of a mixing matrix. Unit: dimensionless. Value: 212° (PMNS), 1.147 rad (CKM). Status: STANDARD.
- $J_{\mathrm{CP}}$: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: −0.01777 (PMNS), $3.117 \times 10^{-5}$ (CKM). Status: STANDARD.

with the NuFIT 6.0 values $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{23} = 0.470$, $\sin^2\theta_{13} = 0.02215$, $\delta_{\mathrm{CP}} = 212°$ ($J_{\mathrm{CP}} = -0.01777$). A neutrino produced by the charged current with the charged lepton $\ell_\alpha^-$ is the flavour state $|\nu_\alpha\rangle = \sum_i U_{\alpha i}^* |\nu_i\rangle$ ($\alpha = e, \mu, \tau$). The oscillation phases are differences of squared $x_0$ eigenvalues, so oscillation experiments measure the $x_0$ spectrum directly; they are the quantities of UM-2 most sensitive to the block scale ($\mathrm{d}\ln\Delta m_{21}^2/\mathrm{d}\ln f = -77.5$, $\mathrm{d}\ln\Delta m_{31}^2/\mathrm{d}\ln f = -74.0$; DERIVED). The light neutrino states also mix with the Kaluza-Klein excitations of the lepton fields, which makes the $3 \times 3$ PMNS matrix non-unitary at relative order $(m_i/m_{\mathrm{KK}})^2 \leq 1.7 \times 10^{-31}$ (ESTIMATE): unobservable. The angles and the phase are inputs of UM-2 (NuFIT 6.0), not predictions.

## 16.2 The flavour evolution

At energies far below $m_{\mathrm{W}}$ the charged and neutral currents act as four-fermion interactions (written out again)

$$\mathcal{L}_{\mathrm{CC}} = -2\sqrt{2}\,G_{\mathrm{F}}\big(\bar\nu_\alpha\gamma^\mu P_L\,\ell_\alpha^-\big)\big(\bar\ell_\beta^-\gamma_\mu P_L\,\nu_\beta\big), \qquad \mathcal{L}_{\mathrm{NC}} = -2\sqrt{2}\,G_{\mathrm{F}}\big(\bar\nu_\alpha\gamma^\mu P_L\,\nu_\alpha\big)\sum_f \bar\psi_f\gamma_\mu\big(\tilde g_L^f P_L + \tilde g_R^f P_R\big)\psi_f,$$

$$\nu_\alpha = \sum_{i=1}^{3} U_{\alpha i}\,\nu_i, \qquad P_L = \frac{1+\Gamma_4}{2}, \qquad P_R = \frac{1-\Gamma_4}{2}$$

(FE-101)

**Symbols in (FE-101).**

- $\mathcal{L}_{\mathrm{CC}},\ \mathcal{L}_{\mathrm{NC}}$: The charged-current and neutral-current four-fermion Lagrangians (exchange of $W^\pm$ and of $Z$ far below their masses). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788 \times 10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour $\alpha$; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\gamma^a,\ \gamma^M,\ \gamma^\mu$: The Dirac matrices: $\gamma^a$ with a frame index are the author's real $16 \times 16$ matrices of the Clifford algebra of signature $(4, 4)$; $\gamma^M = \mathbf{e}_a{}^M\gamma^a$ along a coordinate ($\gamma^{x_0}$, $\gamma^{x_4}$, … the matrices along $x_0$, $x_4$, …), $\gamma_M = g_{MN}\gamma^N$; $\gamma^\mu$ the matrices of the four observable directions in four-dimensional formulas. Unit: dimensionless. Status: AUTHOR.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $P_L,\ P_R$: The projectors on the four-dimensional chiralities, $P_L = (1+\Gamma_4)/2$ (left-handed) and $P_R = (1-\Gamma_4)/2$ (right-handed). Unit: dimensionless. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.

- $\ell_\alpha^-$, $\bar{\ell}_\alpha^-$: The charged lepton of the flavour $\alpha$ ($e^-$, $\mu^-$, $\tau^-$; the lower member of the lepton doublet) at low energy, and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\beta$: A second flavour index running over $e, \mu, \tau$. Status: STANDARD.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_\mathrm{c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\psi_i$, $\bar{\psi}_i$: A Standard-Model fermion field of eight dimensions: $\psi_i$ is the multiplet $\psi \in \{Q, u_R, d_R, \ell, e_R, \nu_R\}$ of the generation $i$, $\psi_f$ one fermion component $f$ (in four-dimensional formulas its zero mode); $\bar{\psi} = \psi^\dagger C$ the Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$ (eight dimensions), $\mathrm{GeV}^{3/2}$ (four dimensions). Status: ASSUMED.
- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}$. Unit: dimensionless. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_\mathrm{CP} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $\Gamma_4$: The four-dimensional chirality matrix, $\Gamma_4 = \mathrm{i}\gamma^{x_1}\gamma^{x_2}\gamma^{x_3}\gamma^{x_4}$; a doublet zero mode has $\Gamma_4 = +1$ (left-handed), a singlet zero mode $\Gamma_4 = -1$ (right-handed). Unit: dimensionless. Status: DERIVED.

$$\frac{G_\mathrm{F}}{\sqrt{2}} = \frac{g_2^2}{8m_\mathrm{W}^2}, \qquad \tilde{g}_L^f = T_3^f - Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}, \qquad \tilde{g}_R^f = -Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}, \qquad \tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f = T_3^f - 2Q_\mathrm{em}^f \sin^2\theta_\mathrm{W} \tag{FE-102}$$

**Symbols in (FE-102).**

- $G_\mathrm{F}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788 \times 10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $g_2$: The four-dimensional SU(2) coupling ($g_2^2 = 4\pi\alpha_2$). Unit: dimensionless. Value: $g_2^2 = 0.3781$ at $2M_\mathrm{DM}$. Status: DERIVED.
- $m_\mathrm{W}$: The mass of $W^+$ and $W^-$. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}$. Unit: dimensionless. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_\mathrm{c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1 + \Gamma_4)/2$. Status: STANDARD.
- $T_3$: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); $T_3^f$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $Q_\mathrm{em}$: The electric charge in units of the elementary charge, $Q_\mathrm{em} = T_3 + Y$; $Q_\mathrm{em}^f$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $\theta_\mathrm{W}$: The weak mixing angle of the couplings, $\cos\theta_\mathrm{W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_\mathrm{W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\mathrm{EW},0}$. Unit: dimensionless. Value: $\cos^2\theta_\mathrm{W} = 0.776368$. Status: DERIVED.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1 - \Gamma_4)/2$. Status: STANDARD.
- $V$: The label V of the vector coupling of a fermion to the $Z$, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.

with the measured $G_\mathrm{F} = 1.1663788 \times 10^{-5}$ $\mathrm{GeV}^{-2}$. Coherent forward scattering in matter then adds the potentials (written out again)

$$V_\alpha = \sqrt{2}\,G_\mathrm{F}\Big[\delta_{\alpha e}\big(n_{e^-} - n_{e^+}\big) + \sum_f \tilde{g}_V^f \big(n_f - n_{\bar{f}}\big)\Big],$$
$$V = \mathrm{diag}\big(V_\mathrm{CC},\ 0,\ 0\big) + V_\mathrm{NC}\,\mathbf{1}, \qquad V_\mathrm{CC} = \sqrt{2}\,G_\mathrm{F}\,n_\mathrm{e}, \qquad V_\mathrm{NC} = -\frac{G_\mathrm{F}\,n_\mathrm{n}}{\sqrt{2}} \tag{FE-103}$$

**Symbols in (FE-103).**

- $V_\alpha$, $V$: The matter potential of forward scattering: $V_\alpha$ of the flavour $\alpha$ ($V_e$, $V_\mu$), $V$ the $3 \times 3$ matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $G_\mathrm{F}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788 \times 10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1 \cdots N_n}^{M_1 \cdots M_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.
- $n_{e^-}$, $n_{e^+}$: The number densities of electrons and of positrons. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_\mathrm{c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $\tilde{g}_L^f$, $\tilde{g}_R^f$, $\tilde{g}_V^f$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}_L^f = T_3^f - Q_\mathrm{em}^f \sin^2\theta_\mathrm{W}$. Unit: dimensionless. Status: STANDARD.
- $V$: The label V of the vector coupling of a fermion to the $Z$, $\tilde{g}_V^f = \tilde{g}_L^f + \tilde{g}_R^f$. Status: STANDARD.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $n_{\bar{f}}$: The number density of the antiparticles of the fermion $f$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $V_\mathrm{CC}$, $V_\mathrm{NC}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\mathbf{1}$: The identity matrix (of the size fixed by the context: 16 x 16 on spinors, 3 x 3 on the generations or flavours, 2 x 2 for the sigma field). Unit: dimensionless. Status: STANDARD.
- $n_\mathrm{e}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $n_\mathrm{n}$: The number density of neutrons (free and bound). Unit: $\mathrm{GeV}^3$. Status: STANDARD.

to the free propagation, and the flavour amplitudes of a neutrino of energy $E$ evolve as (STANDARD)

$$\mathrm{i}\,\frac{\mathrm{d}\boldsymbol{\psi}}{\mathrm{d}x_4} = H_\mathrm{fl}\,\boldsymbol{\psi}, \qquad H_\mathrm{fl} = \frac{1}{2E}\,U\,\mathrm{diag}\big(m_1^2,\ m_2^2,\ m_3^2\big)\,U^\dagger + V, \qquad \mathcal{U}(L_\mathrm{b}) = \mathrm{e}^{-\mathrm{i}H_\mathrm{fl}L_\mathrm{b}}, \qquad \mathcal{P}(\nu_\alpha \to \nu_\beta) = |\mathcal{U}_{\beta\alpha}|^2 \tag{FE-104}$$

**Symbols in (FE-104).**

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\boldsymbol{\psi}$: The vector of the three flavour amplitudes of a neutrino. Unit: dimensionless. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $H_\mathrm{fl}$: The flavour Hamiltonian of a neutrino of energy $E$ (vacuum mass term plus matter potential). Unit: GeV. Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.

- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\rm CP} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $m_1$, $m_2$, $m_3$: The masses of the neutrino mass states $\nu_1$, $\nu_2$, $\nu_3$ (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV ($m_1$ ASSUMED). Status: DERIVED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $V_\alpha$, $V$: The matter potential of forward scattering: $V_\alpha$ of the flavour $\alpha$ ($V_e$, $V_\mu$), $V$ the $3\times3$ matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $\mathcal{U}$: The flavour evolution operator over the baseline $L_{\rm b}$; $\mathcal{U}_{\beta\alpha}$ the amplitude of $\nu_\alpha \to \nu_\beta$. Unit: dimensionless. Status: STANDARD.
- $L_{\rm b}$: The baseline: the distance travelled by the neutrino. Unit: length. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\beta$: A second flavour index running over $e, \mu, \tau$. Status: STANDARD.

($V_{\rm NC}$ is flavour universal and drops out of the probabilities; for antineutrinos $U \to U^*$ and $V \to -V$). In vacuum

$$\mathcal{P}(\nu_\alpha \to \nu_\beta) = \Big| \sum_{i=1}^{3} U_{\beta i} U_{\alpha i}^* \, \mathrm{e}^{-\mathrm{i} m_i^2 L_{\rm b}/(2E)} \Big|^2, \qquad \mathcal{P}(\nu_\alpha \to \nu_\beta) = \mathcal{P}(\bar\nu_\beta \to \bar\nu_\alpha),$$

$$\mathcal{P}(\nu_\mu \to \nu_e) - \mathcal{P}(\nu_e \to \nu_\mu) = -16 J_{\rm CP} \sin\Delta_{21} \sin\Delta_{31} \sin\Delta_{32}, \qquad \Delta_{ij} = \frac{\Delta m_{ij}^2 L_{\rm b}}{4E}, \qquad J_{\rm CP} = \mathrm{Im}\big(U_{e2} U_{\mu 3} U_{e3}^* U_{\mu 2}^*\big) \tag{FE-105}$$

**Symbols in (FE-105).**

- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.
- $\beta$: A second flavour index running over $e, \mu, \tau$. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\rm CP} = 212^\circ$ (NuFIT 6.0). Status: STANDARD.
- $^*$: Complex conjugation of a number or of each entry of a matrix. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $L_{\rm b}$: The baseline: the distance travelled by the neutrino. Unit: length. Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $\bar\nu_\alpha$: The antineutrino of the flavour $\alpha$; in a Lagrangian the Dirac adjoint of the neutrino field (it annihilates antineutrinos). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.

- $J_{\rm CP}$: The Jarlskog invariant of a mixing matrix (the measure of CP violation). Unit: dimensionless. Value: $-0.01777$ (PMNS), $3.117\times10^{-5}$ (CKM). Status: STANDARD.
- $\Delta_{ij}$: The oscillation phase of the pair of mass states $(i,j)$, $\Delta_{ij}=\Delta m^2_{ij}L_{\rm b}/(4E)$. Unit: dimensionless. Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21}=m^2_2-m^2_1$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0}\geq\hbar/2$). Status: STANDARD.
- $j$: A second generation index ($j=1,2,3$). Status: STANDARD.
- Im: The imaginary part of a complex number. Status: STANDARD.

In a dense neutrino gas (the early Universe, a supernova core) the state of the neutrinos of momentum $\vec{k}$ is a $3\times3$ density matrix, which obeys the quantum kinetic equation (STANDARD)

$$\begin{aligned}&{\rm i}\Big(\partial_{x_4}+\frac{\vec{k}}{|\vec{k}|}\cdot\vec{\nabla}\Big)\boldsymbol{\rho}_{\vec{k}}=\big[H^{\rm vac}_{\vec{k}}+V+V_{\nu\nu},\ \boldsymbol{\rho}_{\vec{k}}\big]+{\rm i}\,C_{\rm coll}[\boldsymbol{\rho}],\qquad H^{\rm vac}_{\vec{k}}=\frac{U\,{\rm diag}\big(m_1^2,\ m_2^2,\ m_3^2\big)\,U^\dagger}{2|\vec{k}|},\\&V_{\nu\nu}=\sqrt{2}\,G_{\rm F}\int\frac{{\rm d}^3\vec{k}_2}{(2\pi)^3}\Big(1-\frac{\vec{k}\cdot\vec{k}_2}{|\vec{k}||\vec{k}_2|}\Big)\big(\boldsymbol{\rho}_{\vec{k}_2}-\bar{\boldsymbol{\rho}}_{\vec{k}_2}\big),\qquad\partial_{x_4}\to\partial_{x_4}-\mathcal{H}|\vec{k}|\,\partial_{|\vec{k}|}\ \ \text{(expanding Universe)}\end{aligned}\tag{FE-110}$$

**Symbols in (FE-110).**

- i: The imaginary unit, ${\rm i}^2=-1$. Unit: dimensionless. Status: STANDARD.
- $\partial$: A partial derivative: $\partial_M$ is the derivative with respect to the coordinate $X^M$, $\partial_y$ with respect to $y$, $\partial_{x_4}$ with respect to the time $x_4$, $\partial X/\partial Y$ the derivative of $X$ with respect to $Y$ at fixed other variables. Status: STANDARD.
- $x_4$: The time; dots denote ${\rm d}/{\rm d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_{\rm lat}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $\vec{\nabla}$: The gradient in ordinary 3-space, $(\partial/\partial x_1,\partial/\partial x_2,\partial/\partial x_3)$. Unit: 1/length. Status: STANDARD.
- $\boldsymbol{\rho}_{\vec{k}}$, $\bar{\boldsymbol{\rho}}_{\vec{k}}$: The $3\times3$ flavour density matrices of the neutrinos and of the antineutrinos of momentum $\vec{k}$. Unit: dimensionless. Status: STANDARD.
- $H^{\rm vac}_{\vec{k}}$: The vacuum flavour Hamiltonian of a neutrino of momentum $\vec{k}$. Unit: GeV. Status: STANDARD.
- $V_\alpha$, $V$: The matter potential of forward scattering: $V_\alpha$ of the flavour $\alpha$ ($V_e$, $V_\mu$), $V$ the $3\times3$ matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $V_{\nu\nu}$: The potential of neutrino-neutrino forward scattering (a matrix in flavour space). Unit: GeV. Status: STANDARD.
- $C_{\rm coll}$: The collision term of the quantum kinetic equations. Unit: GeV. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha=\sum_i U_{\alpha i}\nu_i$ (rows $\alpha=e,\mu,\tau$, columns the mass states $i$); in UM-2 $U=\mathcal{K}_e^\dagger\mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12}=0.308$, $\sin^2\theta_{13}=0.02215$, $\sin^2\theta_{23}=0.470$, $\delta_{\rm CP}=212^\circ$ (NuFIT 6.0). Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0,x_1,\ldots,x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- $m_1$, $m_2$, $m_3$: The masses of the neutrino mass states $\nu_1$, $\nu_2$, $\nu_3$ (normal ordering). Unit: GeV (quoted in meV). Value: 1.000, 8.712, 50.140 meV ($m_1$ ASSUMED). Status: DERIVED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $G_{\rm F}$: The Fermi constant. Unit: ${\rm GeV}^{-2}$. Value: $1.1663788\times10^{-5}$ ${\rm GeV}^{-2}$ (measured). Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of $X$, ${\rm d}X/{\rm d}Y$ the derivative of $X$ with respect to $Y$, ${\rm d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $k_2$, $\vec{k}_2$: A second momentum (variable of integration). Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.

## 16.3 Vacuum, matter, the Sun and accelerators

The computed evolution (exact matrix exponentials, the ordered products of thin layers for varying densities) reproduces the vacuum formula to $3.9 \times 10^{-15}$, unitarity, CPT and the T-odd difference (DERIVED). In matter the mixing is resonant (the MSW effect) where the potential cancels the vacuum splitting of a sector:

$$2E\,V_{\mathrm{CC}} = \Delta m^2_{ee}\cos 2\theta_{13}, \qquad \Delta m^2_{ee} = c^2_{12}\,\Delta m^2_{31} + s^2_{12}\,\Delta m^2_{32}, \qquad 2E\,V_{\mathrm{CC}}\,c^2_{13} = \Delta m^2_{21}\cos 2\theta_{12} \tag{FE-106}$$

**Symbols in (FE-106).**

- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $V_{\mathrm{CC}}$, $V_{\mathrm{NC}}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\Delta m^2_{ee}$: The effective squared-mass difference of electron-neutrino disappearance, $\Delta m^2_{ee} = c^2_{12}\Delta m^2_{31} + s^2_{12}\Delta m^2_{32}$. Unit: GeV$^2$. Status: STANDARD.
- $\theta_{ij}$: The mixing angles of the standard parametrisation of $U$ (or of $V_{\mathrm{CKM}}$): $\theta_{12}$, $\theta_{13}$, $\theta_{23}$; in two-flavour formulas $\theta_{ij}$ is the angle of the sector $(i, j)$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- $c_{12}$, $c_{13}$, $c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $\Delta m^2_{21}$, $\Delta m^2_{31}$, $\Delta m^2_{32}$: The differences of the squared neutrino masses, $\Delta m^2_{ij} = m^2_i - m^2_j$. Unit: GeV$^2$ (quoted in eV$^2$). Value: $\Delta m^2_{21} = 7.49 \times 10^{-5}$ eV$^2$, $\Delta m^2_{31} = 2.513 \times 10^{-3}$ eV$^2$ (NuFIT 6.0). Status: STANDARD.
- $s_{12}$, $s_{13}$, $s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.

Representative results (DERIVED; normal ordering, $\delta_{\mathrm{CP}} = 212°$): reactor electron antineutrinos of 4 MeV survive with the probability 0.918 at 1.65 km, 0.226 at 52.5 km and 0.267 at 180 km; atmospheric muon neutrinos have their first survival minimum, 0.006, at $L_{\mathrm{b}}/E = 1516$ km/GeV; the MSW resonance of the 1-3 sector lies at 11.135 GeV in rock of 2.8 g/cm$^3$ (equal numbers of protons and neutrons; the exact three-flavour resonance $|U^{\mathrm{m}}_{e3}|^2 = 1/2$, with the mixing matrix in matter, is at 11.133 GeV), and that of the 1-2 sector at the centre of the Sun at 1.93 MeV. The $\nu_\mu \to \nu_e$ appearance at the long-baseline experiments (matter of constant density):

| experiment | $L_{\mathrm{b}}$ | $E$ | density (g/cm$^3$) | $\mathcal{P}(\nu_\mu \to \nu_e)$ | $\mathcal{P}(\bar\nu_\mu \to \bar\nu_e)$ | vacuum $\mathcal{P}(\nu_\mu \to \nu_e)$ |
|---|---|---|---|---|---|---|
| T2K | 295 km | 0.6 GeV | 2.6 | 0.0517 | 0.0313 | 0.0478 |
| NOvA | 810 km | 2.0 GeV | 2.84 | 0.0481 | 0.0236 | 0.0404 |
| DUNE | 1300 km | 2.5 GeV | 2.848 | 0.0701 | 0.0195 | 0.0491 |

Solar electron neutrinos arrive at the Earth as an incoherent mixture of mass states, with the adiabatic survival probability

$$\mathcal{P}_{ee} = c^4_{13}\Big(\frac{1}{2} + \frac{1}{2}\cos 2\theta_{12}\cos 2\theta^{\mathrm{m}}_{12}\Big) + s^4_{13}, \qquad A_{\mathrm{MSW}} = \frac{2E\,V_{\mathrm{CC}}\,c^2_{13}}{\Delta m^2_{21}},$$
$$\cos 2\theta^{\mathrm{m}}_{12} = \frac{\cos 2\theta_{12} - A_{\mathrm{MSW}}}{\sqrt{\big(\cos 2\theta_{12} - A_{\mathrm{MSW}}\big)^2 + \sin^2 2\theta_{12}}} \tag{FE-107}$$

**Symbols in (FE-107).**

- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.

- $c_{12},\ c_{13},\ c_{23}$: The cosines of the mixing angles, $c_{ij} = \cos\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $\theta_{ij}$: The mixing angles of the standard parametrisation of $U$ (or of $V_{\mathrm{CKM}}$): $\theta_{12},\ \theta_{13},\ \theta_{23}$; in two-flavour formulas $\theta_{ij}$ is the angle of the sector $(i,j)$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- $\theta_{12}^{\mathrm{m}}$: The effective 1-2 mixing angle in matter at the production point of a solar neutrino. Unit: dimensionless. Status: STANDARD.
- $s_{12},\ s_{13},\ s_{23}$: The sines of the mixing angles, $s_{ij} = \sin\theta_{ij}$. Unit: dimensionless. Status: STANDARD.
- $A_{\mathrm{MSW}}$: The ratio of the matter term to the vacuum term in the 1-2 sector. Unit: dimensionless. Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $V_{\mathrm{CC}},\ V_{\mathrm{NC}}$: The charged-current potential of electron neutrinos and the neutral-current potential common to all flavours. Unit: GeV. Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\Delta m^2_{21},\ \Delta m^2_{31},\ \Delta m^2_{32}$: The differences of the squared neutrino masses, $\Delta m^2_{ij} = m_i^2 - m_j^2$. Unit: $\mathrm{GeV}^2$ (quoted in $\mathrm{eV}^2$). Value: $\Delta m^2_{21} = 7.49\times10^{-5}\ \mathrm{eV}^2$, $\Delta m^2_{31} = 2.513\times10^{-3}\ \mathrm{eV}^2$ (NuFIT 6.0). Status: STANDARD.

evaluated at the production point in the electron density $n_{\mathrm{e}} = N_{\mathrm{A}}\min(245\,\mathrm{e}^{-10.54\,r/R_\odot}, 100)\ \mathrm{cm}^{-3}$ ($N_{\mathrm{A}}$ Avogadro's number, $R_\odot$ the solar radius; ASSUMED profile): $\mathcal{P}_{ee} = 0.541$ (pp, 0.3 MeV), 0.520 ($^7$Be), 0.498 (pep), 0.377, 0.319, 0.306 ($^8$B at 5, 10, 15 MeV); the numerical three-flavour evolution through the solar density agrees with the adiabatic formula to $1.3\times10^{-3}$: the conversion is vacuum dominated at low energy and matter dominated (MSW) for $^8$B.

## 16.4 The early Universe and $N_{\mathrm{eff}}$

The electron-positron plasma gives the electron neutrinos a thermal potential (STANDARD)

$$V_e - V_\mu = -\frac{8\sqrt{2}\,G_{\mathrm{F}}|\vec{k}|}{3m_{\mathrm{W}}^2}\,\rho_{\mathrm{e}}, \qquad \rho_{\mathrm{e}} \simeq \frac{7\pi^2}{60}\,T^4 \quad (T \gg m_{\mathrm{e}}) \tag{FE-109}$$

**Symbols in (FE-109).**

- $V_\alpha,\ V$: The matter potential of forward scattering: $V_\alpha$ of the flavour $\alpha$ ($V_e$, $V_\mu$), $V$ the $3\times3$ matrix of the potentials in the flavour basis; antineutrinos have $-V$. Unit: GeV. Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788\times10^{-5}\ \mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $\vec{k},\ \vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $m_{\mathrm{W}}$: The mass of $W^+$ and $W^-$. Unit: GeV. Value: 80.3625 GeV (measured, input). Status: STANDARD.
- $\rho_{\mathrm{e}},\ p_{\mathrm{e}}$: The energy density and the pressure of the electrons and positrons together (ideal gas). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $m_{\mathrm{e}}$: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.

which suppresses the mixing until it falls below the vacuum term; at the mean momentum $|\vec{k}| = 3.15T$ the atmospheric oscillations start at $T = 4.99$ MeV and the solar ones at $T = 2.39$ MeV (DERIVED): flavour conversion is active while the neutrinos decouple. Decoupling is computed from the temperature equations (STANDARD form, Escudero 2020; MeV units)

$$\frac{\mathrm{d}T_\gamma}{\mathrm{d}x_4} = -\frac{4\mathcal{H}\rho_\gamma + 3\mathcal{H}\big(\rho_\mathrm{e}+p_\mathrm{e}\big) + 3\mathcal{H}T_\gamma\,\frac{\mathrm{d}p_\mathrm{QED}}{\mathrm{d}T_\gamma} + C_{\nu_e} + 2C_{\nu_\mu}}{\frac{\mathrm{d}\rho_\gamma}{\mathrm{d}T_\gamma} + \frac{\mathrm{d}\rho_\mathrm{e}}{\mathrm{d}T_\gamma} + T_\gamma\,\frac{\mathrm{d}^2 p_\mathrm{QED}}{\mathrm{d}T_\gamma^2}}, \qquad \frac{\mathrm{d}T_{\nu_\alpha}}{\mathrm{d}x_4} = -\mathcal{H}T_{\nu_\alpha} + \frac{C_{\nu_\alpha}}{\mathrm{d}\rho_{\nu_\alpha}/\mathrm{d}T_{\nu_\alpha}},$$
$$\mathcal{H} = \frac{\sqrt{8\pi\rho_\mathrm{tot}/3}}{M_\mathrm{Pl}}, \qquad \rho_\gamma = \frac{\pi^2}{15}\,T_\gamma^4, \qquad \rho_{\nu_\alpha} = \frac{7\pi^2}{120}\,T_{\nu_\alpha}^4 \tag{FE-111}$$

**Symbols in (FE-111).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_\mathrm{Pl}$). Unit: GeV. Status: DERIVED.
- $\rho_\gamma$: The energy density of the photons, $\rho_\gamma = \pi^2 T_\gamma^4/15$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\rho_\mathrm{e}$, $p_\mathrm{e}$: The energy density and the pressure of the electrons and positrons together (ideal gas). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $p_\mathrm{QED}$, $p_\mathrm{QED}^{(2)}$, $p_\mathrm{QED}^{(3)}$: The finite-temperature QED correction to the pressure of the electron-photon plasma, and its parts of order $e^2$ and $e^3$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $C_{\nu_\alpha}$, $C_{\nu_e}$, $C_{\nu_\mu}$: The energy transfer rate per volume into the neutrinos of the flavour $\alpha$ by collisions. Unit: $\mathrm{GeV}^5$. Status: STANDARD.
- $T_{\nu_\alpha}$, $T_{\nu_e}$, $T_{\nu_\mu}$: The temperature of the neutrinos of the flavour $\alpha$ (of $\nu_e$, of $\nu_\mu$ and $\nu_\tau$) during decoupling. Unit: GeV. Status: STANDARD.
- $\rho_\nu$, $\rho_{\nu_\alpha}$: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature $T_\nu$, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; $\rho_{\nu_\alpha}$ that of the flavour $\alpha$ at its temperature $T_{\nu_\alpha}$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\rho_\mathrm{tot}$, $p_\mathrm{tot}$: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $M_\mathrm{Pl}$: The Planck mass, $M_\mathrm{Pl} = G_\mathrm{N}^{-1/2}$. Unit: GeV. Value: $1.22091\times10^{19}$ GeV. Status: STANDARD.

with the energy transfer by collisions (the radiatively corrected couplings 0.727, 0.233, −0.273, 0.233 are used; the tree-level ones change $N_\mathrm{eff}$ by $4\times10^{-4}$)

$$\begin{aligned}
C_{\nu_e} &= \frac{G_\mathrm{F}^2}{\pi^5}\Big[4\Big(\big(\tilde g_L^{\nu_e e}\big)^2 + \big(\tilde g_R^{\nu_e e}\big)^2\Big)\mathcal{F}_\mathrm{tr}(T_\gamma, T_{\nu_e}) + 2\mathcal{F}_\mathrm{tr}(T_{\nu_\mu}, T_{\nu_e})\Big],\\
C_{\nu_\mu} &= \frac{G_\mathrm{F}^2}{\pi^5}\Big[4\Big(\big(\tilde g_L^{\nu_\mu e}\big)^2 + \big(\tilde g_R^{\nu_\mu e}\big)^2\Big)\mathcal{F}_\mathrm{tr}(T_\gamma, T_{\nu_\mu}) - \mathcal{F}_\mathrm{tr}(T_{\nu_\mu}, T_{\nu_e})\Big],\\
\mathcal{F}_\mathrm{tr}(T_\mathrm{A}, T_\mathrm{B}) &= 32 f_\mathrm{a}\big(T_\mathrm{A}^9 - T_\mathrm{B}^9\big) + 56 f_\mathrm{s} T_\mathrm{A}^4 T_\mathrm{B}^4\big(T_\mathrm{A} - T_\mathrm{B}\big),\\
\tilde g_L^{\nu_e e} &= \frac{1}{2} + \sin^2\theta_\mathrm{W}, \qquad \tilde g_R^{\nu_e e} = \tilde g_R^{\nu_\mu e} = \sin^2\theta_\mathrm{W}, \qquad \tilde g_L^{\nu_\mu e} = -\frac{1}{2} + \sin^2\theta_\mathrm{W}
\end{aligned} \tag{FE-112}$$

**Symbols in (FE-112).**

- $C_{\nu_\alpha}$, $C_{\nu_e}$, $C_{\nu_\mu}$: The energy transfer rate per volume into the neutrinos of the flavour $\alpha$ by collisions. Unit: $\mathrm{GeV}^5$. Status: STANDARD.
- $G_\mathrm{F}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788\times10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde g_L^{\nu_e e}$, $\tilde g_R^{\nu_e e}$: The effective left- and right-handed couplings of electron neutrinos to electrons (charged and neutral current), $\tilde g_L^{\nu_e e} = \frac{1}{2} + \sin^2\theta_\mathrm{W}$, $\tilde g_R^{\nu_e e} = \sin^2\theta_\mathrm{W}$. Unit: dimensionless. Value: 0.727 and 0.233 (radiatively corrected). Status: STANDARD.
- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $R$: The chirality label R: the right-handed (singlet) zero mode or field, or the projection $P_R = (1-\Gamma_4)/2$. Status: STANDARD.

- $\mathcal{F}_{\rm tr}$: The energy-transfer function of two species with the temperatures $T_{\rm A}$ and $T_{\rm B}$ (Maxwell-Boltzmann with the corrections $f_{\rm a}$, $f_{\rm s}$). Unit: GeV$^9$. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $T_{\nu_\alpha}$, $T_{\nu_e}$, $T_{\nu_\mu}$: The temperature of the neutrinos of the flavour $\alpha$ (of $\nu_e$, of $\nu_\mu$ and $\nu_\tau$) during decoupling. Unit: GeV. Status: STANDARD.
- $\tilde{g}_L^{\nu_\mu e}$, $\tilde{g}_R^{\nu_\mu e}$: The effective couplings of muon (and tau) neutrinos to electrons (neutral current only), $\tilde{g}_L^{\nu_\mu e} = -\frac{1}{2} + \sin^2\theta_{\rm W}$, $\tilde{g}_R^{\nu_\mu e} = \sin^2\theta_{\rm W}$. Unit: dimensionless. Value: $-0.273$ and 0.233 (radiatively corrected). Status: STANDARD.
- $T_{\rm A}$, $T_{\rm B}$: The temperatures of the two species that exchange energy in $\mathcal{F}_{\rm tr}$. Unit: GeV. Status: STANDARD.
- $f_{\rm a}$, $f_{\rm s}$: The Fermi-Dirac corrections of the annihilation and scattering parts of the energy transfer. Unit: dimensionless. Value: 0.884 and 0.829. Status: STANDARD.
- $\theta_{\rm W}$: The weak mixing angle of the couplings, $\cos\theta_{\rm W} = \hat{g}_2/\hat{g}_Z$, $\sin\theta_{\rm W} = \hat{g}_1/\hat{g}_Z$; fixed by the measured $W$ and $Z$ masses with $\rho_{\rm EW,0}$. Unit: dimensionless. Value: $\cos^2\theta_{\rm W} = 0.776368$. Status: DERIVED.

and the finite-temperature QED pressure of the electron-photon plasma (STANDARD; Bennett et al. 2020)

$$\begin{aligned} &p_{\rm QED} = p_{\rm QED}^{(2)} + p_{\rm QED}^{(3)}, \qquad p_{\rm QED}^{(2)} = -\frac{e^2T^2}{12\pi^2}\,\mathcal{Q}_1 - \frac{e^2}{8\pi^4}\,\mathcal{Q}_1^2 + \frac{e^2 m_{\rm e}^2}{16\pi^4}\,\mathcal{Q}_2, \\ &p_{\rm QED}^{(3)} = \frac{e^3T}{12\pi^4}\,\mathcal{Q}_3^{3/2}, \qquad \rho_{\rm QED} = -p_{\rm QED} + T\,\frac{{\rm d}p_{\rm QED}}{{\rm d}T} \end{aligned} \tag{FE-113}$$

**Symbols in (FE-113).**

- $p_{\rm QED}$, $p_{\rm QED}^{(2)}$, $p_{\rm QED}^{(3)}$: The finite-temperature QED correction to the pressure of the electron-photon plasma, and its parts of order $e^2$ and $e^3$. Unit: GeV$^4$. Status: STANDARD.
- $e$: The elementary charge (the charge of the positron), $e^2 = 4\pi\alpha_{\rm em}$; $W^+$ has the charge $+e$ and $W^-$ the charge $-e$. Unit: dimensionless. Value: $e^2 = 4\pi/137.036$ at low energy. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\mathcal{Q}_1$, $\mathcal{Q}_2$, $\mathcal{Q}_3$: The momentum integrals of the QED pressure. Unit: GeV$^2$. Status: STANDARD.
- $m_{\rm e}$: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.
- $\rho_{\rm QED}$: The QED correction to the energy density of the plasma. Unit: GeV$^4$. Status: STANDARD.
- d: The differential: d$X$ is an infinitesimal change of $X$, d$X$/d$Y$ the derivative of $X$ with respect to $Y$, d$^n X$ the volume element of $n$ variables. Status: STANDARD.

$$\begin{aligned} &\mathcal{Q}_1 = \int_0^\infty {\rm d}k\,\frac{k^2}{E}\,n_{\rm D}(k), \qquad \mathcal{Q}_2 = \int_0^\infty\!\!\int_0^\infty {\rm d}k\,{\rm d}k_2\,\frac{kk_2}{E(k)E(k_2)}\,\ln\left|\frac{k+k_2}{k-k_2}\right|\,n_{\rm D}(k)\,n_{\rm D}(k_2), \\ &\mathcal{Q}_3 = \int_0^\infty {\rm d}k\,\frac{k^2+E^2}{E}\,n_{\rm D}(k), \qquad n_{\rm D}(k) = \frac{2}{{\rm e}^{E/T}+1}, \qquad E = \sqrt{k^2+m_{\rm e}^2} \end{aligned} \tag{FE-114}$$

**Symbols in (FE-114).**

- $\mathcal{Q}_1$, $\mathcal{Q}_2$, $\mathcal{Q}_3$: The momentum integrals of the QED pressure. Unit: GeV$^2$. Status: STANDARD.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: d$X$ is an infinitesimal change of $X$, d$X$/d$Y$ the derivative of $X$ with respect to $Y$, d$^n X$ the volume element of $n$ variables. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum ${\rm e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2+m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $n_{\rm D}(k)$: The Fermi-Dirac occupation of the electrons and positrons summed over the two spin states. Unit: dimensionless. Status: STANDARD.

- $k_2$, $\vec{k}_2$: A second momentum (variable of integration). Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $m_\mathrm{e}$: The electron mass. Unit: GeV. Value: 0.51099895 MeV. Status: STANDARD.

Then

$$N_\mathrm{eff} = \Big(\frac{11}{4}\Big)^{4/3}\Big[\Big(\frac{T_{\nu_e}}{T_\gamma}\Big)^4 + 2\Big(\frac{T_{\nu_\mu}}{T_\gamma}\Big)^4\Big], \qquad \frac{\mathrm{d}T_\nu}{\mathrm{d}x_4} = -\mathcal{H}T_\nu + \frac{C_{\nu_e} + 2C_{\nu_\mu}}{3\,\mathrm{d}\rho_\nu/\mathrm{d}T_\nu} \tag{FE-115}$$

**Symbols in (FE-115).**

- $N_\mathrm{eff}$: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); $\Delta N_\mathrm{eff}$ an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $T_{\nu_\alpha}$, $T_{\nu_e}$, $T_{\nu_\mu}$: The temperature of the neutrinos of the flavour $\alpha$ (of $\nu_e$, of $\nu_\mu$ and $\nu_\tau$) during decoupling. Unit: GeV. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_\mathrm{Pl}$). Unit: GeV. Status: DERIVED.
- $C_{\nu_\alpha}$, $C_{\nu_e}$, $C_{\nu_\mu}$: The energy transfer rate per volume into the neutrinos of the flavour $\alpha$ by collisions. Unit: $\mathrm{GeV}^5$. Status: STANDARD.
- $\rho_\nu$, $\rho_{\nu_\alpha}$: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature $T_\nu$, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; $\rho_{\nu_\alpha}$ that of the flavour $\alpha$ at its temperature $T_{\nu_\alpha}$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.

The method reproduces the published values of the same method (DERIVED: instantaneous decoupling 3.0002, with the QED corrections 3.0108 and 3.0098; Fermi-Dirac energy transfer 3.0393, with QED 3.0495 and 3.0485; published 3.000, 3.011, 3.010, 3.039, 3.049). For UM-2 it gives $N_\mathrm{eff} = 3.0476$ with separate $\nu_e$ and $\nu_{\mu,\tau}$ temperatures (published with separate temperatures: 3.048) and 3.0485 with one common temperature; $T_{\nu_e}/T_\gamma = 0.71732$, $T_{\nu_\mu}/T_\gamma = 0.71621$, the comoving neutrino temperatures being raised by 0.265 % ($\nu_e$) and 0.109 % ($\nu_{\mu,\tau}$) by electron-positron annihilation. The plasma at MeV temperatures is that of the three-flavour Standard Model; the only extra radiation is that of the fold bands produced by freeze-in through I11, $\Delta N_\mathrm{eff} = 0.001$ (neither fold band is ever thermalised; the dark sector), so UM-2 adopts for the neutrinos the precision Standard-Model value with oscillations and the electron mass in the rates, $N_\mathrm{eff} = 3.0432 \pm 0.0002$ (Cielo, Escudero, Mangano and Pisanti 2023), and $N_\mathrm{eff} = 3.0442$ with the fold bands.

## 16.5 The cosmic neutrino background

After decoupling the flavour states lose their coherence: the CNB of UM-2 is an incoherent mixture of the three MASS eigenstates (DERIVED; instantaneous decoupling; the CMB temperature $T_\gamma = 2.7255$ K today, STANDARD) with the temperature $T_\nu = (4/11)^{1/3}T_\gamma = 1.9454$ K and

$$f_\mathrm{FD}(k) = \frac{1}{\mathrm{e}^{k/T_\nu}+1}, \qquad n_i = 2\int\frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\,f_\mathrm{FD} = \frac{3\zeta(3)}{2\pi^2}\,T_\nu^3, \qquad \rho_i = 2\int\frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\,\sqrt{k^2+m_i^2}\,f_\mathrm{FD},$$
$$\bar{v}_i = \frac{\int_0^\infty k^2\,\frac{k}{\sqrt{k^2+m_i^2}}\,f_\mathrm{FD}\,\mathrm{d}k}{\int_0^\infty k^2\,f_\mathrm{FD}\,\mathrm{d}k}, \qquad \text{flavour content of } \nu_i:\ |U_{\alpha i}|^2 \tag{FE-120}$$

**Symbols in (FE-120).**

- $f_{\mathrm{FD}}$: The Fermi-Dirac distribution of the relic neutrinos, $f_{\mathrm{FD}} = 1/(\mathrm{e}^{k/T_\nu} + 1)$. Unit: dimensionless. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $n$, $n_o$, $n_{\psi i}$, $n_f$, $n_i$: A number density: $n(y)$ the particle density of dirac16complex in the Kohn-Sham equations and $n_o$ the contribution of the orbital $o$, $n_{\psi i}(y)$ the density of the Standard-Model fermion $\psi_i$ (per proper 7-volume of eight dimensions); $n_f$ the density of the fermion component $f$ and $n_i$ that of the neutrino mass state $i$ (per 3-volume of four dimensions). Unit: $\mathrm{GeV}^7$ (eight dimensions), $\mathrm{GeV}^3$ (four dimensions). Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\vec{k}$, $\vec{k}_o$: The wave vector (momentum) in ordinary 3-space; $\vec{k}_o$ that of the orbital $o$, a point of the lattice $k_{\mathrm{lat}}\vec{n}_o$. Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\zeta$: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers $n$. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $\bar{v}_i$: The mean velocity of the relic neutrinos of the mass state $i$. Unit: dimensionless (c = 1). Value: 0.435, 0.0604, 0.0105 today. Status: STANDARD.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: $\mathrm{GeV}^{3/2}$. Status: STANDARD.
- $U$: The PMNS matrix, the unitary 3×3 mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i}\nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- $\alpha$: A flavour index running over $e, \mu, \tau$ (the flavour states of the leptons and neutrinos). Status: STANDARD.

$n_i = 112.0\ \mathrm{cm}^{-3}$ per mass state (neutrinos and antineutrinos), $336\ \mathrm{cm}^{-3}$ in all, and the density parameter

$$\Omega_\nu h_{100}^2 = \frac{\sum_{i=1}^3 \rho_i}{\rho_{\mathrm{crit}}/h_{100}^2} = 6.382 \times 10^{-4}, \qquad \frac{\rho_{\mathrm{crit}}}{h_{100}^2} = 1.05368 \times 10^4\ \mathrm{eV/cm}^3$$

**Symbols in this equation.**

- $\Omega_\nu$: The density parameter of the three neutrino mass states today: their energy density divided by the critical density $\rho_{\mathrm{crit}}$ (with $h_{100}^2$: $\Omega_\nu h_{100}^2$). Unit: dimensionless. Value: $\Omega_\nu h_{100}^2 = 6.382 \times 10^{-4}$ (instantaneous decoupling), $6.4511 \times 10^{-4}$ (heated spectrum, CAMB). Status: DERIVED.
- $h_{100}$: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a

subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.

- $\rho_{\mathrm{crit}}$: The critical density, $3\mathcal{H}_0^2 M_{\mathrm{Pl}}^2/(8\pi)$. Unit: $\mathrm{GeV}^4$ (quoted in $\mathrm{GeV/cm}^3$). Value: $\rho_{\mathrm{crit}}/h_{100}^2 = 1.05368 \times 10^{-5}\ \mathrm{GeV/cm}^3$. Status: STANDARD.

($6.451 \times 10^{-4}$ with the heated spectrum of $N_{\mathrm{eff}} = 3.0432$: the degeneracy $N_{\mathrm{eff}}/3$ of each state corresponds to a Fermi-Dirac spectrum of temperature $(N_{\mathrm{eff}}/3)^{1/4} T_\nu$, so the non-relativistic density is multiplied by $(N_{\mathrm{eff}}/3)^{3/4}$). The mean velocities today and the flavour contents are

| state | mass | mean velocity today | $\nu_e$ content $\lvert U_{ei}\rvert^2$ | $\nu_\mu$ content | $\nu_\tau$ content |
|---|---|---|---|---|---|
| $\nu_1$ | 1.000 meV | $0.435\,c$ | 0.677 | 0.112 | 0.211 |
| $\nu_2$ | 8.712 meV | $0.0604\,c$ | 0.301 | 0.428 | 0.271 |
| $\nu_3$ | 50.140 meV | $0.0105\,c$ | 0.022 | 0.460 | 0.518 |

## 16.6 Supernovae and the dense neutrino gas

**MSW in the envelope.** In the envelope, with the density profile $10^4\,(10^9\,\mathrm{cm}/r)^3$ g/cm$^3$ and equal numbers of protons and neutrons (ASSUMED), a resonance lies at the electron density of FE-108 and is crossed adiabatically when the adiabaticity is large (two-flavour conditions: $\Delta m^2_{31}$ and $\theta_{13}$ for the H resonance, $\Delta m^2_{21}$ and $\theta_{12}$ for the L resonance, without the factor $c^2_{13}$ of the solar condition):

$$n_{\mathrm{e}}^{\mathrm{res}} = \frac{\Delta m^2_{ij}\cos 2\theta_{ij}}{2\sqrt{2}\,G_{\mathrm{F}}E}, \qquad \gamma_{\mathrm{ad}} = \frac{\Delta m^2_{ij}\sin^2 2\theta_{ij}}{2E\cos 2\theta_{ij}\,\lvert \mathrm{d}\ln n_{\mathrm{e}}/\mathrm{d}r\rvert} \gg 1, \qquad \mathcal{P}_{\mathrm{LZ}} = \mathrm{e}^{-\pi\gamma_{\mathrm{ad}}/2} \tag{FE-108}$$

**Symbols in (FE-108).**

- $n_{\mathrm{e}}^{\mathrm{res}}$: The electron density of an MSW resonance. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\theta_{ij}$: The mixing angles of the standard parametrisation of $U$ (or of $V_{\mathrm{CKM}}$): $\theta_{12}$, $\theta_{13}$, $\theta_{23}$; in two-flavour formulas $\theta_{ij}$ is the angle of the sector $(i, j)$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$ (PMNS). Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788 \times 10^{-5}\ \mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $\gamma_{\mathrm{ad}}$: The adiabaticity parameter of a resonance (adiabatic for $\gamma_{\mathrm{ad}} \gg 1$). Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $n_{\mathrm{e}}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $r$: A distance: between the two masses in $V_{\mathrm{N}}(r)$, from the centre of the star in the adiabaticity condition. Unit: length. Status: STANDARD.
- $\mathcal{P}_{\mathrm{LZ}}$: The Landau-Zener probability of a jump between the matter eigenstates at a resonance. Unit: dimensionless. Status: STANDARD.

- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

For 20 MeV the 1-3 (H) resonance lies at 1573 g/cm$^3$ ($r = 1.85 \times 10^4$ km) with $\gamma_{\mathrm{ad}} = 178$ ($\mathcal{P}_{\mathrm{LZ}} = 2.7 \times 10^{-122}$): adiabatic. For the normal ordering an electron neutrino from the core leaves the star as $\nu_3$ and an electron antineutrino as $\bar{\nu}_1$: $\mathcal{P}(\nu_e \to \nu_e) = |U_{e3}|^2 = 0.022$ and $\mathcal{P}(\bar{\nu}_e \to \bar{\nu}_e) = |U_{e1}|^2 = 0.677$ (without collective effects and Earth matter; DERIVED). For neutrinospheres with the luminosities 6, 6 and $4 \times 10^{52}$ erg/s and the mean energies $\bar{E}_{\mathrm{e}} = 11$ MeV, $\bar{E}_{\bar{\mathrm{e}}} = 14$ MeV, $\bar{E}_{\mathrm{x}} = 15.5$ MeV of the electron neutrinos, electron antineutrinos and each heavy-lepton flavour (ASSUMED), with the number emission rates proportional to the luminosity over the mean energy, the mean energies at the Earth are

$$\bar{E}_{\mathrm{e}}^{\mathrm{Earth}} = \frac{\mathcal{P}\,\dot{N}_{\mathrm{e}}\bar{E}_{\mathrm{e}} + (1-\mathcal{P})\,\dot{N}_{\mathrm{x}}\bar{E}_{\mathrm{x}}}{\mathcal{P}\,\dot{N}_{\mathrm{e}} + (1-\mathcal{P})\,\dot{N}_{\mathrm{x}}} = 15.29 \text{ MeV} \qquad \big(\mathcal{P} = \mathcal{P}(\nu_e \to \nu_e)\big)$$

**Symbols in this equation.**

- $\bar{E}_{\mathrm{e}}^{\mathrm{Earth}}$, $\bar{E}_{\bar{\mathrm{e}}}^{\mathrm{Earth}}$: The mean energies of the electron neutrinos and of the electron antineutrinos of a supernova at the Earth, after the MSW conversion in the envelope (the emitted spectra weighted with the survival probability and the number emission rates). Unit: GeV (quoted in MeV). Value: 15.29 MeV and 14.34 MeV. Status: DERIVED.
- $\mathcal{P}$: The probability of a flavour transition, $\mathcal{P}(\nu_\alpha \to \nu_\beta)$; $\mathcal{P}_{ee}$ the survival probability of electron neutrinos. Unit: dimensionless. Status: STANDARD.
- $\dot{N}_{\mathrm{e}}$, $\dot{N}_{\bar{\mathrm{e}}}$, $\dot{N}_{\mathrm{x}}$, $\dot{N}_{\mathrm{pair}}$: The number emission rates of a supernova neutrinosphere (neutrinos per unit time, proportional to the luminosity divided by the mean energy): $\dot{N}_{\mathrm{e}}$ of the electron neutrinos, $\dot{N}_{\bar{\mathrm{e}}}$ of the electron antineutrinos, $\dot{N}_{\mathrm{x}}$ of each heavy-lepton flavour; $\dot{N}_{\mathrm{pair}} = \frac{2}{3}(\dot{N}_{\bar{\mathrm{e}}} - \dot{N}_{\mathrm{x}})$ is the rate of electron neutrino-antineutrino pairs turned into heavy-lepton pairs by lepton-number-conserving pair conversion. Unit: 1/time (relative values are used). Status: ASSUMED.
- $\bar{E}_{\mathrm{e}}$, $\bar{E}_{\bar{\mathrm{e}}}$, $\bar{E}_{\mathrm{x}}$: The mean energies of the neutrinos emitted by a supernova neutrinosphere: of the electron neutrinos, of the electron antineutrinos and of each heavy-lepton flavour. Unit: GeV (quoted in MeV). Value: 11, 14, 15.5 MeV. Status: ASSUMED.
- $\nu_\alpha$, $\nu_i$: A left-handed neutrino: $\nu_\alpha$ of the flavour $\alpha = e, \mu, \tau$ ($\nu_e$, $\nu_\mu$, $\nu_\tau$), $\nu_i$ of the mass state $i$ (mass $m_i$). Unit: GeV$^{3/2}$. Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.

and, with $\mathcal{P}(\bar{\nu}_e \to \bar{\nu}_e)$, $\dot{N}_{\bar{\mathrm{e}}}$ and $\bar{E}_{\bar{\mathrm{e}}}$, $\bar{E}_{\bar{\mathrm{e}}}^{\mathrm{Earth}} = 14.34$ MeV for the electron antineutrinos.

**The fast flavour instability.** Near the neutrinosphere the neutrino gas is dense enough for the neutrino-neutrino potential to dominate. For an axially symmetric gas the electron-lepton-number angular distribution and its moments are

$$G_{\mathrm{ELN}}(u) = \sqrt{2}\,G_{\mathrm{F}}\big[n_{\nu_e}\,g_{\nu_e}(u) - n_{\bar{\nu}_e}\,g_{\bar{\nu}_e}(u)\big], \qquad \int_{-1}^{1} g_{\nu_e}\,\frac{\mathrm{d}u}{2} = \int_{-1}^{1} g_{\bar{\nu}_e}\,\frac{\mathrm{d}u}{2} = 1, \qquad \mathcal{I}_l = \int_{-1}^{1} \frac{\mathrm{d}u}{2}\,G_{\mathrm{ELN}}(u)\,u^l \tag{FE-116}$$

**Symbols in (FE-116).**

- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- $u$: The cosine of the angle between a momentum and the symmetry axis $x_3$ of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: GeV$^{-2}$. Value: $1.1663788 \times 10^{-5}$ GeV$^{-2}$ (measured). Status: STANDARD.
- $n_{\nu_e}$, $n_{\bar{\nu}_e}$: The number densities of electron neutrinos and electron antineutrinos. Unit: GeV$^3$. Status: STANDARD.
- $g_{\nu_e}(u)$, $g_{\bar{\nu}_e}(u)$: The normalised angular distributions of electron neutrinos and antineutrinos. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.

- $\mathcal{I}_l$: The $l$-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- $l$: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.

the linearised equations for the flavour coherence give plane-wave modes along the symmetry axis $x_3$ with the dispersion relation (STANDARD)

$$\mathcal{C}(u) \propto \mathrm{e}^{-\mathrm{i}(\omega_\mathrm{f} x_4 - k_\mathrm{f} x_3)}, \qquad \bar{\omega} = \omega_\mathrm{f} - \mathcal{I}_0, \qquad \bar{k} = k_\mathrm{f} - \mathcal{I}_1 : \qquad \det \Pi(\bar{\omega}, \bar{k}) = 0,$$
$$\Pi = \mathrm{diag}(1,\ -1) + \int_{-1}^{1} \frac{\mathrm{d}u}{2} \frac{G_\mathrm{ELN}(u)}{\bar{\omega} - \bar{k}u} \begin{pmatrix} 1 & u \\ u & u^2 \end{pmatrix} \tag{FE-117}$$

**Symbols in (FE-117).**

- $\mathcal{C}(u)$: The flavour coherence (off-diagonal element of the density matrix) of the neutrinos moving in the direction $u$. Unit: dimensionless. Status: STANDARD.
- $u$: The cosine of the angle between a momentum and the symmetry axis $x_3$ of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\omega_\mathrm{f}$, $k_\mathrm{f}$: The frequency and the wavenumber (along $x_3$) of a flavour wave. Unit: GeV. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $x_3$: The third coordinate of ordinary 3-space (also the symmetry axis of the fast-flavour analysis). Unit: length. Status: AUTHOR.
- $\bar{\omega}$, $\bar{k}$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar{\omega} = \omega_\mathrm{f} - \mathcal{I}_0$, $\bar{k} = k_\mathrm{f} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar{\omega} > 0$ grows. Unit: GeV. Status: STANDARD.
- $\mathcal{I}_l$: The $l$-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- $\Pi$: The $2 \times 2$ dispersion matrix of the fast flavour instability. Unit: dimensionless. Status: STANDARD.
- diag: A diagonal matrix with the listed entries on its diagonal (in the order of the coordinates $x_0, x_1, \ldots, x_7$ when it is a tensor in eight dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $G_\mathrm{ELN}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.

whose axisymmetric and azimuthal branches are

$$\big(1 + \mathcal{J}_0\big)\big(-1 + \mathcal{J}_2\big) - \mathcal{J}_1^2 = 0, \qquad \mathcal{J}_l = \int_{-1}^{1} \frac{\mathrm{d}u}{2} \frac{G_\mathrm{ELN}(u)\, u^l}{\bar{\omega} - \bar{k}u}; \qquad -1 + \int_{-1}^{1} \frac{\mathrm{d}u}{2} \frac{G_\mathrm{ELN}(u)\,(1 - u^2)}{2(\bar{\omega} - \bar{k}u)} = 0 \tag{FE-118}$$

**Symbols in (FE-118).**

- $\mathcal{J}_l$: The $l$-th moment of the dispersion integral of the fast flavour instability. Unit: dimensionless. Status: STANDARD.
- $l$: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $u$: The cosine of the angle between a momentum and the symmetry axis $x_3$ of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- $G_\mathrm{ELN}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- $\bar{\omega}$, $\bar{k}$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar{\omega} = \omega_\mathrm{f} - \mathcal{I}_0$, $\bar{k} = k_\mathrm{f} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar{\omega} > 0$ grows. Unit: GeV. Status: STANDARD.

and the homogeneous mode and the mode with $k_\mathrm{f} = \mathcal{I}_1$ obey

$$
\begin{aligned}
&\mathrm{i}\,\frac{\mathrm{d}\mathcal{C}(u)}{\mathrm{d}x_4} = \big(\mathcal{I}_0 - u\,\mathcal{I}_1\big)\mathcal{C}(u) - \int_{-1}^{1}\frac{\mathrm{d}\tilde{u}}{2}\,G_{\mathrm{ELN}}(\tilde{u})\,(1-u\tilde{u})\,\mathcal{C}(\tilde{u});\\
&\bar{\omega}^2 - \big(\mathcal{I}_2 - \mathcal{I}_0\big)\bar{\omega} - \big(\mathcal{I}_0\mathcal{I}_2 - \mathcal{I}_1^2\big) = 0, \qquad \mathrm{Im}\,\bar{\omega} = \frac{1}{2}\sqrt{4\mathcal{I}_1^2 - \big(\mathcal{I}_0 + \mathcal{I}_2\big)^2}
\end{aligned}
\tag{FE-119}
$$

**Symbols in (FE-119).**

- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\mathcal{C}(u)$: The flavour coherence (off-diagonal element of the density matrix) of the neutrinos moving in the direction $u$. Unit: dimensionless. Status: STANDARD.
- $u$: The cosine of the angle between a momentum and the symmetry axis $x_3$ of the neutrino gas. Unit: dimensionless. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{I}_l$: The $l$-th angular moment of the electron-lepton-number distribution. Unit: GeV. Status: STANDARD.
- $\tilde{u}$: A second direction cosine (variable of integration). Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{ELN}}(u)$: The angular distribution of the electron lepton number of a dense neutrino gas (in units of a potential). Unit: GeV. Status: STANDARD.
- $\bar{\omega}$, $\bar{k}$: The frequency and wavenumber of a flavour wave shifted by the moments, $\bar{\omega} = \omega_{\mathrm{f}} - \mathcal{I}_0$, $\bar{k} = k_{\mathrm{f}} - \mathcal{I}_1$; a mode with $\mathrm{Im}\,\bar{\omega} > 0$ grows. Unit: GeV. Status: STANDARD.
- Im: The imaginary part of a complex number. Status: STANDARD.

A root with $\mathrm{Im}\,\bar{\omega} > 0$ is an instability. In units of $\sqrt{2}\,G_{\mathrm{F}} n_{\nu_{\mathrm{e}}}$ (DERIVED): with an ELN crossing (forward-peaked antineutrinos, $n_{\bar{\nu}_{\mathrm{e}}}/n_{\nu_{\mathrm{e}}} = 0.8$) the largest growth rate is 0.088 (homogeneous mode 0.018); with linear distributions ($g_{\nu_{\mathrm{e}}} = 1 + 0.2u$, $g_{\bar{\nu}_{\mathrm{e}}} = 1 + 0.6u$, $n_{\bar{\nu}_{\mathrm{e}}}/n_{\nu_{\mathrm{e}}} = 0.9$) it is 0.098 although the homogeneous mode is stable; without a crossing no mode grows, as the general theorem (Morinaga 2022) states. At $n_{\nu_{\mathrm{e}}} = 10^{32}$ to $10^{33}$ cm$^{-3}$ the instability grows by the factor e in $6\times10^{-10}$ to $6\times10^{-11}$ s over 18 to 1.8 cm, a million times faster than the hydrodynamics.

**Flavour conversion and the explosions.** Conversion moves energy from the electron-flavour neutrinos, which heat the matter behind the shock through charged-current absorption, to the heavy-lepton neutrinos, which do not. A schematic estimate of the heating rate (Fermi-Dirac-like spectra with the pinching parameter 3, so that the mean squared energy is $\frac{5}{4}$ of the squared mean energy, a common factor) compares full flavour equipartition and lepton-number-conserving pair conversion of $\dot{N}_{\mathrm{pair}}$ pairs with no conversion:

$$
\begin{aligned}
&\dot{E}_{\mathrm{h}} \propto \dot{N}_{\mathrm{e}}\bar{E}_{\mathrm{e}}^2 + \dot{N}_{\bar{\mathrm{e}}}\bar{E}_{\bar{\mathrm{e}}}^2, \qquad \dot{E}_{\mathrm{h}}^{\mathrm{eq}} \propto \tfrac{1}{3}\big(\dot{N}_{\mathrm{e}}\bar{E}_{\mathrm{e}}^2 + 2\dot{N}_{\mathrm{x}}\bar{E}_{\mathrm{x}}^2\big) + \tfrac{1}{3}\big(\dot{N}_{\bar{\mathrm{e}}}\bar{E}_{\bar{\mathrm{e}}}^2 + 2\dot{N}_{\mathrm{x}}\bar{E}_{\mathrm{x}}^2\big),\\
&\dot{E}_{\mathrm{h}}^{\mathrm{pair}} \propto \big(\dot{N}_{\mathrm{e}} - \dot{N}_{\mathrm{pair}}\big)\bar{E}_{\mathrm{e}}^2 + \big(\dot{N}_{\bar{\mathrm{e}}} - \dot{N}_{\mathrm{pair}}\big)\bar{E}_{\bar{\mathrm{e}}}^2, \qquad \dot{N}_{\mathrm{pair}} = \tfrac{2}{3}\big(\dot{N}_{\bar{\mathrm{e}}} - \dot{N}_{\mathrm{x}}\big)\,:
\end{aligned}
$$

**Symbols in this equation.**

- $\dot{E}_{\mathrm{h}}$, $\dot{E}_{\mathrm{h}}^{\mathrm{eq}}$, $\dot{E}_{\mathrm{h}}^{\mathrm{pair}}$: The rate at which the neutrinos heat the matter behind a supernova shock by charged-current absorption (up to a common factor): $\dot{E}_{\mathrm{h}}$ without flavour conversion, $\dot{E}_{\mathrm{h}}^{\mathrm{eq}}$ with full flavour equipartition, $\dot{E}_{\mathrm{h}}^{\mathrm{pair}}$ with lepton-number-conserving pair conversion (schematic estimate). Unit: energy per time (relative values are used). Value: ratios 0.884 and 0.760 to the unconverted rate. Status: DERIVED.
- $\dot{N}_{\mathrm{e}}$, $\dot{N}_{\bar{\mathrm{e}}}$, $\dot{N}_{\mathrm{x}}$, $\dot{N}_{\mathrm{pair}}$: The number emission rates of a supernova neutrinosphere (neutrinos per unit time, proportional to the luminosity divided by the mean energy): $\dot{N}_{\mathrm{e}}$ of the electron neutrinos, $\dot{N}_{\bar{\mathrm{e}}}$ of the electron antineutrinos, $\dot{N}_{\mathrm{x}}$ of each heavy-lepton flavour; $\dot{N}_{\mathrm{pair}} = \frac{2}{3}(\dot{N}_{\bar{\mathrm{e}}} - \dot{N}_{\mathrm{x}})$ is the rate of electron neutrino-antineutrino pairs turned into heavy-lepton pairs by lepton-number-conserving pair conversion. Unit: 1/time (relative values are used). Status: ASSUMED.
- $\bar{E}_{\mathrm{e}}$, $\bar{E}_{\bar{\mathrm{e}}}$, $\bar{E}_{\mathrm{x}}$: The mean energies of the neutrinos emitted by a supernova neutrinosphere: of the electron neutrinos, of the electron antineutrinos and of each heavy-lepton flavour. Unit: GeV (quoted in MeV). Value: 11, 14, 15.5 MeV. Status: ASSUMED.

$\dot{E}_{\mathrm{h}}^{\mathrm{eq}}/\dot{E}_{\mathrm{h}} = 0.884$ and $\dot{E}_{\mathrm{h}}^{\mathrm{pair}}/\dot{E}_{\mathrm{h}} = 0.760$ for the neutrinospheres above (DERIVED, schematic). In 195 collapse simulations (9 to 120 solar masses) with instantaneous flavour equipartition below a baryon density, Gogilashvili and Tamborra (2026) find the fraction of failed explosions 25.6 % without conversion and 50.8, 77.4, 93.3, 96.4 % for the threshold densities $10^9$, $10^{10}$, $10^{11}$, $10^{13}$ g/cm$^3$ (27.0 % and 47.6, 61.9, 73.9, 88.3 % weighted with

the Salpeter mass function), most strongly between 16 and 30 solar masses, with lighter neutron stars from the successful explosions (STANDARD, cited).

## 16.7 The $Z$ boson: invisible width and the widths of the three generations

With three neutrinos and no tip $Z$ portal the invisible width of the $Z$ counts $N_\nu = 3$ neutrino species, against the measured $2.9963 \pm 0.0074$ (PDG 2025; 0.5 standard deviations); a fold band of $Z$ coupling $r_{\mathrm{Z}}$ times a neutrino's would add $2r_{\mathrm{Z}}^2$, bounded by $r_{\mathrm{Z}}^2 < 0.0056$ (the section on the interactions). The tree-level partial widths (STANDARD)

$$\Gamma_{\mathrm{Z}}^{f} = \frac{N_{\mathrm{col}}\, G_{\mathrm{F}} m_{\mathrm{Z}}^3}{6\sqrt{2}\,\pi}\Big( \big(\tilde{g}_V^{f}\big)^2 + \big(T_3^{f}\big)^2 \Big), \qquad \tilde{g}_V^{f} = T_3^{f} - 2Q_{\mathrm{em}}^{f} \sin^2\theta_{\overline{\mathrm{MS}}},$$

**Symbols in this equation.**

- $\Gamma_{\mathrm{Z}}^{f}$: The decay width of the $Z$ boson; with the fermion label $f$ it is the partial width into the pair of $f$ and its antiparticle (tree level, massless fermion). Unit: GeV (quoted in MeV). Value: total 2485.2 MeV with $1 + \alpha_3/\pi$ on the quarks (measured 2495.5 MeV). Status: DERIVED.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\mathrm{col}}$: The number of colour components of a fermion multiplet: 3 for a quark, 1 for a lepton. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{F}}$: The Fermi constant. Unit: $\mathrm{GeV}^{-2}$. Value: $1.1663788 \times 10^{-5}$ $\mathrm{GeV}^{-2}$ (measured). Status: STANDARD.
- $m_{\mathrm{Z}}$: The mass of the $Z$. Unit: GeV. Value: 91.1879 GeV (reproduced). Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\tilde{g}_L^{f}$, $\tilde{g}_R^{f}$, $\tilde{g}_V^{f}$: The neutral-current couplings of the fermion $f$ to the $Z$ (left-handed, right-handed and vector), $\tilde{g}_L^{f} = T_3^{f} - Q_{\mathrm{em}}^{f} \sin^2\theta_{\mathrm{W}}$. Unit: dimensionless. Status: STANDARD.
- $V$: The label V of the vector coupling of a fermion to the $Z$, $\tilde{g}_V^{f} = \tilde{g}_L^{f} + \tilde{g}_R^{f}$. Status: STANDARD.
- $T_3$: The third component of the weak isospin ($+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower member of a doublet, 0 for a singlet); $T_3^{f}$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $Q_{\mathrm{em}}$: The electric charge in units of the elementary charge, $Q_{\mathrm{em}} = T_3 + Y$; $Q_{\mathrm{em}}^{f}$ that of the fermion $f$. Unit: dimensionless. Status: STANDARD.
- $\theta_{\overline{\mathrm{MS}}}$: The weak mixing angle in the $\overline{\mathrm{MS}}$ scheme at $m_{\mathrm{Z}}$. Unit: dimensionless. Value: $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$. Status: STANDARD.

with the $\overline{\mathrm{MS}}$ value $\sin^2\theta_{\overline{\mathrm{MS}}} = 0.23122$ give (DERIVED) 165.9 MeV per neutrino (497.7 MeV invisible; measured 499.1 MeV), 83.4 MeV per charged lepton, 285.4 MeV for each of the up and charm quarks and 367.9 MeV for each of the down, strange and bottom quarks (the top quark is too heavy), a hadronic width of 1674.4 MeV (1737.3 MeV with the QCD factor $1 + \alpha_3/\pi$, $\alpha_3 = 0.1180$) and a total of 2422.3 MeV (2485.2 MeV with the QCD factor) against the measured 2495.5 MeV; the ratio of the hadronic to the leptonic width is 20.83 against 20.77. The remaining difference (0.41 % of the total) is the size of the electroweak loop corrections, which UM-2 does not compute; the angle of the couplings of the $x_0$ mass fit, $\sin^2\theta_{\mathrm{W}} = 0.223632$, would change the total by +0.82 % (0.83 % with the QCD factor; residual analysis).

## 16.8 Dirac or Majorana (DERIVED)

UM-2 gives the neutrinos Dirac masses: the overlap at the tip of the lepton-doublet zero mode with three sharply localised right-handed singlets ($|c_R| = 31$ to 35) with tip couplings of order one. The alternative without right-handed neutrinos is a Majorana mass from the gauge-invariant tip operator (the tip form of the Weinberg operator; the upper component of $\Sigma^\dagger \ell_i$ is neutral, since $\Sigma \to U_L \Sigma U_Y^\dagger$ absorbs both the $SU(2)_L$ and the hypercharge transformations of $\ell_i$)

$$S_{\mathrm{M}} = -\frac{1}{2}\int_{y=-L} \mathrm{d}^7X_{\mathrm{tip}} \sqrt{|g_{\mathrm{tip}}|} \sum_{i,j=1}^{3} (\mathcal{K}_{\mathrm{M}})_{ij}\, \tilde{\nu}_i \tilde{\nu}_j + \mathrm{h.c.}, \qquad \tilde{\nu}_i = \big(\Sigma^\dagger \ell_i\big)_1.$$

**Symbols in this equation.**

- $S_{\mathrm{M}}$: The action of the gauge-invariant Majorana mass operator on the tip, the alternative to the Dirac neutrinos of UM-2 without right-handed neutrinos (not part of UM-2). Unit: dimensionless (an action, $\hbar = 1$). Status: INVENTED.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X_{\mathrm{tip}}$: The seven coordinates $x_1, \ldots, x_7$ of the tip $y = -L$ of the hidden direction; $\mathrm{d}^7 X_{\mathrm{tip}}$ is their coordinate volume element. Unit: length (each coordinate). Status: DERIVED.
- $g_{\mathrm{tip}}$: The metric induced on the seven-dimensional tip $y = -L$ (the metric $g_{MN}$ restricted to the tip directions $x_1, \ldots, x_7$); $|g_{\mathrm{tip}}|$ is the absolute value of its determinant. Unit: dimensionless. Status: DERIVED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $j$: A second generation index ($j = 1, 2, 3$). Status: STANDARD.
- $\mathcal{K}_{\mathrm{M}}^{(i)}$: The dimensionless Majorana tip coefficients of the alternative tip operator $S_{\mathrm{M}}$: the complex symmetric $3 \times 3$ matrix $(\mathcal{K}_{\mathrm{M}})_{ij}$ (generation indices $i$, $j$; dimensionless because the tip operator is integrated over seven dimensions and $\tilde{\nu}_i \tilde{\nu}_j$ has mass dimension 7); its eigenvalues $\mathcal{K}_{\mathrm{M}}^{(i)}$ give the four-dimensional masses $m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \mathcal{Z}(c_L)^2/(fL)$ to first order in the tip value of the doublet zero mode. Unit: dimensionless. Value: $5.63 \times 10^{-13}$, $4.90 \times 10^{-12}$, $2.82 \times 10^{-11}$ for $i = 1, 2, 3$. Status: DERIVED.
- $\tilde{\nu}_i$: The neutral upper component $(\Sigma^\dagger \ell_i)_1$ of the lepton doublet of generation $i$ dressed with the tip sigma field: a left-handed field that is neutral under $SU(2)_L \times U(1)_Y$; $\tilde{\nu}_i \tilde{\nu}_j$ is the Lorentz-invariant (Majorana) product of two such fields. Unit: $\mathrm{GeV}^{7/2}$ (an eight-dimensional spinor field at the tip). Status: INVENTED.
- $\Sigma$: The tip sigma field: a unitary $2 \times 2$ matrix field on the tip ($\Sigma^\dagger \Sigma = \mathbf{1}$) that replaces the Higgs doublet; its three modes are the longitudinal $W^+$, $W^-$ and $Z$, and $\Sigma = \mathbf{1}$ is the unitary gauge. Unit: dimensionless. Status: INVENTED.
- $^\dagger$: Hermitian conjugation: the complex conjugate of the transpose of a matrix or spinor (for field operators the adjoint operator). Status: STANDARD.
- $\ell_i$, $\bar{\ell}_i$: The lepton doublet of the generation $i$ (left-handed neutrino and charged lepton) and its Dirac adjoint. Unit: $\mathrm{GeV}^{7/2}$. Status: ASSUMED.

On the zero modes it gives, to first order in the tip value of the doublet zero mode,

$$m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \frac{\mathcal{Z}(c_L)^2}{fL}, \qquad \mathcal{Z}(c_L)^2 = 8.04 \times 10^{-5}, \qquad \frac{1}{fL} = 22.1 \text{ TeV},$$

**Symbols in this equation.**

- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\mathcal{K}_{\mathrm{M}}^{(i)}$: The dimensionless Majorana tip coefficients of the alternative tip operator $S_{\mathrm{M}}$: the complex symmetric $3 \times 3$ matrix $(\mathcal{K}_{\mathrm{M}})_{ij}$ (generation indices $i$, $j$; dimensionless because the tip operator is integrated over seven dimensions and $\tilde{\nu}_i \tilde{\nu}_j$ has mass dimension 7); its eigenvalues $\mathcal{K}_{\mathrm{M}}^{(i)}$ give the four-dimensional masses $m_i = \mathcal{K}_{\mathrm{M}}^{(i)}\, \mathcal{Z}(c_L)^2/(fL)$ to first order in the tip value of the doublet zero mode. Unit: dimensionless. Value: $5.63 \times 10^{-13}$, $4.90 \times 10^{-12}$, $2.82 \times 10^{-11}$ for $i = 1, 2, 3$. Status: DERIVED.
- $\mathcal{Z}(c)$: The tip amplitude of a normalised zero mode with the localisation $c$, $\mathcal{Z}(c)^2 = c\,\mathrm{e}^{-c}/\sinh c$. Unit: dimensionless. Status: DERIVED.
- $c$, $c_L$, $c_R$: The localisation of a zero mode, $c = f\mu L$ (bulk mass times the proper depth); $c_L$ of a doublet (localised at the fold for $c_L > 0$), $c_R$ of a singlet (localised at the fold for $c_R < 0$). Unit: dimensionless. Value: doublets $c_L = 3.38230$ (quarks) and 5.95288 (leptons); singlets $c_R$: up $-15.4300$, down $-14.6258$, strange $-11.5158$, charm $-8.7683$, bottom $-7.5005$, top $-3.3823$, electron $-14.5539$, muon $-8.9809$, tau $-5.9529$, neutrinos $-35.0452$, $-32.8481$, $-31.0702$ (fixed by the measured masses). Status: DERIVED.

- $L$: The chirality label L: the left-handed (doublet) zero mode, or the left-handed projection $P_L = (1+\Gamma_4)/2$. Status: STANDARD.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.

so the measured masses need the tip couplings $\mathcal{K}_{\mathrm{M}}^{(i)} = 5.6 \times 10^{-13}$, $4.9 \times 10^{-12}$, $2.8 \times 10^{-11}$: an unexplained hierarchy, whereas the Dirac masses follow from the localisation with couplings of order one; the $x_0$ mechanism favours Dirac neutrinos. Observable differences: neutrinoless double-beta decay, with

$$m_{\beta\beta} = \Big| \sum_{i=1}^{3} U_{ei}^2 \, m_i \, \mathrm{e}^{\mathrm{i}\varphi_i^{\mathrm{M}}} \Big|$$

**Symbols in this equation.**

- $m_{\beta\beta}$: The effective Majorana mass of neutrinoless double-beta decay, $|\sum_i U_{ei}^2 m_i \mathrm{e}^{\mathrm{i}\varphi_i^{\mathrm{M}}}|$. Unit: eV. Value: 0.84 to 4.41 meV over all Majorana phases. Status: DERIVED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $U$: The PMNS matrix, the unitary $3\times3$ mixing matrix of the leptons, $\nu_\alpha = \sum_i U_{\alpha i} \nu_i$ (rows $\alpha = e, \mu, \tau$, columns the mass states $i$); in UM-2 $U = \mathcal{K}_e^\dagger \mathcal{K}_\nu$. Unit: dimensionless. Value: $\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\mathrm{CP}} = 212°$ (NuFIT 6.0). Status: STANDARD.
- $e$: The electron flavour as a value of a flavour index ($\delta_{\alpha e}$, $U_{e2}$, $V_e$, $\mathcal{P}_{ee}$). Status: STANDARD.
- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- i: The imaginary unit, $\mathrm{i}^2 = -1$. Unit: dimensionless. Status: STANDARD.
- $\varphi_i^{\mathrm{M}}$: The Majorana phase of the neutrino mass state $i$ in the Majorana alternative (a free parameter; all values are scanned). Unit: dimensionless (radians). Status: ASSUMED.

between 0.84 and 4.41 meV for all Majorana phases ($|U_{ei}|^2 m_i = 0.677$, 2.624, 1.111 meV; a numerical scan of the phases agrees), below the reach of present experiments; and the capture of the non-relativistic CNB, twice as frequent for Majorana as for Dirac neutrinos (both helicities active). The flavour conversion is the same in both cases (the Majorana phases cancel in the oscillation probabilities).

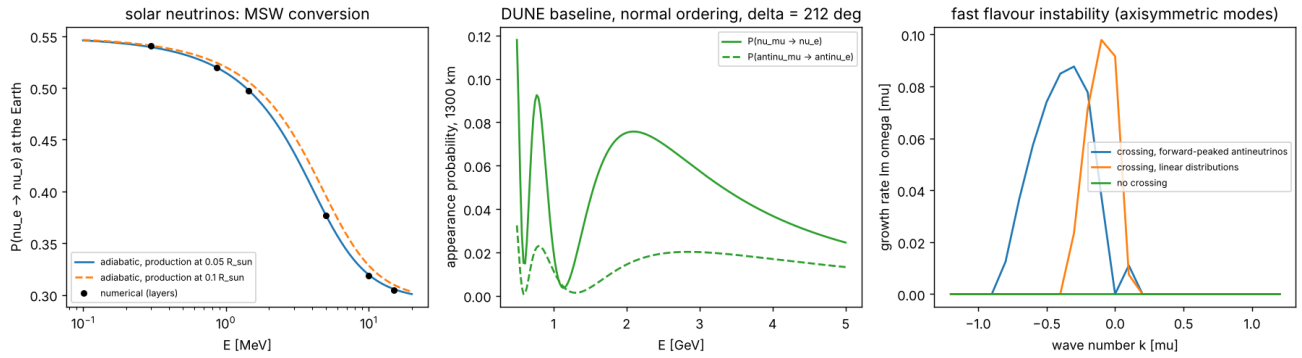



Figure 3: Neutrino flavour conversion in UM-2. Left: the survival probability of solar electron neutrinos at the Earth against the energy (adiabatic formula FE-107 for two production radii, and the numerical three-flavour evolution through the solar density). Middle: the appearance probabilities $\mathcal{P}(\nu_\mu \to \nu_e)$ and $\mathcal{P}(\bar\nu_\mu \to \bar\nu_e)$ at the DUNE baseline of 1300 km (normal ordering, $\delta_{\mathrm{CP}} = 212°$). Right: the growth rate $\mathrm{Im}\,\bar\omega$ of the axisymmetric fast-flavour modes against the wavenumber, in units of $\sqrt{2}\,G_{\mathrm{F}} n_{\nu_e}$ (labelled mu in the figure), for the three angular distributions of the text.

# 17 The dark sector

## 17.1 Dark matter: the candidate (DERIVED)

The $x_0$ spectrum of dirac16complex on the static block (written out again)

$$\begin{aligned}&\text{even: } \chi = \left(\mathrm{e}^{c_\Psi \xi},\ 0\right)^T,\ \varepsilon = 0 \text{ (all eight blocks)}, \qquad \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi_n}{fL}\right)^2},\ \varpi_n = \pi,\ 2\pi,\ \ldots;\\ &\text{odd: } \varepsilon = \pm\sqrt{m_\Psi^2 + \left(\frac{\varpi}{fL}\right)^2}, \quad \tan\varpi = -\frac{\varpi}{c_\Psi}; \qquad c_\Psi = f\, m_\Psi L; \qquad \text{fold band: } \varepsilon = \mathrm{e}^{-a_4}|k|\end{aligned} \tag{FE-62}$$

**Symbols in (FE-62).**

- $\chi$, $\chi_o$, $\chi_1$, $\chi_2$: The two-component spinor of one block of the reduction (a function of $y$ and $x_4$), $\chi = (\chi_1, \chi_2)^T = (a, \mathrm{i}b)^T$; $\chi_o$ the Kohn-Sham orbital $o$, normalised as $\int_{-L}^{0} \chi_o^\dagger \chi_o \,\mathrm{d}y = 1$. Unit: $\mathrm{GeV}^{1/2}$ (normalised orbital). Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $c_\Psi$: The localisation parameter of dirac16complex, $c_\Psi = f\, m_\Psi L$. Unit: dimensionless. Value: 0.646667 at the stabilised block. Status: DERIVED.
- $\xi$: The scaled hidden coordinate, $\xi = y/L$, running from $-1$ (tip) to 0 (fold) on each half of a period. Unit: dimensionless. Status: DERIVED.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $\varepsilon$, $\varepsilon_o$: The energy of a fermion mode (an eigenvalue of the block Hamiltonian; at $k = 0$ the four-dimensional mass); $\varepsilon_o$ the Kohn-Sham level of the orbital $o$. Unit: GeV. Status: DERIVED.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $\varpi$, $\varpi_n$: The wave number of an oscillating $x_0$ profile on $-1 \le \xi \le 0$ (of a boson after the rescaling, or of a bulk level of dirac16complex); $\varpi_n$ that of the $n$-th even level ($\varpi_n = n\pi$ for $\beta = 0$). Unit: dimensionless. Value: 1.89901 for the lowest odd level of dirac16complex. Status: DERIVED.
- $n$: The level number of the Kaluza-Klein tower ($n = 1, 2, 3, \ldots$; $n = 0$ the flat zero mode). Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4} b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926 \times 10^{-21}$ m at the stabilised block). Status: ASSUMED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

has, above the massless fold bands, the lowest odd level: the lightest odd states of $\Psi_A$ and of $\Psi_B$ (the same orbital in all eight blocks), two degenerate Dirac fermions of mass

$$M_{\mathrm{DM}} = \varepsilon_{\mathrm{DM}} = 3.102204\, m_\Psi = 44.35 \text{ TeV} \qquad \left(m_\Psi = H = 14.296 \text{ TeV}\right).$$

**Symbols in this equation.**

- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\varepsilon_{\mathrm{DM}}$: The energy of the dark-matter level at rest (its mass $M_{\mathrm{DM}}$ on the static block). Unit: GeV. Value: $3.102204\, m_\Psi = 44.35$ TeV. Status: DERIVED.

- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.

They are cold (the equation of state of a Kohn-Sham state of quanta at rest is $w_{\mathrm{KS}} = 0$, and $2.8 \times 10^{-5}$ with $\lambda_{\mathrm{test}}$), electrically neutral and $Z$-neutral (UM-2 has no tip operator; a left-handed tip current would not reach $\Psi_B$ at all), and stable as long as the fold parity is exact (UM-2 has no oriented tip operator and the bulk is reflection symmetric) and the numbers of quanta of the two halves are conserved (both hold in the good sector; the three portals commute with $\mathcal{B}$). Their hidden wave number carries 90 % of $\varepsilon^2$ ($\varpi^2/(\varepsilon_{\mathrm{DM}} fL)^2 = 0.896$): their mass is the author's $x_0$ mechanism in the uncertainty sense ($m/m_{\mathrm{Heis}} = 2.29$). Since every portal is diagonal in the halves, each species freezes out separately and must supply half of $\Omega_{\mathrm{DM}} h^2_{100} = 0.120$.

## 17.2 The thermal history and the relic (DERIVED)

**Degrees of freedom.** The particle content of UM-2 at the temperature $T$, with its $x_0$ masses: photon, gluons, $W^+$, $W^-$, $Z$ (no Higgs particle), the six quarks, three charged leptons, three neutrinos; the right-handed neutrinos and the fold bands are never thermalised, the radion and the Kaluza-Klein modes not below 10 TeV. Every species in equilibrium contributes the ideal-gas densities

$$\rho_s = \frac{d_s}{2\pi^2}\int_0^\infty \frac{k^2 E\,\mathrm{d}k}{\mathrm{e}^{E/T}\pm 1}, \qquad p_s = \frac{d_s}{6\pi^2}\int_0^\infty \frac{k^4\,\mathrm{d}k}{E\,(\mathrm{e}^{E/T}\pm 1)}, \qquad s_s = \frac{\rho_s + p_s}{T}, \qquad E = \sqrt{k^2 + m_s^2} \qquad \text{(FE-121)}$$

**Symbols in (FE-121).**

- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $s$: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, $Z$, quarks, leptons, nucleons, dark matter); a sum over $s$ runs over the species present. Status: STANDARD.
- $d_o$, $d_s$: A degeneracy: $d_o = 4r_3(|\vec{n}_o|^2)$ the number of states of the momentum shell of the orbital $o$; $d_s$ the number of internal states (spin, colour, particle and antiparticle) of the species $s$. Unit: dimensionless. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2 + m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $s$, $s_s$: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; $s_s$ that of the species $s$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.

- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.

which give the effective degrees of freedom and the Hubble rate of the hot Universe

$$g_\rho = \frac{30}{\pi^2 T^4}\sum_s \rho_s, \qquad g_s = \frac{45}{2\pi^2 T^3}\sum_s s_s, \qquad \mathcal{H} = \Big(\frac{8\pi^3 g_\rho}{90}\Big)^{1/2}\frac{T^2}{M_{\rm Pl}} \tag{FE-122}$$

**Symbols in (FE-122).**

- $g_\rho$, $g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $s$: The label of a particle species of the plasma or of matter (photon, gluons, $W^\pm$, $Z$, quarks, leptons, nucleons, dark matter); a sum over $s$ runs over the species present. Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: GeV$^8$ (eight dimensions), GeV$^4$ (four dimensions). Status: STANDARD.
- $s$, $s_s$: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; $s_s$ that of the species $s$. Unit: GeV$^3$. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $M_{\rm Pl}$: The Planck mass, $M_{\rm Pl} = G_{\rm N}^{-1/2}$. Unit: GeV. Value: $1.22091\times10^{19}$ GeV. Status: STANDARD.

with the QCD crossover modelled by the weight of the quark-gluon plasma against a gas of the lightest hadrons ($T_{\rm c} = 155$ MeV, $T_{\rm w} = 15$ MeV; ASSUMED model; masses in GeV and internal states: pion 0.1380, 3; kaon 0.4956, 4; eta 0.5479, 1; rho 0.7753, 9; omega 0.7827, 3; $K^*$ 0.8917, 12; nucleon 0.9389, 8; eta' 0.9578, 1; phi 1.0195, 3; Lambda 1.1157, 4; Sigma 1.1932, 12; Delta 1.232, 32; from memory) and neutrinos that decouple at 2 MeV:

$$w_{\rm QGP}(T) = \frac12\Big[1+\tanh\frac{T-T_{\rm c}}{T_{\rm w}}\Big], \qquad \frac{T_\nu}{T} = \Big(\frac{g_{s,e\gamma}(T)}{11/2}\Big)^{1/3}, \qquad \frac{{\rm d}\ln T}{{\rm d}x_4} = -\frac{\mathcal{H}}{1+\frac13\,{\rm d}\ln g_s/{\rm d}\ln T} \tag{FE-123}$$

**Symbols in (FE-123).**

- $w_{\rm QGP}(T)$: The weight of the quark-gluon plasma in the model of the QCD crossover (1 above, 0 below). Unit: dimensionless. Status: ASSUMED.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $T_{\rm c}$, $T_{\rm w}$: The temperature and the width of the QCD crossover. Unit: GeV. Value: 155 MeV and 15 MeV. Status: ASSUMED.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $g_{s,e\gamma}$: The entropy degrees of freedom of the electrons, positrons and photons only (after neutrino decoupling). Unit: dimensionless. Value: 11/2 before and 2 after the annihilation of $e^\pm$. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of $X$, ${\rm d}X/{\rm d}Y$ the derivative of $X$ with respect to $Y$, ${\rm d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $x_4$: The time; dots denote ${\rm d}/{\rm d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $g_\rho$, $g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

$g_\rho = 105.75$ above the top threshold (the Standard Model of three generations without a Higgs particle), 86.2 at 10 GeV, 76.3 at 1 GeV, 10.75 at a few MeV, and today $g_s = 43/11 = 3.9091$, $g_\rho = 3.3626$, $s_0 = 2891.3\ \mathrm{cm}^{-3}$ ($T_\gamma = 2.7255$ K). (Residual analysis: the thermal history decouples the neutrinos instantaneously with $N_{\mathrm{eff}} = 3$, while the CMB computation uses 3.0442, a difference of 0.59 % in $g_\rho$ today that enters only the entropy today; the dark matter, left out of $g_\rho$, carries $1.1 \times 10^{-2}$ of the energy density at the start of the Boltzmann integration, $x = 5$, where it is in equilibrium: harmless.)

**Boltzmann equation.** For one Dirac species

$$\frac{\mathrm{d}n_{\mathrm{DM}}}{\mathrm{d}x_4} + 3\mathcal{H}n_{\mathrm{DM}} = -\langle\sigma v_{\mathrm{rel}}\rangle\big(n_{\mathrm{DM}}\bar{n}_{\mathrm{DM}} - n^2_{\mathrm{DM,eq}}\big) \;\Rightarrow\; \frac{\mathrm{d}Y_{\mathrm{DM}}}{\mathrm{d}x} = -\Big(1 + \frac{1}{3}\frac{\mathrm{d}\ln g_s}{\mathrm{d}\ln T}\Big)\frac{s\,\langle\sigma v_{\mathrm{rel}}\rangle}{2x\mathcal{H}}\big(Y^2_{\mathrm{DM}} - Y^2_{\mathrm{eq}}\big),$$
$$Y_{\mathrm{DM}} = \frac{n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}}{s}, \qquad x = \frac{M_{\mathrm{DM}}}{T}, \qquad s = \frac{2\pi^2}{45}\,g_s T^3, \qquad Y_{\mathrm{eq}} = \frac{4M^2_{\mathrm{DM}}TK_2(x)}{2\pi^2 s} \tag{FE-124}$$

**Symbols in (FE-124).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $n_{\mathrm{DM}}$, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\sigma$: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $Y_{\mathrm{DM}}$, $Y_{\mathrm{eq}}$, $Y_\infty$: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- $x$: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.
- $g_\rho$, $g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $s$, $s_s$: The entropy density of the plasma, $s = 2\pi^2 g_s T^3/45$; $s_s$ that of the species $s$. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $K_1$, $K_2$: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable $K$). Unit: dimensionless. Status: STANDARD.

with the exact relativistic thermal average (Gondolo and Gelmini)

$$\langle\sigma v_{\mathrm{rel}}\rangle = \frac{1}{8M^4_{\mathrm{DM}}TK_2(x)^2}\int_{4M^2_{\mathrm{DM}}}^{\infty}\sigma\left(E^2_{\mathrm{cm}} - 4M^2_{\mathrm{DM}}\right)E_{\mathrm{cm}}\,K_1\Big(\frac{E_{\mathrm{cm}}}{T}\Big)\,\mathrm{d}\big(E^2_{\mathrm{cm}}\big) \tag{FE-125}$$

**Symbols in (FE-125).**

- $\sigma$: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.

- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $K_1,\ K_2$: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable $K$). Unit: dimensionless. Status: STANDARD.
- $x$: The ratio $x = M_{\rm DM}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\rm f} = 31.23$ at the freeze-out. Status: STANDARD.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- $E_{\rm cm}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.

and the relic density

$$\Omega_{\rm DM}h^2_{100} = \frac{M_{\rm DM}\,Y_\infty\,s_0}{\rho_{\rm crit}/h^2_{100}}, \qquad \frac{\rho_{\rm crit}}{h^2_{100}} = 1.05368\times10^{-5}\ \mathrm{GeV/cm^3} \tag{FE-126}$$

**Symbols in (FE-126).**

- $\Omega_{\rm DM}$: The dark-matter density today in units of the critical density. Unit: dimensionless. Value: $\Omega_{\rm DM}h^2_{100} = 0.120$ (both species). Status: DERIVED.
- $h_{100}$: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.
- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $Y_{\rm DM},\ Y_{\rm eq},\ Y_\infty$: The abundance (number density per entropy density) of one dark-matter species (particles and antiparticles), its equilibrium value and its final value. Unit: dimensionless. Status: STANDARD.
- $s_0$: The entropy density today. Unit: $\mathrm{GeV}^3$ (quoted per $\mathrm{cm}^3$). Value: 2891.3 $\mathrm{cm}^{-3}$. Status: STANDARD.
- $\rho_{\rm crit}$: The critical density, $3\mathcal{H}_0^2M_{\rm Pl}^2/(8\pi)$. Unit: $\mathrm{GeV}^4$ (quoted in $\mathrm{GeV/cm^3}$). Value: $\rho_{\rm crit}/h^2_{100} = 1.05368\times10^{-5}$ $\mathrm{GeV/cm^3}$. Status: STANDARD.

Checks: for an s-wave cross-section $4.4\times10^{-26}$ $\mathrm{cm^3/s}$ it gives $\Omega_{\rm DM}h^2_{100} = 0.1275$, against 0.1306 from the freeze-out formula (STANDARD, Kolb and Turner, for a Dirac fermion with the constants $c = 1/2$ and two internal states inserted, $\langle\sigma v_{\rm rel}\rangle = a_{\rm sw} + 6b_{\rm pw}/x$, $M_{\rm Pl} = 1.22091\times10^{19}$ GeV)

$$x_{\rm f} = \ln\Big[\frac{0.095\,M_{\rm Pl}M_{\rm DM}\langle\sigma v_{\rm rel}\rangle}{\sqrt{g_\rho x_{\rm f}}}\Big], \qquad \Omega_{\rm DM}h^2_{100} = \frac{2\times1.07\times10^9\ \mathrm{GeV^{-1}}\,x_{\rm f}}{\sqrt{g_\rho}\,M_{\rm Pl}\left(a_{\rm sw}+3b_{\rm pw}/x_{\rm f}\right)},$$

**Symbols in this equation.**

- $x_{\rm f}$: The freeze-out point $x_{\rm f} = M_{\rm DM}/T$ at which the dark matter leaves chemical equilibrium. Unit: dimensionless. Value: 30.38 (freeze-out formula, I12), 31.23 (Boltzmann equation, I12 + I13). Status: DERIVED.
- $M_{\rm Pl}$: The Planck mass, $M_{\rm Pl} = G_{\rm N}^{-1/2}$. Unit: GeV. Value: $1.22091\times10^{19}$ GeV. Status: STANDARD.
- $M_{\rm DM}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\sigma$: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $v_{\rm rel}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\rm rel}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $g_\rho,\ g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $\Omega_{\rm DM}$: The dark-matter density today in units of the critical density. Unit: dimensionless. Value: $\Omega_{\rm DM}h^2_{100} = 0.120$ (both species). Status: DERIVED.
- $h_{100}$: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.

- $a_{\mathrm{sw}}$, $b_{\mathrm{pw}}$: The s-wave and p-wave coefficients of the non-relativistic expansion of the thermally averaged annihilation cross-section, $\langle\sigma v_{\mathrm{rel}}\rangle = a_{\mathrm{sw}} + 6b_{\mathrm{pw}}/x$. Unit: $\mathrm{GeV}^{-2}$ (quoted in $\mathrm{cm}^3/\mathrm{s}$). Value: for I12 alone at the relic strength $a_{\mathrm{sw}} = 0$, $b_{\mathrm{pw}} = 8.741 \times 10^{-8}\ \mathrm{GeV}^{-2}$. Status: STANDARD.

and the late-time integral is reproduced to $1.3 \times 10^{-8}$. The cross-sections of the two portals that annihilate the dark matter and their partial-wave unitarity bounds are (written out again)

$$\begin{aligned}
\sigma_{\mathrm{I12}} &= \frac{G_{\mathrm{I12}}^2\, E_{\mathrm{cm}}^3 \left(E_{\mathrm{cm}}^2 - 4M_{\mathrm{DM}}^2\right)^{1/2}}{8\pi}, \qquad G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2, \\
\sigma_{\mathrm{I13}} &= \sum_f \frac{N_{\mathrm{c}}\, G_f^2\, E_{\mathrm{cm}}^2 \left(1 + 2M_{\mathrm{DM}}^2/E_{\mathrm{cm}}^2\right)}{24\pi\beta_{\mathrm{cm}}}, \qquad G_f = G_{\mathrm{A}}\, q_f^B\, \tilde{I}_{of}^{(b)}, \\
\beta_{\mathrm{cm}} &= \sqrt{1 - \frac{4M_{\mathrm{DM}}^2}{E_{\mathrm{cm}}^2}}, \qquad k_{\mathrm{cm}} = \sqrt{\frac{E_{\mathrm{cm}}^2}{4} - M_{\mathrm{DM}}^2}
\end{aligned} \tag{FE-127}$$

**Symbols in (FE-127).**

- $\sigma_{\mathrm{I12}}$, $\sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $G_{\mathrm{I12}}$: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: $\mathrm{GeV}^{-3}$. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}\ \mathrm{GeV}^{-3}$. Status: DERIVED.
- $E_{\mathrm{cm}}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\mathrm{g}}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $6.09 \times 10^{-15}\ \mathrm{GeV}^{-3}$ (benchmark). Status: DERIVED.
- $G_{\mathrm{W}}$: The coupling of a dark-matter pair to canonically normalised SU(2) gauge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $2.96 \times 10^{-15}\ \mathrm{GeV}^{-3}$. Status: DERIVED.
- $G_{\mathrm{B}}$: The coupling of a dark-matter pair to canonically normalised hypercharge pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $1.08 \times 10^{-15}\ \mathrm{GeV}^{-3}$. Status: DERIVED.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\mathrm{c}}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- $G_f$: The I13 coupling of the dark matter to the quark component $f$, $G_f = G_{\mathrm{A}} q_f^B \tilde{I}_{of}^{(b)}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\beta_{\mathrm{cm}}$: The velocity of each particle of the pair in the centre-of-mass frame. Unit: dimensionless. Status: STANDARD.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278 \times 10^{-9}\ \mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $q_\psi^B$, $q_f^B$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}_{of}^{(b)} = H^{-1} \int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy} \mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $k_{\mathrm{cm}}$: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

$$\sigma_{\mathrm{I12}} \le \frac{\pi}{4k_{\mathrm{cm}}^2}\ \text{(total angular momentum 0)}, \qquad \sigma_{\mathrm{I13}} \le \frac{3\pi}{4k_{\mathrm{cm}}^2}\ \text{(total angular momentum 1)} \tag{FE-128}$$

**Symbols in (FE-128).**

- $\sigma_{\mathrm{I12}}$, $\sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $k_{\mathrm{cm}}$: The momentum of each particle of the pair in the centre-of-mass frame. Unit: GeV. Status: STANDARD.

for I12 per abelian channel $\sigma_{\mathrm{I12}} v_{\mathrm{rel}} \to G_{\mathrm{I12}}^2 M_{\mathrm{DM}}^4 v_{\mathrm{rel}}^2/\pi$ (p-wave, from ${}^3P_0$ only; its thermal average exceeds the non-relativistic $6b_{\mathrm{pw}}/x$ by about $1+3/x$), and for I13 (s-wave, from ${}^3S_1$, the quark masses neglected)

$$\sigma_{\mathrm{I13}} v_{\mathrm{rel}} \to \sum_f \frac{N_{\mathrm{c}}\, G_f^2 M_{\mathrm{DM}}^2}{2\pi} \qquad \big(v_{\mathrm{rel}} \to 0\big),$$

**Symbols in this equation.**

- $\sigma_{\mathrm{I12}}$, $\sigma_{\mathrm{I13}}$: The spin-averaged annihilation cross-sections of a dark-matter pair through I12 (into gauge-boson pairs) and through I13 (into quark pairs). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless ($\mathrm{c} = 1$). Status: STANDARD.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\mathrm{c}}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- $G_f$: The I13 coupling of the dark matter to the quark component $f$, $G_f = G_{\mathrm{A}} q_f^B \tilde{I}_{of}^{(b)}$. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

with $\sum_f N_{\mathrm{c}} (q_f^B \tilde{I}_{of}^{(b)})^2 = 16.35$ over the quark Weyl components of three generations for the dark-matter orbital.

**I12 alone** (the result of version 2, recomputed). The effective operator would need $G_{\mathrm{I12}} = (15.86\ \mathrm{TeV})^{-3}$ for $\Omega_{\mathrm{DM}} h_{100}^2 = 0.060$ per species ($x_{\mathrm{f}} = 31.7$, $\langle \sigma v_{\mathrm{rel}} \rangle(x_{\mathrm{f}}) = 1.89 \times 10^{-25}\ \mathrm{cm}^3/\mathrm{s}$). It annihilates only from ${}^3P_0$ (total angular momentum 0), and the bound FE-128 gives, for non-relativistic pairs,

$$\sigma v_{\mathrm{rel}} \le \frac{\pi}{M_{\mathrm{DM}}^2 v_{\mathrm{rel}}}, \qquad \left\langle \frac{\pi}{M_{\mathrm{DM}}^2 v_{\mathrm{rel}}} \right\rangle = \frac{\sqrt{\pi x}}{M_{\mathrm{DM}}^2}.$$

**Symbols in this equation.**

- $\sigma$: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle \sigma v_{\mathrm{rel}} \rangle$ the thermal average. Unit: dimensionless ($\mathrm{c} = 1$). Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $x$: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.

In version 2 the freeze-out formula (with $a_{\mathrm{sw}} = 0$, $g_\rho = 105.75$) gave $G_{\mathrm{I12}} = (15.5\ \mathrm{TeV})^{-3}$ at $x_{\mathrm{f}} = 30.4$ and a requirement of $2.01 \times 10^{-25}\ \mathrm{cm}^3/\mathrm{s}$ against the bound $5.8 \times 10^{-26}\ \mathrm{cm}^3/\mathrm{s}$, 3.5 times. With the numerical solution the effective cross-section crosses the bound at $E_{\mathrm{cm}} = 89\,510$ GeV, i.e. at the relative velocity $v_{\mathrm{rel}} = 2\beta_{\mathrm{cm}} = 0.269$ (0.274 from the non-relativistic formula)

$$v_{\mathrm{c}} = \left(\frac{\pi}{G_{\mathrm{I12}} M_{\mathrm{DM}}^3}\right)^{2/3};$$

**Symbols in this equation.**

- $v_{\mathrm{c}}$: The relative velocity above which the effective I12 cross-section at the relic strength exceeds the ${}^3P_0$ unitarity bound (non-relativistic formula). Unit: dimensionless (units of c). Value: 0.263. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\mathrm{I12}}$: The combined coupling of I12 to all gauge-boson pairs, $G_{\mathrm{I12}}^2 = 8G_{\mathrm{g}}^2 + 3G_{\mathrm{W}}^2 + G_{\mathrm{B}}^2$. Unit: $\mathrm{GeV}^{-3}$. Value: benchmark $\pi/(2M_{\mathrm{DM}}^3) = 1.801 \times 10^{-14}$ $\mathrm{GeV}^{-3}$. Status: DERIVED.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.

at freeze-out 96 % of the annihilation rate comes from collisions above the crossing, and 75 % of the pairs of the non-relativistic Maxwell distribution are faster. A cross-section at the ${}^3P_0$ limit at every energy leaves $\Omega_{\mathrm{DM}}h_{100}^2 = 0.058$ per species (at its freeze-out, $x_{\mathrm{f}} = 30.8$, $\langle\sigma v_{\mathrm{rel}}\rangle = 5.3 \times 10^{-26}$ $\mathrm{cm}^3/\mathrm{s}$ against $\sqrt{\pi x}/M_{\mathrm{DM}}^2 = 5.8 \times 10^{-26}$ $\mathrm{cm}^3/\mathrm{s}$ at the same $x$), and the effective operator capped at the limit leaves 0.154, 0.108, 0.084, 0.065, 0.059 for $G_{\mathrm{I12}}^2$ equal to 1, 3, 10, 100 and $10^4$ times the relic value: within perturbative couplings I12 overproduces the dark matter, and only a strongly coupled ${}^3P_0$ channel at the unitarity limit could give the relic. The benchmark $G_{\mathrm{I12}} = \pi/(2M_{\mathrm{DM}}^3)$ alone leaves $\Omega_{\mathrm{DM}}h_{100}^2 = 8.0$ per species.

**With I13.** I12 at its benchmark and I13 together give each species $\Omega_{\mathrm{DM}}h_{100}^2 = 0.060$ for $G_{\mathrm{A}} = 1.278 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ ($x_{\mathrm{f}} = 31.23$; $\langle\sigma v_{\mathrm{rel}}\rangle(x_{\mathrm{f}}) = 9.5 \times 10^{-26}$ $\mathrm{cm}^3/\mathrm{s}$ from I13, $9.9 \times 10^{-28}$ $\mathrm{cm}^3/\mathrm{s}$ from I12), with the I13 cross-section capped at the unitarity limit of total angular momentum 1, which it reaches only above $v_{\mathrm{rel}} = 0.51$ (without the cap $G_{\mathrm{A}} = 1.269 \times 10^{-9}$ $\mathrm{GeV}^{-2}$, and the uncapped cross-section would cross the bound above $v_{\mathrm{rel}} = 0.52$, 25 % of the rate). With the cap, 22 % of the freeze-out rate comes from collisions at the bound and 78 % from collisions below it, so the relic of UM-2 is a thermal freeze-out with most annihilations perturbative; the contact form of I13 is an effective description (for the quark doublet $(G_{\mathrm{A}} q_Q^B \tilde{I}_Q^{(b)})^{-1/2} = 30$ TeV, below the energies $E_{\mathrm{cm}} \approx 89$ TeV of the annihilations), whose completion by a mediator is OPEN. It assumes that the radion oscillation left over from the deflation does not dominate and dilute the Universe after the freeze-out (the moduli problem, the last section). The residual analysis adds two neglected effects of definite size: the Hubble rate of the thermal history leaves out the fold bands of the freeze-in ($3.5 \times 10^{-4}$ of the energy density at the freeze-out; the four-dimensional Friedmann equation with all species has a Hubble rate larger by $1.7 \times 10^{-4}$, and the relic moves by about as much), and the ideal-gas thermal history leaves out the order-$\alpha$ interaction pressure of the plasma (−5.4 % at the freeze-out: QCD −4.7 %, $SU(2)$ −0.7 %), which would raise $\Omega_{\mathrm{DM}}h_{100}^2$ by about 2.7 % and $G_{\mathrm{A}}$ by about 1.4 % (ESTIMATE): the physics accuracy of $G_{\mathrm{A}}$ is about 1 %, not the $10^{-6}$ of its root finding.

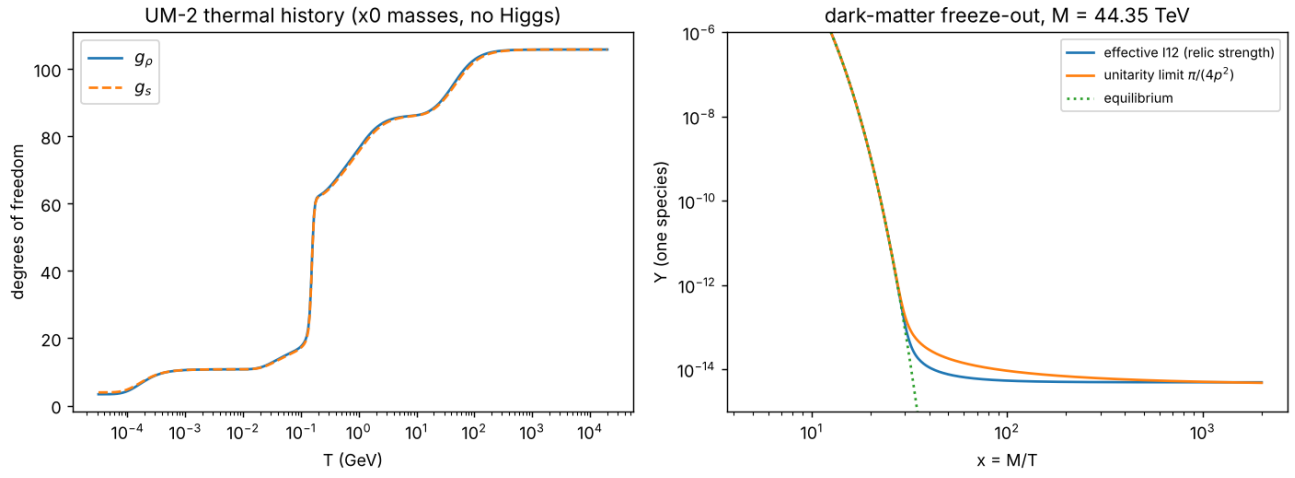



Figure 4: The thermal history of UM-2 and the dark-matter relic.

**The fold bands (freeze-in through I11).** The plasma produces fold-band pairs (Maxwell-Boltzmann, massless;

the sum runs over the fifteen Standard-Model multiplets of three generations in the plasma, $\sum_f N_{\mathrm{c}}(q_f\tilde{I}_{of})^2 = 126.85$ for the fold band; the right-handed neutrinos are not in the plasma):

$$\Gamma_{\mathrm{fi}} = \frac{T}{32\pi^4}\int_0^\infty \mathrm{d}\big(E_{\mathrm{cm}}^2\big)\Big(\sum_{\text{spins}}\sigma\Big)E_{\mathrm{cm}}^3 K_1\Big(\frac{E_{\mathrm{cm}}}{T}\Big) = \frac{4}{\pi^5}\,N_{\mathrm{fold}}\,G_{\mathrm{V}}^2\,T^8\sum_f N_{\mathrm{c}}\big(q_f\tilde{I}_{of}\big)^2, \qquad \frac{\mathrm{d}\big(\rho_{\mathrm{fold}}\,\mathrm{e}^{4a_4}\big)}{\mathrm{d}x_4} = 8T\,\Gamma_{\mathrm{fi}}\,\mathrm{e}^{4a_4} \tag{FE-129}$$

**Symbols in (FE-129).**

- $\Gamma_{\mathrm{fi}}$: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $E_{\mathrm{cm}}$: The total energy of the colliding pair in its centre-of-mass frame. Unit: GeV. Status: STANDARD.
- $\sigma$: A cross-section (of the dark-matter annihilation or production process named in the equation; summed over the spins where marked). Unit: $\mathrm{GeV}^{-2}$. Status: STANDARD.
- $K_1$, $K_2$: The modified Bessel functions of the second kind of orders 1 and 2 (not related to the block variable $K$). Unit: dimensionless. Status: STANDARD.
- $N_{\mathrm{fold}}$: The number of fold bands: the massless fold zero modes of $\Psi_A$ and of $\Psi_B$ (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\mathrm{c}}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\mathrm{c}}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^0 (a_o^2+b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $\rho_{\mathrm{fold}}$: The energy density of the fold bands. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

integrated from the highest temperature $T_{\max} = 10$ TeV of the hot Universe (ASSUMED); today

$$\Delta N_{\mathrm{eff}} = \frac{\rho_{\mathrm{fold}}}{\rho_\nu}, \qquad \rho_\nu = \frac{7}{8}\,\frac{\pi^2}{15}\,T_\nu^4, \qquad \frac{2\Gamma_{\mathrm{fi}}}{n_{\mathrm{fold,eq}}\,\mathcal{H}} \ll 1, \qquad n_{\mathrm{fold,eq}} = N_{\mathrm{fold}}\cdot 4\cdot\frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\,T^3 \tag{FE-130}$$

**Symbols in (FE-130).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m^2_{21} = m^2_2 - m^2_1$ a difference of squared masses, $\Delta N_{\rm eff}$ an addition to $N_{\rm eff}$, $\Delta S_{\rm m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $N_{\rm eff}$: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); $\Delta N_{\rm eff}$ an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $\rho_{\rm fold}$: The energy density of the fold bands. Unit: GeV$^4$. Status: DERIVED.
- $\rho_\nu$, $\rho_{\nu_\alpha}$: The energy density of one neutrino flavour (neutrinos and antineutrinos) at the common temperature $T_\nu$, $\frac{7}{8}\frac{\pi^2}{15}T_\nu^4$ when massless; $\rho_{\nu_\alpha}$ that of the flavour $\alpha$ at its temperature $T_{\nu_\alpha}$. Unit: GeV$^4$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $\Gamma_{\rm fi}$: The rate per volume of the production of fold-band pairs from the plasma through I11 (freeze-in). Unit: GeV$^4$. Status: DERIVED.
- $n_{\rm fold,eq}$: The number density the fold bands would have in thermal equilibrium. Unit: GeV$^3$. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.
- $N_{\rm fold}$: The number of fold bands: the massless fold zero modes of $\Psi_A$ and of $\Psi_B$ (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- $\zeta$: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers $n$. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

which gives $\Delta N_{\rm eff} = 8.0\times10^{-6}$, $1.25\times10^{-4}$, $1.0\times10^{-3}$, $8.0\times10^{-3}$ for $T_{\rm max} = 2$, 5, 10, 20 TeV at the benchmark $G_{\rm V} = 4.38\times10^{-17}$ GeV$^{-2}$; their production rate per quantum is 0.012 of the Hubble rate at $T_{\rm max} = 10$ TeV, so they never reach equilibrium. I12 and I13 do not act on the fold bands.

**The right-handed neutrinos.** If I13 coupled to the $B-L$ current instead of $J^B$, the relic would need a different strength and the right-handed neutrinos would be produced in pairs:

$$
\begin{aligned}
&G_{\rm A}^{B-L} = G_{\rm A}\Big[\frac{\mathcal{S}_B}{\mathcal{S}_{B-L}}\Big]^{1/2}, \qquad \mathcal{S}_B = \sum_f N_{\rm c}\big(q_f^B \tilde I_{of}^{(b)}\big)^2, \qquad \mathcal{S}_{B-L} = \sum_f N_{\rm c}\big(q_f \tilde I_{of}^{(b)}\big)^2,\\
&\frac{{\rm d}n_{\nu_R}}{{\rm d}x_4} + 3\mathcal{H}n_{\nu_R} = \Gamma_\nu\Big(1 - \frac{n_{\nu_R}^2}{(n_{\nu_R}^{\rm eq})^2}\Big), \qquad \Gamma_\nu = 2\,n_{\rm DM,eq}\,\bar n_{\rm DM,eq}\,\frac{G_\nu^2 M_{\rm DM}^2}{2\pi}, \qquad G_\nu = G_{\rm A}^{B-L}\,\tilde I_{\nu_R}^{(b)}, \qquad \text{(FE-131)}\\
&n_{\nu_R}^{\rm eq} = \frac{3}{4}\,\frac{\zeta(3)}{\pi^2}\,T^3, \qquad n_{\rm DM,eq} = \bar n_{\rm DM,eq} = 2\Big(\frac{M_{\rm DM}T}{2\pi}\Big)^{3/2}{\rm e}^{-x}
\end{aligned}
$$

**Symbols in (FE-131).**

- $G_{\rm A}^{B-L}$: The strength I13 would need if it coupled to the $B-L$ current (the same relic density). Unit: GeV$^{-2}$. Status: DERIVED.
- $G_{\rm A}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\rm A} = c_{\rm A}H/(2f^4 v_{\rm t}\Lambda_\star^6)$. Unit: GeV$^{-2}$. Value: $1.278\times10^{-9}$ GeV$^{-2}$ (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B-L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.
- $f$: The label of a fermion species: a Standard-Model multiplet of one generation (its $N_{\rm c}$ Weyl components share one zero-mode profile; the zero mode of each chirality) or a fermion of a medium (electron, proton, neutron); a sum over $f$ runs over these species. Status: STANDARD.
- $N_{\rm c}$: The number of Weyl components of a fermion multiplet that share its zero-mode profile and overlap (colour times isospin): 6 for a quark doublet, 3 for a quark singlet, 2 for a lepton doublet, 1 for a charged-lepton singlet. Unit: dimensionless. Status: STANDARD.

- $q^B_\psi$, $q^B_f$: The baryon number of the multiplet $\psi$ or of the fermion component $f$ ($\frac{1}{3}$ for quarks, 0 for leptons). Unit: dimensionless. Status: STANDARD.
- $\tilde{I}^{(b)}_{of}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}^{(b)}_{of} = H^{-1}\int_{-L}^{0} b_o^2 \varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}^{(b)}_{Q}$, $\tilde{I}^{(b)}_{u_R}$, $\tilde{I}^{(b)}_{d_R}$, $\tilde{I}^{(b)}_{\nu_R}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}^{(b)}_{Q} = 2.6135$. Status: DERIVED.
- $o$: The label of a Kohn-Sham orbital (a state of dirac16complex in the cell: block, momentum shell, fold parity and level); a sum over $o$ runs over the orbitals. Status: RECORD.
- $q_\psi$, $q_f$: The $B-L$ charge: $q_\psi$ of the multiplet $\psi$ ($q_Q = q_{u_R} = q_{d_R} = \frac{1}{3}$, $q_\ell = q_{e_R} = q_{\nu_R} = -1$), $q_f$ of the fermion component $f$. Unit: dimensionless. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $n_{\nu_R}$, $n^{\mathrm{eq}}_{\nu_R}$: The number density of the right-handed neutrinos (three generations) and its equilibrium value. Unit: $\mathrm{GeV}^3$. Status: DERIVED.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $\Gamma_\nu$: The production rate per volume of right-handed neutrinos in the counterfactual $B-L$ variant of I13. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $G_\nu$: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\zeta$: The Riemann zeta function; only its value $\zeta(3)$ occurs, the sum of $1/n^3$ over the positive integers $n$. Unit: dimensionless. Value: $\zeta(3) = 1.2020569$. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $x$: The ratio $x = M_{\mathrm{DM}}/T$ of the dark-matter mass to the temperature. Unit: dimensionless. Value: $x_{\mathrm{f}} = 31.23$ at the freeze-out. Status: STANDARD.

while elastic scattering on the dark matter keeps them at the plasma temperature as long as (ESTIMATE)

$$\Gamma_{\mathrm{el}} = 2\big(n_{\mathrm{DM}} + \bar{n}_{\mathrm{DM}}\big)\,\sigma_{\mathrm{el}} > \mathcal{H}, \qquad \sigma_{\mathrm{el}} = \frac{G_\nu^2 E^2}{2\pi}, \qquad E \simeq 3T \tag{FE-132}$$

**Symbols in (FE-132).**

- $\Gamma_{\mathrm{el}}$, $\sigma_{\mathrm{el}}$: The rate of elastic scattering of a right-handed neutrino on the dark matter, and its cross-section. Unit: GeV; $\mathrm{GeV}^{-2}$. Status: ESTIMATE.
- $n_{\mathrm{DM}}$, $\bar{n}_{\mathrm{DM}}$: The number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.
- $G_\nu$: The four-dimensional coupling of the dark matter to a right-handed neutrino in the $B-L$ variant of I13. Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $E$: The energy of a particle: of the neutrino in the flavour equations, $E = (k^2+m^2)^{1/2}$ of a particle of momentum $k$ in the thermal integrals. Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $T$: The temperature of the plasma (of the photons and of every species in equilibrium with them); in the Mermin occupations the temperature of the Kohn-Sham state. Unit: GeV. Status: STANDARD.

From zero abundance they reach 0.9997 of the equilibrium number by $M_{\mathrm{DM}}/T = 40$; the number-changing rate falls below the Hubble rate at $M_{\mathrm{DM}}/T = 16.1$, the elastic rate at $M_{\mathrm{DM}}/T = 24.4$ ($T = 1.82$ TeV); they would decouple as three Weyl species, a species that decouples early adding (written out again)

$$\Delta N_{\mathrm{eff}} = N_{\mathrm{W}}\Big(\frac{10.75}{g_s}\Big)^{4/3} F_{\mathrm{h}}, \qquad F_{\mathrm{h}} = \Big[\frac{T_\nu}{T_\gamma}\Big(\frac{11}{4}\Big)^{1/3}\Big/\big(1+h_\nu\big)\Big]^4 = 1.0095 \tag{FE-133}$$

**Symbols in (FE-133).**

- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_{\mathrm{eff}}$ an addition to $N_{\mathrm{eff}}$, $\Delta S_{\mathrm{m}}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $N_{\mathrm{eff}}$: The effective number of neutrino species (the radiation beyond the photons in units of one neutrino species with $T_\nu/T_\gamma = (4/11)^{1/3}$); $\Delta N_{\mathrm{eff}}$ an addition to it. Unit: dimensionless. Value: 3.0432 (neutrinos) + 0.0010 (fold bands). Status: STANDARD.
- $N_{\mathrm{W}}$: The number of Weyl species that decouple early. Unit: dimensionless. Status: DERIVED.
- $g_\rho$, $g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.
- $F_{\mathrm{h}}$: The correction factor of the early-decoupled radiation for the heating of the neutrinos. Unit: dimensionless. Value: 1.0095. Status: DERIVED.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $h_\nu$: The relative heating of the neutrinos by the electron-positron annihilation (beyond instantaneous decoupling). Unit: dimensionless. Value: 0.00165. Status: DERIVED.

($N_{\mathrm{W}} = 3$, $g_s = 105.7$), i.e. $\Delta N_{\mathrm{eff}} = 0.144$, $N_{\mathrm{eff}} = 3.19$, 2.5 standard deviations above the ACT DR6 value $2.86 \pm 0.13$. With $J^B$ the right-handed neutrinos keep only their tip coupling and their I11 coupling ($q_{\nu_R} = -1$): the dark matter annihilates into right-handed neutrino pairs through I11 alone, at the fraction

$$r_{\nu_R} = \frac{G_{\mathrm{V}}^2 \sum_{i=1}^{3} \tilde{I}_{\nu_R,i}^2}{G_{\mathrm{A}}^2\, \mathcal{S}_B} \tag{FE-134}$$

**Symbols in (FE-134).**

- $r_{\nu_R}$: The ratio of the I11 production of right-handed neutrinos to the I13 annihilation rate of the dark matter. Unit: dimensionless. Value: $2.0 \times 10^{-17}$. Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38 \times 10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0} (a_o^2 + b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2 f^4 v_{\mathrm{t}} \Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $\mathcal{S}_B$, $\mathcal{S}_{B-L}$: The colour-weighted sums of the squared I13 overlaps of the dark matter with the baryon-number and with the $B - L$ charges. Unit: dimensionless. Value: $\mathcal{S}_B = 16.351$, $\mathcal{S}_{B-L} = 22.92$. Status: DERIVED.

$r_{\nu_R} = 2.0 \times 10^{-17}$ of the I13 rate into quarks, and they are never thermalised (the tip mixing rate is suppressed by $(m_i/T)^2$, STANDARD).

**The self-interaction channel.** The contact self-interaction $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ contains a term that couples a dark-matter pair to a fold-band pair through the transition scalar density, which is odd under the fold reflection while its square is even, so the dark matter can annihilate into the fold bands:

$$\begin{aligned}
&\mathcal{L}_4^\lambda = -\lambda\left(\bar\Psi_{\mathrm{DM}}\Psi_{\mathrm{fold}}\right)\left(\bar\Psi_{\mathrm{fold}}\Psi_{\mathrm{DM}}\right)\left(n_{\mathrm{DF}}^{\mathrm{sc}}\right)^2, \qquad n_{\mathrm{DF}}^{\mathrm{sc}} = a_{\mathrm{DM}}b_{\mathrm{fold}} + b_{\mathrm{DM}}a_{\mathrm{fold}}, \qquad b_{\mathrm{fold}} = 0,\\
&G_\lambda = \frac{\lambda H \tilde I_\lambda}{2f^4 v_{\mathrm{t}}}, \qquad \tilde I_\lambda = \frac{1}{H}\int_{-L}^{0}\left(n_{\mathrm{DF}}^{\mathrm{sc}}\right)^2 \mathrm{e}^{-3Hy}\,\mathrm{d}y, \qquad \sigma_\lambda v_{\mathrm{rel}} = \frac{G_\lambda^2 M_{\mathrm{DM}}^2}{2\pi} \quad \text{(per final fold species)}
\end{aligned} \tag{FE-135}$$

**Symbols in (FE-135).**

- $\mathcal{L}_4^\lambda$: The four-dimensional term of the contact self-interaction that couples a dark-matter pair to a fold-band pair. Unit: $\mathrm{GeV}^4$. Status: ESTIMATE.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7\times10^{-5}H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4\times10^{-10}H^{-6}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: RECORD.
- $\Psi_{\mathrm{DM}}$, $\bar\Psi_{\mathrm{DM}}$: The four-dimensional field of a dark-matter quantum (the lowest odd level of dirac16complex on the static block, in either half) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- $\Psi_{\mathrm{fold}}$, $\bar\Psi_{\mathrm{fold}}$: The four-dimensional field of a fold-band quantum (an even zero mode of dirac16complex, massless) and its Dirac adjoint. Unit: $\mathrm{GeV}^{3/2}$. Status: DERIVED.
- $n_{\mathrm{DF}}^{\mathrm{sc}}$: The transition scalar density between the dark-matter orbital and a fold-band orbital. Unit: GeV (normalised orbitals). Status: DERIVED.
- $a_{\mathrm{DM}}$, $b_{\mathrm{DM}}$: The upper and lower components of the dark-matter orbital (block spinor $(a, \mathrm{i}b)$, normalised on one half). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- $a_{\mathrm{fold}}$, $b_{\mathrm{fold}}$: The upper and lower components of a fold-band orbital ($b_{\mathrm{fold}} = 0$ for the even zero mode). Unit: $\mathrm{GeV}^{1/2}$. Status: DERIVED.
- $G_\lambda$: The four-dimensional strength of the self-interaction channel of the dark matter into fold-band pairs. Unit: $\mathrm{GeV}^{-2}$. Value: $\le 2.3\times10^{-15}$ $\mathrm{GeV}^{-2}$ (for $\Delta N_{\mathrm{eff}} \le 0.001$). Status: ESTIMATE.
- $H$: The author's constant (an inverse length) that fixes the period $\pi/(6H)$ of $x_0$, the Lovelock couplings, the cosmological constant $\Lambda = -10H^2$ and the mass parameter $m_\Psi = H$. Unit: GeV (1/length). Value: 14.296 TeV (from the fit of $m_{\mathrm{W}}$ and $\rho_{\mathrm{EW},0}$). Status: AUTHOR.
- $\tilde I_\lambda$: The overlap of the squared transition scalar density, $\tilde I_\lambda = H^{-1}\int_{-L}^{0}(n_{\mathrm{DF}}^{\mathrm{sc}})^2\mathrm{e}^{-3Hy}\mathrm{d}y$. Unit: dimensionless. Value: 14.38. Status: DERIVED.
- $f$: The block scale: the common scale factor of the hidden direction and the extra times, $f = \mathrm{e}^{-a_4}b_4^{1/2}$ (the block is $f^2$ times a four-dimensional space of constant curvature $-H^2$). Unit: dimensionless. Status: DERIVED.
- $v_{\mathrm{t}}$: The coordinate volume of the compact extra times $x_5, x_6, x_7$. Unit: $\mathrm{GeV}^{-3}$. Value: not fixed; $v_{\mathrm{t}} = H^{-3}$ in the quoted conversions. Status: ASSUMED.
- $L$: The depth of the regular tip: each half of a period of $x_0$ ends at $y = -L$; $HL = 3$. Unit: length. Value: $L = 3/H$ ($fL = 8.926\times10^{-21}$ m at the stabilised block). Status: ASSUMED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $y$: The hidden coordinate of the $y$ chart, $y = \ln|\sin z|/(6H)$, with $y = 0$ at the fold ($z = \pi/2$) and $y = -L$ at the regular tip; each half of a period of $x_0$ is the interval $-L \le y \le 0$. Unit: length. Status: DERIVED.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $\sigma_\lambda$: The cross-section of the self-interaction channel (per final fold species). Unit: $\mathrm{GeV}^{-2}$. Status: ESTIMATE.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

with $\tilde I_\lambda = 14.38$ (DERIVED; the cross-section is the value of one vector channel, an ESTIMATE, the Fierz coefficients are not computed). While the dark matter is in equilibrium ($T_{\max} = 10$ TeV to $x_{\mathrm{f}} = 31.23$) each annihilation puts $2M_{\mathrm{DM}}$ into the fold bands:

$$\frac{\mathrm{d}\left(\rho_{\mathrm{fold}}\,\mathrm{e}^{4a_4}\right)}{\mathrm{d}x_4} = 2M_{\mathrm{DM}}\,\Gamma_\lambda\,\mathrm{e}^{4a_4}, \qquad \Gamma_\lambda = 2\,n_{\mathrm{DM,eq}}\,\bar{n}_{\mathrm{DM,eq}}\,\sigma_\lambda v_{\mathrm{rel}}\,N_{\mathrm{fold}}, \qquad \frac{2\Gamma_\lambda}{n_{\mathrm{fold,eq}}\,\mathcal{H}} \ll 1 \tag{FE-136}$$

**Symbols in (FE-136).**

- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\rho_{\mathrm{fold}}$: The energy density of the fold bands. Unit: $\mathrm{GeV}^4$. Status: DERIVED.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.
- $M_{\mathrm{DM}}$: The mass of a dark-matter particle (the lowest odd level of dirac16complex). Unit: GeV. Value: 44.35 TeV. Status: DERIVED.
- $\Gamma_\lambda$: The rate per volume of the self-interaction channel. Unit: $\mathrm{GeV}^4$. Status: ESTIMATE.
- $n_{\mathrm{DM,eq}}$, $\bar{n}_{\mathrm{DM,eq}}$: The equilibrium number densities of dark-matter particles and antiparticles of one species. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\sigma_\lambda$: The cross-section of the self-interaction channel (per final fold species). Unit: $\mathrm{GeV}^{-2}$. Status: ESTIMATE.
- $v_{\mathrm{rel}}$: The relative velocity of the two annihilating particles; $\langle\sigma v_{\mathrm{rel}}\rangle$ the thermal average. Unit: dimensionless (c = 1). Status: STANDARD.
- $N_{\mathrm{fold}}$: The number of fold bands: the massless fold zero modes of $\Psi_A$ and of $\Psi_B$ (two Dirac fermions). Unit: dimensionless. Value: 2. Status: DERIVED.
- $n_{\mathrm{fold,eq}}$: The number density the fold bands would have in thermal equilibrium. Unit: $\mathrm{GeV}^3$. Status: DERIVED.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H} = \dot{a}_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H} = (8\pi^3 g_\rho/90)^{1/2}T^2/M_{\mathrm{Pl}}$). Unit: GeV. Status: DERIVED.

which gives $\Delta N_{\mathrm{eff}} = 1.83\times10^{26}\,G_\lambda^2$ $\mathrm{GeV}^4$ as long as the fold bands stay out of equilibrium. Hence $\Delta N_{\mathrm{eff}} \le 0.001$ requires $G_\lambda \le 2.3\times10^{-15}$ $\mathrm{GeV}^{-2}$ ($1.8\times10^{-6}\,G_{\mathrm{A}}$; the production rate per fold quantum is then at most 0.027 of the Hubble rate). The self-coupling $\lambda_{\mathrm{test}} = 7\times10^{-5}H^{-6}$ of the numerical Kohn-Sham tests would, with $v_{\mathrm{t}} = H^{-3}$, give $G_\lambda = 1.1\times10^{-9}$ $\mathrm{GeV}^{-2}$, keep the fold bands in equilibrium down to $T = 2.6$ TeV (production rate up to $6\times10^9$ times the Hubble rate) and add $\Delta N_{\mathrm{eff}} = 0.19$ (four Weyl species in FE-133; $N_{\mathrm{eff}} = 3.23$, 2.9 standard deviations above ACT DR6). UM-2 therefore ASSUMES $G_\lambda \le 2.3\times10^{-15}$ $\mathrm{GeV}^{-2}$, i.e. $\lambda \le 1.4\times10^{-10}H^{-6}$ for $v_{\mathrm{t}} = H^{-3}$, or $v_{\mathrm{t}} \ge 4.9\times10^5 H^{-3}$ for $\lambda = \lambda_{\mathrm{test}}$ (DERIVED bound, 1 check); with such a $\lambda$ the self-interaction mean fields of every Kohn-Sham state of the cosmology are negligible, and $\lambda_{\mathrm{test}}$ is only the value with which the Kohn-Sham machinery was tested.

## 17.3 Detection (DERIVED and ESTIMATE)

I13 and I11 couple the dark matter coherently to the vector currents of the quarks (unpolarised quarks, each chirality carrying half of the density):

$$G_{\mathrm{u}} = \frac{G_{\mathrm{A}}}{3}\,\frac{\tilde{I}_Q^{(b)} + \tilde{I}_{u_R}^{(b)}}{2} + \frac{G_{\mathrm{V}}}{3}\,\frac{\tilde{I}_Q + \tilde{I}_{u_R}}{2}, \qquad G_{\mathrm{d}} = \frac{G_{\mathrm{A}}}{3}\,\frac{\tilde{I}_Q^{(b)} + \tilde{I}_{d_R}^{(b)}}{2} + \frac{G_{\mathrm{V}}}{3}\,\frac{\tilde{I}_Q + \tilde{I}_{d_R}}{2},$$

**Symbols in this equation.**

- $G_{\mathrm{u}}$, $G_{\mathrm{d}}$: The four-dimensional vector couplings of the dark matter to the up and to the down quark (I13 and I11, unpolarised quarks with each chirality carrying half of the density). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $G_{\mathrm{A}}$: The four-dimensional strength of the portal I13 on the zero modes, $G_{\mathrm{A}} = c_{\mathrm{A}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: $1.278\times10^{-9}$ $\mathrm{GeV}^{-2}$ (dark-matter relic density). Status: DERIVED.
- $\tilde{I}_{of}^{(b)}$: The overlap of the lower component of the state $o$ with the zero mode of $f$, $\tilde{I}_{of}^{(b)} = H^{-1}\int_{-L}^{0} b_o^2\varrho_f\mathrm{e}^{-3Hy}\mathrm{d}y$ (the overlap of I13); $\tilde{I}_Q^{(b)}$, $\tilde{I}_{u_R}^{(b)}$, $\tilde{I}_{d_R}^{(b)}$, $\tilde{I}_{\nu_R}^{(b)}$ those of the dark matter. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q^{(b)} = 2.6135$. Status: DERIVED.
- $G_{\mathrm{V}}$: The four-dimensional strength of the vector portal I11 on the zero modes, $G_{\mathrm{V}} = c_{\mathrm{V}}H/(2f^4 v_{\mathrm{t}}\Lambda_\star^6)$. Unit: $\mathrm{GeV}^{-2}$. Value: benchmark $4.38\times10^{-17}$ $\mathrm{GeV}^{-2}$ ($\Delta N_{\mathrm{eff}} = 0.001$). Status: DERIVED.

- $\tilde{I}_{of}$: The overlap of the state $o$ of dirac16complex with the zero mode of the fermion component $f$, $\tilde{I}_{of} = H^{-1}\int_{-L}^{0}(a_o^2+b_o^2)\varrho_f \mathrm{e}^{-3Hy}\mathrm{d}y$ (the integral in units of $H$); $\tilde{I}_Q$, $\tilde{I}_{u_R}$, $\tilde{I}_{d_R}$, $\tilde{I}_\ell$, $\tilde{I}_{e_R}$ those of the dark matter with $Q$, $u_R$, $d_R$, $\ell$, $e_R$, $\tilde{I}_{\nu_R,i}$ with the right-handed neutrino of the generation $i$. Unit: dimensionless. Value: dark matter: $\tilde{I}_Q = 8.2722$, $\tilde{I}_\ell = 1.2524$, $\tilde{I}_{u_R} = 0.3854$, $\tilde{I}_{d_R} = 0.3961$, $\tilde{I}_{e_R} = 0.3972$. Status: DERIVED.

$$f_\mathrm{p} = 2G_\mathrm{u} + G_\mathrm{d}, \qquad f_\mathrm{n} = G_\mathrm{u} + 2G_\mathrm{d}, \qquad f_\mathrm{N} = \frac{Z_\mathrm{Xe} f_\mathrm{p} + \left(A_\mathrm{Xe} - Z_\mathrm{Xe}\right) f_\mathrm{n}}{A_\mathrm{Xe}}, \qquad \sigma_\mathrm{SI} = \frac{\mu_\mathrm{red}^2 f_\mathrm{N}^2}{\pi}$$

**Symbols in this equation.**

- $f_\mathrm{N}$, $f_\mathrm{p}$, $f_\mathrm{n}$: The spin-independent effective couplings of the dark matter to a nucleon: $f_\mathrm{p}$ to the proton, $f_\mathrm{n}$ to the neutron, and $f_\mathrm{N}$ the average per nucleon of the target (for the portal I12, the coupling through the gluon content of a nucleon). Unit: $\mathrm{GeV}^{-2}$. Value: $f_\mathrm{p} = 1.915 \times 10^{-9}$, $f_\mathrm{n} = 1.917 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (I13 and I11). Status: ESTIMATE.
- $G_\mathrm{u}$, $G_\mathrm{d}$: The four-dimensional vector couplings of the dark matter to the up and to the down quark (I13 and I11, unpolarised quarks with each chirality carrying half of the density). Unit: $\mathrm{GeV}^{-2}$. Status: DERIVED.
- $Z_\mathrm{Xe}$, $A_\mathrm{Xe}$: The atomic number and the mass number of the xenon target. Unit: dimensionless. Value: 54 and 131. Status: STANDARD.
- $\sigma_\mathrm{SI}$: The spin-independent elastic scattering cross-section of a dark-matter particle on one nucleon. Unit: $\mathrm{cm}^2$. Value: $4.01 \times 10^{-46}$ $\mathrm{cm}^2$ (xenon, I13 and I11); $6.3 \times 10^{-53}$ $\mathrm{cm}^2$ through I12 alone. Status: ESTIMATE.
- $\mu_\mathrm{red}$: The reduced mass of the dark-matter particle and a nucleon, $M_\mathrm{DM} m_\mathrm{N}/(M_\mathrm{DM} + m_\mathrm{N})$. Unit: GeV. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.

(xenon, $Z_\mathrm{Xe} = 54$, $A_\mathrm{Xe} = 131$): $f_\mathrm{p} = 1.915 \times 10^{-9}$ $\mathrm{GeV}^{-2}$, $f_\mathrm{n} = 1.917 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ and $\sigma_\mathrm{SI} = 4.0 \times 10^{-46}$ $\mathrm{cm}^2$, 0.16 of the LZ limit extrapolated linearly in the mass from $2.2 \times 10^{-48}$ $\mathrm{cm}^2$ at 40 GeV to 44.35 TeV ($2.44 \times 10^{-45}$ $\mathrm{cm}^2$; STANDARD, from memory; the extrapolation an ESTIMATE): allowed, and within reach of the next generation of xenon detectors. I11 alone gives $3.9 \times 10^{-60}$ $\mathrm{cm}^2$; I12 through the trace anomaly of QCD,

$$f_\mathrm{N} = G_\mathrm{g} \frac{8\pi}{9\alpha_3} m_\mathrm{N} f_\mathrm{TG} \qquad \left(f_\mathrm{TG} = 0.92,\ \alpha_3 = 0.3\right),$$

**Symbols in this equation.**

- $f_\mathrm{N}$, $f_\mathrm{p}$, $f_\mathrm{n}$: The spin-independent effective couplings of the dark matter to a nucleon: $f_\mathrm{p}$ to the proton, $f_\mathrm{n}$ to the neutron, and $f_\mathrm{N}$ the average per nucleon of the target (for the portal I12, the coupling through the gluon content of a nucleon). Unit: $\mathrm{GeV}^{-2}$. Value: $f_\mathrm{p} = 1.915 \times 10^{-9}$, $f_\mathrm{n} = 1.917 \times 10^{-9}$ $\mathrm{GeV}^{-2}$ (I13 and I11). Status: ESTIMATE.
- $G_\mathrm{g}$: The coupling of a dark-matter pair to canonically normalised gluon pairs through I12. Unit: $\mathrm{GeV}^{-3}$. Value: $6.09 \times 10^{-15}$ $\mathrm{GeV}^{-3}$ (benchmark). Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\alpha_3$: The strong fine-structure constant, $\alpha_3 = g_3^2/(4\pi)$, at the scale given in brackets. Unit: dimensionless. Value: $\alpha_3(m_\mathrm{Z}) = 0.1180$; $\alpha_3(m_\mathrm{KK}) = 0.0606$. Status: STANDARD.
- $m_\mathrm{N}$: The mass of a nucleon. Unit: GeV. Value: 0.939 GeV. Status: STANDARD.
- $f_\mathrm{TG}$: The fraction of the nucleon mass carried by the gluons through the trace anomaly of QCD. Unit: dimensionless. Value: 0.92 (the value used in the computation; approximate). Status: STANDARD.

at most $6 \times 10^{-53}$ $\mathrm{cm}^2$. Indirect detection: the s-wave annihilation into quark pairs continues today with $\sigma_\mathrm{I13} v_\mathrm{rel} = \sum_f N_\mathrm{c} G_f^2 M_\mathrm{DM}^2/(2\pi) = 9.8 \times 10^{-26}$ $\mathrm{cm}^3/\mathrm{s}$ (per pair of particle and antiparticle), the thermal value for a heavy relic; the I12 annihilation is p-wave, suppressed by $v_\mathrm{rel}^2 \sim 10^{-6}$ in galaxies today. Colliders: 44 TeV is beyond present and planned machines.

## 17.4 Dark energy (NOT SUPPORTED)

No positive-energy state of $\Psi$ has a negative equation of state on the static block (DERIVED): eight dark-matter quanta at rest have $w_\mathrm{KS} = 0$ ($2.8 \times 10^{-5}$ with $\lambda_\mathrm{test}$), the fold-band gas of 592 quanta $w_\mathrm{KS} = 1/3$. A homogeneous scalar condensate of $\Psi$ with the self-interaction energy $U_\mathrm{cond}$ has

$$\rho = m_\Psi n^{\mathrm{sc}} + U_{\mathrm{cond}}, \qquad p = n^{\mathrm{sc}}\,\frac{\mathrm{d}U_{\mathrm{cond}}}{\mathrm{d}n^{\mathrm{sc}}} - U_{\mathrm{cond}}, \qquad U_{\mathrm{cond}} = \frac{\lambda}{2}\big(n^{\mathrm{sc}}\big)^2,$$

**Symbols in this equation.**

- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- $n^{\mathrm{sc}}$, $n^{\mathrm{sc}}_o$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and $n^{\mathrm{sc}}_o$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $U_{\mathrm{cond}}$: The self-interaction energy density of a homogeneous scalar condensate of dirac16complex, $U_{\mathrm{cond}} = \frac{\lambda}{2}(n^{\mathrm{sc}})^2$ (the contact term of the action evaluated on the condensate). Unit: $\mathrm{GeV}^8$ (an eight-dimensional energy density). Status: DERIVED.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.

so, with $n^{\mathrm{sc}} \propto \mathrm{e}^{-3a_4}$,

$$w_{\mathrm{de}} = \frac{\mathcal{X}}{1+\mathcal{X}}, \qquad \mathcal{X} = \frac{\lambda n^{\mathrm{sc}}}{2m_\Psi} = \mathcal{X}_0\,\mathrm{e}^{-3a_4}, \qquad \mathcal{X}_0 = \frac{w_{\mathrm{de},0}}{1-w_{\mathrm{de},0}}, \qquad w_{\mathrm{de},a} = \frac{3\mathcal{X}_0}{\big(1+\mathcal{X}_0\big)^2}.$$

**Symbols in this equation.**

- $w_{\mathrm{de}}$, $w_{\mathrm{de},0}$, $w_{\mathrm{de},a}$: The equation of state of a candidate dark energy, the ratio of its pressure to its energy density. In the parametrisation $w_{\mathrm{de}} = w_{\mathrm{de},0} + w_{\mathrm{de},a}(1 - \mathrm{e}^{a_4})$ used by supernova fits, $w_{\mathrm{de},0}$ is the value today ($a_4 = 0$) and $w_{\mathrm{de},a} = -\mathrm{d}w_{\mathrm{de}}/\mathrm{d}\mathrm{e}^{a_4}$ at $a_4 = 0$ (the tangent at today). Unit: dimensionless. Status: STANDARD.
- $\mathcal{X}$, $\mathcal{X}_0$: The ratio of the self-interaction energy density to the rest-energy density of a homogeneous scalar condensate of dirac16complex, $\mathcal{X} = \lambda n^{\mathrm{sc}}/(2m_\Psi)$; it scales as $\mathrm{e}^{-3a_4}$, and $\mathcal{X}_0$ is its value today ($a_4 = 0$). Unit: dimensionless. Status: DERIVED.
- $\lambda$: The contact self-coupling of dirac16complex (the term $-\frac{\lambda}{2}(\bar\Psi\Psi)^2$ of the record). Unit: $\mathrm{GeV}^{-6}$. Value: not fixed; Kohn-Sham states computed for $\lambda = 0$ and $\pm 7 \times 10^{-5} H^{-6}$; the fold-band radiation requires $|\lambda| \lesssim 1.4 \times 10^{-10} H^{-6}$ (for $\Delta N_{\mathrm{eff}} \leq 0.001$). Status: RECORD.
- $n^{\mathrm{sc}}$, $n^{\mathrm{sc}}_o$: The scalar density of dirac16complex (the Kohn-Sham expectation of $\bar\Psi\Psi$, per proper 7-volume) and $n^{\mathrm{sc}}_o$ the contribution of the orbital $o$. Unit: $\mathrm{GeV}^7$. Status: RECORD.
- $m_\Psi$: The mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M} = H$. Unit: GeV. Value: 14.296 TeV. Status: RECORD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.

A negative equation of state needs $\lambda < 0$, and then the energy density $\rho = m_\Psi n^{\mathrm{sc}}(1+\mathcal{X})$ crosses zero at $\mathrm{e}^{a_4} = (-\mathcal{X}_0)^{1/3}$: a condensate tuned to $w_{\mathrm{de},0} = -1$ runs away ($w_{\mathrm{de},a} = -6$) and its energy density turns negative at the redshift $z_{\mathrm{red}} = 0.26$. A natural vacuum energy of the cell exceeds the dark-energy density by $5.5 \times 10^{64}$; in UM-2 the four-dimensional cosmological term is the residual of the stabiliser balance (FE-48), zero at the static point, and the observed dark energy needs $\Lambda_4/H^2 \sim 1.0 \times 10^{-92}$: the cosmological-constant problem, unsolved (OPEN). The dark energy of UM-2 is this tuned residual $\Lambda_4$; a slowly deflating block cannot provide a dynamical dark energy instead, because it would make $\alpha_{\mathrm{em}}$, $G_{\mathrm{N}}$ and the masses drift (the bounds of the section on dimensional reduction).

### 17.5 Assessment: is dirac16complex the dark matter, and the dark energy?

The association of dirac16complex with dark matter is valid in UM-2 under stated conditions: (i) the good sector (no extra-time momenta; outside it the halves mix and modes are tachyonic, OPEN); (ii) the exact fold parity of both halves, which UM-2 secures by containing no oriented tip operator (a left-handed tip current would not even reach $\Psi_B$); (iii) fold bands that never thermalise (I11 weak, I12 and I13 blind to them, a weak self-interaction, no dipole portal, no significant gravitational production, OPEN); (iv) an annihilation channel that produces the relic within unitarity: I12 alone cannot, the INVENTED s-wave portal I13 does, and it predicts a direct-detection signal at 0.16 of the present LZ limit; (v) no late dilution by the radion (the moduli problem). The candidate is falsifiable: a xenon exposure about six times that of LZ would test it. It is the heavy odd part of the field, not the field as a whole, that is dark matter; the massless fold bands are radiation ($\Delta N_{\text{eff}} = 0.001$). The association of dirac16complex with dark energy is NOT SUPPORTED: its positive-energy states are matter or radiation, and its only accelerating configuration is unstable and reaches negative energy density at $z_{\text{red}} \approx 0.2$ to 0.6.

## 18 Cosmology: the CMB, the CNB and the BAO

### 18.1 The content and the equations (STANDARD equations, DERIVED content)

The four-dimensional effective theory of UM-2 is general relativity (FE-49) with photons; the three neutrino mass states (1.000, 8.712 and 50.140 meV, with the heated spectrum of the electron-positron annihilation: $N_{\text{eff}} = 3.0432$); the fold bands produced by freeze-in (massless, $\Delta N_{\text{eff}} = 0.0010$), so $N_{\text{eff}} = 3.0442$ in all; the baryons; the dark matter (the odd quanta of $\Psi_A$ and $\Psi_B$, cold); and the cosmological term $\Lambda_4$. The order-1 terms of the metric series, the perturbations of this content, obey the standard Einstein-Boltzmann equations in the Newtonian gauge (STANDARD, Ma and Bertschinger). They are written here for one Fourier mode of the comoving wave number $k$, with the conformal time $\tau_{\text{c}}$ (a prime is $\mathrm{d}/\mathrm{d}\tau_{\text{c}}$ and nothing else). The metric:

$$g^{(4)}_{\mu\nu}\,\mathrm{d}X^\mu \mathrm{d}X^\nu = \mathrm{e}^{2a_4}\Big[-\big(1+2\Phi\big)\,\mathrm{d}\tau_{\text{c}}^2 + \big(1-2\Psi_{\text{N}}\big)\,\mathrm{d}\vec{x}^{\,2}\Big], \qquad \mathrm{d}x_4 = \mathrm{e}^{a_4}\,\mathrm{d}\tau_{\text{c}} \tag{FE-89}$$

**Symbols in (FE-89).**

- $g^{(4)}_{\mu\nu}$: The four-dimensional metric of the observable coordinates $x_1, \ldots, x_4$ (the metric of the effective four-dimensional general relativity on the stabilised block). Unit: dimensionless. Status: DERIVED.
- $\mu$: A four-dimensional coordinate index running over the observable coordinates $x_1, x_2, x_3, x_4$ (signature $+,+,+,-$); on the tip it runs over the tip directions; a repeated index is summed. Status: STANDARD.
- $\nu$: A four-dimensional coordinate index running over $x_1, \ldots, x_4$ (as $\mu$). Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $X = \{x_0, \ldots, x_7\}$: The author's coordinates of the eight-dimensional spacetime, $X = \{x_0, x_1, \ldots, x_7\}$ with the hidden direction first; $X^M$ is the coordinate with the index $M$, $\mathrm{d}^8X = \mathrm{d}x_0\,\mathrm{d}x_1 \cdots \mathrm{d}x_7$ the coordinate volume element and $X^\mu$ (four-dimensional index $\mu$) the observable coordinates $x_1, \ldots, x_4$. Unit: length (each coordinate). Status: AUTHOR.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\tau_{\text{c}}$: The conformal time, $\mathrm{d}x_4 = \mathrm{e}^{a_4}\mathrm{d}\tau_{\text{c}}$; primes denote $\mathrm{d}/\mathrm{d}\tau_{\text{c}}$. Unit: length. Status: STANDARD.
- $\Psi_{\text{N}}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_{\text{N}}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\vec{x}$: The position in ordinary 3-space, $\vec{x} = (x_1, x_2, x_3)$; $\mathrm{d}\vec{x}^{\,2} = \mathrm{d}x_1^2 + \mathrm{d}x_2^2 + \mathrm{d}x_3^2$. Unit: length. Status: STANDARD.
- $x_4$: The time; dots denote $\mathrm{d}/\mathrm{d}x_4$. Unit: length (= time, c = 1). Status: AUTHOR.

The Einstein equations, summed over all species:

$$
\begin{aligned}
&k^2\Psi_\mathrm{N} + 3a_4'\bigl(\Psi_\mathrm{N}' + a_4'\Phi\bigr) = -4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\,\Delta\rho, \qquad k^2\bigl(\Psi_\mathrm{N}' + a_4'\Phi\bigr) = 4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\bigl(\rho_\mathrm{tot} + p_\mathrm{tot}\bigr)\theta_\mathrm{tot},\\
&\Psi_\mathrm{N}'' + a_4'\bigl(\Phi' + 2\Psi_\mathrm{N}'\bigr) + \bigl(2a_4'' + a_4'^2\bigr)\Phi + \frac{k^2}{3}\bigl(\Psi_\mathrm{N} - \Phi\bigr) = 4\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\,\Delta p,\\
&k^2\bigl(\Psi_\mathrm{N} - \Phi\bigr) = 12\pi G_\mathrm{N}\,\mathrm{e}^{2a_4}\bigl(\rho_\mathrm{tot} + p_\mathrm{tot}\bigr)\sigma_\mathrm{an}
\end{aligned}
\qquad \text{(FE-90)}
$$

**Symbols in (FE-90).**

- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_\mathrm{N}$: Newton's constant of four-dimensional gravity, $8\pi G_\mathrm{N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_\mathrm{Pl} = G_\mathrm{N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\Delta$: The change (difference or small perturbation) of the quantity that follows: $\Delta K$, $\Delta\rho$, $\Delta p_3$ are first-order changes about the static block, $\Delta\varepsilon_o$ a level shift, $\Delta m_{21}^2 = m_2^2 - m_1^2$ a difference of squared masses, $\Delta N_\mathrm{eff}$ an addition to $N_\mathrm{eff}$, $\Delta S_\mathrm{m}$ the change of an action under a change $\Delta g^{MN}$ of the metric. In an uncertainty relation it is the spread, the root-mean-square deviation from the mean value in a state: $\Delta x_0$, $\Delta k_{x_0}$, $\Delta\xi$, $\Delta(fy)$ (as in the author's $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$). Status: STANDARD.
- $\rho$: An energy density: in eight dimensions $\rho = -\mathcal{T}^{x_4}{}_{x_4}$ (of the total source, or of the species named by a subscript; $\rho_s$ of the species $s$, $\rho_i$ of the neutrino mass state $i$); in the cosmological perturbation equations $\Delta\rho$ is the density perturbation summed over the species. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $\rho_\mathrm{tot}$, $p_\mathrm{tot}$: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\theta_\mathrm{tot}$: The velocity divergence of the total matter, weighted with $\rho + p$ of each species. Unit: 1/length. Status: STANDARD.
- $p$: A pressure: $p_M = \mathcal{T}^M{}_M$ (no sum) for the direction $M \neq x_4$, $p_s$ the pressure of the species $s$; $\Delta p$ the pressure perturbation in the cosmological perturbation equations; in the Feynman-Hellmann relation the pressure of a gauge plasma. Unit: $\mathrm{GeV}^8$ (eight dimensions), $\mathrm{GeV}^4$ (four dimensions). Status: STANDARD.
- $\sigma_\mathrm{an}$: The anisotropic stress (shear) of the total matter (the quantity $\sigma$ of Ma and Bertschinger), weighted with $\rho + p$. Unit: dimensionless. Status: STANDARD.

The dark matter, cold (the Kohn-Sham states of quanta at rest have $w_\mathrm{KS} = 0$):

$$
\delta_\mathrm{DM}' = -\theta_\mathrm{DM} + 3\Psi_\mathrm{N}', \qquad \theta_\mathrm{DM}' = -a_4'\,\theta_\mathrm{DM} + k^2\Phi
\qquad \text{(FE-91)}
$$

**Symbols in (FE-91).**

- $\delta_\mathrm{DM}$, $\theta_\mathrm{DM}$: The density contrast and the velocity divergence of the dark matter. Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.

- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.

The baryons, coupled to the photons by Thomson scattering:

$$\delta_\mathrm{b}' = -\theta_\mathrm{b} + 3\Psi_\mathrm{N}', \qquad \theta_\mathrm{b}' = -a_4'\,\theta_\mathrm{b} + c_\mathrm{s}^2 k^2 \delta_\mathrm{b} + R_\mathrm{b}\,\mathrm{e}^{a_4} n_\mathrm{e} \sigma_\mathrm{T} \big(\theta_\gamma - \theta_\mathrm{b}\big) + k^2 \Phi, \qquad R_\mathrm{b} = \frac{4\rho_\gamma}{3\rho_\mathrm{b}} \tag{FE-92}$$

**Symbols in (FE-92).**

- $\delta_\mathrm{b}$, $\theta_\mathrm{b}$: The density contrast and the velocity divergence of the baryons (nucleons and electrons). Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $c_\mathrm{s}$: The sound speed of the baryon fluid. Unit: dimensionless (c = 1). Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $R_\mathrm{b}$: The photon-to-baryon momentum ratio of the Thomson drag, $R_\mathrm{b} = 4\rho_\gamma/(3\rho_\mathrm{b})$. Unit: dimensionless. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $n_\mathrm{e}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\sigma_\mathrm{T}$: The Thomson cross-section. Unit: $\mathrm{GeV}^{-2}$. Value: $6.6524587 \times 10^{-29}$ $\mathrm{m}^2$. Status: STANDARD.
- $\delta_\gamma$, $\theta_\gamma$: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $\rho_\gamma$: The energy density of the photons, $\rho_\gamma = \pi^2 T_\gamma^4/15$. Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\rho_\mathrm{b}$: The energy density of the baryons. Unit: $\mathrm{GeV}^4$. Status: STANDARD.

The photons: density and velocity, the temperature hierarchy and the polarisation hierarchy:

$$\delta_\gamma' = -\frac{4}{3}\,\theta_\gamma + 4\Psi_\mathrm{N}', \qquad \theta_\gamma' = k^2\Big(\frac{1}{4}\,\delta_\gamma - \sigma_{\mathrm{an},\gamma}\Big) + k^2\Phi + \mathrm{e}^{a_4} n_\mathrm{e} \sigma_\mathrm{T} \big(\theta_\mathrm{b} - \theta_\gamma\big) \tag{FE-93}$$

**Symbols in (FE-93).**

- $\delta_\gamma$, $\theta_\gamma$: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals

the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.

- $\sigma_{\mathrm{an},\gamma}$: The anisotropic stress (shear) of the photons, $F_2^\gamma/2$. Unit: dimensionless. Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1+2\Phi$). Unit: dimensionless. Value: at most $2.4\times10^{-5}$. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $n_\mathrm{e}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\sigma_\mathrm{T}$: The Thomson cross-section. Unit: $\mathrm{GeV}^{-2}$. Value: $6.6524587\times10^{-29}$ $\mathrm{m}^2$. Status: STANDARD.
- $\delta_\mathrm{b}$, $\theta_\mathrm{b}$: The density contrast and the velocity divergence of the baryons (nucleons and electrons). Unit: dimensionless; 1/length. Status: STANDARD.

$$\begin{aligned} &F_l^{\gamma\prime} = \frac{k}{2l+1}\big[l\,F_{l-1}^\gamma - (l+1)F_{l+1}^\gamma\big] - \mathrm{e}^{a_4}n_\mathrm{e}\sigma_\mathrm{T}\Big[F_l^\gamma - \frac{\delta_{l2}}{10}\big(F_2^\gamma + G_0^\gamma + G_2^\gamma\big)\Big] \quad (l\geq 2),\\ &F_0^\gamma = \delta_\gamma, \qquad F_1^\gamma = \frac{4\theta_\gamma}{3k}, \qquad F_2^\gamma = 2\sigma_{\mathrm{an},\gamma} \end{aligned} \tag{FE-94}$$

**Symbols in (FE-94).**

- $F_l^\gamma$: The multipole $l$ of the photon temperature perturbation (the brightness function of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- $l$: The multipole (angular moment) number $l=0,1,2,\ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l\cdots$; as a coefficient it is the same number. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar=1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $n_\mathrm{e}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\sigma_\mathrm{T}$: The Thomson cross-section. Unit: $\mathrm{GeV}^{-2}$. Value: $6.6524587\times10^{-29}$ $\mathrm{m}^2$. Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta_N^M=1$ if $M=N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1\cdots N_n}^{M_1\cdots M_n}$, the determinant of the $n\times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.
- $G_l^\gamma$: The multipole $l$ of the photon polarisation perturbation. Unit: dimensionless. Status: STANDARD.
- $\delta_\gamma$, $\theta_\gamma$: The density contrast and the velocity divergence of the photons. Unit: dimensionless; 1/length. Status: STANDARD.
- $\sigma_{\mathrm{an},\gamma}$: The anisotropic stress (shear) of the photons, $F_2^\gamma/2$. Unit: dimensionless. Status: STANDARD.

$$G_l^{\gamma\prime} = \frac{k}{2l+1}\big[l\,G_{l-1}^\gamma - (l+1)G_{l+1}^\gamma\big] + \mathrm{e}^{a_4}n_\mathrm{e}\sigma_\mathrm{T}\Big[-G_l^\gamma + \frac{1}{2}\big(F_2^\gamma + G_0^\gamma + G_2^\gamma\big)\Big(\delta_{l0} + \frac{\delta_{l2}}{5}\Big)\Big] \tag{FE-95}$$

**Symbols in (FE-95).**

- $G_l^\gamma$: The multipole $l$ of the photon polarisation perturbation. Unit: dimensionless. Status: STANDARD.
- $l$: The multipole (angular moment) number $l=0,1,2,\ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l\cdots$; as a coefficient it is the same number. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.

- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $n_\mathrm{e}$: The number density of electrons (net of positrons); in the Thomson terms of the photon and baryon equations only free (ionised) electrons count. Unit: $\mathrm{GeV}^3$. Status: STANDARD.
- $\sigma_\mathrm{T}$: The Thomson cross-section. Unit: $\mathrm{GeV}^{-2}$. Value: $6.6524587 \times 10^{-29}$ $\mathrm{m}^2$. Status: STANDARD.
- $F_l^\gamma$: The multipole $l$ of the photon temperature perturbation (the brightness function of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- $\delta$: The Kronecker delta: with two indices $\delta_N^M = 1$ if $M = N$ and 0 otherwise (likewise $\delta_{\alpha e}$, $\delta_{l0}$, $\delta^{I3}$); with upper and lower index strings the generalised Kronecker delta $\delta_{N_1 \cdots N_n}^{M_1 \cdots M_n}$, the determinant of the $n \times n$ matrix of Kronecker deltas $\delta_{N_b}^{M_a}$ (+1 or -1 if the upper indices are an even or odd permutation of the lower ones, 0 otherwise). Status: STANDARD.

The cosmic neutrino background: one hierarchy for each neutrino mass state, whose unperturbed distribution is the heated spectrum of the section on the three neutrino flavours; the fold bands produced by freeze-in obey its massless limit $E_i^\mathrm{c} = k_\nu$:

$$\begin{aligned} \mathcal{N}_0^{(i)\prime} &= -\frac{k_\nu k}{E_i^\mathrm{c}}\mathcal{N}_1^{(i)} - \Psi_\mathrm{N}'\frac{\mathrm{d}\ln f_\mathrm{FD}}{\mathrm{d}\ln k_\nu}, \qquad \mathcal{N}_1^{(i)\prime} = \frac{k_\nu k}{3E_i^\mathrm{c}}\left(\mathcal{N}_0^{(i)} - 2\mathcal{N}_2^{(i)}\right) - \frac{E_i^\mathrm{c} k}{3k_\nu}\Phi\frac{\mathrm{d}\ln f_\mathrm{FD}}{\mathrm{d}\ln k_\nu}, \\ \mathcal{N}_l^{(i)\prime} &= \frac{k_\nu k}{(2l+1)E_i^\mathrm{c}}\left[l\,\mathcal{N}_{l-1}^{(i)} - (l+1)\mathcal{N}_{l+1}^{(i)}\right] \quad (l \geq 2), \qquad E_i^\mathrm{c} = \sqrt{k_\nu^2 + \mathrm{e}^{2a_4}m_i^2} \end{aligned} \tag{FE-96}$$

**Symbols in (FE-96).**

- $\mathcal{N}_l^{(i)}$: The multipole $l$ of the perturbation of the distribution of the neutrino mass state $i$ (a function of $k$, $k_\nu$ and $\tau_\mathrm{c}$; the quantity $\Psi_l$ of Ma and Bertschinger). Unit: dimensionless. Status: STANDARD.
- $i$: The generation index ($i = 1, 2, 3$), which also labels the neutrino mass states $\nu_1, \nu_2, \nu_3$; a repeated index is not summed unless a sum sign is written. Status: STANDARD.
- $'$: The prime on a function of the conformal time $\tau_\mathrm{c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_\mathrm{c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $k_\nu$: The comoving momentum of a neutrino (the physical momentum times $\mathrm{e}^{a_4}$). Unit: GeV. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $E_i^\mathrm{c}$: The comoving energy of a neutrino of the mass state $i$, $E_i^\mathrm{c} = (k_\nu^2 + \mathrm{e}^{2a_4}m_i^2)^{1/2}$. Unit: GeV. Status: STANDARD.
- $\Psi_\mathrm{N}$: The curvature potential of the Newtonian gauge (the perturbation of the 3-space metric, $1 - 2\Psi_\mathrm{N}$); not related to the field $\Psi$. Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $f_\mathrm{FD}$: The Fermi-Dirac distribution of the relic neutrinos, $f_\mathrm{FD} = 1/(\mathrm{e}^{k/T_\nu} + 1)$. Unit: dimensionless. Status: STANDARD.
- $\Phi$: The lapse potential of the Newtonian gauge (the perturbation of the time component, $1 + 2\Phi$). Unit: dimensionless. Value: at most $2.4 \times 10^{-5}$. Status: STANDARD.
- $l$: The multipole (angular moment) number $l = 0, 1, 2, \ldots$ of a Boltzmann hierarchy or of an angular moment $\int u^l \cdots$; as a coefficient it is the same number. Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.

- $m$, $m_s$, $m_i$: A four-dimensional mass: $m$ the mass of a mode (the eigenvalue of an $x_0$ problem) or of a fermion; $m_s$ the mass of the species $s$; $m_i$ the mass of the neutrino mass state $i$. Unit: GeV. Status: DERIVED.

The background expansion, the neutrino temperature and the primordial spectrum, which is an input of UM-2 (Planck 2018), not a prediction:

$$a_4'^2 = \frac{8\pi G_{\rm N}}{3}\,\mathrm{e}^{2a_4}\rho_{\rm tot} + \frac{\Lambda_4}{3}\,\mathrm{e}^{2a_4}, \qquad T_\nu = \Big(\frac{4}{11}\Big)^{1/3} T_\gamma, \qquad P_{\mathcal{R}}(k) = A_{\rm s}\Big(\frac{k}{k_0}\Big)^{n_{\rm s}-1} \tag{FE-97}$$

**Symbols in (FE-97).**

- $a_4$: The author's metric function of the time $x_4$ that scales ordinary 3-space: the scale factor of cosmology is $\mathrm{e}^{a_4}$. Unit: dimensionless. Status: AUTHOR.
- $'$: The prime on a function of the conformal time $\tau_{\rm c}$ is its derivative $\mathrm{d}/\mathrm{d}\tau_{\rm c}$ (it is used only in the cosmological perturbation equations; two primes are the second derivative). Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $G_{\rm N}$: Newton's constant of four-dimensional gravity, $8\pi G_{\rm N} = \kappa/(\mathcal{G}(K_*)V_\perp)$. Unit: $\mathrm{GeV}^{-2}$. Value: $6.709 \times 10^{-39}$ $\mathrm{GeV}^{-2}$ ($M_{\rm Pl} = G_{\rm N}^{-1/2} = 1.22091 \times 10^{19}$ GeV). Status: STANDARD.
- e: Euler's number, the base of the natural logarithm; $\mathrm{e}^X$ is the exponential of $X$. Unit: dimensionless. Value: 2.718281828. Status: STANDARD.
- $\rho_{\rm tot}$, $p_{\rm tot}$: The total background energy density and pressure of four-dimensional cosmology (all species). Unit: $\mathrm{GeV}^4$. Status: STANDARD.
- $\Lambda_4$: The four-dimensional cosmological constant of the effective theory on the block, $\Lambda_4 = [6H^2K(1 - 2w_{(2)}K) + \Lambda + \kappa\rho_{\rm stab}]/\mathcal{G}(K)$; zero at the static point, the observed dark energy a tuned residual. Unit: $\mathrm{GeV}^2$. Value: 0 at the static point; observed $\Lambda_4/H^2 \sim 10^{-92}$. Status: DERIVED.
- $T_\nu$: The temperature of the relic neutrinos (one common temperature). Unit: GeV. Value: 1.9454 K today. Status: STANDARD.
- $T_\gamma$: The temperature of the photons (equal to the plasma temperature $T$; the subscript is written where the neutrinos have their own temperature). Unit: GeV. Value: 2.7255 K today. Status: STANDARD.
- $P_{\mathcal{R}}(k)$: The dimensionless power spectrum of the primordial curvature perturbation $\mathcal{R}$. Unit: dimensionless. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $A_{\rm s}$: The amplitude of the primordial spectrum at $k_0$. Unit: dimensionless. Value: $2.099 \times 10^{-9}$ ($\ln(10^{10}A_{\rm s}) = 3.044$, Planck 2018). Status: STANDARD.
- $k_0$: The pivot wavenumber of the primordial spectrum. Unit: 1/length. Value: 0.05 $\mathrm{Mpc}^{-1}$. Status: STANDARD.
- $n_{\rm s}$: The spectral index of the primordial spectrum. Unit: dimensionless. Value: 0.9649 (Planck 2018). Status: STANDARD.

## 18.2 The solution (DERIVED, 8 checks)

The equations were solved with the Einstein-Boltzmann code CAMB 2.0.5 (default accuracy, about $10^{-3}$; lensing included) at the Planck 2018 values (STANDARD) $\Omega_{\rm b}h^2_{100} = 0.02237$, $\Omega_{\rm DM}h^2_{100} = 0.1200$, $100\,\theta_{\rm MC} = 1.04092$ (the angular size of the sound horizon at last scattering, held fixed, so that $\mathcal{H}_0$ is an output), the optical depth to reionisation 0.0544, $\ln(10^{10}A_{\rm s}) = 3.044$ and $n_{\rm s} = 0.9649$. Each neutrino mass state is evolved separately with the degeneracy $N_{\rm eff}/3 = 1.0144$; CAMB gives each the density of a Fermi-Dirac gas of the temperature $T_\nu$ times the degeneracy, so its masses were iterated (13 iterations, to $7.8 \times 10^{-12}$) until the late-time density equals that of the heated spectrum. The comparison model is ΛCDM with the Planck 2018 parameters (one neutrino of 0.06 eV, $N_{\rm eff} = 3.044$).

| model | $\mathcal{H}_0$ (km/s/Mpc) | $\Omega_\mathrm{m}$ | $N_\mathrm{eff}$ | age (Gyr) | $r_\mathrm{d}$ (Mpc) | $\chi^2_\mathrm{BAO}$ (13 points) | $\sigma_8$ |
|---|---|---|---|---|---|---|---|
| ΛCDM | 67.34 | 0.3154 | 3.044 | 13.799 | 147.103 | 32.55 | 0.8109 |
| UM-2 | 67.38 | 0.3150 | 3.0442 | 13.796 | 147.104 | 31.49 | 0.8118 |
| UM-2 without the freeze-in fold bands | 67.37 | 0.3151 | 3.0432 | 13.797 | 147.109 | 31.72 | 0.8119 |

**The CMB.** The TT and EE spectra of UM-2 agree with Planck ΛCDM to $2.5 \times 10^{-4}$ and $2.7 \times 10^{-4}$ (multipoles 30 to 2500; for TT at most 0.007 of the cosmic variance $\sqrt{2/(2l+1)}$ of a single multipole $l$; the binned cosmic-variance $\chi^2$, the sum over bins of 50 multipoles of the squared mean relative deviation over the cosmic variance of the bin, is 0.020). By themselves, the three mass states instead of one state of 60 meV move TT and EE by $2.6 \times 10^{-4}$ and $2.3 \times 10^{-4}$, and $N_\mathrm{eff} = 3.0432$ instead of 3.044 by $1.3 \times 10^{-4}$; without the freeze-in fold bands the deviations would be $2.8 \times 10^{-4}$ (TT and EE), and the fold bands themselves move TT by at most $1.6 \times 10^{-4}$. UM-2 reproduces the CMB of ΛCDM. Its $N_\mathrm{eff} = 3.0442$ lies 1.4 standard deviations above the ACT DR6 value $2.86 \pm 0.13$ (STANDARD, from memory). A fold band thermalised in the early Universe, which UM-2 does not contain, would be excluded: decoupled above the electroweak scale ($\Delta N_\mathrm{eff} = 0.0958$) or just above the QCD crossover (0.1963) it moves TT by up to 1.5 % and 3.1 % at fixed Planck parameters (0.8 and 1.5 cosmic-variance units of a single multipole; binned $\chi^2 = 293$ and 1208) and lies 2.1 and 2.9 standard deviations above ACT DR6; its BAO $\chi^2_\mathrm{BAO}$ would be lower (15.4 and 10.7), but these variants are not fits to the CMB.

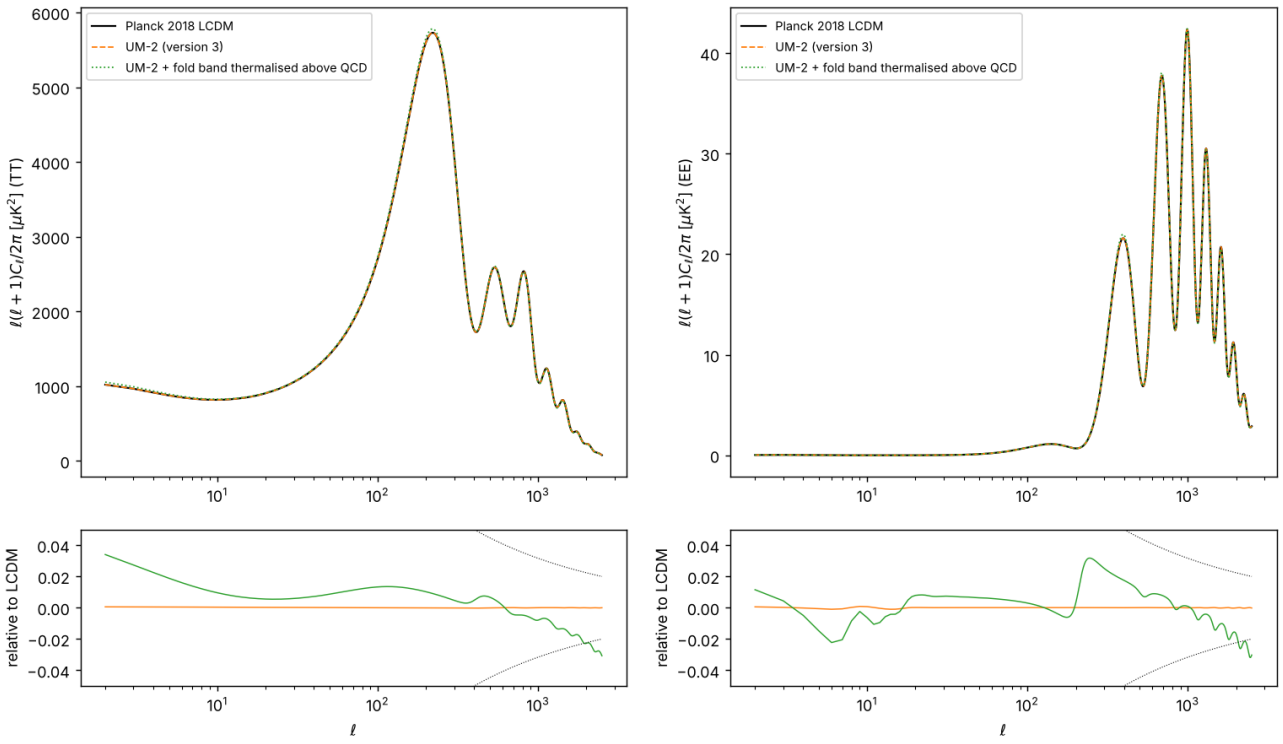



Figure 5: CMB temperature (left) and E-mode (right) spectra of UM-2 (three massive neutrinos and the freeze-in fold bands), of Planck 2018 ΛCDM and of the variant with a fold band thermalised above the QCD crossover (not part of UM-2); the figure labels UM-2 "version 3", whose physics version 4 keeps; lower panels: the relative difference to ΛCDM, with the cosmic variance of one multipole (dotted).

**The CNB.** The cosmic neutrino background enters through $N_\mathrm{eff} = 3.0432$ and the three masses. The neutrino density parameter of UM-2 in CAMB, $\Omega_\nu h^2_{100} = 6.4511 \times 10^{-4}$, equals the value $6.382 \times 10^{-4}$ of the CNB at the

temperature of instantaneous decoupling times $(N_{\rm eff}/3)^{3/4}$ $(6.4510\times10^{-4})$; the mass sum 0.0599 eV lies below the DESI DR2 bound 0.064 eV (STANDARD, from memory), which would allow a lightest mass up to 4.1 meV.

**The BAO.** The distances (flat space, $c=1$) at the redshift $z_{\rm red}$,

$$d_{\rm M}(z_{\rm red})=\int_0^{z_{\rm red}}\frac{{\rm d}\tilde z_{\rm red}}{\mathcal{H}(\tilde z_{\rm red})},\qquad d_{\rm H}(z_{\rm red})=\frac{1}{\mathcal{H}(z_{\rm red})},\qquad d_{\rm V}(z_{\rm red})=\left(z_{\rm red}\,d_{\rm M}^2 d_{\rm H}\right)^{1/3},$$

**Symbols in this equation.**

- $d_{\rm M}$, $d_{\rm H}$, $d_{\rm V}$: The distances of the BAO analysis at the redshift $z_{\rm red}$ (flat space, $c=1$): the comoving transverse distance $d_{\rm M}$, the Hubble distance $d_{\rm H}=1/\mathcal{H}$ at that redshift, and the volume-averaged distance $d_{\rm V}=(z_{\rm red}d_{\rm M}^2d_{\rm H})^{1/3}$ (written with a lower-case $d$ because $D_M$ is the covariant derivative). Unit: length (Mpc). Status: STANDARD.
- $z_{\rm red}$, $\tilde z_{\rm red}$: The cosmological redshift, $1+z_{\rm red}={\rm e}^{-a_4}$ with $a_4=0$ today (not $z=6Hx_0$, the angle variable of the hidden direction); $\tilde z_{\rm red}$ is the redshift as the variable of an integral. Unit: dimensionless. Status: STANDARD.
- d: The differential: ${\rm d}X$ is an infinitesimal change of $X$, ${\rm d}X/{\rm d}Y$ the derivative of $X$ with respect to $Y$, ${\rm d}^nX$ the volume element of $n$ variables. Status: STANDARD.
- $\mathcal{H}$: The expansion rate of ordinary 3-space, $\mathcal{H}=\dot a_4$ (the Hubble rate of the observer; in the hot Universe $\mathcal{H}=(8\pi^3g_\rho/90)^{1/2}T^2/M_{\rm Pl}$). Unit: GeV. Status: DERIVED.

in units of the sound horizon at the drag epoch $r_{\rm d}$, are compared with the DESI DR2 measurements (STANDARD, arXiv:2503.14738; the last column is the published correlation coefficient of the pair $d_{\rm M}/r_{\rm d}$, $d_{\rm H}/r_{\rm d}$):

| tracer | $z_{\rm red}$ | $d_{\rm V}/r_{\rm d}$ | $d_{\rm M}/r_{\rm d}$ | $d_{\rm H}/r_{\rm d}$ | correlation |
|---|---|---|---|---|---|
| BGS | 0.295 | $7.942\pm0.075$ | | | |
| LRG1 | 0.510 | | $13.588\pm0.167$ | $21.863\pm0.425$ | $-0.459$ |
| LRG2 | 0.706 | | $17.351\pm0.177$ | $19.455\pm0.330$ | $-0.404$ |
| LRG3+ELG1 | 0.934 | | $21.576\pm0.152$ | $17.641\pm0.193$ | $-0.416$ |
| ELG2 | 1.321 | | $27.601\pm0.318$ | $14.176\pm0.221$ | $-0.434$ |
| QSO | 1.484 | | $30.512\pm0.760$ | $12.817\pm0.516$ | $-0.500$ |
| Ly$\alpha$ | 2.330 | | $38.988\pm0.531$ | $8.632\pm0.101$ | $-0.431$ |

through

$$\chi^2_{\rm BAO}=\sum_{\rm tracers}\left(\mathbf{d}-\mathbf{d}^{\rm th}\right)^T\mathbf{C}^{-1}\left(\mathbf{d}-\mathbf{d}^{\rm th}\right),\qquad \mathbf{d}=\left(\frac{d_{\rm M}}{r_{\rm d}},\ \frac{d_{\rm H}}{r_{\rm d}}\right)$$

**Symbols in this equation.**

- $\chi^2_{\rm BAO}$: The chi-squared of the model distances against the 13 DESI DR2 BAO measurements (the pairs of the anisotropic tracers with their correlations, and one point for BGS). Unit: dimensionless. Value: 31.49 for UM-2 (32.55 for ΛCDM). Status: DERIVED.
- $\mathbf{d}$, $\mathbf{d}^{\rm th}$: The measured pair $(d_{\rm M}/r_{\rm d}, d_{\rm H}/r_{\rm d})$ of an anisotropic BAO tracer, and the pair $\mathbf{d}^{\rm th}$ computed in the model. Unit: dimensionless. Status: STANDARD.
- $^T$: The transpose: a row written with $^T$ is a column. Status: STANDARD.
- $\mathbf{C}$: The covariance matrix of the measured pair of one BAO tracer, built from the published errors and the published correlation coefficient of the pair. Unit: dimensionless. Status: STANDARD.
- $d_{\rm M}$, $d_{\rm H}$, $d_{\rm V}$: The distances of the BAO analysis at the redshift $z_{\rm red}$ (flat space, $c=1$): the comoving transverse distance $d_{\rm M}$, the Hubble distance $d_{\rm H}=1/\mathcal{H}$ at that redshift, and the volume-averaged distance $d_{\rm V}=(z_{\rm red}d_{\rm M}^2d_{\rm H})^{1/3}$ (written with a lower-case $d$ because $D_M$ is the covariant derivative). Unit: length (Mpc). Status: STANDARD.
- $r_{\rm d}$: The comoving sound horizon at the drag epoch, the standard ruler of the BAO. Unit: length (Mpc). Value: 147.104 Mpc (UM-2). Status: DERIVED.

for the anisotropic tracers (the covariance $\mathbf{C}$ of each tracer built from the published errors and correlation), plus the squared deviation over the variance of $d_\mathrm{V}/r_\mathrm{d}$ for BGS: $\chi^2_\mathrm{BAO} = 31.49$ for the 13 measurements ($\Lambda$CDM: 32.55; without the freeze-in fold bands 31.72) (DERIVED).

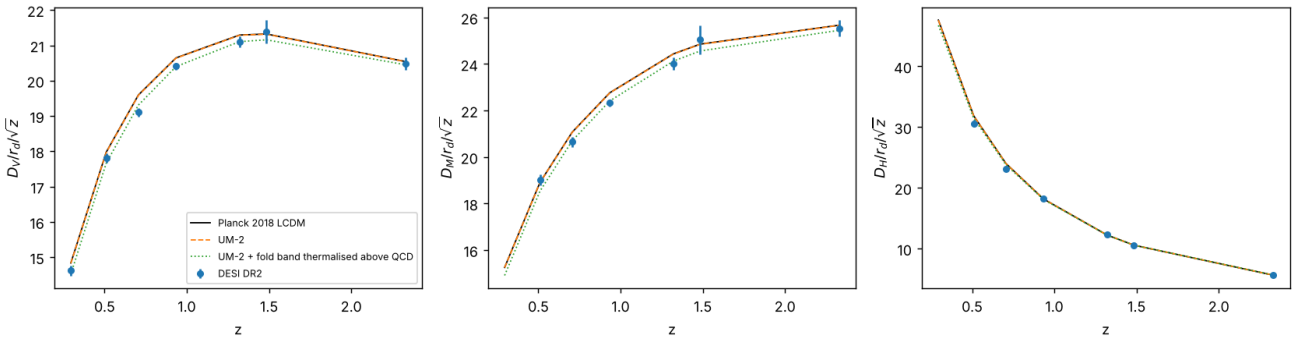



Figure 6: BAO distances divided by the square root of the redshift, against the DESI DR2 measurements (the figure writes $D$ for the distances $d$ of the text); $\chi^2_\mathrm{BAO}$ uses $d_\mathrm{V}$ for BGS and $d_\mathrm{M}$, $d_\mathrm{H}$ with their correlation for the other tracers (the $d_\mathrm{V}$ points of the anisotropic tracers are shown for illustration).

**The matter fluctuations.** A separate CAMB run of each model gives the linear matter power spectrum and

$$\sigma_8^2 = \int_0^\infty \frac{\mathrm{d}k}{k}\,\frac{k^3 P_\mathrm{m}(k)}{2\pi^2}\,W_\mathrm{th}(kR_8)^2, \qquad W_\mathrm{th}(kR_8) = \frac{3\big[\sin(kR_8) - kR_8\cos(kR_8)\big]}{(kR_8)^3}, \qquad R_8 = 8\,h_{100}^{-1}\,\mathrm{Mpc} \quad \text{(FE-99)}$$

**Symbols in (FE-99).**

- $\sigma_8$: The root-mean-square linear matter fluctuation in spheres of radius $R_8$ today. Unit: dimensionless. Value: 0.8118. Status: DERIVED.
- $\infty$: Infinity: as a limit of an integral or a sum it means that the range has no upper bound. Status: STANDARD.
- d: The differential: $\mathrm{d}X$ is an infinitesimal change of $X$, $\mathrm{d}X/\mathrm{d}Y$ the derivative of $X$ with respect to $Y$, $\mathrm{d}^n X$ the volume element of $n$ variables. Status: STANDARD.
- $k$: The magnitude of the wave vector (the momentum, $\hbar = 1$) of a mode: in the reduction, the Kohn-Sham and the perturbation equations the comoving wavenumber (physical momentum $\mathrm{e}^{-a_4}k$); in the thermal integrals the momentum of a particle at the time considered (the variable of integration). Unit: GeV (1/length). Status: STANDARD.
- $P_\mathrm{m}(k)$: The linear matter power spectrum today. Unit: length$^3$. Status: STANDARD.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $W_\mathrm{th}$: The Fourier transform of a spherical top-hat window. Unit: dimensionless. Status: STANDARD.
- $R_8$: The radius of the spheres of $\sigma_8$, $R_8 = 8\,h_{100}^{-1}$ Mpc. Unit: length. Status: STANDARD.
- $h_{100}$: The reduced Hubble constant, $\mathcal{H}_0/(100\ \mathrm{km\ s^{-1}\ Mpc^{-1}})$. Unit: dimensionless. Value: 0.6738. Status: STANDARD.

$\sigma_8 = 0.8118$ for UM-2 (0.8119 without the freeze-in fold bands) against 0.8109 for $\Lambda$CDM (Planck 2018: 0.811, STANDARD, from memory); the product of the growth rate $\mathrm{d}\ln D_+/\mathrm{d}a_4$ and $\sigma_8$ at $z_\mathrm{red} = 0.5$ is 0.4761 (0.4762 without the fold bands) against 0.4756. The sizes of the first-order potentials quoted in the section on the metric series come from the same solution.

# 19 Parameters

Every parameter of UM-2, with its value and its status (the measured inputs of the Standard Model, the masses, the couplings and the mixing angles, are STANDARD; the cosmological inputs are the Planck 2018 values of the section on cosmology).

| parameter | value | status |
|---|---|---|
| $H$ (the warp rate of the hidden direction) | 14.296 TeV | DERIVED from $m_{\mathrm{KK}}$, $HL$ and $f_*$ |
| Lovelock weights $w_{(1)}$, $w_{(2)}$, $w_{(3)}$; cosmological constant $\Lambda$ | 1, 1/32, 1/1152; $-10H^2$ | AUTHOR |
| tip cutoff $HL$ | 3 | ASSUMED (record) |
| mass parameter of dirac16complex, $m_\Psi = 3H\mathcal{M}$ | $\mathcal{M} = 1/3$, so $m_\Psi = H$ | ASSUMED (record) |
| stabiliser law | $\rho_{\mathrm{stab}} \propto f^{-8}$, $p_0 = p_5 = \rho_{\mathrm{stab}} = -p_3$ | ASSUMED |
| $K_*$, $f_*$, $\mathcal{G}(K_*)$ | 21.5219, 0.215556, 13.8081 | DERIVED |
| radion mass $\omega_{\mathrm{r}}$ | $3.38705H = 48.42$ TeV | DERIVED |
| Kaluza-Klein scale $m_{\mathrm{KK}}$ | 121.32 TeV | DERIVED (fit to $\rho_{\mathrm{EW},0} - 1 = 0.00038$ and $m_{\mathrm{W}}$) |
| Robin parameters $\beta_{\mathrm{W}}$, $\beta_{\mathrm{Z}}$; $\cos^2\theta_{\mathrm{W}}$ | 0.011912, 0.015343; 0.776368 | DERIVED (fit; $\beta_{\mathrm{Z}} = \beta_{\mathrm{W}}/\cos^2\theta_{\mathrm{W}}$, the angle of the couplings fixed by $m_{\mathrm{W}}/m_{\mathrm{Z}}$ and $\rho_{\mathrm{EW},0}$) |
| tip matrices $\mathcal{K}_e$, $\mathcal{K}_u$; $\mathcal{K}_\nu$; $\mathcal{K}_d$ | $\mathbf{1}$; $U$; $V_{\mathrm{CKM}}$ (unitary) | ASSUMED (the mixing values STANDARD, measured) |
| doublet localisations $c_L$ (leptons; quarks) | 5.9529; 3.3823 | DERIVED (fit to the tau and top masses with $c_R = -c_L$) |
| singlet localisations $c_R$ | neutrinos $-35.045$, $-32.848$, $-31.070$; electron, muon, tau $-14.554$, $-8.981$, $-5.953$; up, charm, top $-15.430$, $-8.768$, $-3.382$; down, strange, bottom $-14.626$, $-11.516$, $-7.501$ | DERIVED (fit to the masses) |
| lightest neutrino mass $m_1$; ordering | 1.000 meV; normal | ASSUMED |
| $\Delta m^2_{21}$, $\Delta m^2_{31}$; $\sin^2\theta_{12}$, $\sin^2\theta_{23}$, $\sin^2\theta_{13}$, $\delta_{\mathrm{CP}}$ of $U$ | $7.49 \times 10^{-5}$, $2.513 \times 10^{-3}$ eV$^2$; 0.308, 0.470, 0.02215, 212° | STANDARD (NuFIT 6.0) |
| $N_{\mathrm{eff}}$ | 3.0432 (neutrinos) +0.0010 (freeze-in fold bands) = 3.0442 | STANDARD for the neutrinos (precision value; UM-2 computes 3.0476 to 3.0485), DERIVED for the fold bands |
| $c_{\mathrm{Z}}$, $\tilde{c}_{\mathrm{Z}}$ (the chiral tip $Z$ portal of version 1) | 0 (removed) | ASSUMED (bounded by the invisible $Z$ width, disfavoured by $N_{\mathrm{eff}}$) |
| I11 strength $G_{\mathrm{V}} = c_{\mathrm{V}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$ | $4.38 \times 10^{-17}$ GeV$^{-2}$ (freeze-in $\Delta N_{\mathrm{eff}} = 0.001$ from 10 TeV) | ASSUMED benchmark (version 3; version 2: 0) |
| I12 strength $G_{\mathrm{I12}}$, with $G^2_{\mathrm{I12}} = 8G^2_{\mathrm{g}} + 3G^2_{\mathrm{W}} + G^2_{\mathrm{B}}$ | $\pi/(2M^3_{\mathrm{DM}}) = (38.2\ \mathrm{TeV})^{-3}$; universal $c_{\mathrm{I12}}/\Lambda^3_\star = 1.106 \times 10^{-14}$ GeV$^{-3}$ | ASSUMED benchmark (unitarity safe) |
| I13 strength $G_{\mathrm{A}} = c_{\mathrm{A}} H/(2f^4 v_{\mathrm{t}} \Lambda_\star^6)$ | $1.278 \times 10^{-9}$ GeV$^{-2}$ | DERIVED from $\Omega_{\mathrm{DM}} h^2_{100} = 0.120$ (physics accuracy about 1 %) |
| highest temperature of the hot Universe $T_{\max}$ | 10 TeV | ASSUMED |
| self-interaction strength $G_\lambda = \lambda H \tilde{I}_\lambda/(2f^4 v_{\mathrm{t}})$ | $\leq 2.3 \times 10^{-15}$ GeV$^{-2}$ ($\lambda \leq 1.4 \times 10^{-10} H^{-6}$ for $v_{\mathrm{t}} = H^{-3}$) | ASSUMED (bound from $\Delta N_{\mathrm{eff}} \leq 0.001$) |
| dark-matter mass $M_{\mathrm{DM}}$ | $3.102204\, m_\Psi = 44.35$ TeV | DERIVED |

# 20 Assumptions, open problems and conflicts with data

## 20.1 Assumptions (ASSUMED and INVENTED)

**ASSUMED**: the regular tip cutoff $HL = 3$; the good sector (no dependence on $x_5$, $x_6$, $x_7$) and compact extra times of coordinate volume $v_{\mathrm{t}}$; $m_\Psi = H$; the block stabiliser and its $f^{-8}$ law; one doublet bulk mass per sector, common to the three generations, with $c_R = -c_L$ for the tau and the top quark; unitary tip matrices $\mathcal{K}_e = \mathcal{K}_u = \mathbf{1}$, $\mathcal{K}_\nu = U$, $\mathcal{K}_d = V_{\mathrm{CKM}}$; the normal ordering with $m_1 = 1.000$ meV; the solar density profile and the supernova envelope profile of the flavour computations; the model of the QCD crossover in the thermal history; the record's particle convention for the Krein metric of dirac16complex; the highest temperature of the hot Universe, $T_{\max} = 10$ TeV; the benchmark strengths of I11 and I12; the bound $G_\lambda \leq 2.3 \times 10^{-15}$ GeV$^{-2}$ on the self-interaction channel.

**INVENTED**: the boundary assignments of the Standard-Model fields (doublets and singlets with opposite $x_0$-chiralities at the fold); the tip sigma field with its Yukawa terms, which breaks the electroweak symmetry without a Higgs particle; the portals I11, I12 and I13 (the chiral tip $Z$ portal I9 of version 1 is removed). The strength of I13 follows from the relic abundance (DERIVED).

## 20.2 Open problems (OPEN)

- The microscopic origin of the block stabiliser.
- The tachyonic extra-time modes outside the good sector, and the closed time-like curves implied by compact extra times.
- The radion oscillation left over from the deflation (ESTIMATE). Its Planck-suppressed decay rate and the temperature of the plasma it would produce are

$$\Gamma_{\mathrm{r}} = \frac{C_{\mathrm{r}}\,\omega_{\mathrm{r}}^3}{4\pi \bar{M}_{\mathrm{Pl}}^2}, \qquad T_{\mathrm{RH}} = \Big(\frac{90}{\pi^2 g_\rho}\Big)^{1/4} \sqrt{\Gamma_{\mathrm{r}} \bar{M}_{\mathrm{Pl}}} = 1.85 \sqrt{C_{\mathrm{r}}}\ \mathrm{MeV} \qquad \big(g_\rho = 10.75\big),$$

**Symbols in this equation.**

- $\Gamma_{\mathrm{r}}$: The decay rate of the radion (the quantum of the block-size oscillation) into lighter particles, a Planck-suppressed estimate. Unit: GeV. Value: $1.52 \times 10^{-24}\, C_{\mathrm{r}}$ GeV (lifetime $0.43\,\mathrm{s}/C_{\mathrm{r}}$). Status: ESTIMATE.
- $C_{\mathrm{r}}$: The order-one factor of the radion decay rate that collects its unknown couplings to the decay products. Unit: dimensionless. Value: not computed (of order one). Status: ESTIMATE.
- $\omega_{\mathrm{r}}$: The angular frequency of the oscillation of the block about the static point (the radion), equal to the radion mass. Unit: GeV. Value: $3.38705\,H = 48.42$ TeV. Status: DERIVED.
- $\pi$: The number pi, the ratio of the circumference of a circle to its diameter. Unit: dimensionless. Value: 3.141592654. Status: STANDARD.
- $\bar{M}_{\mathrm{Pl}}$: The reduced Planck mass, $\bar{M}_{\mathrm{Pl}} = (8\pi G_{\mathrm{N}})^{-1/2} = M_{\mathrm{Pl}}/\sqrt{8\pi}$. Unit: GeV. Value: $2.435 \times 10^{18}$ GeV. Status: STANDARD.
- $T_{\mathrm{RH}}$: The reheating temperature: the temperature of the plasma produced when the radion oscillation left over from the primordial deflation decays. Unit: GeV. Value: $1.85\sqrt{C_{\mathrm{r}}}$ MeV. Status: ESTIMATE.
- $g_\rho$, $g_s$: The effective numbers of relativistic degrees of freedom of the energy density and of the entropy density of the plasma. Unit: dimensionless. Value: 105.75 when every species of UM-2 in equilibrium is relativistic (no Higgs particle), 10.75 near 1 MeV, $g_s = 43/11$ today. Status: STANDARD.

with $\bar{M}_{\mathrm{Pl}} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor $C_{\mathrm{r}}$: a lifetime of $0.43\,\mathrm{s}/C_{\mathrm{r}}$, and a reheating temperature below the 4.7 MeV that nucleosynthesis needs unless $C_{\mathrm{r}} > 6.5$. This is the cosmological moduli problem of UM-2, which also conditions the dark-matter relic (a late radion domination would dilute it).

- The Kaluza-Klein graviton spectrum with the Lovelock corrections; the shape mode of the block, driven by $p_0 - p_5$ (the residual analysis quantifies its source: about 4 times the kept $\bar{p}$ for non-relativistic dark matter, with an absolute size of order $10^{-32}$ of the stabiliser at the freeze-out; its stability is open).
- The $S$ parameter of the Higgsless tip, and the anomalies of the eight-dimensional bulk theory.
- The cosmological constant: $\Lambda_4$ is tuned to $1.0 \times 10^{-92} H^2$.
- The electroweak sector beyond tree level: the angle of the couplings of the $W^\pm$, $Z$ mass fit, $\sin^2\theta_{\mathrm{W}} = 0.223632$, differs from the $\overline{\mathrm{MS}}$ value 0.23122 used in the $Z$ couplings by the Standard-Model loop corrections, which UM-2 does not compute; the total tree-level $Z$ width changes by 0.82 % between the two (0.83 % with the QCD factor), and the computed total, 2485.2 MeV, lies 0.41 % below the measured value.

- The absolute neutrino mass scale and the mass ordering (only bounded, and preferred, by data); the collective flavour evolution of supernova neutrinos beyond the linear analysis.
- The microscopic origin of the portals I11, I12 and I13: I12 alone could give the relic only as a strongly coupled ${}^3P_0$ channel at the unitarity limit; I13, whose contact form needs a mediator at or below about 90 TeV, gives it with mostly perturbative annihilations. The Fierz coefficients of the self-interaction channel.
- The order-2 terms of the metric in detail, and the primordial spectrum ($A_\mathrm{s}$, $n_\mathrm{s}$, the tensor modes), an input of UM-2.

## 20.3 The findings of the residual analysis

The residual analysis (52 checks, all passing; 287 checks in all) substituted the stored solution into every coupled field equation and compared every quantity that appears in two equations. The approximate solution satisfies its equations to the stated accuracy; the findings that are not small, all stated where they arise in this document, are:

- the weak angle has two values, 0.223632 on shell in the $x_0$ problem of $W^\pm$ and $Z$ and 0.23122 ($\overline{\mathrm{MS}}$) in the widths and couplings; $\Gamma_\mathrm{Z}$ moves by 0.83 %, within the electroweak loop corrections that UM-2 does not compute;
- the Hubble rate of the thermal history leaves out the fold bands of the freeze-in ($3.5 \times 10^{-4}$ of the energy density at the freeze-out): the Hubble rate is larger by $1.7 \times 10^{-4}$, and the relic changes by about as much;
- the thermal history uses $N_\mathrm{eff} = 3$ after the neutrino decoupling while the CMB computation uses 3.0442: 0.59 % in $g_\rho$ today;
- the energy density of the plasma at the freeze-out differs by 0.13 % between the thermal history and the coupled block sources (the trace anomaly, which only the latter contain, by design);
- the ideal-gas thermal history leaves out the order-$\alpha$ interaction pressure of the plasma ($-5.4$ % at the freeze-out: QCD $-4.7$ %, $SU(2)$ $-0.7$ %), so $\Omega_\mathrm{DM} h^2_{100}$ would rise by about 2.7 % and $G_\mathrm{A}$ by about 1.4 % (ESTIMATE): the physics accuracy of $G_\mathrm{A}$ is about 1 %, not the $10^{-6}$ of its root finding;
- the block-shape source $p_0 - p_5$ is about 4 times the kept $\bar{p}$ for non-relativistic dark matter (OPEN, now quantified).

## 20.4 Conflicts with data

**Inherent in the author's order**: there is no Higgs boson, while one is observed at 125 GeV. Without it the longitudinal $W^+W^-$ scattering violates perturbative unitarity near $\sqrt{8\pi} \times 246$ GeV $\approx 1.2$ TeV, far below the Kaluza-Klein modes (above 100 TeV) that could restore it. (The one-generation conflict of version 1, a total $Z$ width of 1.23 GeV, is removed by the three generations.)

**Not conflicts, but tensions to watch**: $N_\mathrm{eff} = 3.0442$ lies 1.4 standard deviations above the ACT DR6 value $2.86 \pm 0.13$; the tree-level total $Z$ width lies 0.41 % below the measured value (the size of the loop corrections); the direct-detection cross-section of the dark matter, $4.0 \times 10^{-46}$ cm$^2$, is 0.16 of the present LZ limit at 44.35 TeV (extrapolated), so the next generation of xenon detectors will test the candidate.